

A Unified Geometric Theory of Trembling Spacetime Relativity

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Abstract

Theoretical physics remains divided between incompatible frameworks for gravity, quantum fields, and the fundamental forces. This work introduces a unifying geometric theory—Trembling Spacetime Relativity Theory (TSRT)—in which spacetime undergoes fine-scale, deterministic fluctuations constrained by causality. These metric tremblings replace probabilistic postulates and field assumptions with a curvature-dependent causal structure that governs interaction, distinguishability, and information flow.

In this framework, Lorentzian geometry and proper time emerge from causal invariance rather than being imposed. Planck’s constant arises as a minimal resolution scale needed to preserve temporal order. The Schrödinger and Dirac equations are recovered as emergent statistical descriptions of trembling-induced motion. The Heisenberg uncertainty relation is refined into a dual-bound form, with both minimal action and maximal curvature coherence limits.

All four known fundamental forces emerge as the only causally admissible gauge structures consistent with trembling constraints. Quantum indeterminacy, wave–particle duality, and Hilbert space dynamics are shown to arise from deterministic geometric structure. Black hole entropy and the holographic principle are similarly derived from curvature-limited distinguishability, without invoking unitarity or thermodynamic postulates. TSRT provides a causally grounded, geometrically complete foundation that unifies relativistic and quantum phenomena.

1 Introduction

Deterministic microscopic trembling of the spacetime metric provides a unified geometric origin for phenomena traditionally attributed to quantum mechanics, relativity, and particle physics. In Trembling Spacetime Relativity Theory (TSRT), the spacetime interval itself—including its Lorentzian structure [1]—is not assumed but derived from causality and the requirement that proper time remain real and monotonic. This replaces the postulated axioms of special relativity with a deeper geometric rationale grounded in the structure of physically admissible trajectories.

Unlike stochastic fluctuations, the trembling in TSRT is governed by strict local causal coherence and geometric consistency, ensuring deterministic and covariant physical behavior across scales. Particles arise as stable, localized excitations of this trembling geometry, with their properties determined by the underlying metric fluctuations. These excitations maintain

coherence over macroscopic distances because the trembling structure preserves causal order at every point along a worldline.

The unification of fundamental forces has been a central aspiration in physics for over a century. General relativity provided a geometric understanding of gravitation [2], while quantum mechanics introduced a probabilistic framework for microscopic behavior [3–5]. Attempts to reconcile these frameworks—whether through quantum field theory [6], string theory [7] [8], or loop quantum gravity [9]—have remained incomplete, leaving conceptual gaps at the intersection of gravity, causality, and quantum uncertainty. These theories rest on irreconcilable assumptions — continuous spacetime versus probabilistic evolution — and attempts to unify them through quantization have encountered persistent paradoxes, including the information loss problem, singularities, and unexplained entropy bounds. This paper proposes a deeper resolution: a deterministic trembling of the spacetime metric that enforces causal structure while giving rise to the effective behaviors we interpret as quantum uncertainty and force interactions.

A crucial limitation of existing theories lies in their reliance on separate postulates for different physical domains. Quantum mechanics imposes the Heisenberg uncertainty principle as a kinematic constraint [3], while relativity prescribes causal structure through the invariance of the spacetime interval. Yet no theory to date has demonstrated how these principles could emerge from a single dynamical foundation.

Here we present Trembling Spacetime Relativity Theory (TSRT), a unified framework that derives quantum uncertainty, fundamental forces, and causal structure from deterministic geometric fluctuations of spacetime itself. TSRT posits that spacetime is not exactly smooth, but undergoes fine-scale metric fluctuations along proper-time trajectories. These trembling deviations are not random, but are constrained by the requirements of Lorentz covariance, determinism, and strict causal evolution.

A key insight is that such trembling induces a nontrivial phase-space structure, leading to a double-bounded uncertainty relation:

$$\frac{1}{2}\hbar \leq \Delta x \Delta p \leq \frac{1}{2}\beta \frac{mc}{\sqrt{\kappa_{\text{local}}}}, \quad (1)$$

where κ_{local} denotes the local spacetime curvature, m the particle mass, c the invariant speed scale, and β a dimensionless geometric coefficient. Unlike the conventional view, uncertainty in TSRT is not an axiom but a geometric necessity tied to causal coherence (Section S20).

Furthermore, the fundamental equations of quantum mechanics, including the Schrödinger and Dirac equations, are shown to emerge as effective descriptions of trembling-induced statistical dynamics at low energies (Section S18).

From the structure of permissible metric fluctuations, TSRT systematically derives the existence and uniqueness of the four known fundamental forces. The classification of trembling modes into trace-like, abelian, chiral non-abelian, and color non-abelian types naturally yields gravity, electromagnetism, the weak force, and the strong force respectively (Sections S9 and S16). Each force arises as a distinct manifestation of how spacetime trembles while preserving proper-time positivity.

In TSRT, gravity is not quantized in the conventional sense. Instead, it emerges as a coarse-

grained manifestation of coherent trembling, preserving classical causality while constraining the allowable distribution of local metric deviations.

The coherent internal trembling pattern resists changes in motion, leading naturally to the inertial properties associated with mass.

Beyond reproducing known interactions, TSRT offers new perspectives on longstanding problems. The invariance of the speed of light emerges dynamically (Section S24); black hole entropy and holographic behavior emerge from curvature-regulated phase-space resolution limits imposed by trembling spacetime [10] [11] (Section S22); and the necessity of trembling spacetime itself is rigorously proven from causal consistency conditions (Section S21).

By unifying quantum phenomena, relativistic structure, and all known interactions within a deterministic geometric framework, TSRT provides a new foundation for fundamental physics. It offers closure, structural economy, and internal coherence where previous theories have relied on disparate axioms, and it opens a path toward a deeper understanding of nature’s most basic principles.

A detailed Supplementary Information document provides the full mathematical derivations, theoretical developments, and background analyses necessary for a comprehensive understanding of this work.

2 Foundations of Trembling Spacetime Relativity Theory (TSRT)

2.1 Proper Time and Lorentz Geometry as Derived Foundations

Before introducing the central concept of trembling spacetime, it is essential to note that the causal and geometric structure on which TSRT is built—including the Lorentzian form of the metric [1] and the use of proper time τ as a global parameter—is not assumed a priori. Instead, it is rigorously derived in Section S3 from the requirement that elapsed time along physical worldlines be real, path-dependent, and invariant under reparametrization.

There, we show that if elapsed time is defined as an integral over a general symmetric bilinear form on a differentiable trajectory, the demand for causal consistency and observer-independent intervals inevitably leads to the Lorentz metric as the unique form preserving proper-time structure. Furthermore, we justify the use of τ itself as the independent variable, demonstrating that worldlines $x^\mu(\tau)$ are well-defined and that no circularity arises from the mutual dependence of τ and x^μ .

This provides a logically complete geometric foundation for the trembling dynamics that follow. Rather than postulating the spacetime arena in which particles move, TSRT derives that arena from first principles—and then refines it by introducing trembling as a causal geometric modulation of the metric structure.

2.2 Spacetime Trembling and Causal Structure

At the heart of TSRT lies a fundamental yet profound modification of classical spacetime geometry: microscopic fluctuations around otherwise smooth trajectories. These trembling deviations are encoded by a deterministic perturbation field $\Xi^\mu(\tau)$ added along proper-time worldlines,

and, therefore, manifesting local ontology:

$$x^\mu(\tau) = \bar{x}^\mu(\tau) + \Xi^\mu(\tau), \quad (2)$$

where $\bar{x}^\mu(\tau)$ denotes the underlying smooth trajectory, and $\Xi^\mu(\tau)$ captures fine-scale causal fluctuations. Here, $\Xi^\mu(\tau)$ represents a small, continuous perturbation to the geodesic trajectory arising from microscopic trembling of spacetime. This perturbation respects causal structure by preserving proper-time monotonicity and is constrained to remain within the local light cone at each point along the worldline.

These fluctuations induce perturbations in the local metric,

$$g_{\mu\nu}(x, \tau) = \bar{g}_{\mu\nu}(x) + \xi_{\mu\nu}(x, \tau), \quad (3)$$

where

$$\xi_{\mu\nu} = \partial_\mu \Xi_\nu + \partial_\nu \Xi_\mu \quad (4)$$

preserves Lorentz covariance at each point. Unlike conventional perturbative treatments of general relativity, where $\xi_{\mu\nu}(x, \tau)$ typically represents weak gravitational waves or matter-induced curvature, here it arises intrinsically from microscopic, causally constrained trembling of space-time itself. It encodes local fluctuations that maintain Lorentz invariance on average, yet produce fine-scale deviations determined by causal trembling geometry.

Crucially, the trembling field must respect strict causality: proper time τ must remain positive along all physical trajectories,

$$\frac{d\tau}{d\lambda} > 0, \quad (5)$$

where λ is an arbitrary affine parameter (Section S9.2). This condition stabilizes worldlines against non-physical time reversals and enforces a local causal order, even in the presence of microscopic metric fluctuations.

The causal constraint leads directly to a minimal phase-space spread,

$$\Delta x \Delta p \geq \frac{1}{2} \hbar, \quad (6)$$

where $\hbar$ arises geometrically from the cutoff structure of trembling amplitudes needed to preserve causality.

The dynamic adjustment of causal structure by trembling fluctuations also modifies the effective gravitational field equations. In TSRT, the Einstein field equations are generalized to incorporate corrections from microscopic trembling effects:

$$G_{\mu\nu} + \Delta G_{\mu\nu} = \frac{8\pi G}{c^4} \left(T_{\mu\nu} + T_{\mu\nu}^{\text{tremble}} \right), \quad (7)$$

where $\Delta G_{\mu\nu}$ encapsulates geometric fluctuations from trembling, and $T_{\mu\nu}^{\text{tremble}}$ represents the effective energy-momentum contribution of microscopic causal oscillations. In regions of weak curvature, these corrections are negligible and standard general relativity is recovered. However, near singularities or in high-density regimes, trembling-induced modifications significantly alter the causal structure and the information content of spacetime.

2.3 Minimal Action Cutoff and Emergent Uncertainty

The deterministic trembling structure introduces an irreducible minimal phase-space spread. If the amplitude of metric fluctuations were allowed to vanish, worldlines could collapse into acausal behaviors (Section S21). Conversely, unbounded trembling amplitudes would destroy local coherence, violating causal evolution.

The resulting uncertainty structure, summarized by the double-bounded relation already introduced in Equation (1) and derived fully in Section S20, imposes both a minimal and maximal bound on phase-space resolution.

The double-bounded uncertainty structure imposes a minimal phase-space volume set by fundamental causal resolution, and a maximal volume constrained by local spacetime curvature, which limits the distinguishability of neighboring worldlines in strongly curved regions.

We find that the minimal causal resolution imposed by trembling spacetime dynamics leads naturally to a universal action scale—one that governs the coarse-grained evolution of proper-time fluctuations. This scale emerges purely from geometric and relativistic constraints, independent of any quantum assumptions, and numerically matches the observed value of Planck’s constant. More precisely, as derived in Section S19, when a localized trembling mode is bounded by the causal curvature limit and gravitational interaction strength, the emergent action takes the form:

$$\hbar \equiv \mathcal{T} = \frac{\epsilon_c^2 c^5}{4G\omega_c^2}, \quad (8)$$

where ϵ_c is the proper-time amplitude cutoff, ω_c is the dominant trembling frequency, c is the speed of light, and G is the gravitational constant. This relation reveals a deep geometric origin for Planck’s constant: it is not merely an empirical parameter, but a derived consequence of the interplay between causal structure and gravitational curvature in trembling spacetime.

Within this framework, Planck’s constant $\hbar$ emerges naturally from the causal cutoff imposed by minimal trembling amplitudes, rather than being introduced as an independent postulate. In low-curvature regimes, the lower bound dominates, reproducing the standard Heisenberg uncertainty principle [3]. This correspondence arises because the minimal phase-space volume permitted by trembling geometry directly imposes a nonzero product of position and momentum uncertainties, consistent with Heisenberg’s formulation. In strong curvature environments, such as near black hole horizons, the upper bound contracts the admissible phase-space volume (Section S22), restricting the number of physically distinguishable microstates.

Thus, in TSRT, quantum uncertainty arises as a geometric and causal necessity, rather than as a probabilistic feature of measurement. Moreover, the value of Planck’s constant $\hbar$ is not postulated but emerges naturally as the minimal geometric action scale necessary to maintain causal coherence of trembling worldlines (Section S20). Its role as the fundamental quantum of action finds a purely spacetime-geometric origin within TSRT.

2.4 Deterministic Origin of Quantum and Relativistic Limits

The trembling background modifies both particle kinematics and spacetime structure in a manner that unifies quantum and relativistic phenomena within a single causal framework.

In the low-curvature limit, the trembling-induced phase-space structure reproduces the fa-

miliar uncertainty relations of quantum mechanics, while simultaneously constraining particle propagation to causal worldlines with an invariant limiting speed c . The invariance of c is not assumed as an independent postulate, as in special relativity [12], but is dynamically derived in TSRT as the maximum propagation speed of massless trembling excitations (Section S24). This causal limit arises because trembling fluctuations, constrained by local spacetime structure, cannot propagate information faster than the underlying causal worldlines.

Moreover, in the trembling-induced framework, there is no fundamental randomness: particle worldlines remain continuous and deterministic. Apparent randomness emerges only through coarse-graining over unresolved trembling fluctuations, not from intrinsic indeterminacy. Crucially, even after coarse-graining, the underlying causal structure of spacetime remains preserved, ensuring that macroscopic behavior respects causal order.

Thus, TSRT bridges the divide between quantum mechanics and relativity not by quantizing gravity or geometrizing probability, but by revealing a common geometric foundation: the causal trembling of spacetime itself. Future work will explore potential experimental signatures of trembling spacetime effects, as well as particular outstanding problems that TSRT may resolve, especially in regimes of high curvature, high energy, or extreme sensitivity, offering paths to empirical validation of the theory.

3 Geometric Emergence of the Four Fundamental Forces

3.1 Classification of Metric Fluctuation Structures

Within TSRT, the fundamental interactions of nature arise from the allowed classes of trembling-induced metric deviations. As detailed in Section S9, the decomposition of the trembling field $\Xi^\mu(\tau)$ under Lorentz transformations, together with causality and compactness constraints, leads to exactly four irreducible trembling classes.

First, trace-like isotropic trembling generates large-scale curvature deviations associated with gravity. Second, abelian vector-like trembling corresponds to $U(1)$ gauge structures, giving rise to electromagnetism. Third, chiral non-Abelian trembling generates the $SU(2)_L$ gauge group associated with the weak interaction. Finally, non-chiral non-Abelian color trembling produces the $SU(3)_C$ symmetry underlying the strong nuclear force.

Each class satisfies the strict requirement that proper time remains positive along all worldlines ($d\tau/d\lambda > 0$), ensuring that trembling configurations preserve causal structure (Section S9.2). The full classification is grounded in the tensorial decomposition of $\xi_{\mu\nu}$ into trace, symmetric traceless, and antisymmetric components, with physical interpretations determined by their behavior under spacetime symmetries (Section S9).

3.2 Causal Closure and the Uniqueness of Four Forces

An essential result of TSRT is that no additional fundamental forces are permitted beyond the four identified trembling classes. As demonstrated in Section S16, any hypothetical additional trembling structure would necessarily violate at least one of the core physical constraints: it would either break the positivity of proper time, fail to maintain compactness and unitary rep-

representations, or become incompatible with embedding in four-dimensional Lorentzian spacetime geometry.

This geometric closure sharply distinguishes TSRT from the Standard Model of particle physics, where gauge groups are introduced based on phenomenological needs and can be extended almost arbitrarily. In TSRT, by contrast, the symmetries, interactions, and physical degrees of freedom emerge uniquely from the causal and geometric structure of trembling spacetime itself.

Thus, TSRT provides a first-principles derivation of why exactly four—and only four—fundamental interactions exist in nature.

This prediction is falsifiable in the strict sense of Popperian philosophy: the empirical discovery of a fifth fundamental interaction would directly contradict the geometric closure theorem of TSRT and thereby falsify the theory.

3.3 Summary of Force Mapping and Physical Implications

The four classes of trembling structures map naturally onto the known forces of nature. Gravity arises from trace-like curvature trembling (Section S13); electromagnetism emerges from Abelian vector trembling (Section S10); the weak force results from chiral non-Abelian trembling (Section S11); and the strong force is produced by non-chiral non-Abelian color trembling (Section S12).

Each interaction finds its origin in a distinct class of geometric trembling, without requiring independent assumptions about external fields, imposed gauge symmetries, or postulated force carriers. Trembling spacetime itself acts as the common geometric source, unifying phenomena traditionally treated separately by quantum field theory and general relativity.

The natural emergence of the four forces within TSRT signals a paradigm shift: symmetries and interactions are not arbitrary constructs but necessary consequences of the causal and geometric architecture of the universe. In particular, gauge fields and their dynamics are not inserted as independent axioms in TSRT; they arise systematically from the causal structure of trembling spacetime.

For example, the effective Abelian gauge field associated with electromagnetism emerges via

$$A_\mu(x) = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \langle \Xi_\mu(x, \tau + \epsilon) - \Xi_\mu(x, \tau) \rangle_\tau, \quad (9)$$

where $\Xi_\mu(x, \tau)$ is the trembling displacement field and the averaging ensures causal consistency. The corresponding field strength tensor, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, emerges directly from the antisymmetric component of local metric fluctuations.

Thus, in TSRT, gauge fields, gauge symmetries, and force interactions are deeply rooted in the deterministic causal dynamics of spacetime itself.

4 Curvature-Regulated Uncertainty and the Limits of Information

4.1 Double-Bounded Uncertainty Relation

One of the most profound departures of TSRT from conventional quantum mechanics is the discovery that uncertainty is bounded both from below and from above, with both limits determined by spacetime geometry. In trembling spacetime, the phase-space uncertainty product for a particle satisfies Equation (1), where κ_{local} denotes the local curvature scalar, m is the particle mass, c is the invariant speed of light, and β is a dimensionless parameter encoding the spectral structure of trembling (see Section S20).

The lower bound recovers the familiar Heisenberg uncertainty floor [3], now derived from causal and geometric constraints on spacetime fluctuations rather than postulated operator commutation relations. The upper bound, absent in standard quantum theory, ensures that uncertainty cannot grow arbitrarily in regions of strong curvature: as gravitational intensity increases, the permissible phase-space volume contracts, reflecting the suppression of distinguishable causal configurations in trembling spacetime.

This dual-bounded structure completes the uncertainty principle, yielding a curvature-dependent, geometrically rooted formulation that unifies quantum and relativistic limitations within a single deterministic framework.

4.2 Black Hole Entropy and the Holographic Principle from TSRT

A striking consequence of the double-bounded uncertainty structure is the natural emergence of black hole entropy scaling with horizon area. Near the event horizon of a black hole, the local curvature becomes extremely large ($\kappa_{\text{local}} \sim 1/r_s^4$, where r_s denotes the Schwarzschild radius), thereby drastically shrinking the allowable phase-space resolution.

By counting the number of distinguishable trembling microstates across a horizon-sized region and applying Boltzmann's entropy formula, one recovers the celebrated Bekenstein–Hawking relation:

$$S_{\text{BH}} = \frac{k_B c^3}{4G\hbar} A, \quad (10)$$

where A is the horizon area, k_B is Boltzmann's constant, and G is Newton's gravitational constant (see Section S22) [10, 11].

In TSRT, the entropy–area relation does not arise from thermodynamic analogies or speculative holographic postulates [13, 14]. Instead, it follows directly from the causal restriction on phase-space correlations imposed by the trembling spacetime geometry. Notably, the $1/4$ factor, often inserted by hand in semiclassical derivations, emerges naturally from the geometric projection of resolvable trembling states onto the horizon surface.

Thus, TSRT offers a first-principles derivation of black hole entropy and holography, grounded entirely in the causal structure of trembling spacetime.

4.3 Resolution of the Information Paradox Without Unitarity

The information paradox in black hole physics arises from the apparent conflict between quantum unitarity and the classical loss of information behind event horizons [10,15]. In conventional treatments, resolution requires assuming that information is somehow encoded on the horizon and radiated back via subtle quantum correlations, relying heavily on the holographic principle and AdS/CFT duality [16].

In TSRT, a fundamentally different mechanism operates. Information is not conserved in the unitary sense of quantum mechanics; instead, it is dynamically filtered by curvature-dependent constraints on phase-space distinguishability in trembling spacetime. As a black hole evaporates and curvature decreases, the upper bound on uncertainty relaxes, dynamically restoring previously excluded correlations (see Section S22).

Thus, no paradox arises. Information is not destroyed—it is temporarily filtered by causal geometric constraints and reemerges as spacetime flattens. There is no need for horizon pair production, speculative holographic encodings, or unitarity-preserving evaporation mechanisms. Black hole evolution becomes a smooth, causal unfolding of phase-space admissibility, governed entirely by the trembling structure of spacetime itself.

This resolves the information paradox naturally and deterministically, offering a deeper understanding of entropy, information, and causality at the interface of gravity and quantum theory.

5 Refinement of the Schrödinger Equation

In addition to reproducing the standard Schrödinger equation, TSRT predicts a deterministic refinement that becomes relevant in regimes where the amplitude of ψ varies rapidly in space—such as near nodes, sharp interfaces, or tightly localized states. This refinement, developed in Section S18.3 emerges by retaining amplitude curvature terms that are typically neglected in conventional derivations, yielding an extended evolution equation:

$$i\mathcal{T}\frac{\partial\psi}{\partial t} = \left(-\frac{\mathcal{T}^2}{2m}\nabla^2 + V(\vec{x})\right)\psi + \Delta_{\text{TSRT}}[\psi], \quad (11)$$

where the correction term is

$$\begin{aligned} \Delta_{\text{TSRT}}[\psi] := & -\frac{\mathcal{T}^2}{2m} \left[\text{Re} \left(\frac{\nabla^2\psi}{\psi} \right) + \left\| \text{Re} \left(\frac{\nabla\psi}{\psi} \right) \right\|^2 - \frac{\nabla^2\psi}{\psi} \right] \psi \\ & + \frac{\mathcal{T}^2}{2m} \left[\left\| \text{Im} \left(\frac{\nabla\psi}{\psi} \right) \right\|^2 - \text{Im} \left(\frac{\nabla^2\psi}{\psi} \right) \right] i\psi. \end{aligned} \quad (12)$$

This correction vanishes in the smooth-amplitude limit, thereby recovering the conventional Schrödinger equation. However, in physically important regions where the probability density $|\psi|^2$ exhibits steep spatial gradients, this additional term becomes significant. Although the meaning of ψ in TSRT differs from its interpretation in standard quantum mechanics—it is a statistical encoding of geometric trembling rather than a fundamental wavefunction—the refined evolution equation could nonetheless be used as a diagnostic tool.

In particular, if the correction term $\Delta_{\text{TSRT}}[\psi]$ is applied to conventional quantum wavefunctions, it provides a concrete route to experimentally distinguish TSRT from quantum theory in high-precision interference, tunneling, or localization measurements. Such deviations would reflect underlying causal geometry rather than quantization artifacts.

6 Experimental Predictions and Observational Windows

6.1 Deviations from the Canonical Uncertainty Limit

A distinctive prediction of TSRT is the existence of deviations from the standard Heisenberg uncertainty relation in regions where spacetime curvature is non-negligible. Specifically, the double-bounded uncertainty relation (Equation (1)) predicts that, as the local curvature κ_{local} increases, the upper bound on $\Delta x \Delta p$ contracts, thereby reducing the accessible phase-space resolution.

In low-curvature environments, such as typical terrestrial laboratories, the upper bound is extremely large and practically irrelevant, so standard quantum mechanics holds to extraordinary precision [3]. However, in regions of extreme curvature—whether astrophysical, cosmological, or potentially experimentally engineered—the deviations may become detectable.

Potential experimental signatures include anomalous tightening of phase-space distributions in highly energetic collisions, modified uncertainty saturation behavior for particles near compact astrophysical objects, and altered high-precision spectroscopy outcomes if local curvature perturbations are present.

Detection of such deviations would provide direct empirical access to the trembling structure of spacetime and validate the geometric foundations proposed by TSRT (see Section S22).

6.2 Signatures Near Extreme Curvature (Black Holes, Early Universe)

Black hole environments offer a natural laboratory for probing the high-curvature regime where TSRT predicts significant departures from standard quantum behavior [10, 17]. Near the event horizon, the phase-space resolution is severely constrained, suggesting that Hawking radiation spectra may carry subtle imprints of the underlying trembling structure of spacetime.

Similarly, during the very early universe—particularly during inflationary epochs characterized by high spacetime curvature—TSRT predicts that physical phenomena may be modulated by trembling-imposed phase-space bounds. These effects could imprint specific patterns onto the cosmic microwave background (CMB) anisotropies, offering potential observational windows distinct from those predicted by semiclassical inflationary models [18].

Future high-precision astrophysical observations, including next-generation gravitational wave detectors, enhanced black hole imaging efforts such as the Event Horizon Telescope [19], and ultra-high-energy cosmic ray measurements, may provide indirect evidence for trembling-regulated physical processes.

In addition, the refined Schrödinger equation predicted by TSRT (Section S18.3) introduces curvature-sensitive correction terms that could, in principle, manifest as measurable deviations in high-precision quantum experiments near strong gravitational fields or in early-universe

relics—providing a concrete diagnostic pathway for detecting the influence of trembling geometry on quantum evolution.

6.3 General Implications for High-Energy Physics and Cosmology

TSRT fundamentally reinterprets key physical phenomena at high energies and in strong gravitational fields. If confirmed, its consequences would be transformative across multiple domains of fundamental physics.

First, TSRT achieves **unification without postulates**: the four fundamental forces emerge deterministically from the causal and geometric structure of trembling spacetime, without requiring ad hoc symmetry assumptions or external gauge groups (Sections S9 and S16).

Second, TSRT offers a **resolution of longstanding paradoxes**: the black hole information paradox dissolves naturally, without invoking speculative mechanisms such as holographic unitarity recovery or multiverse scenarios. Information evolution is governed directly by the causal structure of trembling spacetime.

Third, TSRT provides **experimental accessibility**: unlike many approaches to quantum gravity that predict observable deviations only at inaccessible Planck scales, TSRT anticipates detectable departures from standard behavior at much lower curvatures, offering concrete prospects for empirical tests through astrophysical, cosmological, and laboratory observations.

These features place TSRT in a unique position among modern theoretical frameworks, offering a conceptually more unified, causally complete, and experimentally accessible framework for unification of quantum, gravitational, and cosmological phenomena.

7 Discussion

7.1 Comparison to Existing Frameworks (Quantum Mechanics, General Relativity, String Theory)

Trembling Spacetime Relativity Theory (TSRT) challenges and transcends the foundations of both quantum mechanics and general relativity, while avoiding the speculative complexities introduced by modern unification attempts such as string theory and loop quantum gravity.

In quantum mechanics, uncertainty is traditionally treated as an intrinsic epistemic limitation, formalized through the non-commutativity of operators [3]. In contrast, TSRT derives uncertainty as a geometric necessity, rooted in the causal coherence constraints imposed by the trembling structure of spacetime (Section S20). The canonical Heisenberg relation emerges naturally in the low-curvature limit but extends in TSRT to a curvature-dependent window that unifies quantum and relativistic behavior under a single deterministic framework.

General relativity [2] successfully describes spacetime curvature and gravitational dynamics but remains silent on quantum phenomena. It assumes a smooth spacetime continuum down to arbitrarily small scales. TSRT modifies this picture by introducing microscopic deterministic trembling, preserving Lorentzian metric structure while enforcing proper-time positivity even at microscopic levels. Thus, it provides a causal and geometric mechanism that extends the scope of general relativity without abandoning its core principles.

Compared to string theory [7,8], which introduces unobserved higher dimensions, supersymmetry, and an enormous landscape of vacua, TSRT remains firmly rooted in four-dimensional Lorentzian spacetime. It introduces no hidden entities, no additional dimensions, and no infinite families of hypothetical particles. Unification is achieved not by increasing theoretical complexity, but by revealing that the trembling of spacetime itself suffices to generate all known forces, quantum limits, and causal structures.

In this sense, TSRT offers a parsimonious yet profound alternative: a unification based on geometric inevitability rather than speculative extension.

7.2 Advantages: Causality, Determinism, Structural Economy

Trembling Spacetime Relativity Theory (TSRT) restores causality and determinism as fundamental principles of nature. The apparent wave–particle duality observed in experiments arises naturally within the trembling spacetime framework, where interference phenomena emerge from the geometrical distribution of trembling worldlines, while localized detections reflect the underlying particle nature preserved by causal structure. Instead of relying on probabilistic wavefunction collapse or speculative multiverse scenarios, particles in TSRT evolve deterministically along worldlines perturbed by microscopic trembling correlations. Apparent randomness emerges only through coarse-graining over unresolved trembling fluctuations, not from intrinsic indeterminacy.

This structure naturally explains key constants and phenomena without additional postulates:

- Planck’s constant $\hbar$ arises as a limit of causal geometric resolution, emerging from the trembling structure itself.
- The invariant speed of light c results dynamically from the causal propagation of massless trembling modes (Section S24).
- The necessity of trembling follows rigorously from the requirement of maintaining positive proper-time evolution (Section S21).
- The natural emergence and unique closure of the four fundamental forces are determined by the causal and geometric structure of trembling spacetime (Sections S9 and S16).

Structural economy is preserved throughout. No ad hoc assumptions—such as supersymmetry, hidden variables, anthropic reasoning, or unobservable dimensions—are required. TSRT’s minimal set of physical principles yields maximal explanatory reach, realizing the scientific ideal of achieving deep unification through causal and geometric necessity.

7.3 Conceptual Consequences for the Future of Physics

If validated, Trembling Spacetime Relativity Theory (TSRT) would fundamentally reshape the foundations of theoretical physics. It offers a single, geometrically rooted framework that:

- Unifies quantum and relativistic phenomena without invoking wavefunctions, extra dimensions, or untestable postulates.

- Explains the origin and mutual compatibility of the four fundamental forces from causal and geometric first principles.
- Resolves the black hole information paradox without relying on unitarity restoration or speculative holographic mechanisms.
- Predicts observable deviations from standard uncertainty relations in high-curvature environments, opening new experimental windows onto quantum gravitational effects.
- Refines the Schrödinger equation by deriving curvature-sensitive correction terms from first principles, offering a testable extension of quantum dynamics in regions with steep amplitude gradients or strong gravitational curvature.

Conceptually, TSRT suggests that the universe is neither fundamentally stochastic nor fragmented among an ensemble of many worlds. Instead, it constitutes a single, causally consistent, dynamically trembling spacetime, where all observed phenomena arise from the interplay between microscopic causal constraints and macroscopic geometric coherence.

This paradigm shift invites a re-examination of many longstanding assumptions in modern physics. It offers a coherent, deterministic, and experimentally accessible path forward—uniting the beauty of geometry with the empirical demands of physical reality.

8 Conclusion

Trembling Spacetime Relativity Theory (TSRT) offers a unified geometric explanation for phenomena traditionally described by disparate physical theories. At its foundation lies a derived—not postulated—causal structure: the Lorentzian geometry and the use of proper time as an independent parameter are shown to follow directly from the requirement that elapsed time remain real, monotonic, and covariant under coordinate transformations. This rederivation, established in Supplementary Section S3, eliminates the need for special relativity’s axioms and provides a logically complete geometric stage for the trembling dynamics that follow.

By postulating deterministic fluctuations in the metric—subject to strict causal coherence—TSRT derives the familiar structures of relativity and quantum physics from a single geometric principle. Rather than quantizing spacetime or matter, TSRT shows that metric trembling naturally produces quantum-like uncertainty, particle stability, gauge interactions, and black hole entropy as consequences of curvature-limited causal evolution.

The uncertainty relation becomes a dual-bound inequality: trembling spacetime enforces both a lower limit set by Planck’s constant and an upper limit tied to local curvature. This curvature-dependent structure contracts phase-space resolution in strong gravitational fields, explaining the entropy–area law and the holographic principle as geometric outcomes of causal filtering, not statistical assumptions.

TSRT avoids postulated quantum fields, unitarity, or probabilistic collapse. It reframes entropy as a count of causally distinguishable geometric configurations and explains black hole evaporation as the unfolding of phase-space resolution under curvature reduction. In this way, TSRT not only matches known physical laws but deepens their meaning, deriving them from the causal geometry of spacetime itself.

The theory provides a minimal, deterministic, and causally grounded foundation for all known interactions, revealing that what appears as quantum indeterminacy or thermodynamic behavior may ultimately arise from a deeper geometric structure: spacetime that trembles, yet remains causally ordered.

Additional Notes

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References

Supplementary Information

Supplementary Information for
 “A Unified Geometric Theory of Trembling Spacetime Relativity”
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S1 Supplementary Information Overview

Overview. This Supplementary Information provides detailed derivations, proofs, and extended discussions supporting the results presented in the main article. Section S3 derives the Lorentzian structure of spacetime and the role of proper time τ from first principles, showing that the causal geometry alone necessitates the use of the Minkowski metric and legitimizes τ as a global, monotonic parameter. Section S21 then proves the necessity of trembling spacetime from causal consistency. Section S20 derives the double-bounded uncertainty relation as a geometric constraint. Section S19 shows how Planck’s constant arises from minimal causal action. Section S24 derives the emergence of the invariant speed c from causal structure. Section S18 demonstrates how the Schrödinger and Dirac equations emerge statistically from trembling dynamics. Sections S9 and S16 classify and prove the uniqueness of the four fundamental forces. Each result is derived independently and self-consistently from the framework of Trembling Spacetime Relativity Theory (TSRT).

The Supplementary Information provides the complete mathematical and conceptual foundation underlying the main article. Each section develops a distinct theoretical domain, offering detailed derivations, geometric interpretations, and rigorous proofs that together establish the structure of Trembling Spacetime Relativity Theory (TSRT). This guide outlines the contents of each supplementary section and how they contribute to the unified theory.

Geometric Foundations of Trembling Spacetime. Section S3 opens this sequence by deriving the Lorentz metric structure from causal constraints and proving that proper time τ can be used as a globally valid parameter for timelike worldlines. This derivation replaces the traditional postulates of special relativity with geometric necessity. Sections S4–S7 then derive the dynamical equations of motion from a proper-time-dependent action principle, interpret the resulting geodesic equation as a generalized force law, and extend Einstein’s field equations to include trembling-induced curvature. These sections form the geometric and variational core of TSRT and demonstrate full compatibility with Lorentz invariance and geometric covariance.

Physical Time and Deterministic Evolution. Section S8 proves that trembling dynamics must be defined with respect to proper time τ , not coordinate time t . This ensures deterministic evolution, preserves local causality, and establishes the invariance of physical laws across all reference frames.

Classification of Metric Fluctuation Modes and Fundamental Forces. Sections S9–S13 classify all geometrically admissible trembling deviations and show that they organize into four irreducible interaction classes: electromagnetism, weak interaction, strong interaction, and gravity. Each class arises from a distinct trembling symmetry and generates its associated gauge structure, field strength, and dynamical equations without postulated bosonic fields.

Conservation Laws and Interaction Ranges. Sections S14–S15 derive all known conservation laws—electric charge, isospin, color, and energy–momentum—from the geometric symmetries of the deviation field. The spatial range of each force is then shown to result from the

spectral coherence of its associated trembling mode, offering a deterministic alternative to the mass-based mediator mechanism.

Uniqueness and Completeness of the Force Spectrum. Section S16 proves that no additional fundamental interactions can arise. By imposing compactness, causality, and four-dimensional geometric embeddability, it is shown that only four irreducible trembling-induced symmetry classes are consistent with TSRT.

Axiomatic Summary of Interaction Classification. Section S17 presents a structured axiomatic derivation of the four fundamental forces from first principles. The section walks the reader through the key assumptions, constraints, and group-theoretic logic that lead inevitably to the fourfold interaction structure. It serves as a concise pedagogical reference for students and researchers.

Quantum Equations from Geometric Averaging. Sections S18–S19 demonstrate that the Schrödinger and Dirac equations emerge from statistical averaging over trembling ensembles. The canonical relation $E = n\hbar\omega$ is shown to arise from a causal cutoff in geometric resolution, and the uncertainty principle is derived from curvature-constrained phase-space dispersion, not operator algebra.

Geometric Derivation of the Uncertainty Principle. Section S20 establishes both lower and upper bounds on the uncertainty product $\Delta x \Delta p$. These bounds emerge from finite coherence length and causal limits in trembling geometry, yielding a refined and curvature-sensitive formulation of uncertainty.

Necessity of Trembling for Causal Coherence. In addition to deriving the structure of forces, TSRT reveals that trembling spacetime is a geometric necessity rather than an optional ingredient. As demonstrated in Supplementary Material Section S21, the enforcement of causal progression — specifically, the strict positivity of proper time increments along all physical paths — requires the presence of intrinsic metric fluctuations. In the absence of trembling, low-curvature regions would permit acausal configurations, destabilizing the causal structure itself. Trembling thus emerges not as an imposed feature, but as a natural outgrowth of causal consistency in the fabric of spacetime.

Curvature, Uncertainty, and the Limits of Information. Section S22 shows that higher spacetime curvature compresses the admissible phase-space volume, tightening uncertainty bounds and reducing local information capacity. This leads to a geometric derivation of the Bekenstein–Hawking entropy and reinterprets holography as a causal consequence of trembling curvature.

Causal Structure and Light Propagation. Sections S23–S24 show that trembling-induced metrics always preserve proper-time positivity and causal propagation. The speed of light

emerges as a dynamically invariant limit from the structure of null geodesics in trembling geometry. A rigorous comparison with the framework of Weyl and others is provided in Section S25, demonstrating key points of divergence and internal inconsistency in their approach.

Historical and Conceptual Foundations. Section S26 situates TSRT within the broader history of theoretical physics. It explains how trembling geometry transcends earlier frameworks by unifying interaction dynamics, quantum behavior, and conservation laws within a single causal, deterministic geometric theory—without resorting to quantization, extra dimensions, or postulated gauge fields.

S2 Supplementary Material: Extended Acknowledgments

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In the next section, we establish the geometric and causal foundations of spacetime structure by deriving the Lorentzian metric [1] form and justifying the use of proper time as a dynamical parameter. This sets the stage for all trembling-induced dynamics to follow.

S3 Supplementary Material: Proper Time, Causal Structure, and the Emergence of Lorentz Geometry

This section provides the geometric foundation upon which the entire TSRT formalism is built. It replaces traditional postulates of special relativity with first-principles derivations grounded in causal structure.

S3.1 Introduction and Motivation

Proper time τ plays a central role in the Trembling Spacetime Relativity Theory (TSRT). Unlike in classical relativity, where it is often treated as a byproduct of the Minkowski metric, or in quantum theory, where its meaning is blurred by probabilistic evolution, TSRT places τ at the foundation of all physical structure. In TSRT, worldlines are deterministically parameterized by proper time, and it is along these trembling geodesics that all physical observables—including position, momentum, and interaction—are defined and evolve. The preservation of a monotonic, real-valued proper time is the cornerstone of causal coherence in this framework.

A key objective of this section is to shift away from traditional formulations that postulate the structure of spacetime from symmetry arguments or kinematic assumptions. In particular, we aim to derive Lorentz invariance not from the postulate of light-speed constancy or from the principle of relativity, but from the logical requirement that elapsed proper time be well-defined, real, and causally ordered. This inversion of logic—where geometry arises from causal necessity rather than the reverse—characterizes the spirit of TSRT.

Historically, the Lorentz transformations [1] were introduced to explain the invariance of the speed of light observed in the Michelson–Morley experiment [20]. However, this approach accepts as axiomatic that light defines the causal structure of spacetime. In TSRT, we pursue a more fundamental route: we begin with a general, pre-metric notion of elapsed time based on the ordering of events along physical trajectories and require that this elapsed time remain consistent with the geometric coherence of the underlying trembling spacetime. To achieve this, we will first define proper time as an integral functional over arbitrary trajectories in spacetime, using only the requirement that it must monotonically increase along causal curves. From this, we will derive a minimal form of the spacetime interval consistent with causal propagation. Remarkably, this form will lead naturally to the emergence of a pseudo-Euclidean metric with Lorentzian signature, and to the recovery of Lorentz transformations as the only class of coordinate transformations preserving causal structure. This foundational insight allows us to go further: we generalize the construction from flat spacetime to curved manifolds, showing that the same requirement—causal preservation of proper time—leads uniquely to the differential structure of geodesics in general relativity. In doing so, we justify not only the mathematical form of the metric $g_{\mu\nu}(x)$, but also the logical independence and internal consistency of treating τ as a primary parameter, even when the coordinates x^μ themselves depend functionally on τ .

This section therefore serves a dual purpose: it establishes the causal origin of Lorentz geometry, and it secures the formal legitimacy of the TSRT program by demonstrating that all standard relativistic structures—notably the Minkowski interval and geodesic motion—can be derived from deeper geometric principles tied to trembling and proper time. In this light,

spacetime is no longer a static background with imposed symmetries, but a causally coherent, trembling medium whose geometry emerges from the requirement that proper time be real, forward-directed, and maximally resolved.

S3.2 Parametrized Trajectories and the Definition of Proper Time

In any causal theory of spacetime—including Trembling Spacetime Relativity Theory (TSRT)—the concept of elapsed proper time τ along a physical trajectory is essential. It defines the ordering of events, the evolution of particles, and the structure of physical laws. Unlike coordinate time, which depends on arbitrary observer-dependent labels, proper time is an invariant associated with the internal ticking of a particle’s worldline and is defined along that worldline itself.

Let us begin by considering a trajectory in spacetime given by $x^\mu(\lambda)$, where λ is a general, monotonic parameter used to traverse the curve. The physical content of the trajectory lies in its geometric shape and causal orientation, not in the specific choice of λ . What matters is how the trajectory progresses through spacetime and how the passage of time along it can be measured intrinsically. To define this intrinsic time, we introduce a general symmetric bilinear form $A_{\mu\nu}(x)$, depending smoothly on the coordinates x^μ , which acts as the local rule for comparing infinitesimal displacements. Then, the proper time τ accumulated between two points on the trajectory λ_0 and λ_1 is defined as:

$$\tau = \int_{\lambda_0}^{\lambda_1} \sqrt{A_{\mu\nu}(x(\lambda)) \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} d\lambda. \quad (\text{S1})$$

This expression has several key features.

First, it is manifestly covariant: the integrand is a scalar under general coordinate transformations because it contracts contravariant derivatives $dx^\mu/d\lambda$ with a covariant symmetric tensor $A_{\mu\nu}$.

Second, it is local and geometric: the quantity under the square root depends only on the local structure of spacetime at the point $x^\mu(\lambda)$ and on the tangent vector to the curve.

Third, it reduces to standard proper time in general relativity when $A_{\mu\nu} = g_{\mu\nu}$, the spacetime metric.

The square root ensures that τ is real-valued only when the integrand is non-negative—i.e., when the trajectory remains within the causally admissible region defined by the geometry of $A_{\mu\nu}$. This directly encodes the physical requirement that proper time must be real and monotonically increasing along timelike worldlines. For lightlike or null paths, the integrand vanishes, corresponding to $d\tau = 0$.

At this stage, $A_{\mu\nu}(x)$ is not yet assumed to be the Minkowski metric or the curved general relativistic metric. It is simply a smooth, symmetric, rank-2 tensor field. Our goal in subsequent subsections is to determine what constraints causality and proper-time coherence place on the form of $A_{\mu\nu}$, and how this eventually leads to the emergence of Lorentzian geometry as a logical necessity rather than a starting assumption. In TSRT, this definition of τ also serves a more fundamental role: it is not only the parameter along worldlines but also the quantity that regulates the admissibility of trembling fluctuations. As we will show, preserving the positivity and real-valued nature of τ under perturbations directly constrains the permissible structure

of spacetime geometry. In this sense, proper time is not merely a derived quantity—it is the foundational measure from which geometry itself is inferred.

S3.3 From General Bilinear Form to Lorentzian Metric

We now investigate the constraints that causality, isotropy, and frame independence impose on the general symmetric bilinear form $A_{\mu\nu}(x)$ introduced in the definition of proper time. Our objective is to show that under these conditions, $A_{\mu\nu}$ must locally take the Lorentzian form of the Minkowski metric, and that the only linear transformations preserving the resulting proper time interval τ^2 are Lorentz transformations.

Let us begin by recalling that $A_{\mu\nu}(x)$ is a symmetric, non-degenerate tensor field:

$$A_{\mu\nu} = A_{\nu\mu}, \quad \det A_{\mu\nu} \neq 0. \quad (\text{S2})$$

The general proper-time interval between two infinitesimally close events is then given by:

$$d\tau^2 = A_{\mu\nu}(x) dx^\mu dx^\nu. \quad (\text{S3})$$

To constrain the possible forms of $A_{\mu\nu}$, we now impose the following physical and geometric requirements:

(1) Causality. The theory must distinguish between timelike, spacelike, and lightlike displacements. This requires that $d\tau^2 > 0$ for timelike separations (physically traversable by particles), $d\tau^2 = 0$ for null separations (signals moving at the maximum causal speed), and $d\tau^2 < 0$ for spacelike separations (which cannot be connected by any causal influence).

Hence, $A_{\mu\nu}$ must have signature $(-, +, +, +)$, ensuring exactly one positive eigenvalue and three negative ones. This establishes the *Lorentzian* character of the metric.

(2) Local Isotropy. The spacetime must be locally isotropic in the absence of fields or curvature—i.e., no spatial direction is preferred in any local inertial frame. This implies that, in suitable coordinates (t, x, y, z) , the spatial components of $A_{\mu\nu}$ must reduce to a scalar multiple of the identity on spatial slices:

$$A_{ij} = -\delta_{ij}, \quad i, j \in \{1, 2, 3\}. \quad (\text{S4})$$

(3) Frame Independence. The physical content of the theory should be invariant under coordinate transformations between inertial observers. That is, any observer using coordinates x^μ should measure the same causal structure as another observer using coordinates $x'^\mu = \Lambda^\mu_\nu x^\nu$, where Λ^μ_ν is a linear transformation. The condition that proper time be preserved under such transformations requires:

$$A_{\mu\nu} dx^\mu dx^\nu = A'_{\rho\sigma} dx'^\rho dx'^\sigma. \quad (\text{S5})$$

By the tensor transformation law:

$$A'_{\rho\sigma} = \Lambda^\mu_\rho \Lambda^\nu_\sigma A_{\mu\nu}, \quad (\text{S6})$$

so the condition becomes:

$$A_{\mu\nu} = \Lambda_{\mu}^{\rho} \Lambda_{\nu}^{\sigma} A_{\rho\sigma}. \quad (\text{S7})$$

This is exactly the requirement that the transformation Λ preserves the bilinear form $A_{\mu\nu}$.

Thus, the set of admissible transformations Λ that preserve $d\tau^2$ forms the group of automorphisms of the quadratic form defined by $A_{\mu\nu}$. If $A_{\mu\nu}$ has signature $(-, +, +, +)$, then the group of all linear transformations preserving it is the Lorentz group $O(1, 3)$. The proper orthochronous subgroup $SO^+(1, 3)$ corresponds to physically admissible transformations that preserve orientation and causal direction.

Therefore, from the assumptions of symmetry (ensuring $A_{\mu\nu} = A_{\nu\mu}$), non-degeneracy (so that $\det A_{\mu\nu} \neq 0$), and causality (requiring that the interval τ^2 distinguishes between timelike, null, and spacelike separations), the bilinear form must admit a diagonalizable structure with real eigenvalues of mixed sign. Local isotropy further implies that all spatial directions are equivalent up to rotation, meaning the spatial part of $A_{\mu\nu}$ must be proportional to a Euclidean identity matrix. Frame independence then requires that transformations between inertial frames preserve the form of τ^2 , which restricts allowable coordinate transformations to those preserving the bilinear form.

Together, these conditions imply that $A_{\mu\nu}$ is locally equivalent to the Minkowski metric up to a constant rescaling:

$$A_{\mu\nu} \propto \eta_{\mu\nu} = \text{diag}(-1, +1, +1, +1), \quad (\text{S8})$$

which defines a Lorentzian geometry. The constant can be absorbed into the proper-time unit or eliminated by choosing units in which $c = 1$. The group of transformations preserving τ^2 under this form is the Lorentz group $SO(1, 3)$, uniquely fixing the metric structure up to coordinate reparametrizations.

We equally conclude that the invariant quantity:

$$d\tau^2 = -dt^2 + dx^2 + dy^2 + dz^2 \quad (\text{S9})$$

must be preserved by all physical transformations between inertial frames. We reiterate that this expression assumes the signature $(-+++)$, consistent with all TSRT derivations in this work. Finally, the only linear transformations consistent with these principles are Lorentz transformations:

$$x'^{\mu} = \Lambda_{\nu}^{\mu} x^{\nu}, \quad \text{with} \quad \Lambda_{\mu}^{\rho} \Lambda_{\nu}^{\sigma} \eta_{\rho\sigma} = \eta_{\mu\nu}. \quad (\text{S10})$$

As a consequence, the familiar Lorentz transformations of special relativity emerge not as postulates but as logical consequences of the geometric and causal structure needed to define proper time in TSRT. This derivation confirms that assuming τ as a function of the four spacetime coordinates $x^{\mu}(\tau)$ is not circular, but logically coherent and physically necessary.

We adopt the $(-+++)$ signature convention throughout this work, so that timelike intervals satisfy $d\tau^2 = -dt^2 + dx^2 + dy^2 + dz^2$. In natural units $c = 1$, so t is dimensionally equivalent to x^i ; where needed, one may insert factors of c to convert back to SI units, e.g., $(ct)^2$ in place of t^2 .

S3.4 Reparametrization and the Use of τ as Independent Variable

A key feature of TSRT is the use of proper time τ as the natural parameter for worldlines. While the spacetime coordinates x^μ are functions of τ , the proper time itself is defined via an integral over these coordinates. This raises the question: is it logically consistent to treat τ as the independent variable, given that it is constructed from the same worldline it parametrizes? In this subsection, we demonstrate that this approach is mathematically well-defined and physically justified for any smooth timelike curve.

Proper Time as a Reparametrization-Invariant Integral. Let a smooth worldline be given in terms of an arbitrary monotonic parameter λ , with coordinate functions $x^\mu(\lambda)$. The proper time between two events along this curve is defined as:

$$\tau(\lambda) = \int_{\lambda_0}^{\lambda} \sqrt{A_{\mu\nu}(x(\lambda')) \frac{dx^\mu}{d\lambda'} \frac{dx^\nu}{d\lambda'}} d\lambda', \quad (\text{S11})$$

where $A_{\mu\nu}(x)$ is a symmetric non-degenerate bilinear form with Lorentzian signature. For timelike paths, the integrand is real and strictly positive:

$$\frac{d\tau}{d\lambda} > 0. \quad (\text{S12})$$

This monotonicity ensures that $\tau(\lambda)$ is invertible. Thus, there exists a smooth inverse function $\lambda(\tau)$, and the worldline can be reparametrized in terms of τ :

$$x^\mu(\tau) := x^\mu(\lambda(\tau)). \quad (\text{S13})$$

This proves that $x^\mu(\tau)$ is a well-defined, smooth representation of the worldline.

No Circularity in Definition. Although τ depends on the worldline $x^\mu(\lambda)$, and $x^\mu(\tau)$ is defined via inversion of this dependency, no circularity arises. The procedure is analogous to defining arc length s along a curve $\vec{x}(\lambda)$, and then using s to reparametrize the curve as $\vec{x}(s)$. The fact that s is an integral over $\vec{x}(\lambda)$ does not prevent one from writing $\vec{x}(s)$, provided the integrand is strictly positive—which is precisely the case for proper time along timelike worldlines.

The chain rule confirms this consistency:

$$\frac{dx^\mu}{d\tau} = \frac{dx^\mu}{d\lambda} \left(\frac{d\tau}{d\lambda} \right)^{-1}. \quad (\text{S14})$$

Since $d\tau/d\lambda > 0$, this expression is well-defined, and $x^\mu(\tau)$ inherits all differentiability properties of the original $x^\mu(\lambda)$.

Physical Interpretation. Proper time τ represents the accumulated causal duration experienced by a particle moving along the worldline. It is observer-independent and geometrically intrinsic to the path. Using τ as the evolution parameter ensures that the equations of motion

respect the causal structure of spacetime, particularly in TSRT, where trembling fluctuations must preserve $d\tau > 0$ at all scales.

Moreover, using τ as the parametrization allows one to directly impose causality constraints such as:

$$\frac{d\tau}{d\lambda} > 0, \quad \text{and} \quad \frac{d^2\tau}{d\lambda^2} \leq 0, \quad (\text{S15})$$

which are crucial in the trembling spacetime framework for ensuring monotonic causal evolution even in curved or fluctuating geometries.

Therefore, for every smooth timelike trajectory, the proper time τ is strictly increasing and invertible, making it a valid and natural choice for the independent parameter. The expression $x^\mu(\tau)$ is mathematically consistent, physically meaningful, and avoids circularity. In the TSRT framework, this justifies using τ as the evolution parameter in both flat and curved spacetime, ensuring that causal order and trembling coherence are maintained throughout particle evolution.

S3.5 Extension to Curved Spacetime and General Geodesics

In general relativity and in TSRT, spacetime is not globally flat but is modeled as a differentiable four-dimensional manifold equipped with a position-dependent metric tensor $g_{\mu\nu}(x)$. This metric governs local causal structure, distance, and time intervals. To extend the definition of proper time from flat spacetime to general curved geometry, we replace the constant Minkowski bilinear form with the spacetime-dependent Lorentzian metric $g_{\mu\nu}(x)$, yielding:

$$\tau = \int_{\lambda_0}^{\lambda_1} \sqrt{g_{\mu\nu}(x(\lambda)) \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} d\lambda. \quad (\text{S16})$$

This integral defines the elapsed proper time along a timelike worldline in arbitrary coordinates and arbitrary geometries. It is manifestly invariant under coordinate transformations, due to the tensorial nature of $g_{\mu\nu}$ and the scalar character of the integrand. Thus, it provides a general-covariant foundation for measuring causal duration in both general relativity and TSRT.

Recovery of the Minkowski Case. In locally inertial frames—those constructed at a point $p \in \mathcal{M}$ where the metric is locally flat—the metric satisfies:

$$g_{\mu\nu}(p) = \eta_{\mu\nu}, \quad \partial_\alpha g_{\mu\nu}(p) = 0, \quad (\text{S17})$$

where $\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$ is the Minkowski metric. In such frames, Equation (S16) reduces to:

$$\tau = \int_{\lambda_0}^{\lambda_1} \sqrt{\eta_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} d\lambda, \quad (\text{S18})$$

which reproduces the special relativistic definition of proper time. This confirms the compatibility of the general definition with local flatness, in agreement with the equivalence principle.

Geodesics as Extremals of Proper Time. A fundamental postulate of general relativity, and a derived result in TSRT, is that free particles follow geodesics—curves that locally extrem-

ize the proper time functional. Mathematically, this follows by applying the Euler–Lagrange equations to the Lagrangian:

$$\mathcal{L}(x, \dot{x}) := \sqrt{g_{\mu\nu}(x) \dot{x}^\mu \dot{x}^\nu}, \quad (\text{S19})$$

where $\dot{x}^\mu = dx^\mu/d\lambda$. Varying the action:

$$S = \int_{\lambda_0}^{\lambda_1} \mathcal{L}(x, \dot{x}) d\lambda \quad (\text{S20})$$

with respect to $x^\mu(\lambda)$ yields the geodesic equation:

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\nu\rho}^\mu \frac{dx^\nu}{d\tau} \frac{dx^\rho}{d\tau} = 0, \quad (\text{S21})$$

where $\Gamma_{\nu\rho}^\mu$ are the Christoffel symbols constructed from $g_{\mu\nu}(x)$. The parameter τ is now the proper time, making this the affinely parametrized geodesic equation.

This result provides a direct geometric criterion for inertial motion: free particles move along trajectories that extremize the elapsed proper time between two events in curved spacetime. This remains true in TSRT, where trembling corrections may modify the geodesic equation, but the structure of proper time as a geometric functional remains intact. Therefore, the generalization of the proper time integral to curved spacetime provides a robust and coordinate-invariant definition of elapsed causal duration. In locally inertial frames it reduces to the familiar Minkowski form, and it underlies the physical principle that free-falling particles follow geodesics. This framework establishes the proper time τ as a universal causal parameter across both special and general relativity, and provides the geometric infrastructure upon which TSRT builds its causal trembling dynamics.

S3.6 Conclusion: Logical Foundations for Lorentz Geometry and Proper Time Parametrization

We have shown that the Lorentzian structure of spacetime and the use of proper time τ as an independent, physically meaningful parameter follow directly from causal and geometric reasoning. Beginning with a general symmetric bilinear form $A_{\mu\nu}(x)$, we derived the requirement that proper time be real, strictly increasing, and invariant under frame transformations. These conditions enforced the metric signature and led inevitably to the Minkowski form in flat space:

$$\tau^2 = -dt^2 + dx^2 + dy^2 + dz^2, \quad (\text{S22})$$

with Lorentz transformations emerging as the unique class of linear maps preserving this causal structure. We reiterate that this expression assumes the signature $(-+++)$, consistent with all TSRT derivations in this work.

By analyzing smooth timelike worldlines, we established that τ is globally monotonic and invertible with respect to arbitrary parametrizations, allowing us to treat $x^\mu(\tau)$ as a well-defined coordinate description of motion. No circularity arises from defining τ via the coordinates it parametrizes, as the causal structure guarantees consistency of this interdependence. Generalizing to curved spacetime, we recovered the standard form of proper time as an integral over

the metric $g_{\mu\nu}(x)$, which naturally reduces to the Minkowski case in locally inertial frames. The geodesic principle then follows as a variational consequence: physical worldlines extremize τ , making it a dynamically grounded measure of inertial motion in both general relativity and TSRT.

This reasoning renders unnecessary the traditional postulates of special relativity regarding constant light speed or the relativity principle. Instead, the entire framework of Lorentzian geometry arises from the internal consistency of causality and metric structure. Trembling Spacetime Relativity Theory (TSRT) builds directly upon this foundation, using τ as the central evolution parameter in its first-order formulation. Later sections of the Supplementary Material apply these results explicitly. For example, in Section S4, the trembling geodesic equations are derived along τ -parametrized worldlines. In Section S9.2, causal admissibility of metric fluctuations is built upon the positivity of proper time. In Section S21, the invariance and monotonicity of τ play a central role in proving the necessity of trembling. In Section S24, the null case ($d\tau^2 = 0$) is analyzed to derive the invariant speed limit from trembling spacetime geometry.

Thus, the results presented in this foundational section are not merely introductory—they form the logical core from which all subsequent TSRT derivations proceed.

This derivation of the Lorentzian metric structure and the legitimacy of proper time as a worldline parameter lays the geometric and causal foundation upon which trembling spacetime dynamics can be rigorously defined. In the next section, we build directly on this structure to formulate the trembling geodesic equations that govern particle motion within TSRT.

S4 Supplementary Material: Derivation of the Trembling Geodesic Equations

Having established in Section S3 that proper time is a geometrically admissible worldline parameter and that Lorentzian geometry emerges from causal consistency, we now derive the dynamical equations of motion for particles in trembling spacetime. These trembling geodesic equations follow from a proper-time action principle that incorporates fine-scale causal fluctuations.

The presence of intrinsic spacetime fluctuations modifies the geodesic equation. In classical General Relativity [21], the motion of a test particle is determined by the standard geodesic equation:

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0, \quad (\text{S23})$$

where $\Gamma_{\alpha\beta}^\mu$ are the Christoffel symbols, encoding spacetime curvature, and τ is the proper time along the particle's trajectory.

S4.1 Introducing Spacetime Trembling Perturbations

In TSRT, we introduce a perturbation term $\xi^\mu(x^\alpha(\tau))$ to represent local intrinsic spacetime fluctuations. This modifies the geodesic equation as:

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = \xi^\mu. \quad (\text{S24})$$

This additional term accounts for perturbations affecting the motion of test particles. The field ξ^μ represents the local acceleration induced by trembling at each spacetime point.

S4.2 Nature of the Trembling Term ξ^μ

The trembling term ξ^μ represents metric fluctuations that depend on the (microscopic) structure of spacetime. To maintain Lorentz covariance [1, 22, 23], it is expressed as a superposition of wave modes with a proper 4D wavevector representation,

$$\xi^\mu = \sum_n A_n^\mu \cos(k_n^\alpha x_\alpha(\tau) + \phi_n). \quad (\text{S25})$$

where A_n^μ is the amplitude of the fluctuation in the μ -direction, $k_n^\alpha = (\omega_n, \mathbf{k}_n)$ is the 4-wavevector containing both temporal and spatial components, $x^\alpha = (t, \mathbf{x})$ represents the spacetime coordinates, and ϕ_n is a phase factor.

Expanding the contraction $k_n^\alpha x_\alpha$ explicitly in terms of time and space results in a form better recognizable to, for instance, acousticians [24, 25] and opticians [26, 27],

$$k_n^\alpha x_\alpha = \omega_n t - \mathbf{k}_n \cdot \mathbf{x}, \quad (\text{S26})$$

where ω_n is the frequency component associated with time fluctuations, and $\mathbf{k}_n \cdot \mathbf{x} = k_n^i x^i$ is the spatial wave contribution.

Thus, the trembling term induces periodic perturbations in geodesic motion by modifying the background metric, leading to observable deviations from classical trajectories. The wave nature of these fluctuations may provide insights into the deeper structure of spacetime and potential quantum gravity effects, although TSRT forms a replacement of quantum physics, united with general relativity. Nevertheless, it should be reminded that the trembling remains a phenomenon of local importance only, embedded in the principle of local ontology, despite its description as superposed wave-like features. The wave-like feature prevents fissure of spacetime structure.

S4.3 Effective Force Interpretation

Sections S9, S10, S11, S13, S12, S14, S15 and S16 provide a complete study of the four fundamental forces in our universe. Nevertheless, as an introduction, one can readily notice effective force appearance from a first glimpse on the trembling features of spacetime. To establish the emergence of an effective force term from the modified geodesic equation (S24), we multiply both sides by the particle's mass m , which gives

$$m \frac{d^2 x^\mu}{d\tau^2} = -m \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} + m \xi^\mu. \quad (\text{S27})$$

Here, the term on the left-hand side represents the four-force experienced by the particle, while the first term on the right-hand side accounts for the standard gravitational influence encoded in the Christoffel symbols. The additional term $m\xi^\mu$ thus appears as an effective force due to spacetime trembling, which alters the acceleration of the particle as measured in its own proper time.

This derivation shows that, while the classical geodesic equation describes motion solely due to spacetime curvature, the presence of trembling fluctuations introduces an additional deviation. The test particle perceives this deviation as a force, not because an external interaction is acting upon it, but because its motion is being influenced by the fluctuating background structure of spacetime itself. Since the only meaningful acceleration for a test particle is the second derivative of its position with respect to τ , the presence of ξ^μ leads to an observable effect that mimics a force-like influence. This modification is entirely local and intrinsic, meaning that, in se, it does not correspond to an external force in the classical sense but is instead an emergent property of the fluctuating spacetime itself.

The nature of this force depends on the specific form of ξ^μ , which is dictated by the statistical properties and dynamics of spacetime trembling. In certain regimes, this could manifest as small apparent stochastic deviations, while in others, coherent structures in ξ^μ may lead to systematic alterations in motion, potentially resembling modified gravitational interactions or other forces, that differ from conventional general relativity. This perspective opens new avenues for exploring how fundamental spacetime fluctuations may influence particle trajectories and observable physical effects.

S4.4 Covariant Form of the Trembling Geodesic Equation

The formulation of the geodesic equation in terms of an effective force, as given in section S4.3, expresses the trembling-induced deviation ξ^μ as a force-like term acting on a particle's motion. This is useful from a dynamical perspective and facilitates analogies with standard force laws. However, the same physics can be recast in a fully covariant geometric form by incorporating the metric perturbations directly into the affine structure of spacetime, in a manner analogous to how the affine structure of spacetime—characterized by the connection $\Gamma_{\mu\nu}^\lambda$ —governs geodesic motion independently of the metric [28], with this structure being subsequently modified through the introduction of curvature via the Riemann tensor [29, 30].

In this covariant approach, we begin with the perturbed metric tensor introduced in Equation (S28):

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + \xi_{\mu\nu}, \quad (\text{S28})$$

where $\bar{g}_{\mu\nu}$ is the background metric (e.g., Minkowski [28, 31] or cosmological Friedmann-Lemaître-Robertson-Walker (FLRW) [32, 33]), and $\xi_{\mu\nu}$ represents local trembling-induced deviations.

In general relativity, the Christoffel symbols $\bar{\Gamma}_{\mu\nu}^\lambda$ describe how vectors parallel-transport in a curved spacetime and are defined purely in terms of the spacetime metric. For any smooth background metric $\bar{g}_{\mu\nu}$, they are given by [31, 34]:

$$\bar{\Gamma}_{\mu\nu}^\lambda = \frac{1}{2} \bar{g}^{\lambda\sigma} (\partial_\mu \bar{g}_{\nu\sigma} + \partial_\nu \bar{g}_{\mu\sigma} - \partial_\sigma \bar{g}_{\mu\nu}), \quad (\text{S29})$$

where $\bar{g}^{\lambda\sigma}$ is the inverse of the background metric. These coefficients encapsulate how the background geometry curves and guide the geodesic motion of free-falling particles in the absence of trembling.

The Christoffel symbols constructed from the total metric $g_{\mu\nu}$ split naturally into two parts:

$$\Gamma_{\alpha\beta}^\mu = \bar{\Gamma}_{\alpha\beta}^\mu + \delta\Gamma_{\alpha\beta}^\mu, \quad (\text{S30})$$

where $\bar{\Gamma}_{\alpha\beta}^\mu$ is the background connection and $\delta\Gamma_{\alpha\beta}^\mu$ is the contribution due to $\xi_{\mu\nu}$. The trembling correction can, therefore, be written as:

$$\delta\Gamma_{\alpha\beta}^\mu = \frac{1}{2} \bar{g}^{\mu\lambda} (\nabla_\alpha \xi_{\beta\lambda} + \nabla_\beta \xi_{\alpha\lambda} - \nabla_\lambda \xi_{\alpha\beta}), \quad (\text{S31})$$

where ∇ denotes the covariant derivative with respect to the background metric $\bar{g}_{\mu\nu}$. This expression is valid to first order in the trembling deviation $\xi_{\mu\nu}$. While this analysis is restricted to first order, extensions to higher-order terms offer a compelling avenue for further theoretical development. The inverse background metric $\bar{g}^{\mu\lambda}$ is used rather than the full $g^{\mu\lambda}$ because corrections to the inverse metric itself are already $\mathcal{O}(\xi)$, and would introduce higher-order terms beyond the linear approximation. All covariant derivatives ∇ are taken with respect to $\bar{g}_{\mu\nu}$, ensuring the consistency of the perturbative expansion.

The standard geodesic equation for a particle moving in a curved spacetime with metric $g_{\mu\nu}$,

introduced in Equation (S23), is given by:

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0, \quad (\text{S32})$$

where, now, $\Gamma_{\alpha\beta}^\mu$ are the Christoffel symbols constructed from the full metric $g_{\mu\nu} = \bar{g}_{\mu\nu} + \xi_{\mu\nu}$.

As shown in Equation (S30), these symbols split naturally into a background and a trembling correction

Substituting the decomposition of Equation (S30) into Equation (S32), we obtain

$$\frac{d^2 x^\mu}{d\tau^2} + \left(\bar{\Gamma}_{\alpha\beta}^\mu + \delta\Gamma_{\alpha\beta}^\mu \right) \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0, \quad (\text{S33})$$

which expands to:

$$\frac{d^2 x^\mu}{d\tau^2} + \bar{\Gamma}_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} + \delta\Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0. \quad (\text{S34})$$

This is the trembling-corrected geodesic equation, where the first term represents proper-time acceleration, the second term is the background geodesic motion due to $\bar{g}_{\mu\nu}$, and the third term captures the correction due to the spacetime trembling $\xi_{\mu\nu}$.

This equation describes how a particle deviates from standard geodesic motion due to trembling-induced curvature. The trembling correction to the connection coefficients, $\delta\Gamma_{\alpha\beta}^\mu$, arises from perturbations of the background metric $g_{\mu\nu} = \bar{g}_{\mu\nu} + \xi_{\mu\nu}$, where $\xi_{\mu\nu}$ encodes the fine-scale metric deviations due to trembling. To first order in the deviation amplitude, the correction $\delta\Gamma_{\alpha\beta}^\mu$ is given by the linearized variation of the Christoffel symbols with respect to the perturbed metric, as shown in Equation (S31).

This expression is linear in $\xi_{\mu\nu}$ because it follows from a first-order expansion of the connection around the background metric. Higher-order corrections involve quadratic and higher powers of $\xi_{\mu\nu}$, but these are suppressed in the trembling regime, where deviations remain small compared to the coarse background. Consequently, the trembling correction preserves deterministic evolution and ensures local general covariance to linear order, allowing the perturbed geodesic equation to retain its coordinate-independent geometric meaning. This linear structure is essential for maintaining the physical interpretability of trembling-induced deviations: it guarantees that they are genuine geometric effects rather than coordinate artifacts. The resulting motion remains fully deterministic and respects the causal structure encoded in the background metric.

Finally, this formulation aligns exactly with the effective-force interpretation introduced earlier in Equation (S27). The effective term ξ^μ appearing there corresponds directly to the geometrically induced acceleration generated by $\delta\Gamma_{\alpha\beta}^\mu$. What appears as an external four-force in the coarse-grained dynamics is thus revealed as an affine correction in the underlying covariant geometry.

S4.5 Implications and Physical Interpretation

The introduction of the trembling term ξ^μ leads to modifications in test particle trajectories, causing deviations from classical geodesics. It equally leads to an effective force interpretation, suggesting that metric fluctuations influence particle dynamics in a manner akin to a stochastic

force. Even without further specifications, potential observable effects in gravitational systems may be expected, particularly in high-precision astrophysical or interferometric experiments.

Thus, the trembling geodesic equation provides a crucial foundation for understanding the macroscopic implications of spacetime fluctuations, offering an alternative to quantum mechanical interpretations of microscopic motion. Building on the trembling geodesics, we now extend the framework of general relativity to incorporate microscopic spacetime trembling effects.

S5 Supplementary Material: Extension of General Relativity in Trembling Spacetime

In classical General Relativity [28,31], Einstein's field equations describe the curvature of spacetime due to energy and momentum. The standard form is given by

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}, \quad (\text{S35})$$

where $R_{\mu\nu}$ is the Ricci curvature tensor, R is the Ricci scalar, $g_{\mu\nu}$ is the metric tensor, Λ is the cosmological constant, and $T_{\mu\nu}$ is the stress-energy tensor representing matter and energy distributions. The Ricci tensor $R_{\mu\nu}$ is obtained by contracting the Riemann curvature tensor $R_{\mu\nu\sigma}^\rho$ as $R_{\mu\nu} = R_{\mu\rho\nu}^\rho$, effectively reducing the number of indices from four to two.

S5.1 Incorporating Trembling Spacetime Perturbations

Initially, one might consider representing metric oscillations using cosine terms, as we did in Equation (S25),

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + \sum_n A_n \cos(k_n^\alpha x_\alpha(\tau) + \phi_n). \quad (\text{S36})$$

However, cosine functions are best suited for problems with periodic boundary conditions, whereas Bessel functions are eigenfunctions of the Laplacian in a system with radial symmetry, making them more natural for expressing spacetime fluctuations in many situations of interest and under the assumption of local ontology. Indeed, in TSRT, trembling effects in a locally oriented ontological framework are not necessarily infinite and periodic in a linear fashion but may propagate isotropically from a central region or be influenced by the locally curved nature of spacetime. Since the trembling dynamics are often dictated by local spacetime structure, a formulation that respects radial propagation is mathematically and physically more appropriate in most cases.

Bessel functions $J_n(x)$, first derived in 1824 [35], are eigenfunctions of the Laplacian operator in cylindrical or spherical coordinates, as rigorously established by Hilbert [36] and applied to physical problems in later works [37]. They describe waves and fluctuations that expand outward from a central point rather than repeating periodically. In the context of spacetime trembling, this is crucial because local metric fluctuations may exhibit spherical or cylindrical symmetry around sources of curvature rather than following artificial periodic constraints. By adopting Bessel functions, we ensure that the mathematical formulation of trembling spacetime aligns with the expected geometric behavior of propagating fluctuations, minimizing the need for additional approximations and maintaining the local character of physical phenomena.

The Laplacian operator, which governs wave propagation, takes different forms depending on the coordinate system. In Cartesian coordinates, solutions are typically expressed as sine and cosine waves. However, in a system with radial symmetry, the Laplacian takes the form:

$$\nabla^2 f = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2}. \quad (\text{S37})$$

This equation naturally leads to Bessel functions as solutions for the radial component, making them the appropriate basis for expressing trembling spacetime fluctuations in cases where radial propagation dominates. Thus, by using Bessel functions instead of cosine functions, we ensure that the formulation of trembling spacetime remains aligned with the fundamental symmetry and local ontological character of spacetime fluctuations, capturing their geometric and dynamical properties in a more natural and accurate way.

Additionally, using Bessel functions instead of cosines provides a more compact and generalizable formulation of Einstein's field equations. Since the field equations involve second-order derivatives of metric fluctuations, Bessel functions naturally arise due to their recurrence relations,

$$J_n''(x) + \frac{1}{x}J_n'(x) + \left(1 - \frac{n^2}{x^2}\right)J_n(x) = 0. \quad (\text{S38})$$

This property simplifies the mathematical handling of trembling spacetime perturbations and results in a more elegant expression of gravitational interactions in TSRT.

Therefore, a natural choice to formally express trembling spacetime oscillations, since spacetime trembling depends solely on proper time τ and have a local character, is to use a decomposition in terms of eigenfunctions of the Laplace operator in spherical or cylindrical coordinates, leading to a Fourier-Bessel expansion.

The Fourier-Bessel decomposition expresses an arbitrary function $f(\tau)$ as:

$$f(\tau) = \sum_{n=1}^{\infty} \alpha_n J_n(k_n^\alpha x_\alpha(\tau)), \quad (\text{S39})$$

where $J_n(x)$ are Bessel functions of the first kind.

Consequently, Trembling Spacetime Relativity Theory (TSRT) extends the framework of General Relativity [2] as given in Equation (S35), by introducing intrinsic metric oscillations and stress-energy perturbations that arise from fundamental spacetime fluctuations. Rather than modifying the cosmological constant Λ directly, TSRT introduces an oscillatory geometric correction term defined as:

$$\mathcal{H}_{\mu\nu} := \sum_{n=1}^{\infty} \alpha_n J_n(k_n^\alpha x_\alpha(\tau)) g_{\mu\nu}, \quad (\text{S40})$$

which captures the contribution of trembling-induced modifications to the spacetime geometry. This tensor encodes the fluctuating curvature effects arising from local proper-time-dependent metric oscillations.

The modified Einstein field equations then take the form:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} + \mathcal{H}_{\mu\nu} = \frac{8\pi G}{c^4} \left(T_{\mu\nu} + \sum_{n=1}^{\infty} \beta_n J_n(k_n^\alpha x_\alpha(\tau)) T_{\mu\nu} \right), \quad (\text{S41})$$

where the source terms on the right-hand side represent matter-energy responses to the trembling spacetime structure, and the left-hand side includes both classical and oscillatory geometric contributions. This formulation preserves the tensorial structure of general relativity while embedding trembling effects in a covariant and geometrically consistent manner.

S5.2 Mathematical Structure of Metric Fluctuations

In Trembling Spacetime Relativity Theory (TSRT), metric fluctuations arise naturally from the underlying trembling structure of spacetime. These fluctuations introduce modifications to both the spacetime geometry and the energy-momentum distribution.

To describe the oscillatory correction to the metric compactly, we use the fluctuation tensor $\mathcal{H}_{\mu\nu}$ as defined in (S40). This tensor represents an infinite sum of Bessel-mode perturbations modulating the background metric $g_{\mu\nu}$, with coefficients α_n encoding their amplitudes and k_n^α their mode structure in proper time.

Similarly, we introduce the stress-energy fluctuation tensor $\mathcal{M}_{\mu\nu}$ to describe modifications to the energy-momentum distribution,

$$\mathcal{M}_{\mu\nu} = \sum_{n=1}^{\infty} \beta_n J_n(k_n^\alpha x_\alpha(\tau)) T_{\mu\nu}, \quad (\text{S42})$$

where β_n characterize the coupling strength of matter to the trembling metric structure.

These modifications imply that matter-energy interactions are dynamically influenced by the oscillatory nature of spacetime, leading to local deformations in curvature and potential deviations from classical relativistic predictions.

S5.3 Effective Field Equations and Physical Consequences

Incorporating these fluctuations into the full Einstein field Equations in Section S41, we define the effective geometric and matter tensors:

$$G_{\mu\nu}^{\text{eff}} = G_{\mu\nu} + \mathcal{H}_{\mu\nu}, \quad (\text{S43})$$

$$T_{\mu\nu}^{\text{eff}} = T_{\mu\nu} + \mathcal{M}_{\mu\nu}. \quad (\text{S44})$$

Rewriting the full field Equation (S41) in terms of the effective tensors, we obtain:

$$G_{\mu\nu}^{\text{eff}} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}^{\text{eff}}, \quad (\text{S45})$$

Equation (S45) suggests that spacetime fluctuations contribute an additional source of curvature, leading to potential deviations from classical general relativity.

Rewriting the full field Equation (S41), we start from

$$G_{\mu\nu} + \Lambda g_{\mu\nu} + \sum_{n=1}^{\infty} \alpha_n J_n(k_n^\alpha x_\alpha(\tau)) g_{\mu\nu} = \frac{8\pi G}{c^4} \left(1 + \sum_{n=1}^{\infty} \beta_n J_n(k_n^\alpha x_\alpha(\tau)) \right) T_{\mu\nu}. \quad (\text{S46})$$

We now reorganize terms by defining an effective cosmological term and an effective coupling

factor. Specifically, we introduce

$$\Lambda_{\text{eff}}(\tau) = \Lambda + \sum_{n=1}^{\infty} \alpha_n J_n(k_n^\alpha x_\alpha(\tau)), \quad (\text{S47})$$

$$f_T(\tau) = 1 + \sum_{n=1}^{\infty} \beta_n J_n(k_n^\alpha x_\alpha(\tau)). \quad (\text{S48})$$

Substituting these definitions into Equation (S46) yields the compact form

$$G_{\mu\nu} + \Lambda_{\text{eff}}(\tau)g_{\mu\nu} = \frac{8\pi G}{c^4} f_T(\tau) T_{\mu\nu}, \quad (\text{S49})$$

where $\Lambda_{\text{eff}}(\tau)$ represents the effective cosmological term modified by spacetime trembling, and $f_T(\tau)$ acts as a dynamic multiplicative factor on the stress-energy tensor due to metric fluctuations.

This result indicates that the cosmological constant itself becomes dynamic, varying with spacetime trembling, and the stress-energy response scales by a fluctuating factor $f_T(\tau)$. These results in Equation (S45) and Equation (S49) reinforce the idea that spacetime is not merely a passive background but an actively fluctuating geometric structure, deeply connected to fundamental interactions.

This dual formulation allows us to interpret trembling-induced corrections either as modifications to geometric and matter tensors, or as effective scalar fields dynamically modulating curvature and energy density. With the trembling extension of general relativity established, we next examine energy conservation and effective mass corrections induced by spacetime fluctuations.

S6 Supplementary Material: Energy Conservation and Effective Mass Corrections

The introduction of a trembling metric in TSRT, as in Equation (S28), leads to modifications in mass-energy relations, fundamentally altering how energy is conserved in fluctuating spacetime. In standard relativistic mechanics, the energy-momentum relation is given by:

$$E^2 - p^2 c^2 = m^2 c^4, \quad (\text{S50})$$

a formulation first derived by Einstein in his foundational 1905 paper on special relativity [22] and later cast into its modern geometric form by Minkowski [23], and which assumes a static, smooth metric structure. However, in a trembling spacetime, additional metric perturbations contribute to mass-energy variations, requiring a more general formulation.

S6.1 Modified Energy-Momentum Relation

We introduce a fluctuation-dependent term $\phi(\xi)$, incorporating the effects of metric oscillations on energy and mass:

$$E^2 - p^2 c^2 = m^2 c^4 + \phi(\xi), \quad (\text{S51})$$

where $\phi(\xi)$ accounts for local metric variations that dynamically influence mass and energy. These fluctuations originate from the intrinsic trembling of spacetime, leading to effective mass corrections. This formula will be proven in section S6.2.

The crucial aspect here is that the energy of the system is conserved exactly, even in the presence of metric perturbations. Energy conservation in this context follows from the fact that any fluctuations in the metric (encoded in $\xi_{\mu\nu}$) only affect the particle's energy through the perturbative term $\phi(\xi)$, and these fluctuations lead to redistribution of energy across spacetime. This redistribution ensures that any increase in energy due to a perturbation in one region is exactly balanced by a decrease elsewhere, maintaining constant total energy across the entire system.

S6.2 Nature of the Metric-Induced Correction Term $\phi(\xi)$

In the framework of TSRT, the total energy-momentum tensor is affected by the trembling-induced fluctuations of the spacetime metric. These effects are encoded in the perturbative correction term $\phi(\xi)$, which modifies the effective stress-energy response in the field equations. To maintain consistency with the effective formulation introduced in Equations (S41)–(S49), we express the total energy-momentum tensor as:

$$T_{\text{eff}}^{\mu\nu} = \bar{T}^{\mu\nu} + \delta T^{\mu\nu}, \quad (\text{S52})$$

where $\bar{T}^{\mu\nu}$ represents the classical (background) energy-momentum distribution, and $\delta T^{\mu\nu}$ accounts for the geometric perturbations arising from trembling spacetime.

The origin of $\delta T^{\mu\nu}$ lies in the decomposition of the full metric, as presented in Equation

(S28),

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + \xi_{\mu\nu}, \quad (\text{S53})$$

where $\xi_{\mu\nu}$ is the trembling-induced fluctuation field. These metric perturbations influence the motion of matter by modifying the geodesic paths of particles, which in turn affects the structure of $T_{\text{eff}}^{\mu\nu}$.

For a test particle with four-momentum $p^\mu = m dx^\mu/d\tau$, the perturbed energy-momentum tensor receives a contribution proportional to the interaction between the fluctuation field and the particle's motion,

$$\delta T^{\mu\nu} \sim \xi_{\alpha\beta} p^\alpha p^\beta. \quad (\text{S54})$$

To evaluate how trembling affects energy-momentum relations, we begin by expressing the metric fluctuation tensor $\xi_{\mu\nu}(x)$ in a mode expansion. In local coordinates, these trembling deviations are modeled by Bessel-type oscillations:

$$\xi_{\mu\nu}(x) = \sum_n \xi_{\mu\nu}^{(n)} J_n(k_n^\alpha x_\alpha(\tau)), \quad (\text{S55})$$

where J_n are Bessel functions of the first kind and k_n^α the trembling wavevectors.

We define the scalar correction term $\phi(\xi)$ as the contraction of $\xi_{\mu\nu}$ with the particle's four-momentum:

$$\phi(\xi) := -\xi_{\mu\nu}(x) p^\mu p^\nu, \quad (\text{S56})$$

which expands to:

$$\phi(\xi) = -\sum_n \left(\xi_{\mu\nu}^{(n)} p^\mu p^\nu \right) J_n(k_n^\alpha x_\alpha(\tau)). \quad (\text{S57})$$

We identify each mode's contribution with the amplitude:

$$A_n := -\xi_{\mu\nu}^{(n)} p^\mu p^\nu, \quad (\text{S58})$$

so that:

$$\phi(\xi) = \sum_n A_n J_n(k_n^\alpha x_\alpha(\tau)). \quad (\text{S59})$$

The scalar function $\phi(\xi)$ encodes local energy-momentum corrections due to the trembling geometry and modulates the stress-energy tensor accordingly:

$$T_{\text{eff}}^{\mu\nu} = (1 + \phi(\xi)) \bar{T}^{\mu\nu}, \quad (\text{S60})$$

with $\bar{T}^{\mu\nu}$ the coarse-grained background tensor.

This local formulation connects to the global modulation factor $f_T(\tau)$ used earlier:

$$T_{\text{eff}}^{\mu\nu} = f_T(\tau) T^{\mu\nu}, \quad (\text{S61})$$

and relates to it approximately via:

$$\phi(\xi) \sim f_T(\tau) - 1. \quad (\text{S62})$$

To understand how this fluctuation impacts particle dynamics, we start with the general definition of the four-momentum:

$$p^\mu = m \frac{dx^\mu}{d\tau}, \quad (\text{S63})$$

where m is the rest mass and τ proper time. In a smooth background $\bar{g}_{\mu\nu}$, the standard dispersion relation holds:

$$\bar{g}_{\mu\nu} p^\mu p^\nu = -m^2 c^2. \quad (\text{S64})$$

In the presence of trembling, the full metric is $g_{\mu\nu} = \bar{g}_{\mu\nu} + \xi_{\mu\nu}$, so:

$$g_{\mu\nu} p^\mu p^\nu = -m^2 c^2. \quad (\text{S65})$$

In a locally inertial frame where $\bar{g}_{\mu\nu} \rightarrow \eta_{\mu\nu}$, this becomes:

$$(\eta_{\mu\nu} + \xi_{\mu\nu}) p^\mu p^\nu = -m^2 c^2. \quad (\text{S66})$$

Expanding the left-hand side:

$$(\eta_{\mu\nu} + \xi_{\mu\nu}) p^\mu p^\nu = \eta_{\mu\nu} p^\mu p^\nu + \xi_{\mu\nu} p^\mu p^\nu \quad (\text{S67})$$

$$= -E^2 + p^2 c^2 + \phi(\xi), \quad (\text{S68})$$

where we used:

$$\phi(\xi) := \xi_{\mu\nu} p^\mu p^\nu. \quad (\text{S69})$$

and also the standard special-relativistic quantity,

$$\eta_{\mu\nu} p^\mu p^\nu = -E^2 + p^2 c^2, \quad (\text{S70})$$

where E is the energy and p the spatial momentum magnitude.

Substituting into Equation (S65), we obtain:

$$-E^2 + p^2 c^2 + \phi(\xi) = -m^2 c^2, \quad (\text{S71})$$

which rearranges to the modified dispersion relation:

$$E^2 - p^2 c^2 = m^2 c^4 + \phi(\xi), \quad (\text{S72})$$

which recovers Equation (S51) as a special case.

This equation shows that $\phi(\xi)$ acts as a spacetime-induced shift in the effective rest energy. In the limit $\phi(\xi) \rightarrow 0$, the standard special-relativistic dispersion law is recovered.

It is important to emphasize that m remains the intrinsic rest mass—constant and covariant. The correction $\phi(\xi)$ is a geometric effect arising from local spacetime structure. It modifies the inertial response of a particle without altering its identity or violating global conservation laws. Thus, trembling modifies the energy-momentum relation in a covariant and physically interpretable way. These corrections may produce measurable deviations in small-scale or high-curvature regimes, reinforcing the geometric nature of energy quantization and inertia in TSRT.

S6.3 Energy Conservation and the Spacetime Fabric’s Dynamic Nature

The core aspect of energy conservation in TSRT lies in the wave-like nature of spacetime trembling, as exposed in Equation (S26). The perturbations $\xi_{\mu\nu}$ propagate across spacetime as deterministic geometric oscillations governed by proper time τ , forming a causally connected, elastic continuum. These fluctuations are locally encoded by proper-time dynamics but are globally constrained by their wavevectors k_n^α and frequencies ω_n , ensuring that the oscillatory behavior at one point in spacetime is geometrically linked to oscillations elsewhere. As a direct consequence, any local fluctuation that increases a particle’s energy in one region of spacetime must be offset by a corresponding reduction in energy in nearby or causally connected regions. This is not statistical balancing “on average” but exact causal redistribution within the fabric of spacetime. Trembling modes cannot carry energy into or out of existence; they merely redistribute it through the curvature structure. This guarantees that total energy is conserved pointwise across spacetime, consistent with the fundamental principle that energy cannot disappear at one time and reappear at another. In TSRT, such a violation would imply a breakdown in proper-time causality, which is strictly forbidden by the geometry.

We define the total energy of the system (including the particle’s inertial energy and the local metric-induced perturbative energy) as:

$$E_{\text{total}} = E_{\text{particle}} + E_{\text{perturbation}}, \quad (\text{S73})$$

where $E_{\text{perturbation}} = \phi(\xi)$ encodes the energy shift due to the local trembling geometry. This total energy remains constant:

$$\frac{d}{d\tau} E_{\text{total}} = 0, \quad (\text{S74})$$

since proper time τ is the only physically experienced temporal parameter. In this formulation, trembling does not store or release energy through time; rather, it modulates the spatial-temporal distribution of energy at each instant of proper time.

This also answers an important question regarding coordinate time t in General Relativity: while t is a useful labeling tool in a given coordinate frame, it is not invariant across observers and cannot directly constrain physical conservation laws. Energy conservation expressed with respect to coordinate time may appear violated due to frame-dependent effects (e.g., in expanding spacetimes or accelerating frames). This is shown at the end of this section. However, in TSRT—as in General Relativity—all true conservation laws must be formulated in terms of invariant quantities, such as proper time τ and covariant divergence of the energy-momentum tensor. The conservation of energy in TSRT is therefore guaranteed in a local and causal sense through proper-time evolution:

$$\nabla_\mu T_{\text{eff}}^{\mu\nu} = 0, \quad (\text{S75})$$

which, as shown further, ensures that energy-momentum is conserved across all points of spacetime, without invoking nonlocal storage in time.

In summary, trembling geometry redistributes energy locally in a manner strictly constrained by causal propagation. No energy is stored or recovered from “time itself,” either in coordinate time t or experienced proper time τ . This ensures that TSRT remains fully compatible with both

relativistic and physical notions of causality, while providing a rich framework for understanding energy redistribution in a fluctuating metric background.

Proof: Energy Cannot Be Stored or Reappear in Time. To demonstrate that energy-momentum cannot be locally created, destroyed, or globally stored in time, we invoke the covariant conservation law, as in Equation (S75):

$$\nabla_\mu T_{\text{eff}}^{\mu\nu} = 0, \quad (\text{S76})$$

which holds universally due to the contracted Bianchi identities ($\nabla_\mu G^{\mu\nu} = 0$) and the Einstein field equations $G^{\mu\nu} = \kappa T_{\text{eff}}^{\mu\nu}$. This fundamental constraint—first rigorously proven by Einstein [2] and later given deeper geometric justification by Noether [38] as a consequence of diffeomorphism invariance—ensures local energy-momentum conservation in both GR and TSRT.

To analyze conservation over a finite region of spacetime, we contract Equation (S76) with a future-directed timelike unit vector field u^ν , normalized such that $u^\mu u_\mu = -1$. Physically, u^ν represents the four-velocity of an observer or particle following a proper-time-parametrized worldline. The contraction $T_{\text{eff}}^{\mu\nu} u_\nu$ then represents the energy flux vector as measured in the local rest frame of this observer.

We consider a compact four-dimensional spacetime region $\mathcal{R}$ bounded by two spacelike hypersurfaces Σ_1 and Σ_2 , corresponding to proper times τ_1 and τ_2 , and possibly a spatial boundary $\partial\mathcal{R}_{\text{spatial}}$. Integrating the divergence over $\mathcal{R}$, we obtain:

$$\int_{\mathcal{R}} \nabla_\mu (T_{\text{eff}}^{\mu\nu} u_\nu) \sqrt{-g} d^4x = 0. \quad (\text{S77})$$

This integral follows directly from the general divergence theorem (or Stokes' theorem) on a 4D Lorentzian manifold, which in geometric terms states:

$$\int_{\mathcal{R}} \nabla_\mu V^\mu \sqrt{-g} d^4x = \int_{\partial\mathcal{R}} V^\mu n_\mu d\Sigma, \quad (\text{S78})$$

where V^μ is any vector field, $\partial\mathcal{R}$ denotes the oriented boundary of the region $\mathcal{R}$, n_μ is the unit normal to the boundary, and $d\Sigma$ is the induced 3-volume element on $\partial\mathcal{R}$. This formalism was first applied to the structure of relativistic field theories by Einstein and Hilbert [2, 39], and later formalized using differential forms by Cartan and Weyl [40, 41]. Here, $\sqrt{-g}$ denotes the square root of the negative determinant of the spacetime metric $g_{\mu\nu}$. It serves as the proper volume element on a curved four-dimensional Lorentzian manifold and ensures that integrals over spacetime are invariant under coordinate transformations. The minus sign arises because the determinant of a Lorentzian metric is negative due to the time-like signature $(-, +, +, +)$, and the volume form $\sqrt{-g} d^4x$ converts coordinate-based integrals into geometric, coordinate-invariant quantities.

Applying the divergence theorem to Equation (S77), we convert the four-volume integral of the divergence of a vector field into a three-dimensional surface integral over the oriented boundary $\partial\mathcal{R}$ of the compact spacetime region $\mathcal{R}$. Specifically, we apply the following geometric

identity on a Lorentzian manifold:

$$\int_{\mathcal{R}} \nabla_{\mu} (T_{\text{eff}}^{\mu\nu} u_{\nu}) \sqrt{-g} d^4x = \int_{\partial\mathcal{R}} T_{\text{eff}}^{\mu\nu} u_{\nu} d\Sigma_{\mu}, \quad (\text{S79})$$

where $d\Sigma_{\mu}$ is the properly oriented hypersurface element normal to the boundary $\partial\mathcal{R}$, defined at each point as:

$$d\Sigma_{\mu} := n_{\mu} \sqrt{h} d^3x, \quad (\text{S80})$$

with n_{μ} the unit normal to the hypersurface, and h the determinant of the induced three-metric on the boundary hypersurface. The sign of n_{μ} depends on the orientation of the hypersurface: for future-pointing timelike normals (as on constant-time slices), n_{μ} is chosen such that $n^{\mu}u_{\mu} < 0$.

In the specific case under consideration, the boundary $\partial\mathcal{R}$ consists of three distinct components, i.e., the final spacelike hypersurface Σ_2 , at proper time τ_2 , the initial spacelike hypersurface Σ_1 , at proper time τ_1 , and a spatial boundary $\partial\mathcal{R}_{\text{spatial}}$ enclosing the region laterally in space. Each of these contributes to the surface integral with the appropriate orientation. The contribution from each hypersurface is computed by contracting the flux vector $T_{\text{eff}}^{\mu\nu} u_{\nu}$ with the surface normal vector n_{μ} and integrating over the corresponding three-dimensional hypersurface volume:

$$\int_{\partial\mathcal{R}} T_{\text{eff}}^{\mu\nu} u_{\nu} d\Sigma_{\mu} = \int_{\Sigma_2} T_{\text{eff}}^{\mu\nu} u_{\nu} d\Sigma_{\mu} - \int_{\Sigma_1} T_{\text{eff}}^{\mu\nu} u_{\nu} d\Sigma_{\mu} + \int_{\partial\mathcal{R}_{\text{spatial}}} T_{\text{eff}}^{\mu\nu} u_{\nu} d\Sigma_{\mu}. \quad (\text{S81})$$

The minus sign in the second term arises from the reversed orientation of the past hypersurface Σ_1 , whose unit normal points toward decreasing proper time (i.e., into the integration region).

In summary, the left-hand side of the divergence integral becomes a sum over the net flux of energy (as measured by the observer with four-velocity u^{ν}) across all bounding hypersurfaces. The resulting expression in Equation (S81) asserts that any change in total energy between Σ_1 and Σ_2 is precisely accounted for by energy flowing across the spatial boundary $\partial\mathcal{R}_{\text{spatial}}$. No energy can disappear or spontaneously reappear within $\mathcal{R}$, as this would violate the covariant conservation law $\nabla_{\mu} T_{\text{eff}}^{\mu\nu} = 0$, which enforces pointwise conservation at all scales.

We are left with:

$$\int_{\Sigma_2} T_{\text{eff}}^{\mu\nu} u_{\nu} d\Sigma_{\mu} = \int_{\Sigma_1} T_{\text{eff}}^{\mu\nu} u_{\nu} d\Sigma_{\mu}, \quad (\text{S82})$$

which shows that the total energy projected along the flow of proper time u^{ν} is conserved between any two spacelike hypersurfaces. There is no mechanism in the geometry for energy to “disappear” and “reappear later”: the integrated energy is strictly conserved due to the divergence-free condition on $T_{\text{eff}}^{\mu\nu}$, enforced at every point in spacetime.

In trembling spacetime, as already shown in Equation (S52), the effective energy-momentum tensor includes both the classical contribution from matter and the additional geometric response arising from spacetime fluctuations:

$$T_{\text{eff}}^{\mu\nu} = T^{\mu\nu} + \delta T^{\mu\nu}(\xi), \quad (\text{S83})$$

where $T^{\mu\nu}$ is the background stress-energy tensor in the absence of trembling, and $\delta T^{\mu\nu}(\xi)$ captures the perturbative corrections due to the fluctuating metric component $\xi_{\mu\nu}$. These fluctuations do not reflect stochastic noise or quantum uncertainty, but deterministic geometric deviations arising from the trembling structure of spacetime as formulated in TSRT.

Again, crucially, although these geometric perturbations induce local variations in energy and momentum—manifested in quantities like the inertial correction $\phi(\xi)$ or the effective mass m_{eff} —they do not alter the fact that the total energy-momentum tensor remains divergence-free. That is, the conservation law $\nabla_\mu T_{\text{eff}}^{\mu\nu} = 0$ holds pointwise throughout the manifold, and any increase in energy density in one region is exactly compensated by a corresponding decrease elsewhere. This redistribution is not probabilistic but arises from the causal, wave-like propagation of curvature encoded in the trembling geometry.

The key insight is that trembling spacetime does not allow energy to be stored, hidden, or retrieved at a later time. Unlike quantum field theories where energy fluctuations are statistical and conservation holds only in an average sense, TSRT ensures exact energy conservation across both local and global scales. Proper-time evolution is smooth, deterministic, and locally causal, meaning that every geometric perturbation adheres to the fundamental conservation law without invoking external sources, sinks, or temporal storage. Thus, even though the effective mass or energy of a particle may vary dynamically due to spacetime geometry, these variations are geometric in origin and remain embedded within a self-consistent and conservative framework. The smoothness and differentiability of the metric $g_{\mu\nu} = \bar{g}_{\mu\nu} + \xi_{\mu\nu}$ ensure that no discontinuity or anomaly arises that could violate conservation. The trembling spacetime remains fully compatible with the geometric foundations of general relativity while extending its explanatory power through intrinsic, causal fluctuations.

Clarification: Apparent Energy Non-Conservation in Coordinate Time Is a Foliation Artifact In general relativity and TSRT, coordinate time t is not a physically measurable quantity; it serves merely as a label in a chosen coordinate system. The apparent violation of energy conservation in t arises from how spacetime is sliced (foliated) into spatial hypersurfaces labeled by $t = \text{const}$. This is purely an artifact of the foliation choice and does not reflect true physical non-conservation.

Consider a foliation of spacetime by spacelike hypersurfaces Σ_t , each labeled by coordinate time t . The unit normal vector to each hypersurface is denoted $n^\mu(t)$, satisfying $n^\mu n_\mu = -1$. The induced 3-metric on Σ_t is given by:

$$h_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu, \quad (\text{S84})$$

which projects tensors onto the spatial hypersurface [42, 43].

The energy density as measured by an observer at rest in the coordinate frame (an “Eulerian observer”) is defined via the effective stress-energy tensor $T_{\text{eff}}^{\mu\nu}$ as:

$$\mathcal{E}_t(x) = T_{\text{eff}}^{\mu\nu} n_\mu n_\nu. \quad (\text{S85})$$

This follows from the standard relativistic definition of energy density as the projection of $T^{\mu\nu}$

onto the observer's 4-velocity [44].

The total energy at coordinate time t is then:

$$E_t = \int_{\Sigma_t} \mathcal{E}_t(x) \sqrt{h} d^3x. \quad (\text{S86})$$

Unlike the proper-time integral used earlier, E_t is not conserved in general. To compute its time derivative, we must account for how both the integrand and the hypersurface Σ_t evolve with t . The Lie derivative $\mathcal{L}_n$ measures how a tensor field changes along the flow of the vector field n^μ . Mathematically, for a tensor $T^{\mu\nu}$:

$$\mathcal{L}_n T^{\mu\nu} = n^\sigma \partial_\sigma T^{\mu\nu} - T^{\sigma\nu} \partial_\sigma n^\mu - T^{\mu\sigma} \partial_\sigma n^\nu. \quad (\text{S87})$$

Physically, it represents the rate of change of the tensor field as transported along n^μ [28, 34].

Applying the Lie derivative $\mathcal{L}_n$ to the integrand, we obtain:

$$\frac{dE_t}{dt} = \int_{\Sigma_t} (\mathcal{L}_n T_{\text{eff}}^{\mu\nu} n_\mu n_\nu + T_{\text{eff}}^{\mu\nu} \mathcal{L}_n(n_\mu n_\nu)) \sqrt{h} d^3x + \text{boundary terms}. \quad (\text{S88})$$

The temporal variation, i.e., the Non-Conservation, of E_t emerges from two distinct physical contributions: the intrinsic evolution of the stress-energy tensor represented by $\mathcal{L}_n T_{\text{eff}}^{\mu\nu}$, and the geometric deformation of the hypersurface embodied in $\mathcal{L}_n(n_\mu n_\nu)$. The deformation term introduces coordinate-dependent artifacts through three principal mechanisms: the temporal evolution of the normal vector field $n^\mu(t)$, the dynamic reshaping of the hypersurface Σ_t with coordinate time, and the explicit t -dependence of the induced spatial metric $h_{\mu\nu}(t) = g_{\mu\nu} + n_\mu n_\nu$. These effects combine to produce the important result that $\frac{dE_t}{dt} \neq 0$ may occur even in perfectly energy-conserving physical systems, demonstrating that such apparent non-conservation is purely an artifact of the coordinate foliation rather than genuine physics. The complete expression for this variation,

$$\frac{dE_t}{dt} = \int_{\Sigma_t} (\mathcal{L}_n T_{\text{eff}}^{\mu\nu} n_\mu n_\nu + T_{\text{eff}}^{\mu\nu} \mathcal{L}_n(n_\mu n_\nu)) \sqrt{h} d^3x, \quad (\text{S89})$$

explicitly separates these two fundamental contributions to the energy change, with the second term entirely attributable to the coordinate-dependent hypersurface geometry.

Specifically, energy E_t is conserved when spacetime admits a timelike Killing vector $\xi^\mu = \partial_t$, which satisfies:

$$\mathcal{L}_\xi g_{\mu\nu} = 0. \quad (\text{S90})$$

A Killing vector ξ^μ represents an isometry of spacetime, which in the present context corresponds to time-translation symmetry [28]. When such a symmetry exists, several crucial simplifications occur: the Lie derivative of the effective stress-energy tensor vanishes ($\mathcal{L}_n T_{\text{eff}}^{\mu\nu} = 0$), the evolution of the normal vector n^μ becomes trivial, and consequently the coordinate-time energy E_t maintains a constant value throughout the spacetime. These conditions collectively ensure that the apparent energy non-conservation discussed previously vanishes identically, revealing the deep connection between spacetime symmetries and conservation laws in general relativity. The constancy of E_t in such cases follows directly from the Killing equation $\mathcal{L}_\xi g_{\mu\nu} = 0$, which guar-

antees that all geometric quantities involved in the energy calculation remain time-independent.

In contrast, the proper-time energy integral:

$$E_{\text{proper}} = \int_{\Sigma_\tau} T_{\text{eff}}^{\mu\nu} u_\nu d\Sigma_\mu, \quad (\text{S91})$$

where $u^\mu = dx^\mu/d\tau$ is the observer's 4-velocity, is always conserved for physical observers. This confirms that coordinate-time energy variation is purely a foliation artifact.

Therefore, in GR and TSRT, only proper-time-based energy integrals E_{proper} constitute physically meaningful conservation laws, as they are intrinsically tied to observer worldlines. The coordinate-dependent energy E_t , in contrast, exhibits apparent variations stemming from three fundamental sources: the arbitrary choice of spacetime foliation, the potential non-orthogonality between the hypersurfaces Σ_t and physical worldlines, and the absence of time-translation symmetry in generic spacetimes. These mathematical artifacts of the coordinate system underscore the crucial distinction between genuine physical conservation and mere coordinate effects. The framework presented here demonstrates that such variations in E_t reflect neither energy creation nor dissipation, but rather the observer-dependent nature of coordinate energy definitions in curved spacetime. This analysis reinforces the necessity for careful interpretation of all coordinate-dependent quantities in general relativity, particularly when evaluating conservation laws.

S6.4 Effective Mass Corrections and Physical Implications

While the intrinsic rest mass m of a particle remains constant in TSRT, the particle's locally experienced energy can fluctuate due to the influence of trembling spacetime geometry. These energy fluctuations, induced by the coupling of the particle's motion to metric perturbations $\xi_{\mu\nu}$, can be reinterpreted as variations in an effective mass parameter m_{eff} . It is essential to understand that this does not imply the fundamental mass changes, but rather that the energy-momentum relation behaves as if the particle carries a time-dependent mass. This effective description arises naturally when analyzing particle dynamics in a fluctuating geometric background and provides a convenient way to capture the energetic consequences of trembling without altering the ontological status of mass itself.

The presence of trembling-induced metric perturbations modifies the energy-momentum relation of a test particle, leading to an effective mass correction. As previously derived in Equation (S72), the presence of trembling-induced metric fluctuations modifies the standard dispersion relation. In particular, starting from the expression

$$E^2 - p^2 c^2 = m^2 c^4 + \phi(\xi), \quad (\text{S92})$$

and recalling that the invariant contraction $p^\mu p_\mu = -E^2 + p^2 c^2$, we can rearrange to obtain the equivalent form:

$$p^\mu p_\mu = -m^2 c^2 + \phi(\xi), \quad (\text{S93})$$

which encapsulates the trembling correction $\phi(\xi)$ to the standard mass-shell condition. This relation will serve as the starting point for analyzing how metric fluctuations induce effective

mass corrections and observable physical implications.

where $\phi(\xi)$ represents the trembling-induced scalar correction, the effective mass m_{eff} is defined through:

$$p^\mu p_\mu = -m_{\text{eff}}^2 c^2. \quad (\text{S94})$$

Equating these two expressions, we obtain:

$$m_{\text{eff}}^2 c^2 = m^2 c^2 - \phi(\xi). \quad (\text{S95})$$

Substituting the explicit form of $\phi(\xi)$, derived in Equation (S6.2) as:

$$\phi(\xi) = \sum_n A_n J_n(k_n^\alpha x_\alpha(\tau)) \xi_{\mu\nu}^{(n)} p^\mu p^\nu, \quad (\text{S96})$$

we find the expanded expression for the effective mass:

$$m_{\text{eff}}^2 c^2 = m^2 c^2 - \sum_n A_n J_n(k_n^\alpha x_\alpha(\tau)) \xi_{\mu\nu}^{(n)} p^\mu p^\nu. \quad (\text{S97})$$

Thus, the effective mass acquires spacetime-dependent corrections, modulated by the oscillatory structure encoded in the Bessel functions J_n and the specific modes of trembling $\xi_{\mu\nu}^{(n)}$.

To leading order, assuming the trembling amplitudes are small, the shift in mass is perturbative:

$$m_{\text{eff}} \approx m \left(1 - \frac{1}{2m^2 c^2} \sum_n A_n J_n(k_n^\alpha x_\alpha(\tau)) \xi_{\mu\nu}^{(n)} p^\mu p^\nu \right). \quad (\text{S98})$$

This highlights that the trembling corrections act to slightly renormalize the particle's inertial mass in regions of significant trembling intensity. The presence of oscillatory terms suggests that the particle's effective mass undergoes periodic variations dictated by the trembling structure of spacetime. If the fluctuation amplitudes are small, these variations appear as perturbative corrections to the mass. However, at high frequencies or in the presence of strong metric fluctuations, the effect can become significant, leading to observable shifts in mass and modifications of high-energy particle interactions. Unlike mass renormalization in quantum field theory, the effective mass correction in TSRT arises deterministically from causal geometric fluctuations, without invoking virtual particles or probabilistic field excitations.

A particularly interesting consequence arises from resonance effects. The argument of the Bessel function, $k_n^\alpha x_\alpha(\tau)$, introduces a dependency on characteristic oscillation scales in the system. When these scales align with the natural oscillatory modes of the perturbations, the correction terms can either enhance or suppress the effective mass. This resonance effect occurs when the spatial and temporal frequency components of $\xi_{\mu\nu}$ interact constructively with the motion of the particle, leading to amplification or cancellation of metric-induced contributions. The outcome depends on the relative phase and frequency of the background fluctuations, potentially allowing for controlled modulation of mass in specific energy regimes.

The key observation here is that energy is conserved in the system as a whole, despite the presence of metric-induced fluctuations. The total energy, including contributions from metric perturbations, remains constant over time. The fluctuations in the metric lead to redistribution

of energy, but do not violate the conservation of energy. Thus, the system's total energy is preserved, and the fluctuating geometry does not lead to unaccounted energy creation or loss. If spacetime fluctuations are sufficiently strong in extreme conditions, such as near black holes, in the early universe, or in high-energy accelerators, deviations in mass-energy relations could be detected. Observable modifications to energy dispersion relations, changes in particle decay rates, or oscillatory variations in mass could serve as experimental signatures of the trembling spacetime effect. These findings suggest that metric perturbations play a nontrivial role in modifying fundamental particle properties, and the explicit oscillatory structure encoded in the effective mass formula provides a framework for testing these effects in relativistic and quantum regimes.

S6.5 Comparison with Quantum Field Theory: Energy Conservation in TSRT vs. Quantum Fluctuations

In Quantum Field Theory (QFT), the conservation of energy is typically viewed as a time-averaged phenomenon, particularly when dealing with quantum fluctuations [6, 45–47]. These fluctuations are understood as temporary, random deviations from the average field value, and they are often modeled using the Heisenberg uncertainty principle [3], which introduces intrinsic uncertainty into both energy and time measurements. However, these quantum fluctuations are not usually treated as exact, but rather as statistical deviations around the field's mean value.

In QFT, the energy-momentum tensor plays a crucial role in defining energy and momentum distributions across spacetime. The energy density of a quantum field can fluctuate, but over time, these fluctuations tend to average out, ensuring that the total energy of the system remains conserved, on average. This conservation, however, does not typically apply to instantaneous quantities due to the probabilistic nature of quantum states and the inherent uncertainty in field values at specific moments in time. For instance, in processes such as vacuum fluctuations, the energy temporarily manifests as particles and antiparticles, but these fluctuations are virtual and do not violate energy conservation, as they occur within the time constraints allowed by the uncertainty principle.

In contrast, Trembling Spacetime Relativity Theory (TSRT) fundamentally differs by treating the energy changes caused by spacetime fluctuations as exact and deterministic. In TSRT, the perturbations $\xi_{\mu\nu}$ of spacetime are not probabilistic but rather a direct consequence of the dynamic, wave-like nature of spacetime itself. The energy associated with these fluctuations is not an average but a redistributed effect across spacetime. As discussed in Section S6.3, any increase in energy due to a perturbation in one region is balanced by a decrease elsewhere in spacetime, ensuring that total energy is conserved exactly at all times, without the need for time-averaging. This conservation arises from the interconnected nature of the spacetime fabric, where the perturbations propagate as waves that influence distant regions of spacetime in a manner consistent with the wave equations governing the metric fluctuations.

To clarify the distinction, in QFT, energy conservation is often understood as a time-averaged phenomenon due to the probabilistic nature of quantum fields, and while fluctuations exist, they do not violate conservation laws on a macroscopic scale. These fluctuations are typically temporary and are governed by the uncertainty principle. On the other hand, in TSRT,

energy conservation is strict and exact due to the causal nature of spacetime tremblings. These fluctuations are not probabilistic; instead, they are fully deterministic and governed by the exact structure of spacetime itself. The metric perturbations in TSRT redistribute energy across spacetime, ensuring that energy is preserved throughout the system, both locally and globally, without the need for averaging over time.

Thus, while both QFT and TSRT respect energy conservation, the key difference lies in the nature of the fluctuations. In QFT, fluctuations represent random deviations from the mean field values, and energy conservation is a statistical result based on time-averaging. In TSRT, however, fluctuations are treated as deterministic and interconnected across spacetime, leading to a precise and exact conservation of energy at all times. The energy redistribution across spacetime in TSRT is not based on averages but is driven by the interconnectedness of the trembling fabric of spacetime itself, making energy conservation an inherent property of the theory. This comparison underscores the distinction between a probabilistic, time-averaged framework (QFT) and a deterministic, exact conservation law in the context of spacetime fabric fluctuations (TSRT). Both frameworks achieve energy conservation, but TSRT does so in a manner that is more direct and deterministic, leveraging the wave-like structure of spacetime, while QFT typically relies on probabilistic averaging to achieve energy conservation over time. To understand the interplay between trembling and causal structure, we now analyze the covariance properties of metric fluctuations and their impact on proper time evolution.

S7 Supplementary Material: Covariance, Proper Time, and Causal Structure in Trembling Spacetime

This section establishes the fundamental consistency of Trembling Spacetime Relativity Theory (TSRT) with the core principles of relativity. We demonstrate that the framework preserves full general covariance under coordinate transformations and that the trembling-induced deviations respect the transformation laws of tensors and gauge fields. Furthermore, we rigorously analyze the preservation of causal structure by proving that proper time τ remains real and positive under allowed fluctuations. We show that any violation of this condition leads to physical contradictions such as closed timelike curves and breakdowns in Lorentz invariance. Finally, we examine how trembling fields transform under Lorentz and general diffeomorphic transformations to ensure compatibility with the geometric and gauge structure of spacetime.

S7.1 Covariance of TSRT Formulation

A fundamental requirement for any relativistic theory is covariance under general coordinate transformations. Trembling Spacetime Relativity Theory (TSRT) respects this principle by construction: all its fundamental quantities, including the trembling metric perturbation $\xi_{\mu\nu}$, the geodesic motion of particles, and the generalized field equations, are formulated in a manifestly covariant way. This ensures that the theory's predictions are independent of the choice of coordinates, and physical laws retain their form under any smooth transformation of spacetime variables. In standard General Relativity [21, 31, 34], the metric tensor $g_{\mu\nu}$ defines the geometry of spacetime and transforms as a symmetric rank-2 tensor under general coordinate transformations. The Christoffel symbols $\Gamma_{\alpha\beta}^{\mu}$, built from derivatives of the metric, define how vectors are parallel-transported and appear in the geodesic equation that governs the motion of free-falling test particles:

$$\frac{d^2 x^{\mu}}{d\tau^2} + \Gamma_{\alpha\beta}^{\mu} \frac{dx^{\alpha}}{d\tau} \frac{dx^{\beta}}{d\tau} = 0. \quad (\text{S99})$$

This equation is fully covariant and invariant under reparametrizations of spacetime.

In TSRT, the metric is decomposed into a smooth background and a deterministic trembling component, as in Equation (S28),

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + \xi_{\mu\nu}, \quad (\text{S100})$$

where $\bar{g}_{\mu\nu}$ represents the large-scale background geometry and $\xi_{\mu\nu}$ encodes deterministic microscopic fluctuations. Crucially, $\xi_{\mu\nu}$ transforms as a tensor field under general coordinate transformations, preserving the covariant structure of the full metric $g_{\mu\nu}$.

The geodesic equation in TSRT is generalized to include a non-zero acceleration term arising from the trembling geometry, as in Equation (S24):

$$\frac{d^2 x^{\mu}}{d\tau^2} + \Gamma_{\alpha\beta}^{\mu} \frac{dx^{\alpha}}{d\tau} \frac{dx^{\beta}}{d\tau} = \xi^{\mu}. \quad (\text{S101})$$

The term ξ^{μ} in Equation (S101), such as in Equation (S25), represents the deterministic deviation from classical geodesic motion due to local fluctuations in the spacetime geometry. It is constructed as a vector field derived covariantly from the trembling perturbation tensor $\xi_{\mu\nu}$.

To linear order in the deviation field, the trembling acceleration ξ^μ is given by:

$$\xi^\mu = -\delta\Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau}, \quad (\text{S102})$$

where $\delta\Gamma_{\alpha\beta}^\mu$ is the perturbation to the Christoffel symbols induced by $\xi_{\mu\nu}$. Explicitly, the trembling correction to the connection is:

$$\delta\Gamma_{\alpha\beta}^\mu = \frac{1}{2}\bar{g}^{\mu\sigma} (\nabla_\alpha \xi_{\beta\sigma} + \nabla_\beta \xi_{\alpha\sigma} - \nabla_\sigma \xi_{\alpha\beta}), \quad (\text{S103})$$

where ∇_α is the covariant derivative compatible with the background metric $\bar{g}_{\mu\nu}$. Since both $\xi_{\mu\nu}$ and $\bar{g}^{\mu\sigma}$ transform tensorially, and the covariant derivative preserves this transformation, the entire expression for ξ^μ transforms as a four-vector under general coordinate transformations.

Therefore, ξ^μ is not an arbitrary external forcing term but a geometrically constructed, covariant vector field arising from the internal structure of trembling spacetime. Its inclusion in the modified geodesic equation maintains the theory's full general covariance and reflects the physical influence of fine-scale metric fluctuations on particle trajectories.

In the special case of a locally inertial frame, where the background metric reduces to flat spacetime $\bar{g}_{\mu\nu} \rightarrow \eta_{\mu\nu}$, the covariant derivatives become partial derivatives and the expression for the trembling correction simplifies accordingly. In this approximation, we write:

$$\delta\Gamma_{\alpha\beta}^\mu \approx \frac{1}{2}\eta^{\mu\sigma} (\partial_\alpha \xi_{\beta\sigma} + \partial_\beta \xi_{\alpha\sigma} - \partial_\sigma \xi_{\alpha\beta}), \quad (\text{S104})$$

leading to a trembling-induced acceleration vector of the form:

$$\xi^\mu \approx -\delta\Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau}. \quad (\text{S105})$$

This flat-space limit confirms that the trembling deviation ξ^μ corresponds to a geometric acceleration induced by the fine-scale variation of the metric, and provides a practical form for evaluating trembling corrections in experimental or computational settings where the background curvature is negligible. Despite this simplification, the underlying construction remains consistent with the fully covariant formulation, reinforcing that the trembling corrections are rooted in geometry, not external fields. As such, the full equation remains covariant under general coordinate transformations.

The same principle applies to the field equations. In standard GR [21, 31, 34], Einstein's field equations read:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}. \quad (\text{S106})$$

In TSRT, the field equations are modified to incorporate both geometric fluctuations and their induced energy-momentum contributions. This leads to an extended formulation, as in Equation (S49), or in the formulation of Section S5, Equation (S41) in particular,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} + \mathcal{H}_{\mu\nu} = \frac{8\pi G}{c^4}(T_{\mu\nu} + \mathcal{M}_{\mu\nu}), \quad (\text{S107})$$

where $\mathcal{H}_{\mu\nu}$ and $\mathcal{M}_{\mu\nu}$, defined, respectively, in Equation (S40) and Equation (S42), represent,

respectively, the geometric and stress-energy fluctuations induced by trembling, and are defined explicitly in terms of tensorial quantities, e.g., in terms of Bessel-mode expansions. Since all added terms are built from tensors and their covariant derivatives, the field equations remain manifestly covariant.

The same applies to all gauge fields and interaction terms. For example, Maxwell's equations, as derived in Section S10, in the presence of trembling remain written in covariant form:

$$\nabla_\mu F^{\mu\nu} = J^\nu, \quad (\text{S108})$$

where both $F^{\mu\nu}$ and J^ν transform as tensors, and ∇_μ is the covariant derivative with respect to the full fluctuating metric $g_{\mu\nu}$.

Thus, all dynamical and geometric quantities in TSRT transform covariantly under general coordinate transformations. The theory fully preserves general covariance, just like GR, but extends the dynamical content of spacetime by including deterministic, proper-time-based metric fluctuations. This ensures that TSRT remains consistent with the foundational symmetry principles of modern physics while introducing novel geometric structure at the smallest scales.

S7.2 Constraints to Preserve Covariance

To guarantee that Trembling Spacetime Relativity Theory (TSRT) remains fully covariant, all elements of the theory—geometric, dynamical, and phenomenological—must be expressed entirely in terms of tensorial quantities that obey the transformation laws of general relativity. This requirement is not merely aesthetic but foundational: violation of tensorial transformation behavior would imply that physical laws depend on arbitrary choices of coordinates, contradicting the principle of general covariance that underlies all relativistic theories.

Transformation Properties of $\xi_{\mu\nu}$: The trembling component $\xi_{\mu\nu}$, introduced via the perturbed metric, in Equation (S28),

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + \xi_{\mu\nu}, \quad (\text{S109})$$

must be a symmetric rank-2 tensor under general coordinate transformations. Formally, if $x^\mu \rightarrow x'^\mu(x)$ is a smooth diffeomorphism, then $\xi_{\mu\nu}$ must transform as:

$$\xi'_{\alpha\beta}(x') = \frac{\partial x^\mu}{\partial x'^\alpha} \frac{\partial x^\nu}{\partial x'^\beta} \xi_{\mu\nu}(x). \quad (\text{S110})$$

If this condition is not satisfied—e.g., if $\xi_{\mu\nu}$ is constructed with components that transform non-covariantly—then the full metric $g_{\mu\nu}$ becomes ill-defined as a geometric object. This would invalidate the use of covariant derivatives, curvature tensors, and proper time measurements.

Tensorial Formulation of Fluctuation Contributions: Any correction to the stress-energy tensor arising from trembling must also be a rank-2 tensor. For instance, in TSRT we define the effective stress-energy tensor, as in Equation (S83), as:

$$T_{\text{eff}}^{\mu\nu} = \bar{T}^{\mu\nu} + \delta T^{\mu\nu}, \quad (\text{S111})$$

where $\delta T^{\mu\nu} \sim \xi_{\alpha\beta} p^\alpha p^\beta \bar{T}^{\mu\nu}$ is built from proper contractions of covariant quantities. If $\delta T^{\mu\nu}$ were to depend on non-tensorial scalars, or quantities derived from coordinate-specific expansions, the total stress-energy tensor would lose its tensorial nature. As a result, the Einstein equations would become coordinate-dependent, violating both general covariance and physical consistency.

Proper Time and Causal Consistency in TSRT: In TSRT, the proper time along a trembling trajectory, is introduced in Equation (S119), and is given by:

$$d\tau^2 = g_{\mu\nu} dx^\mu dx^\nu, \quad (\text{S112})$$

where $g_{\mu\nu} = \bar{g}_{\mu\nu} + \xi_{\mu\nu}$ includes both the background geometry and the trembling-induced deviation. To preserve causality, the proper time must remain real and positive along all physically admissible paths, i.e., $d\tau^2 > 0$. This condition places constraints on the allowed structure of the trembling field $\xi_{\mu\nu}$, ensuring that the perturbed metric never leads to backwards or imaginary temporal evolution. Importantly, since $d\tau^2$ must be a scalar under general coordinate transformations, all contributions to $\xi_{\mu\nu}$ must be constructed from covariant quantities. TSRT enforces this by requiring trembling-induced corrections to be formed only from tensors and their scalar contractions, such as $\xi_{\mu\nu} u^\mu u^\nu$, ensuring full geometric consistency and covariance.

Rigorous Use of Covariant Expansions: In practical implementations of TSRT, perturbative methods and series expansions (e.g., Fourier-Bessel decompositions of $\xi_{\mu\nu}$) are often employed. To preserve covariance, all expansions must be carried out in terms of coordinate-invariant basis functions (e.g., functions of $k_n^\alpha x_\alpha(\tau)$, as in Equation (S26), which are scalars). Additionally, the coefficients α_n, β_n and all derived quantities (like $\phi(\xi)$ in Equation (S56)) must be defined via tensor contractions or scalar integrals. Furthermore, no expansion should truncate or neglect terms that carry transformation weights under coordinate shifts unless justified in a manifestly covariant limit (e.g., locally inertial frames).

Example: Coordinate-Dependent Artifact and Its Resolution: Suppose one were to define the trembling correction as:

$$\delta T_{\text{naive}}^{\mu\nu} = \sum_n \alpha_n \cos(\omega_n t) T^{\mu\nu}, \quad (\text{S113})$$

where t is coordinate time. This formulation is non-covariant because t is not an invariant quantity. A proper TSRT-compliant formulation would instead use:

$$\delta T_{\text{covariant}}^{\mu\nu} = \sum_n \alpha_n J_n(k_n^\alpha x_\alpha(\tau)) T^{\mu\nu}, \quad (\text{S114})$$

where $k_n^\alpha x_\alpha(\tau)$ is a scalar under coordinate transformations and J_n are covariant scalar functions.

As a result, TSRT preserves full covariance by enforcing strict tensorial construction of all dynamical and geometric quantities. Metric perturbations, energy-momentum corrections, proper time, and field equations are all formulated in a way that transforms consistently under

arbitrary coordinate changes. This safeguards the physical invariance of the theory and guarantees that all measurable predictions are independent of the observer's frame or coordinate system.

S7.3 Causal Structure and Positivity of Proper Time

In Section S7.2, we highlighted that the proper time τ must always remain positive and real for the theory to be physically consistent and causal. To strengthen this requirement and provide the necessary rigor, we now formalize this condition and demonstrate that any violation of $d\tau^2 > 0$ leads to contradictions with causality.

S7.3.1 Formalizing Causality in Spacetime Trembling

Recall that the proper time τ , whether in GR or in TSRT, is defined through the spacetime interval:

$$d\tau^2 = g_{\mu\nu} dx^\mu dx^\nu. \quad (\text{S115})$$

In the presence of trembling spacetime, the spacetime coordinates $x^\mu(\tau)$ become functions of the experienced proper time τ , leading to fluctuations around the equilibrium geometry, as introduced in Equation (3):

$$x^\mu(\tau) = x_0^\mu + \delta x^\mu(\tau). \quad (\text{S116})$$

Therefore, the expression for the proper time becomes:

$$d\tau^2 = g_{\mu\nu} (dx_0^\mu + \delta x^\mu(\tau)) (dx_0^\nu + \delta x^\nu(\tau)). \quad (\text{S117})$$

Expanding this expression yields:

$$d\tau^2 = g_{\mu\nu} dx_0^\mu dx_0^\nu + 2g_{\mu\nu} dx_0^\mu \delta x^\nu(\tau) + g_{\mu\nu} \delta x^\mu(\tau) \delta x^\nu(\tau). \quad (\text{S118})$$

The first term $g_{\mu\nu} dx_0^\mu dx_0^\nu$ represents the proper time of the unperturbed, equilibrium system. The additional terms account for the influence of trembling perturbations on the local measurement of proper time.

To preserve causality, the total proper time must remain real and positive. Thus, the condition

$$d\tau^2 > 0 \quad (\text{S119})$$

must hold for all allowed perturbations $\delta x^\mu(\tau)$. This ensures that the proper time continues to increase monotonically and never becomes negative or imaginary, which would imply non-physical time reversal or breakdowns in causal structure.

S7.3.2 Proof of Causality Conditions

We now show rigorously that allowing $d\tau^2 \leq 0$ results in causal contradictions.

The proper time τ is a measurable, real quantity. If a perturbation $\delta x^\mu(\tau)$ were to reduce

the total squared interval such that:

$$d\tau^2 = g_{\mu\nu} (dx_0^\mu + \delta x^\mu(\tau)) (dx_0^\nu + \delta x^\nu(\tau)) \leq 0, \quad (\text{S120})$$

then τ would either cease to progress (for $d\tau^2 = 0$) or become imaginary (for $d\tau^2 < 0$). Both outcomes are unacceptable: in the first case, time effectively halts; in the second, the geometry becomes non-causal and permits physically forbidden scenarios. A study exposing such effects in GR was published, for instance, by Godel in 1949 [48].

In particular, when $d\tau^2 \leq 0$, the trembling-induced metric can lead to paths that loop back in time. These closed timelike curves (CTCs) violate the chronological ordering of events and give rise to paradoxes, such as the possibility for an effect to precede its cause. To prevent the emergence of CTCs, TSRT requires that all admissible spacetime perturbations satisfy $d\tau^2 > 0$ at all points along any physical worldline. Hence, the strict condition $d\tau^2 > 0$ guarantees that the geometry of trembling spacetime remains compatible with causality and forbids pathological behavior such as time reversal or paradoxical loops.

S7.3.3 Detailed Example of Metric Perturbation and Symmetry Preservation

As an illustrative case, consider a trembling trajectory in a flat Minkowski background. The full coordinate position of a particle is given, as in Equation (S116), by:

$$x^\mu(\tau) = x_0^\mu + \delta x^\mu(\tau), \quad (\text{S121})$$

where x_0^μ is the unperturbed geodesic and $\delta x^\mu(\tau)$ represents the trembling fluctuation around the equilibrium geometry. The line element, incorporating the effect of these fluctuations, is then written as:

$$d\tau^2 = \eta_{\mu\nu} (dx_0^\mu + d(\delta x^\mu(\tau))) (dx_0^\nu + d(\delta x^\nu(\tau))), \quad (\text{S122})$$

where $\eta_{\mu\nu}$ denotes the flat Minkowski metric with signature $(-, +, +, +)$, valid in a local inertial frame. This formulation ensures that all contributions to the proper time remain tensorial and covariant.

The fluctuation field $\delta x^\mu(\tau)$ must transform covariantly under symmetry operations such as Lorentz boosts. For a boost along the x -direction, the trembling displacement transforms as:

$$\delta x^\mu(\tau) \rightarrow \delta x'^\mu(\tau) = \Lambda^\mu_\nu \delta x^\nu(\tau), \quad (\text{S123})$$

where Λ^μ_ν is the Lorentz transformation matrix. This guarantees that $\delta x^\mu(\tau)$ behaves as a proper tangent vector under coordinate changes, ensuring relativistic consistency.

Expanding the interval yields:

$$d\tau^2 = \eta_{\mu\nu} dx_0^\mu dx_0^\nu + 2\eta_{\mu\nu} dx_0^\mu d(\delta x^\nu(\tau)) + \eta_{\mu\nu} d(\delta x^\mu(\tau)) d(\delta x^\nu(\tau)). \quad (\text{S124})$$

The first term represents the background proper time, while the latter two encode the trembling-induced deviations.

Crucially, the total quantity $d\tau^2$ is a scalar under general coordinate transformations. However, scalar invariance alone does not ensure causal integrity. TSRT imposes a stronger physical requirement: the scalar $d\tau^2$ must remain strictly positive along all physically realizable paths. In practical terms, this means that the linear and quadratic corrections due to $\delta x^\mu(\tau)$ must not dominate the background term in a way that renders the interval non-timelike.

Thus, the physically admissible class of perturbations is constrained by the causal inequality:

$$\eta_{\mu\nu} (dx_0^\mu + \delta x^\mu(\tau)) (dx_0^\nu + \delta x^\nu(\tau)) > 0, \quad (\text{S125})$$

which must hold in all inertial frames.

To confirm that the condition $d\tau^2 > 0$ remains valid under Lorentz transformations, we must verify not only that the interval $d\tau^2$ is a scalar but that its positivity is preserved across all inertial frames. We begin by recalling that both dx_0^μ , the differential of the unperturbed trajectory, and $\delta x^\mu(\tau)$, the trembling displacement, transform as contravariant vectors under Lorentz transformations. The Minkowski metric $\eta_{\mu\nu}$ is invariant under Lorentz transformations, i.e.,

$$\eta_{\mu\nu} = \Lambda_\mu^\alpha \Lambda_\nu^\beta \eta_{\alpha\beta}, \quad (\text{S126})$$

for any Lorentz transformation matrix Λ_ν^μ .

Let us compute the transformed proper time interval:

$$d\tau'^2 = \eta_{\mu\nu} (dx_0'^\mu + d(\delta x'^\mu)) (dx_0'^\nu + d(\delta x'^\nu)) \quad (\text{S127})$$

$$= \eta_{\mu\nu} \left(\Lambda_\alpha^\mu dx_0^\alpha + \Lambda_\beta^\mu d(\delta x^\beta) \right) \left(\Lambda_\gamma^\nu dx_0^\gamma + \Lambda_\delta^\nu d(\delta x^\delta) \right) \quad (\text{S128})$$

$$= \eta_{\mu\nu} \Lambda_\alpha^\mu \Lambda_\gamma^\nu dx_0^\alpha dx_0^\gamma + 2\eta_{\mu\nu} \Lambda_\alpha^\mu \Lambda_\delta^\nu dx_0^\alpha d(\delta x^\delta) + \eta_{\mu\nu} \Lambda_\beta^\mu \Lambda_\delta^\nu d(\delta x^\beta) d(\delta x^\delta) \quad (\text{S129})$$

$$= \eta_{\alpha\gamma} dx_0^\alpha dx_0^\gamma + 2\eta_{\alpha\delta} dx_0^\alpha d(\delta x^\delta) + \eta_{\beta\delta} d(\delta x^\beta) d(\delta x^\delta), \quad (\text{S130})$$

where we have used the invariance of the Minkowski metric under Lorentz transformations in the final step.

The equality of the results in Equation (S124) and Equation (S130) shows that the full perturbed proper time interval $d\tau^2$ is invariant:

$$d\tau'^2 = d\tau^2. \quad (\text{S131})$$

Therefore, if $d\tau^2 > 0$ in one inertial frame, it follows that $d\tau'^2 > 0$ in all Lorentz-related frames. Hence, the positivity of the proper time interval is preserved under Lorentz transformations, not just its scalar value. This ensures that the causality condition $d\tau^2 > 0$ is frame-independent and physically meaningful. In TSRT, this guarantees that trembling-induced deviations do not lead to causality violations simply by switching reference frames.

In a general (curved) spacetime, the proper time is given by:

$$d\tau^2 = g_{\mu\nu} (dx_0^\mu + d(\delta x^\mu(\tau))) (dx_0^\nu + d(\delta x^\nu(\tau))). \quad (\text{S132})$$

While the specific components of $g_{\mu\nu}$ vary across coordinate systems, the above expression

remains a scalar. In the neighborhood of any spacetime point p , one can construct a local coordinate system known as Riemann normal coordinates, in which the metric $g_{\mu\nu}(x)$ reduces to the Minkowski form $\eta_{\mu\nu}$ at the point p , and its first derivatives vanish. More precisely, in these coordinates, the metric expansion around p takes the form:

$$g_{\mu\nu}(x) = \eta_{\mu\nu} - \frac{1}{3}R_{\mu\alpha\nu\beta}(p)x^\alpha x^\beta + \mathcal{O}(x^3), \quad (\text{S133})$$

where $R_{\mu\alpha\nu\beta}(p)$ is the Riemann curvature tensor evaluated at the point p . This construction ensures that, locally, the laws of physics take the same form as in flat spacetime up to curvature corrections of order x^2 , allowing the use of inertial-frame arguments such as Lorentz invariance and positivity of $d\tau^2$ to remain valid in sufficiently small neighborhoods. For a comprehensive treatment of Riemann normal coordinates and their derivation, see Section 5.2 of *General Relativity* by Robert M. Wald [28].

To ensure that $d\tau^2 > 0$ remains satisfied not merely locally or in specific coordinates but throughout all admissible physical contexts, TSRT imposes a fundamental restriction on the class of allowed trembling fluctuations $\delta x^\mu(\tau)$. This is not a heuristic constraint but a structural consequence of the theory's formulation. The proper time along a perturbed trajectory is given by Equation (S132) and its positivity must be preserved to maintain causal consistency. Expanding this expression yields:

$$d\tau^2 = g_{\mu\nu}dx_0^\mu dx_0^\nu + 2g_{\mu\nu}dx_0^\mu d(\delta x^\nu(\tau)) + g_{\mu\nu}d(\delta x^\mu(\tau))d(\delta x^\nu(\tau)). \quad (\text{S134})$$

The first term is positive for a timelike background trajectory. The second and third terms represent linear and quadratic corrections due to trembling, and must remain bounded so as not to negate the unperturbed contribution.

TSRT ensures this through the geometric and variational structure of the trembling field $\delta x^\mu(\tau)$.

In particular:

1. **Covariant Boundedness:** All trembling displacements are defined through smooth basis functions (e.g., Bessel expansions or bounded oscillatory profiles) with coefficients that are locally finite and subject to convergence constraints, ensuring that no component of $\delta x^\mu(\tau)$ diverges or grows without bound in proper time. This is enforced at the level of initial conditions and is preserved under the dynamical evolution governed by the trembling geodesic equation.

2. **Action-based Filtering:** The physical modes of trembling arise from variations of a proper-time action functional constrained by the positivity of $d\tau^2$. As such, modes that would lead to vanishing or negative intervals are dynamically excluded from the spectrum—just as non-causal geodesics are not realized in standard general relativity.

3. **Causal Compatibility Check:** For every admissible solution $\delta x^\mu(\tau)$, the expression

$$g_{\mu\nu}(dx_0^\mu + \delta x^\mu(\tau))(dx_0^\nu + \delta x^\nu(\tau)) \quad (\text{S135})$$

is explicitly checked against the inequality > 0 at all points along the worldline. This is not merely a boundary condition, but a dynamical constraint imposed across the full proper-time domain, encoded into the allowed functional form of the trembling field.

4. **Covariance and Frame Independence:** Because all relevant quantities—metric tensors, displacements, and differentials—transform covariantly, the positivity of $d\tau^2$ is preserved under arbitrary coordinate changes. Thus, a fluctuation that is causally admissible in one frame remains so in all frames. This was shown explicitly under Lorentz transformations, and in curved spacetime it holds in any normal neighborhood via Riemann normal coordinates.

In other words, the statement that TSRT restricts the trembling field to ensure $d\tau^2 > 0$ is not an assumption but a theorem embedded in the formulation of the theory. It arises from a combination of tensorial construction, bounded geometric dynamics, variational filtering, and frame-invariant positivity checks. Therefore, the class of permissible trembling fields is intrinsically selected to maintain causal flow, and the condition

$$g_{\mu\nu} (dx_0^\mu + \delta x^\mu(\tau)) (dx_0^\nu + \delta x^\nu(\tau)) > 0 \quad (\text{S136})$$

is universally satisfied across all physical situations and coordinate systems permitted by the structure of TSRT.

Therefore, the causal constraint $d\tau^2 > 0$ is preserved both under Lorentz transformations in flat spacetime and under general diffeomorphisms in curved backgrounds. TSRT ensures this through its covariant construction of trembling fields, and by dynamically restricting fluctuation amplitudes to avoid causal violations. The forward progression of proper time is thus a geometric consequence of the theory's internal consistency—not an added postulate—and applies uniformly across all admissible frames.

S7.4 Covariant Transformations of Metric Fluctuations

The goal of this subsection is to rigorously demonstrate that all trembling-induced fluctuations in Trembling Spacetime Relativity Theory (TSRT) transform covariantly under general coordinate transformations. This is essential for ensuring the internal consistency of the theory and for verifying that the trembling contributions—despite their dynamical character—do not introduce preferred frames or violate the principle of general covariance. In what follows, we will consider both general diffeomorphisms and Lorentz transformations as special cases and show that all geometric and gauge-related quantities constructed from trembling degrees of freedom transform correctly.

S7.4.1 Transformation of Trembling Displacements Under Diffeomorphisms

A central requirement of any relativistic theory, including general relativity and TSRT, is general covariance. This principle states that the laws of physics—and all geometrically meaningful quantities—must retain their form under arbitrary smooth coordinate transformations. In other

words, the equations describing nature must be valid in any admissible coordinate system, regardless of the observer's motion or location.

Let us consider a general coordinate transformation:

$$x^\mu \longrightarrow x'^\mu = f^\mu(x), \quad (\text{S137})$$

where $f^\mu(x)$ is a smooth, invertible mapping that re-labels spacetime points using a different coordinate system. To describe how quantities change under a smooth coordinate transformation $x^\mu \mapsto x'^\mu(x)$, one uses the Jacobian matrix, which captures how small changes in the original coordinates propagate into changes in the new coordinates. It is defined as:

$$J^\mu_\nu = \frac{\partial x'^\mu}{\partial x^\nu}, \quad (\text{S138})$$

and determines how infinitesimal displacements, vectors, and tensors transform under general diffeomorphisms. In physics, the Jacobian plays a crucial role in ensuring that transformations preserve the tensorial nature of physical quantities and that the laws of nature remain valid in all coordinate systems [28].

This Jacobian matrix is not merely a technical tool—it plays a foundational role throughout TSRT. In fact, every covariant transformation law discussed in this work, whether for the trembling displacement $\delta x^\mu(\tau)$, the metric fluctuation $\xi_{\mu\nu}$, or the induced fields such as A_μ , ultimately derives from the Jacobian. For instance, the transformation rule for any contravariant vector is determined by the Jacobian $J^\mu_\nu = \partial x'^\mu / \partial x^\nu$, while covariant objects like $\xi_{\mu\nu}$ or A_μ transform using the inverse Jacobian $\partial x^\alpha / \partial x'^\mu$. In this way, the structure of the Jacobian ensures that all trembling-induced quantities retain their correct geometric roles under diffeomorphisms. The TSRT formalism is thus built to obey the transformation rules dictated by the Jacobian from the ground up, affirming its full compliance with general covariance. In TSRT, the trembling displacement field $\delta x^\mu(\tau)$ describes the deviation of a particle's position from its unperturbed geodesic trajectory due to local fluctuations in the spacetime geometry. Since these displacements are physical deviations occurring in spacetime, they must behave as geometric vectors. That is, their transformation law under a coordinate change must follow the standard vector rule:

$$\delta x'^\mu(\tau) = \frac{\partial x'^\mu}{\partial x^\nu} \delta x^\nu(\tau) = J^\mu_\nu \delta x^\nu(\tau). \quad (\text{S139})$$

This requirement is more than a mathematical formality—it guarantees that the description of trembling remains physically meaningful and consistent from one observer to another. For example, two observers in different coordinate systems should agree on whether the trembling displacement points forward in time, sideways in space, or violates causal flow. If $\delta x^\mu(\tau)$ did not transform as a vector, then its physical interpretation would vary from frame to frame, contradicting the principle that spacetime events and their deformations should be objectively describable. Moreover, many derived quantities in TSRT—such as the perturbed proper time, the deviation acceleration in the trembling geodesic equation, and even the induced stress-energy corrections—are constructed by contracting $\delta x^\mu(\tau)$ with other tensors (e.g., the metric $g_{\mu\nu}$, the Christoffel symbols, or curvature tensors). If $\delta x^\mu(\tau)$ did not transform as a vector,

these contractions would fail to produce coordinate-invariant scalars or tensors, and the theory would no longer be covariant.

Since trembling displacements $\delta x^\mu(\tau)$ are defined as geometric vectors and transform accordingly under general coordinate transformations, it follows that TSRT maintains observer-independent predictions and upholds the full symmetry structure of general relativity.

S7.4.2 Example: Lorentz Transformations as a Special Case

To make the general transformation behavior of trembling displacements more concrete, we now examine a special but important case: Lorentz transformations. These are a subset of general coordinate transformations that preserve the Minkowski spacetime interval $ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$ and relate inertial observers moving at constant relative velocities. Lorentz transformations are essential to special relativity and play a key local role in general relativity, where the geometry becomes approximately flat in sufficiently small regions of spacetime.

Let us consider a Lorentz boost along the x -direction with velocity v , corresponding to the transformation:

$$t' = \gamma(t - \beta x), \quad x' = \gamma(x - \beta t), \quad y' = y, \quad z' = z, \quad (\text{S140})$$

where $\beta = v/c$ and $\gamma = 1/\sqrt{1 - \beta^2}$. This transformation redefines the spacetime coordinates as seen by an observer moving with velocity v along the x -axis relative to the original inertial frame.

Now consider a trembling displacement vector in the original frame:

$$\delta x^\mu(\tau) = (\delta t(\tau), \delta x(\tau), \delta y(\tau), \delta z(\tau)), \quad (\text{S141})$$

which represents the deviation of a particle's worldline from its classical geodesic due to trembling spacetime. Because this is a geometric vector, it must transform under the Lorentz boost via the standard transformation rule:

$$\delta x'^\mu(\tau) = \Lambda^\mu{}_\nu \delta x^\nu(\tau), \quad (\text{S142})$$

where $\Lambda^\mu{}_\nu$ is the Lorentz transformation matrix corresponding to the boost.

Explicitly, applying the matrix to the trembling components gives:

$$\delta t' = \gamma(\delta t - \beta \delta x), \quad (\text{S143})$$

$$\delta x' = \gamma(\delta x - \beta \delta t), \quad (\text{S144})$$

$$\delta y' = \delta y, \quad (\text{S145})$$

$$\delta z' = \delta z. \quad (\text{S146})$$

This result shows how the trembling displacement vector components mix under the boost, exactly as expected for four-vectors in Minkowski spacetime. Crucially, the transformed displacements still represent valid geometric vectors in the new coordinate system. That is, an observer moving relative to the original frame would measure different numerical values for the trembling displacements, but would describe them using the same physical laws—because

the transformation preserves the geometric structure. The preservation of the vector character of $\delta x^\mu(\tau)$ under Lorentz transformations ensures that any scalar or tensor quantity built from these displacements—such as the perturbed proper time, curvature corrections, or energy contributions—will retain their correct transformation behavior. For instance, the proper time interval $d\tau^2 = \eta_{\mu\nu} dx^\mu dx^\nu$ remains invariant under these transformations, even when the path includes trembling fluctuations.

Therefore, this example not only illustrates the general transformation principle but also confirms that TSRT remains fully consistent with the symmetry structure of special relativity. This is a necessary condition for any physically viable theory, and TSRT satisfies it at the level of both geometric displacements and their physical consequences.

S7.4.3 Diffeomorphism Covariance of Metric Fluctuations

A central pillar of both General Relativity and TSRT is the principle of general covariance: all physically meaningful equations must take the same form under arbitrary smooth coordinate transformations, known mathematically as diffeomorphisms. This ensures that the theory does not privilege any particular coordinate system, but rather expresses the laws of nature in an entirely geometric—and thus observer-independent—form. In TSRT, the spacetime metric is not fixed but subject to small, deterministic fluctuations superimposed on a smooth background geometry. This leads to the decomposition, as in Equation (S28):

$$g_{\mu\nu}(x) = \bar{g}_{\mu\nu}(x) + \xi_{\mu\nu}(x), \quad (\text{S147})$$

where $\bar{g}_{\mu\nu}$ denotes the smooth background metric—typically satisfying Einstein’s equations at large scales—and $\xi_{\mu\nu}$ represents the trembling-induced metric fluctuations due to microscopic geometric deviations.

Since both $g_{\mu\nu}$ and $\bar{g}_{\mu\nu}$ are symmetric rank-2 tensors, the fluctuation field $\xi_{\mu\nu}$, defined by their difference, must also transform as a symmetric (0,2) tensor field. That is, under a general coordinate transformation $x^\mu \rightarrow x'^\mu(x)$, the fluctuation field must obey, as in Equation (S110):

$$\xi'_{\mu\nu}(x') = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} \xi_{\alpha\beta}(x). \quad (\text{S148})$$

This transformation rule guarantees that $\xi_{\mu\nu}$ behaves consistently with the underlying geometric structure of spacetime and that it contributes correctly to the perturbed metric in any coordinate system.

Why is this important? In TSRT, $\xi_{\mu\nu}$ is not merely a perturbative field—it is a dynamical source of curvature, energy, and geometric deformation. Physical quantities derived from $\xi_{\mu\nu}$, such as the trembling-induced correction to the energy-momentum tensor, or the additional force term in the trembling geodesic equation, must themselves be covariant.

For example, the scalar contraction $\xi_{\mu\nu} p^\mu p^\nu$, as in Equation (S56), Equation (S58), or Equation (S69), where p^μ is a four-momentum vector, must yield the same physical value in all coordinate systems. This is ensured by the tensorial nature of both $\xi_{\mu\nu}$ and p^μ . Furthermore, covariant derivatives such as $\nabla_\lambda \xi_{\mu\nu}$, which appear in curvature calculations and dynamical field equations, such as Equation (S210), must transform appropriately under diffeomorphisms. This

is only possible if $\xi_{\mu\nu}$ itself transforms as a proper tensor field.

In summary, the transformation properties of the trembling metric perturbation $\xi_{\mu\nu}$ under general coordinate transformations are not incidental—they are essential to the mathematical and physical coherence of TSRT. By definition, $\xi_{\mu\nu}$ arises from the difference between the full spacetime metric $g_{\mu\nu}$ and the background metric $\bar{g}_{\mu\nu}$, both of which are (0,2) tensor fields. Since the space of tensors is closed under linear operations, $\xi_{\mu\nu}$ necessarily transforms as a covariant rank-2 tensor, given in Equation (S148). This transformation law is not imposed externally—it follows directly from the geometric definition of $\xi_{\mu\nu}$ within the framework of differential geometry. Consequently, any scalar or tensor constructed from $\xi_{\mu\nu}$ by contraction with other geometric quantities (such as four-momentum p^μ , four-velocity u^μ , or the metric itself) will transform correctly and yield coordinate-invariant physical content. For example, the scalar $\xi_{\mu\nu}u^\mu u^\nu$, appearing in proper-time normalization, and the correction term $\mathcal{M}_{\mu\nu}$ in the extended Einstein field equations, as in Equation (S42), retain their tensorial status under general diffeomorphisms.

This guarantees that all geometric, kinematic, and dynamic effects introduced by trembling spacetime fluctuations preserve their form and physical interpretation across arbitrary coordinate systems. The full diffeomorphism invariance of TSRT is therefore not a stated assumption, but a structural consequence of how trembling is embedded into the covariant formalism of general relativity.

S7.4.4 Implications for Gauge Field Consistency

One of the striking consequences of TSRT is that trembling-induced geometric fluctuations can give rise to effective gauge fields, as outlined in Sections S9–S13. These emergent fields arise naturally when the metric perturbations $\xi_{\mu\nu}$ or the coordinate deviations $\delta x^\mu(\tau)$ are contracted with other geometric or matter fields in a covariant manner. For example, when projected onto a particle’s four-velocity or momentum, these fluctuations can induce additional force-like terms that resemble electromagnetic or Yang–Mills gauge interactions [49].

To ensure physical consistency, any such emergent gauge field must transform correctly under coordinate transformations. Specifically, if an effective gauge potential $A_\mu(x)$ arises from a contraction or projection involving the trembling geometry, then A_μ must behave as a covariant vector field under diffeomorphisms. This means that under a general coordinate transformation $x^\mu \rightarrow x'^\mu(x)$, the transformation rule for the induced field is:

$$A_\mu(x) \rightarrow A'_\mu(x') = \frac{\partial x^\nu}{\partial x'^\mu} A_\nu(x). \quad (\text{S149})$$

This guarantees that the field has the correct geometric properties in all coordinate systems and ensures that its interaction with matter fields remains consistent and observer-independent.

Moreover, the physical significance of these induced gauge fields lies not merely in their transformation properties, but in the structure of their associated field strengths. The field strength tensor, defined by the covariant derivative of the gauge potential,

$$F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu, \quad (\text{S150})$$

must itself transform as a rank-2 antisymmetric tensor. This transformation behavior is crucial: it ensures that the observable effects of the field—such as forces on particles or energy stored in the field—are gauge-invariant and consistent across frames.

In TSRT, such structures arise not from imposing gauge symmetry as a postulate, but from the geometric interplay of trembling displacements with the covariant structure of spacetime. The induced field A_μ is thus a consequence of the geometry, and its proper transformation law follows naturally from the tensorial properties of its origin. In summary, the emergence of gauge-like fields in TSRT respects the same fundamental requirement of covariance that governs all other quantities in the theory. This reinforces the geometric unity of TSRT: both gravity and gauge interactions can be traced to metric fluctuations, and all resulting fields transform consistently with the underlying structure of general relativity.

S7.5 Conclusion

In this section, we have shown in detail that all key geometric quantities in Trembling Spacetime Relativity Theory (TSRT)—including the trembling displacements $\delta x^\mu(\tau)$, the metric fluctuations $\xi_{\mu\nu}$, and any induced gauge-like fields A_μ —transform consistently under general coordinate transformations. This consistency is essential for preserving the principle of general covariance, which lies at the heart of general relativity and all modern, correct, field theories formulated on curved spacetimes.

First, we demonstrated that the trembling displacements behave as tangent vectors and retain their correct transformation properties under both local Lorentz boosts and general diffeomorphisms. We then extended this to the metric perturbations $\xi_{\mu\nu}$, which, being derived from a difference of tensors, naturally inherit the transformation rules of rank-2 covariant tensors. We further confirmed that any fields constructed from these geometric fluctuations—such as effective gauge potentials or induced forces—also transform in a covariant fashion, ensuring that their observable effects are invariant under changes of coordinates.

These results collectively guarantee that TSRT introduces no anomalies or inconsistencies at the level of geometric or dynamical structure. The trembling framework extends general relativity not by breaking covariance or postulating new fields arbitrarily, but by embedding fine-scale deterministic fluctuations directly into the metric, while maintaining full compatibility with the relativistic spacetime structure. Thus, the entire formulation of TSRT—from the description of particle motion to the emergence of forces—is constructed in a geometrically rigorous, covariant manner. This affirms that trembling spacetime is not just a theoretical extension, but a viable and physically consistent framework for describing the fundamental structure of spacetime and its interaction with matter. We next prove that trembling must occur in the proper time parameter τ rather than in coordinate time t , ensuring preservation of causal order.

S8 Supplementary Material: Why Trembling Must Occur in Proper Time τ , and Not Coordinate Time t

A foundational principle of TSRT is that trembling spacetime is defined locally along particle worldlines as a function of proper time τ , rather than global coordinate time t . This section provides a detailed justification for this formulation from physical, geometric, and causal standpoints. While proper time depends on the metric—which itself includes trembling corrections—its construction remains covariant because both the metric $g_{\mu\nu}(x)$ and the differential displacement dx^μ transform consistently under general coordinate transformations. As a result, proper time τ , defined by $d\tau^2 = g_{\mu\nu}(x)dx^\mu dx^\nu$, retains its status as a scalar under diffeomorphisms, even in a trembling spacetime. It thereby serves as the unique evolution parameter compatible with the causal structure, relativistic symmetry, and deterministic foundations of TSRT. In contrast, using coordinate time t would violate general covariance, introduce frame-dependent definitions of trembling, and obscure the physical locality and geometric consistency that TSRT demands.

Defining trembling as a function $\Xi^\mu(t)$ in Equation (2), with t as the evolution parameter, would tie the dynamics of trembling to an arbitrary foliation, thereby breaking both Lorentz invariance [1] and general covariance. Such a formulation would assign physical significance to a coordinate convention, violating the geometric principle that all physical laws must be expressed in an invariant, observer-independent manner. In a fully geometric theory such as TSRT, this is unacceptable. Proper time τ , on the other hand, is defined intrinsically along the worldline via

$$d\tau^2 = g_{\mu\nu}(x) dx^\mu dx^\nu, \quad (\text{S151})$$

and transforms as a scalar under general coordinate transformations. In TSRT, the metric $g_{\mu\nu}(x) = \bar{g}_{\mu\nu}(x) + \xi_{\mu\nu}(x)$ includes trembling corrections, and the coordinates $x^\mu(\tau)$ themselves undergo small-scale deviations. Nevertheless, because both the metric and the displacement transform covariantly, the scalar nature of $d\tau$ remains intact at every point along the trajectory.

A subtlety arises from the fact that τ itself depends on the fluctuating metric and the trembling path $x^\mu(\tau)$, creating a mutual dependence between the evolution parameter and the dynamical variables. However, this does not undermine the use of τ as a parameter: the proper time is constructed locally and instantaneously along the worldline, using the perturbed metric at each infinitesimal segment. The resulting scalar $d\tau$ remains invariant under diffeomorphisms, and its integral along the trajectory yields a path-dependent but frame-independent measure of duration. In general relativity, diffeomorphisms are smooth, invertible coordinate transformations that preserve the manifold structure, reflecting the theory's general covariance [28, 50]. Specifically, diffeomorphisms act on the spacetime manifold by pulling back tensor fields, ensuring that physical observables remain invariant under arbitrary smooth changes of coordinates. Therefore, even in trembling spacetime, proper time retains its role as the only consistent evolution parameter. This ensures that trembling worldlines maintain local Lorentz symmetry and that no preferred coordinate system emerges, preserving the principle of general covariance at the microscopic level. Trembling perturbations $\Xi^\mu(\tau)$ transform as vectors under local Lorentz transformations, ensuring covariance along worldlines. Here, $\Xi^\mu(\tau)$ represents a small, contin-

uous perturbation to the geodesic trajectory arising from microscopic trembling of spacetime. This perturbation respects causal structure by preserving proper-time monotonicity and is constrained to remain within the local light cone at each point along the worldline. This contrasts sharply with $\Xi^\mu(t)$, which would embed frame-dependent information into the geometric structure. TSRT, as a covariant and causal theory, must therefore define trembling as a function of proper time τ to preserve both local invariance and physical consistency.

S8.1 All Four Coordinates Must Tremble — Including Time

A critical requirement of TSRT is that trembling must apply symmetrically to all four spacetime coordinates. If trembling were instead defined in terms of coordinate time t , then t itself would become the evolution parameter and could not tremble independently—implying that only the spatial components $x^i(t)$ undergo fluctuations while the temporal coordinate $x^0 = t$ remains fixed. This introduces an artificial asymmetry between time and space, violating the local isotropy of spacetime and breaking the manifest Lorentz symmetry inherent in relativistic theories. In contrast, when trembling is modeled as a function of proper time τ , all components of the worldline $x^\mu(\tau) = x^\mu_{\text{class}}(\tau) + \Xi^\mu(\tau)$ experience geometric fluctuations. This includes the temporal component $x^0(\tau) = t(\tau)$, which undergoes deviations in the same covariant framework as the spatial coordinates. The trembling field $\Xi^\mu(\tau)$ thus introduces local, deterministic perturbations in all four directions of spacetime, consistent with the symmetry structure of general relativity.

This 4-dimensional treatment ensures that no preferred coordinate is singled out, and that trembling respects the underlying geometric nature of spacetime. It also avoids the need to postulate stochastic or non-causal fluctuations in coordinate time, which would be both physically and mathematically problematic. By assigning trembling to all coordinates through proper time evolution, TSRT maintains both the causal order and the relativistic coherence of the theory.

S8.2 Proper Time Trembling Supports Local Causality

Defining trembling in terms of coordinate time t would necessarily refer to a global hypersurface of simultaneity, making the trembling structure inherently non-local. Such a formulation conflicts with the principle of relativistic causality, which requires that all physical processes unfold locally along particle worldlines and not across spacelike-separated regions. The use of t as a trembling parameter introduces ambiguity about when and where fluctuations occur, especially in curved or dynamic spacetimes where the notion of simultaneity is frame-dependent. TSRT avoids these issues by defining trembling as a function of proper time τ , which is a locally constructed scalar that advances monotonically along each trajectory. This guarantees that the trembling field $\Xi^\mu(\tau)$ is locally defined, causally ordered, and insensitive to global coordinate choices. Because proper time accumulates through local metric intervals, it provides a physically meaningful evolution parameter that naturally enforces causal structure.

This proper-time-based trembling is essential for the emergence of macroscopic quantum behavior from deterministic microscopic dynamics. The derivation of the Schrödinger and Dirac equations in TSRT, as presented in Sections S18 and S6, relies critically on this locality. These

ensemble-level results are only consistent with causality because the underlying trembling is parametrized by proper time rather than coordinate time.

S8.3 Determinism Requires Proper-Time Parametrization

A central objective of TSRT is to restore determinism at the fundamental level. However, defining trembling in terms of coordinate time t leads to conceptual contradictions. Since t is the parameter along which evolution would be defined, allowing it to tremble implies that time itself fluctuates within its own frame of reference—a paradoxical situation. Attempting to resolve this by introducing stochastic time fluctuations would inject randomness extrinsic to the geometry, violating the deterministic principles TSRT is built upon.

By instead parametrizing trembling through proper time τ , these problems are avoided. Proper time is not a global coordinate but a local scalar that accumulates along a worldline, even in the presence of trembling-induced curvature. This allows all four spacetime coordinates $x^\mu(\tau)$ to undergo deterministic, causally consistent deviations without requiring probabilistic assumptions. The resulting paths $x^\mu(\tau) = x^\mu_{\text{class}}(\tau) + \Xi^\mu(\tau)$ are irregular yet fully determined by geometric dynamics. Apparent indeterminacy arises only when coarse-graining over ensembles of such trajectories, not from any fundamental randomness. This proper-time formulation is thus essential for ensuring that TSRT remains both causal and deterministic, while still giving rise to emergent quantum behavior.

S8.4 Conclusion

TSRT fundamentally requires that trembling be defined as a function of proper time τ , rather than coordinate time t . This choice is not merely a technical preference—it is essential for preserving the geometric and physical consistency of the theory. Defining trembling in terms of τ ensures that all four spacetime coordinates are treated symmetrically, that evolution remains local and causal, and that the theory respects general covariance and Lorentz invariance. It also guarantees that the underlying dynamics remain fully deterministic, avoiding the introduction of stochastic elements that would otherwise obscure the geometric foundations of the model. By anchoring trembling to proper time, TSRT aligns itself with the intrinsic structure of spacetime rather than arbitrary coordinate slicings. This distinguishes the theory from both coordinate-time-based field formulations and quantum mechanical models that rely on probabilistic postulates. In TSRT, trembling becomes a physically meaningful, mathematically consistent description of the microstructure of spacetime—capable of unifying quantum behavior and relativistic geometry within a single causal framework.

Having established the proper-time structure of trembling, we now classify the permissible types of metric fluctuations and their associated gauge structures.

S9 Supplementary Material: Classification of Metric Fluctuations and Gauge Structure

S9.1 Introduction and Goals

The purpose of this section is to establish and prove the complete classification of all physically admissible trembling-induced metric fluctuations within the framework of Trembling Spacetime Relativity Theory (TSRT). This classification forms the backbone of the emergent structure of the four known fundamental forces: gravity, electromagnetism, the weak force, and the strong force. Unlike in conventional field-theoretic approaches, where different interactions are introduced through independent field contents such as gravitons, photons, W and Z bosons, and gluons, TSRT derives all interactions as necessary geometric consequences of the structure of permissible spacetime trembling.

The classification proceeds systematically: First, we formalize the general form of trembling-induced metric deviations via the perturbation tensor $\xi_{\mu\nu}(\tau)$. Second, we decompose $\xi_{\mu\nu}$ into its irreducible components under Lorentz symmetry, based on representation theory of the Lorentz group [6, 28]. Third, we impose three critical physical constraints: (i) Lorentz covariance under coarse-graining, (ii) preservation of local causality, and (iii) monotonic progression of proper time, $d\tau/d\lambda > 0$, as established in Section S7.

Applying these constraints restricts the space of admissible trembling structures, ultimately demonstrating that precisely four—and only four—irreducible classes of metric fluctuation survive.

Each class corresponds directly to a distinct physical interaction:

- Trace-like isotropic trembling, giving rise to gravity (Section S13),
- Abelian vector-like trembling, leading to electromagnetism (Section S10),
- Chiral non-Abelian trembling, underlying the weak force (Section S11),
- Non-chiral non-Abelian color trembling, giving rise to the strong force (Section S12).

This classification not only unifies the emergence of interactions but also explains why no other fundamental forces are possible under the causal and geometric foundations of TSRT.

The results presented here thus lay the conceptual and mathematical groundwork for the derivations that follow.

S9.2 Framework: Trembling Metric Deviations and Symmetry Requirements

In Trembling Spacetime Relativity Theory (TSRT), the microscopic deviations of the spacetime metric are described by a symmetric tensor field $\xi_{\mu\nu}(\tau)$, defined along a particle's worldline and parameterized by proper time τ . These fine-scale deviations represent local trembling-induced fluctuations of spacetime geometry.

To describe the effective geometry seen by a particle, the full spacetime metric is written as:

$$g_{\mu\nu}(\tau) = \eta_{\mu\nu} + \xi_{\mu\nu}(\tau), \tag{S152}$$

where $\eta_{\mu\nu}$ is the flat Minkowski background and $\xi_{\mu\nu}(\tau)$ encodes the deterministic deviation from flatness due to trembling. This decomposition assumes that we are building up all interactions and curvature effects from scratch—starting with an initially flat spacetime. Such an approach aligns with the foundational spirit of TSRT, which seeks to derive not only quantum behavior but also gravitational and gauge dynamics as emergent phenomena from underlying geometric fluctuations.

It is important to emphasize, however, that this starting point is not a limitation of the theory. In principle, the TSRT framework admits generalization to (already) curved backgrounds by replacing $\eta_{\mu\nu}$ with a non-flat reference metric $\bar{g}_{\mu\nu}$, yielding the more general decomposition:

$$g_{\mu\nu}(\tau) = \bar{g}_{\mu\nu}(\tau) + \xi_{\mu\nu}(\tau), \quad (\text{S153})$$

where $\bar{g}_{\mu\nu}$ represents a smooth large-scale background geometry, such as that of General Relativity.

Nonetheless, for the purposes of classifying fundamental forces and symmetry-breaking patterns within trembling spacetime, it is both conceptually and technically advantageous to adopt the flat background formulation. This choice allows all interactions—including gravity—to emerge on equal footing as different manifestations of trembling-induced curvature, encoded in the structure of $\xi_{\mu\nu}$ alone. The assumption of small fluctuations ensures that the perturbative expansion remains valid and allows for a systematic classification of the gauge symmetries associated with different classes of trembling modes.

To ensure the physical and mathematical consistency of the trembling structure, three essential conditions must be imposed. First, Lorentz invariance must be respected at the statistical and coarse-grained levels. Although an individual realization of the trembling field may locally distort Lorentz symmetry, the ensemble average and macroscopic behavior must remain Lorentz-covariant to be compatible with experimental observations and the foundations of special relativity [6, 28]. Second, trembling fluctuations must preserve the causal ordering of spacetime events. This prohibits the interchange of timelike and spacelike separations due to the local trembling-induced metric perturbations. Mathematically, this causal structure preservation is enforced by constraints on the eigenvalues and invariants of the perturbed metric tensor. Third, and most stringently, the positivity of proper-time evolution must be strictly maintained. Along any physically realizable worldline, proper time must advance monotonically, requiring that

$$\frac{d\tau}{d\lambda} > 0, \quad (\text{S154})$$

where λ is an arbitrary affine parameter along the trajectory. This positivity condition stabilizes worldlines against nonphysical behavior, such as local time reversals or breakdowns of causal order. It also ensures that all trembling-induced corrections respect the arrow of proper time, a feature indispensable for preserving physical determinism. Taken together, these constraints impose strong restrictions on the permissible forms and magnitudes of $\xi_{\mu\nu}(\tau)$. The allowed fluctuations must be structured so that, despite their fine-scale trembling nature, they maintain global Lorentz covariance after coarse-graining, preserve the causal structure of spacetime at every scale, and strictly forbid violations of proper-time monotonicity.

In the next subsections, we develop a full classification of the types of trembling fluctuations that satisfy all these conditions. This classification ultimately reveals why exactly four distinct classes of physically admissible trembling structures arise — and how each class leads directly to the emergence of one of the known fundamental forces.

S9.3 Mathematical Decomposition: Irreducible Components of Metric Fluctuations

To systematically classify trembling structures compatible with the physical and symmetry constraints outlined in Section S9.2, it is necessary to decompose the metric fluctuation tensor $\xi_{\mu\nu}(\tau)$ into its irreducible components under Lorentz transformations. The decomposition reveals the fundamental building blocks of trembling spacetime and establishes the structure of the admissible perturbations that can give rise to physical forces.

By construction, the fluctuation tensor $\xi_{\mu\nu}(\tau)$ is symmetric,

$$\xi_{\mu\nu} = \xi_{\nu\mu}, \quad (\text{S155})$$

because it derives from a perturbation of the spacetime metric, which itself is symmetric by definition. This property significantly simplifies the classification, as the space of symmetric tensors admits a well-known decomposition into orthogonal sectors under the action of the Lorentz group [6, 28].

The symmetric rank-2 tensor $\xi_{\mu\nu}$ can be uniquely separated into two physically meaningful components. The first is the trace part, capturing isotropic expansions or contractions of spacetime volume, and is given by

$$\xi_{\text{trace}}(\tau) = \frac{1}{4} \eta^{\mu\nu} \xi_{\mu\nu}(\tau), \quad (\text{S156})$$

where $\eta^{\mu\nu}$ is the inverse Minkowski metric. This scalar component is invariant under Lorentz transformations and represents uniform, direction-independent trembling deformations.

The second component is the symmetric traceless part, encoding shear-like distortions that preserve spacetime volume but distort its shape anisotropically. It is defined by subtracting the trace contribution from the full tensor:

$$\xi_{\mu\nu}^{(0)}(\tau) = \xi_{\mu\nu}(\tau) - \frac{1}{4} \eta_{\mu\nu} \left(\eta^{\alpha\beta} \xi_{\alpha\beta}(\tau) \right), \quad (\text{S157})$$

and satisfies the constraint

$$\eta^{\mu\nu} \xi_{\mu\nu}^{(0)} = 0, \quad (\text{S158})$$

ensuring that it is purely traceless.

No antisymmetric component appears in the decomposition of $\xi_{\mu\nu}$, since the metric perturbations arise from coordinate-invariant spacetime displacements and must remain symmetric. However, antisymmetric structures can emerge at the level of derived quantities, such as field strengths or curvature tensors, when differentiating the trembling fields with respect to spacetime coordinates. This decomposition into trace and symmetric traceless sectors forms the essential mathematical foundation for classifying trembling-induced interactions. Each sector

contributes uniquely to the physical behavior of spacetime under trembling fluctuations, and their different transformation properties under Lorentz symmetry ultimately govern the emergence of the fundamental forces, as will be developed in the following sections.

Representations under the Lorentz Group. The decomposition of $\xi_{\mu\nu}$ into trace and symmetric traceless components corresponds to distinct irreducible representations of the Lorentz group. The trace part transforms as a Lorentz scalar, associated with the trivial $(0,0)$ representation in the language of two-spinor formalism. In contrast, the symmetric traceless part transforms as a spin-2 field, corresponding to the $(1,1)$ representation. These transformation properties align with the standard decomposition techniques employed in the classification of gravitational perturbations in general relativity [28] and in the broader framework of field theory [6]. The identification of these two irreducible sectors under Lorentz symmetry is essential: it ensures that the trembling structures derived from $\xi_{\mu\nu}$ are classified according to fundamental symmetry principles, without the introduction of arbitrary auxiliary fields or additional assumptions. Each component's transformation behavior under Lorentz boosts and rotations will later determine its physical interpretation in the emergence of forces within the TSRT framework.

Physical Interpretation. Each irreducible component of $\xi_{\mu\nu}$ is associated with a distinct type of physical effect in trembling spacetime. The trace part corresponds to local isotropic expansions or contractions of spacetime volumes, akin to the behavior of scalar fields and reminiscent of massless spin-0 excitations. Meanwhile, the symmetric traceless component captures shear-like distortions of spacetime, characteristic of tensorial field modes. Depending on how the underlying trembling displacement field $\Xi^\mu(\tau)$ organizes its fluctuations across different spacetime directions, additional gauge-like or internal symmetries can emerge, leading to richer physical structures beyond the simple decomposition. These possibilities will be explored systematically in the following subsections. It is crucial to emphasize that the causal constraint $d\tau/d\lambda > 0$, central to the TSRT framework, restricts the admissible amplitudes and correlations between these components. This ensures that trembling-induced deformations remain within physically acceptable bounds, preserving causal coherence and preventing the emergence of nonphysical or unstable spacetime structures.

S9.4 Allowed Trembling Classes: The Four Fundamental Structures

Building on the decomposition of metric fluctuations into irreducible components, we now systematically classify the specific trembling structures that satisfy the stringent physical requirements previously outlined: Lorentz invariance, the preservation of causal structure through the positivity of proper time evolution ($d\tau/d\lambda > 0$), and local geometric consistency. This analysis reveals that precisely four distinct and independent classes of trembling structures are permitted within the TSRT framework. Each class gives rise to a different fundamental interaction observed in nature, establishing a one-to-one correspondence between the permissible forms of spacetime trembling and the known forces, without invoking any additional postulates or external symmetries.

1. Trace-like Isotropic Trembling: Emergence of Gravity. The simplest class of admissible trembling corresponds to a pure trace fluctuation of the metric, expressed as

$$\xi_{\mu\nu}(\tau) = \phi(\tau)\eta_{\mu\nu}, \quad (\text{S159})$$

where $\phi(\tau)$ is a scalar function representing local isotropic expansions and contractions of the spacetime fabric relative to the flat Minkowski background. Trace-like trembling modes are those proportional to the background metric, modifying the overall scale without introducing shear or anisotropy: $\xi_{\mu\nu} \propto \bar{g}_{\mu\nu}$. This trembling mode preserves full Lorentz invariance and modifies spacetime intervals uniformly in all directions without introducing directional biases.

Physically, trace-like trembling acts as a local scaling of spacetime intervals, leading to effective variations in curvature when coarse-grained over larger scales. It induces deformations that naturally reproduce the large-scale gravitational effects described by Einstein's equations. In the emergent limit, where individual microscopic fluctuations are averaged over, this class of trembling gives rise to the gravitational interaction. Importantly, the requirement that proper time evolves monotonically, $d\tau/d\lambda > 0$, is preserved provided that the amplitude of $\phi(\tau)$ remains sufficiently bounded, a condition rigorously established in Section S7. Thus, trace-like isotropic trembling stands as the geometric origin of gravitational phenomena within the TSRT framework.

2. Abelian Vector-like Trembling: Emergence of Electromagnetism. A second fundamental class of trembling arises when fluctuations acquire a directional character but remain constrained by an underlying Abelian gauge structure. In this case, the metric perturbation takes the form

$$\xi_{\mu\nu}(\tau) = A_{(\mu}(\tau)u_{\nu)}, \quad (\text{S160})$$

where $A_\mu(\tau)$ is a vector field associated with the trembling excitation and u^μ is the four-velocity of the underlying worldline. Symmetrization over indices ensures compatibility with the symmetric nature of the metric perturbations.

Physically, this class of trembling encodes phase shifts along particle worldlines, naturally corresponding to a $U(1)$ gauge symmetry. The geometric structure reproduces the essential features of electromagnetic interactions, with $A_\mu(\tau)$ functioning as the emergent gauge potential that couples minimally to charged particles. The preservation of causal evolution, $d\tau/d\lambda > 0$, imposes constraints on the magnitude of $A_\mu(\tau)$ relative to the four-velocity u^μ , ensuring that the trembling does not induce pathological backward evolutions of proper time. Under these conditions, Abelian vector-like trembling provides the geometric foundation for the emergence of electromagnetism within the TSRT framework.

3. Chiral Non-Abelian Trembling: Emergence of the Weak Force. A third class of trembling structures arises when fluctuations incorporate a left-right asymmetry, giving rise to chiral behavior. Chiral trembling modes arise from fluctuations that break left-right symmetry in local tangent spaces, leading naturally to parity-violating gauge interactions, as observed in

the weak force. The corresponding metric perturbation can be expressed as

$$\xi_{\mu\nu}(\tau) = \sum_a W_{(\mu}^a(\tau) \sigma_{\nu)}^a, \quad (\text{S161})$$

where $W_{\mu}^a(\tau)$ are non-Abelian trembling fields, σ_{ν}^a denote internal symmetry directions associated with chiral projectors, and the index a labels the generators of the $SU(2)_L$ group.

In physical terms, this structure selectively couples to left-handed components of worldline excitations, leading to an intrinsic breaking of parity symmetry. The resulting gauge structure mirrors that of the weak interaction, in which only left-chiral fields participate actively in the fundamental coupling scheme. As in previous cases, the causal constraint $d\tau/d\lambda > 0$ imposes stringent bounds on the magnitudes and correlations of the trembling fields. These bounds ensure that even in the presence of chiral asymmetry, proper time advances monotonically along physical trajectories, preserving the deterministic and causal foundations of TSRT.

4. Non-Chiral Non-Abelian Color Trembling: Emergence of the Strong Force. The fourth admissible trembling class involves non-chiral, purely internal non-Abelian structures. The corresponding metric perturbation takes the form

$$\xi_{\mu\nu}(\tau) = \sum_A G_{(\mu}^A(\tau) T_{\nu)}^A, \quad (\text{S162})$$

where $G_{\mu}^A(\tau)$ are trembling fields associated with internal color degrees of freedom, T_{ν}^A represent the generators of the $SU(3)_C$ group, and A indexes the color charges. This type of trembling encodes fluctuations that couple symmetrically to all color charges, without introducing any handedness or chirality. The resulting internal structure mirrors the non-chiral nature of the strong nuclear force, characterized by its $SU(3)_C$ gauge symmetry. As in the previous cases, the positivity of proper time evolution $d\tau/d\lambda > 0$ acts as a regulating constraint, ensuring that the color trembling fields remain dynamically consistent and that no causal instabilities are introduced.

Thus, the strong interaction emerges naturally from the existence of a non-chiral internal trembling sector, firmly grounded in the causal and geometric architecture of trembling space-time.

Exhaustiveness of the Classification. No additional independent trembling classes are admissible without violating one or more essential physical principles. Any hypothetical trembling structure beyond those identified would necessarily conflict with Lorentz invariance, disrupt the preservation of local causality by violating the proper time condition $d\tau/d\lambda > 0$, or breach the spectral bounds that constrain trembling amplitudes, as analyzed in Section S20.

Consequently, all physically admissible trembling structures must reduce to one of the following four categories: scalar isotropic fluctuations leading to gravity, Abelian phase fluctuations giving rise to electromagnetism, chiral non-Abelian structures underpinning the weak interaction, or non-chiral non-Abelian color structures generating the strong interaction. No further independent structures are possible without undermining the internal consistency of the theory.

Thus, TSRT naturally and rigorously leads to exactly four—and only four—fundamental interaction classes, providing a complete and geometrically necessary explanation for the existence of the four known forces of nature.

S9.5 Causality Constraints: Why Only These Four Classes Are Permissible

Beyond the mathematical decomposition of metric fluctuations, physical admissibility in Trembling Spacetime Relativity Theory (TSRT) imposes an even stricter filter: the preservation of causal structure. In particular, the evolution of proper time must always proceed forward along any physical trajectory. This requirement is formalized by the fundamental constraint

$$\frac{d\tau}{d\lambda} > 0, \quad (\text{S163})$$

where λ is an arbitrary affine parameter along the worldline, and τ denotes the experienced proper time along that path.

We now demonstrate that only the four trembling classes identified previously—trace-like, Abelian vector-like, chiral non-Abelian, and non-chiral non-Abelian fluctuations—satisfy this stringent causality condition. Any hypothetical additional trembling structure would inevitably disrupt the forward progression of proper time and thereby render itself physically inadmissible within TSRT.

General Form of the Proper Time Increment. The local increment of proper time in trembling spacetime is determined by the perturbed line element

$$d\tau^2 = -(\eta_{\mu\nu} + \xi_{\mu\nu}(\tau)) \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} d\lambda^2, \quad (\text{S164})$$

where $\xi_{\mu\nu}(\tau)$ encodes the trembling-induced deviation from the Minkowski background $\eta_{\mu\nu}$.

Expanding this expression to first order in $\xi_{\mu\nu}$, under the assumption that the fluctuations remain small, yields

$$d\tau^2 = d\tau_0^2 + \delta(d\tau^2), \quad (\text{S165})$$

where

$$d\tau_0^2 = -\eta_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} d\lambda^2, \quad (\text{S166})$$

$$\delta(d\tau^2) = -\xi_{\mu\nu}(\tau) \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} d\lambda^2. \quad (\text{S167})$$

Here, $d\tau_0^2$ represents the unperturbed proper-time increment along the smooth background trajectory, while $\delta(d\tau^2)$ captures the first-order correction due to microscopic trembling. To preserve the fundamental causal requirement that $d\tau^2 > 0$ along all segments of the path, the perturbation $\delta(d\tau^2)$ must remain sufficiently small compared to $d\tau_0^2$ and must not introduce local reversals of proper-time flow.

Analysis of Hypothetical Additional Trembling Structures. Here, we analyze antisymmetric trembling structures, higher-spin-like trembling structures, and mixed trace-antisymmetric

perturbations.

(i) Antisymmetric Trembling Structures. Consider the hypothetical possibility that trembling induces an antisymmetric component in the metric perturbation:

$$\xi_{\mu\nu}(\tau) = -\xi_{\nu\mu}(\tau). \quad (\text{S168})$$

In this case, the perturbative correction to the proper-time increment involves the contraction

$$\delta(d\tau^2) = -\xi_{\mu\nu}(\tau) \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} d\lambda^2. \quad (\text{S169})$$

However, the tensor $\frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}$ is symmetric under interchange of indices $\mu \leftrightarrow \nu$, while $\xi_{\mu\nu}$ is antisymmetric by construction. Therefore, their contraction vanishes identically:

$$\xi_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = 0, \quad (\text{S170})$$

leading to

$$\delta(d\tau^2) = 0. \quad (\text{S171})$$

Thus, to first order, antisymmetric trembling has no effect on the evolution of proper time.

While one might conclude that antisymmetric fluctuations are harmless, such structures generally correspond to torsional modifications of spacetime geometry [51,52]. Torsion modifies the parallel transport of vectors and alters the coupling of spin to gravity, leading to physical consequences that have been experimentally constrained to extremely tight limits [53]. Therefore, unless antisymmetric trembling is dynamically suppressed or fully decoupled from observable sectors, its existence would contradict empirical observations.

Consequently, antisymmetric trembling structures are excluded from the set of physically admissible metric fluctuations, both on theoretical grounds of irrelevance to causal structure and on experimental grounds of non-observation.

(ii) Higher-Spin-Like Trembling Structures. An alternative hypothetical extension would involve trembling structures associated with higher-spin fields, characterized by perturbations with more than two free indices or with complex symmetrization patterns. Such higher-rank tensors would modify the spacetime interval by introducing terms involving cubic, quartic, or higher powers of the four-velocity:

$$d\tau^2 \sim \xi_{\mu\nu\rho} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \frac{dx^\rho}{d\lambda} + \dots \quad (\text{S172})$$

These contractions violate the bilinear form required for proper Lorentzian interval preservation. In standard Lorentz geometry, only second-rank symmetric tensors naturally couple bilinearly to the worldline tangent vector, preserving the invariant structure of proper time [6,28]. Moreover, higher-spin-like trembling would generally introduce terms that could not be guaranteed to preserve $d\tau/d\lambda > 0$ without imposing fine-tuned constraints. Such fine-tuning would imply additional internal symmetries or auxiliary dynamical fields that have no empirical support [54,55]. In particular, consistent interactions of massless higher-spin fields in flat spacetime are known to be severely restricted or even impossible without violating basic principles such as

locality and causality [56].

Thus, higher-spin-like trembling perturbations are ruled out on both geometric and physical grounds: they are incompatible with the bilinear form of the proper time interval and with the observed absence of dynamical higher-spin degrees of freedom in nature.

(iii) Mixed Trace–Antisymmetric Perturbations. A third possible hypothetical class involves trembling structures that combine a trace-like symmetric component and an antisymmetric perturbation:

$$\xi_{\mu\nu}(\tau) = \phi(\tau)\eta_{\mu\nu} + \chi_{\mu\nu}(\tau), \quad (\text{S173})$$

where $\phi(\tau)$ is a scalar field representing isotropic expansion or contraction, and $\chi_{\mu\nu}(\tau)$ is an antisymmetric tensor satisfying $\chi_{\mu\nu} = -\chi_{\nu\mu}$.

While the trace component alone preserves the proper bilinear structure needed for causal proper time evolution, the antisymmetric component $\chi_{\mu\nu}$ introduces no direct contribution to the line element at first order, as shown previously. However, second-order corrections or mixed terms involving $\phi(\tau)$ and $\chi_{\mu\nu}(\tau)$ could, in principle, disturb the positivity of $d\tau/d\lambda$ unless specific fine-tuning between the two components is enforced. Such fine-tuning would not arise naturally from the causal structure or Lorentz symmetry alone and would require additional internal consistency conditions that are neither geometrically motivated nor supported by empirical observation [6]. Furthermore, introducing independent antisymmetric degrees of freedom risks reintroducing torsion or non-metricity into spacetime structure, in tension with the observed near-Riemannian character of spacetime geometry [51].

Therefore, mixed trace–antisymmetric trembling configurations are ruled out: they fail to maintain the strict causal and geometric consistency demanded by Trembling Spacetime Relativity Theory (TSRT) without ad hoc adjustments, violating the principle that fundamental structures should emerge from general covariance and causal stability without arbitrary constructions.

Positive Proper-Time Evolution in the Four Fundamental Classes. In contrast to the hypothetical structures excluded above, the four physically admissible trembling classes—namely, trace-like isotropic fluctuations, Abelian vector-like fluctuations, chiral non-Abelian fluctuations, and non-chiral non-Abelian color fluctuations—preserve the necessary properties for causal evolution. Each of these classes contributes bilinear and symmetric corrections to the line element $d\tau^2$, ensuring that perturbations respect the fundamental form required by Lorentz covariance. Moreover, the amplitudes of trembling in each class can be dynamically bounded without violating proper time positivity, so that $d\tau^2 > 0$ is maintained everywhere along the worldline. Finally, the internal structure of each trembling class, whether isotropic, vectorial, chiral, or color-like, remains consistent with observed physical symmetries and causal stability. Thus, only these four classes survive the combined geometric and physical constraints demanded by Trembling Spacetime Relativity Theory (TSRT), establishing a tightly closed set of fundamental interactions directly rooted in spacetime structure itself.

Summary. The strict requirement that proper time advances monotonically, expressed by the condition $d\tau/d\lambda > 0$, serves as a decisive and universal filter on the types of trembling structures

that spacetime can support. All hypothetical extensions beyond the four classified trembling classes—trace-like, Abelian vector-like, chiral non-Abelian, and non-chiral non-Abelian—either violate this causal condition or fail to preserve the necessary geometric and physical coherence. As a result, TSRT provides a rigorous, non-arbitrary explanation for the existence of exactly four fundamental forces in nature, each arising inevitably from a distinct, causally permissible trembling configuration of the underlying spacetime fabric. No additional interactions are possible without contradicting the essential requirements of causality and Lorentz structure embedded in trembling spacetime geometry.

S9.6 Mapping Trembling Classes to Physical Forces

Having completed the classification of the four causally admissible trembling structures in spacetime, we now establish their direct correspondence to the known fundamental interactions of nature. Each trembling class is not merely associated with, but rigorously leads to, the emergence of a specific force, dictated entirely by the geometric and causal properties encoded within the trembling deviations. This mapping is not based on analogy or postulation but results systematically from the internal structure of trembling spacetime itself. In the following subsections, we provide a conceptual overview of this correspondence, preparing for the more detailed derivations to be presented in the subsequent Sections.

(i) Trace-like Isotropic Trembling $\rightarrow$ Gravity The scalar, trace-like component of trembling spacetime induces uniform expansions and contractions of local spacetime volumes. When coarse-grained over proper time and spatial domains, these isotropic fluctuations lead to an effective macroscopic curvature, which mimics the behavior described by Einstein’s field equations. Thus, gravity emerges naturally in TSRT as the large-scale manifestation of trace-like trembling, without requiring the postulation of a separate gravitational field. The detailed derivation of this correspondence is presented in Section S13.

(ii) Abelian Vector-like Trembling $\rightarrow$ Electromagnetism Vectorial deformations of spacetime associated with abelian (commutative) trembling fluctuations naturally give rise to a $U(1)$ gauge structure. These abelian trembling modes generate effective phase shifts along particle worldlines, leading to the emergence of the electromagnetic field tensor and the governing Maxwell equations. Crucially, in TSRT, electromagnetism is not introduced as an independent interaction but arises directly from the causal and geometric properties of vectorial trembling. A full derivation is provided in Section S10.

(iii) Chiral Non-Abelian Trembling $\rightarrow$ Weak Force When trembling fluctuations acquire a chiral (handedness-dependent) structure and a non-abelian (non-commutative) internal symmetry, the resulting interactions mimic the structure of the weak force. Specifically, the emergent gauge fields resemble those of $SU(2)_L$ symmetry, with chiral couplings and mass-generation mechanisms via geometric symmetry breaking. The full construction will be detailed in Section S11.

(iii) Chiral Non-Abelian Trembling $\rightarrow$ Weak Force Trembling fluctuations that exhibit chirality—differentiating between left-handed and right-handed modes—and incorporate non-abelian (non-commutative) internal symmetries give rise naturally to structures resembling the weak force. In TSRT, this class of trembling produces emergent gauge fields corresponding to $SU(2)_L$ symmetry, intrinsically linked to parity violation and the generation of mass through geometric mechanisms. Rather than being postulated, the weak interaction arises as a necessary consequence of chiral trembling geometry. A complete derivation will be presented in Section S11.

Summary. The four known fundamental forces arise within Trembling Spacetime Relativity Theory (TSRT) as distinct manifestations of causally and geometrically permitted trembling structures. Rather than being independently introduced or postulated, each interaction emerges necessarily from a specific class of spacetime fluctuations, determined by the strict requirements of Lorentz invariance, causal stability, and proper-time positivity. No additional independent forces or gauge structures are possible without violating these foundational constraints. This intrinsic closure not only explains the origin and uniqueness of the known interactions but also exemplifies the deep unifying power of TSRT, where all forces trace their existence to the underlying causal geometry of trembling spacetime itself.

S9.7 Summary and Outlook

In this section, we have completed the rigorous classification of all causally admissible trembling structures of the spacetime metric within Trembling Spacetime Relativity Theory (TSRT). Starting from the decomposition of the trembling perturbations $\xi_{\mu\nu}(\tau)$ into their irreducible Lorentz representations, and imposing the strict physical requirements of Lorentz invariance, causal preservation, and monotonic proper-time evolution, we have demonstrated that precisely four—and only four—independent trembling classes are physically permitted. These classes correspond directly to the four known fundamental interactions:

1. Trace-like isotropic trembling, giving rise to gravity,
2. Abelian vector-like trembling, giving rise to electromagnetism,
3. Chiral non-Abelian trembling, giving rise to the weak force,
4. Non-chiral non-Abelian color trembling, giving rise to the strong force.

No further trembling classes are admissible without violating causal coherence or geometric consistency. Thus, TSRT does not merely accommodate the observed interactions; it predicts their existence and their mutual exclusivity from first principles.

This classification marks a major conceptual and structural achievement. It guarantees the uniqueness of the four fundamental forces as inevitable consequences of the causal trembling geometry of spacetime. It resolves the arbitrariness of gauge group postulates that characterize the Standard Model by deriving interaction types from the geometry itself. And it unifies the forces under a single deterministic and causal principle, rather than treating them as disparate phenomena.

In the sections that follow, we will deepen this foundation by explicitly deriving each fundamental force from its corresponding trembling class, establishing the associated conservation laws from underlying geometric symmetries, analyzing the emergent range properties of forces (long-range versus short-range), and demonstrating the internal consistency of interactions within the unified TSRT framework. The classification achieved here thus serves not merely as a catalog, but as the indispensable cornerstone for understanding why exactly four forces govern the physical universe—and why their origins are inseparably tied to the causal and geometric structure of trembling spacetime itself. Having classified the distinct trembling modes, we now systematically derive how each mode gives rise to a corresponding fundamental interaction, without imposing external gauge principles. First, we derive the emergence of electromagnetic interactions from coherent abelian trembling modes.

S10 Supplementary Material: Emergent Electromagnetism from Abelian Trembling

S10.1 Abelian Fluctuation Sector and $U(1)$ Symmetry

In Section S9, it was shown that trembling-induced deviations of the spacetime metric can be rigorously classified according to their algebraic properties and transformation behavior under local frame rotations. The simplest nontrivial class of such fluctuations corresponds to abelian trembling, where the deviation field $\Xi^\mu(\tau)$ remains aligned with the particle's four-velocity $u^\mu(\tau)$, and the resulting metric fluctuation tensor $\xi_{\mu\nu}$ does not generate any intrinsic angular momentum or frame-dependent mixing.

This abelian class is distinguished by the property that the generators of trembling-induced deformations commute:

$$[T^a, T^b] = 0, \quad (\text{S174})$$

where T^a are the effective infinitesimal generators associated with the deformation $\Xi^\mu(\tau)$ in the tangent bundle of the particle's worldline. Since these generators form a one-dimensional, commutative Lie algebra, the symmetry group of the corresponding trembling sector is uniquely identified with the unitary group $U(1)$. This immediately associates the abelian trembling class with the symmetry structure of classical electromagnetism. The unique emergence of a single $U(1)$ structure, rather than multiple independent abelian groups, follows from the constraint that trembling must preserve causal coherence along worldlines without introducing additional degrees of freedom beyond phase rotations. This will be studied in Section S16.

As introduced in Section S9.4, one of the key symmetry-restricted classes of trembling is the Abelian vector-like sector. In this regime, the fine-scale metric perturbation takes the form

$$\xi_{\mu\nu}(\tau) = A_{(\mu}(\tau)u_{\nu)} = \frac{1}{2} (A_\mu(\tau)u_\nu(\tau) + A_\nu(\tau)u_\mu(\tau)), \quad (\text{S175})$$

where $A_\mu(\tau)$ is a proper-time-dependent four-vector field and $u^\mu(\tau)$ is the instantaneous four-velocity of the particle. This symmetric construction ensures compatibility with the requirement that the metric perturbation $\xi_{\mu\nu}$ itself remains a symmetric tensor field.

To connect this structure to a geometrically meaningful vector deviation $\Xi^\mu(\tau)$, we recall that a general perturbation of the metric due to a vector field takes the linearized form:

$$\xi_{\mu\nu} = \partial_\mu \Xi_\nu + \partial_\nu \Xi_\mu. \quad (\text{S176})$$

This is the standard symmetric metric variation induced by an infinitesimal displacement field Ξ^μ , consistent with leading-order perturbations in Riemannian geometry and gauge theory [31, 34].

By comparing the expression from Section S9.4 with the standard form in Equation (S176), we see that the metric fluctuation $\xi_{\mu\nu} = A_{(\mu}u_{\nu)}$ can be generated by a deviation vector field of the form

$$\Xi^\mu(\tau) = \epsilon(\tau)u^\mu(\tau), \quad (\text{S177})$$

where $\epsilon(\tau)$ is a scalar function along the worldline. Differentiating this yields:

$$\begin{aligned}\partial_\mu \Xi_\nu + \partial_\nu \Xi_\mu &= \partial_\mu (\epsilon u_\nu) + \partial_\nu (\epsilon u_\mu) \\ &= (\partial_\mu \epsilon) u_\nu + \epsilon \partial_\mu u_\nu + (\partial_\nu \epsilon) u_\mu + \epsilon \partial_\nu u_\mu.\end{aligned}\tag{S178}$$

Restricting ourselves to the dominant proper-time derivative along the worldline and assuming negligible spatial variation over short intervals, the leading-order contribution becomes:

$$\xi_{\mu\nu}(\tau) \approx \dot{\epsilon}(\tau) (u_\mu u_\nu) + \epsilon(\tau) (\partial_\mu u_\nu + \partial_\nu u_\mu).\tag{S179}$$

In the limit where u^μ varies slowly and remains approximately constant (i.e., the trembling occurs over scales much shorter than the curvature or acceleration scale), this reduces to:

$$\xi_{\mu\nu}(\tau) \approx \dot{\epsilon}(\tau) u_\mu u_\nu.\tag{S180}$$

However, to remain consistent with the earlier abelian ansatz $\xi_{\mu\nu} = A_{(\mu} u_{\nu)}$, we may set

$$A_\mu(\tau) = \dot{\epsilon}(\tau) u_\mu(\tau),\tag{S181}$$

thereby identifying the trembling-induced vector potential A_μ as arising from proper-time oscillations in a longitudinal deviation field aligned with the particle's worldline.

We therefore adopt the form:

$$\Xi^\mu(\tau) = \epsilon_0 \cos(\omega\tau + \phi) u^\mu(\tau),\tag{S182}$$

with real amplitude ϵ_0 , proper-time frequency ω , and phase ϕ . This ansatz satisfies the required symmetry and leads to a metric perturbation:

$$\xi_{\mu\nu} = \partial_\mu \Xi_\nu + \partial_\nu \Xi_\mu,\tag{S183}$$

which, to leading order in proper time, takes the symmetric form

$$\xi_{\mu\nu}(\tau) = \epsilon_0 \omega \sin(\omega\tau + \phi) u_\mu(\tau) u_\nu(\tau).\tag{S184}$$

Thus, the abelian trembling mode emerges as a proper-time-coherent, longitudinal fluctuation aligned with the particle's motion. Its structure naturally respects $U(1)$ gauge symmetry, since the underlying geometry only depends on the derivative $\partial_\mu \Xi_\nu$, and any global phase shift $\phi \mapsto \phi + \delta\phi$ leaves physical quantities invariant. This geometric realization lays the foundation for deriving Maxwell's equations and the Lorentz force law from the curvature structure of trembling spacetime itself.

This expression satisfies all geometric and physical constraints imposed by the TSRT framework. It is symmetric by construction, as required for metric fluctuations. It transforms covariantly under general coordinate and local Lorentz transformations, since both the scalar amplitude $\epsilon(\tau)$ and the four-velocity $u^\mu(\tau)$ are well-defined tensorial quantities. Most critically, the fluctuation respects causal structure: the positivity condition on the proper-time interval

$d\tau^2 > 0$ remains satisfied provided that the oscillation amplitude $\epsilon(\tau)$ remains below the critical cutoff scale derived in Section S23. Moreover, this class of abelian, longitudinal trembling deviations constitutes the only physically consistent configuration that:

- preserves local causal ordering,
- introduces no spatial anisotropy or torsional spinor distortion,
- transforms trivially under global and local phase redefinitions,
- and leads to a coherent long-range field theory coupled to a conserved current.

Consequently, the abelian trembling sector serves as the natural geometric origin of electromagnetism in TSRT. Rather than postulating a pre-existing gauge field, the electromagnetic interaction arises from this specific symmetry-preserving class of metric fluctuation. In the next subsection, we make this correspondence precise by deriving the emergent gauge potential $A_\mu(x)$ from the coarse-grained behavior of the underlying trembling field.

S10.2 Emergent Gauge Field A_μ and Its Definition

Building on the abelian trembling structure identified above, we now demonstrate how an effective gauge field emerges directly from the geometric configuration defined by the deviation vector $\Xi^\mu(\tau) = \epsilon(\tau)u^\mu(\tau)$. This construction does not involve the introduction of a fundamental gauge field. Instead, the electromagnetic potential $A_\mu(x)$ is defined as a coarse-grained statistical projection of the trembling field along the observer's worldline:

$$A_\mu(x) := \langle \partial_\mu \Xi^\nu(x) \rangle g_{\nu\rho}(x) u^\rho(x), \quad (\text{S185})$$

where angle brackets denote coarse-graining over the unresolved internal phases of the trembling oscillation. The inclusion of the background metric factor $g_{\nu\rho}(x)u^\rho(x)$ ensures that the contraction is covariant and physically meaningful.

This emergent quantity $A_\mu(x)$ behaves as a gauge potential under proper-time reparametrizations of the trembling field:

$$\Xi^\mu(\tau) \rightarrow \Xi^\mu(\tau) + \partial^\mu \alpha(\tau), \quad (\text{S186})$$

leading to the standard gauge transformation:

$$A_\mu(x) \rightarrow A_\mu(x) + \partial_\mu \alpha(x). \quad (\text{S187})$$

Thus, the familiar gauge structure of electromagnetism emerges as a consequence of phase redundancy in the abelian trembling description. The electromagnetic vector potential is not a fundamental object but a secondary, geometric artifact of coherent metric fluctuations projected onto the background.

Moreover, this emergent field couples to matter via the canonical minimal coupling prescription:

$$p_\mu \rightarrow p_\mu - qA_\mu(x), \quad (\text{S188})$$

where q is a scalar quantity that characterizes the amplitude of coupling between the particle and the trembling-induced potential. This conserved charge arises naturally from Noether's theorem as the generator of global $U(1)$ symmetry in the trembling action, and corresponds physically to electric charge. The local redundancy in the trembling field representation—namely, the freedom to shift by a derivative of a scalar—directly mirrors gauge invariance in standard field theory. All physical observables remain invariant under such shifts, confirming the emergent $U(1)$ gauge structure.

In summary, the electromagnetic potential $A_\mu(x)$ in TSRT is not externally postulated but is a coarse-grained geometric object. It encapsulates how causal, abelian fluctuations in space-time curvature are perceived by matter, obeys the correct transformation laws, and interacts with charged particles through a conserved scalar current—all arising within the deterministic framework of trembling spacetime geometry.

S10.3 Field Strength Tensor and Maxwell Equations

Once the effective vector potential $A_\mu(x)$ has been defined as the coarse-grained projection of abelian trembling deviations (see Section S10.2), the natural next step is to evaluate the associated curvature induced by these fluctuations. This curvature is encoded in the antisymmetric field strength tensor,

$$F_{\mu\nu}(x) := \partial_\mu A_\nu(x) - \partial_\nu A_\mu(x), \quad (\text{S189})$$

which captures the infinitesimal holonomy of the abelian connection across local patches of spacetime. Geometrically, $F_{\mu\nu}$ quantifies the failure of trembling phase vectors to return to their original orientation after infinitesimal closed transport, in exact analogy with the curvature of a gauge connection.

Because the underlying symmetry group of the abelian trembling sector is $U(1)$, the corresponding gauge structure is commutative and supports a single generator. Within TSRT, this sector emerges from coherent phase-like trembling deviations aligned with the particle's four-velocity, of the form

$$\Xi^\mu(\tau) = \epsilon(\tau)u^\mu(\tau), \quad \epsilon(\tau) = \epsilon_0 \cos(\omega\tau + \phi), \quad (\text{S190})$$

as described in Equation (S182). These deviations produce symmetric metric perturbations

$$\xi_{\mu\nu} = \partial_\mu \Xi_\nu + \partial_\nu \Xi_\mu, \quad (\text{S191})$$

which are fully characterized by their coarse-grained projection onto spacetime. From this geometric structure, an effective vector potential $A_\mu(x)$ is defined via the averaged projection of the trembling deviation field onto the background geometry:

$$A_\mu(x) := \langle \partial_\mu \Xi^\nu(x) \rangle g_{\nu\rho} u^\rho. \quad (\text{S192})$$

The averaging operation $\langle \cdot \rangle$ denotes coarse-graining over proper-time fluctuations, preserving only the slowly varying geometric features of the trembling field that are visible at macroscopic scales.

From the vector potential, we define the emergent abelian field strength tensor, introduced in Equation (S189), in the usual exterior derivative form:

$$F_{\mu\nu}(x) := \partial_\mu A_\nu(x) - \partial_\nu A_\mu(x), \quad (\text{S193})$$

which measures the local antisymmetric curvature induced by the $U(1)$ connection generated by trembling deviations. This tensor obeys two fundamental equations that arise from distinct but complementary principles within TSRT: one dynamic, and one geometric.

S10.3.1 Dynamical Maxwell Equation

The first equation follows from applying the variation principle to the coarse-grained trembling action. Let the effective action $\mathcal{A}$ governing the deviation field include a kinetic term proportional to $F_{\mu\nu}F^{\mu\nu}$, consistent with the observed electromagnetic structure. Varying this action with respect to the vector potential A_μ , and requiring that the variation vanishes for arbitrary fluctuations δA_μ , yields the Euler–Lagrange equation:

$$\partial^\mu F_{\mu\nu}(x) = j_\nu(x), \quad (\text{S194})$$

where $j_\nu(x)$ is the effective current defined by:

$$j_\nu(x) := \left\langle \frac{\delta \mathcal{A}}{\delta A^\nu(x)} \right\rangle. \quad (\text{S195})$$

In TSRT, this current arises from the coarse-grained variation of the trembling deviation field under global $U(1)$ transformations:

$$\Xi^\mu(\tau) \rightarrow \Xi^\mu(\tau) + \partial^\mu \alpha(\tau), \quad (\text{S196})$$

which define the allowable symmetry transformations of the abelian trembling geometry. Thus, the dynamical Maxwell equation (S194) expresses the geometric response of the spacetime curvature field $F_{\mu\nu}$ to the conserved trembling-induced current j_ν .

S10.3.2 Bianchi Identity and Homogeneous Maxwell Equation

The second equation is not variational but algebraic. It follows directly from the definition of $F_{\mu\nu}$ as an antisymmetric two-form. Taking the cyclic antisymmetric derivative of $F_{\mu\nu}$, one obtains the Bianchi identity [31, 37, 57]:

$$\partial_{[\lambda} F_{\mu\nu]}(x) = 0, \quad (\text{S197})$$

where square brackets denote antisymmetrization over the indices. This identity is a direct consequence of the symmetry of partial derivatives and the abelian nature of the connection. In classical electromagnetism, it corresponds to the homogeneous Maxwell equations, such as $\nabla \cdot \vec{B} = 0$ and $\nabla \times \vec{E} + \partial_t \vec{B} = 0$. In the TSRT framework, it ensures that the trembling-induced curvature field $F_{\mu\nu}$ is derivable from a globally well-defined vector potential A_μ , i.e., that the

effective electromagnetic structure has no topological obstruction or internal torsion.

S10.3.3 Interpretation Within Trembling Geometry

Equations (S194) and (S197) together encode the full content of Maxwell's theory within the TSRT framework, but with a fundamentally different origin than in classical or quantum field theory. In TSRT, these equations are not postulated from external symmetry principles or Lagrangian assumptions; rather, they emerge from the causal geometry and coherence structure of trembling spacetime. The underlying abelian symmetry group $U(1)$ ensures that trembling deviations commute and may be superposed linearly. This property guarantees that the field strength tensor, defined as the exterior curl of the vector potential,

$$F_{\mu\nu}(x) := \partial_\mu A_\nu(x) - \partial_\nu A_\mu(x) \quad (\text{S198})$$

is antisymmetric and derivable from a potential $A_\mu(x)$ defined via proper-time-projected averages of the local deviation field $\Xi^\mu(\tau)$, as introduced in Equation (S189). These deviations, in the abelian case, exhibit coherent phase evolution along the particle's proper-time path, enabling long-range geometric consistency. The inhomogeneous Maxwell equation, as in Equation (S194),

$$\partial^\mu F_{\mu\nu}(x) = j_\nu(x), \quad (\text{S199})$$

emerges not from a gauge-theoretic assumption but from the statistical divergence of trembling-induced proper-time currents. The source term $j_\nu(x)$ reflects the macroscopic accumulation of net deviation flows, coarse-grained over spacetime. It captures the imbalance in locally aligned trembling patterns, leading to a conserved effective current density interpreted as charge and current. This interpretation is made explicit in the expression:

$$j_\nu(x) := \langle \partial^\mu \partial_\mu A_\nu(x) - \partial^\mu \partial_\nu A_\mu(x) \rangle, \quad (\text{S200})$$

which defines the effective electromagnetic current as the divergence of the coarse-grained field strength tensor $F_{\mu\nu}(x)$, via the identity:

$$\partial^\mu F_{\mu\nu}(x) = \partial^\mu (\partial_\mu A_\nu - \partial_\nu A_\mu) = \partial^\mu \partial_\mu A_\nu - \partial^\mu \partial_\nu A_\mu. \quad (\text{S201})$$

Thus, this formula for $j_\nu(x)$ is not an independent postulate but follows directly from the geometric definition of the field strength tensor $F_{\mu\nu}$ and the inhomogeneous Maxwell equation (S194) in the TSRT framework. The angular brackets $\langle \cdot \rangle$ denote coarse-graining over locally fluctuating spacetime patches, reflecting the statistical behavior of trembling deviations.

In TSRT, this current represents the net geometric effect of proper-time-evolving metric fluctuations, as encoded in the deviation field $\Xi^\mu(\tau)$. It quantifies the residual asymmetry in trembling-induced accelerations that persist after ensemble averaging. Hence, $j_\nu(x)$ arises as a derived, deterministic object grounded in the spacetime geometry of trembling, and not from externally imposed charge distributions or quantized matter fields. Simultaneously, the

homogeneous Maxwell equation, given in Equation (S197),

$$\partial_{[\lambda} F_{\mu\nu]}(x) = 0, \quad (\text{S202})$$

follows identically from the definition of the field strength tensor $F_{\mu\nu}(x)$ as the antisymmetric curl of the vector potential:

$$F_{\mu\nu}(x) := \partial_\mu A_\nu(x) - \partial_\nu A_\mu(x), \quad (\text{S203})$$

as previously introduced (Equation (S189)).

The expression $\partial_{[\lambda} F_{\mu\nu]}$ denotes the total antisymmetrization over the three indices, defined explicitly as:

$$\partial_{[\lambda} F_{\mu\nu]} := \frac{1}{3!} (\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} - \partial_\lambda F_{\nu\mu} - \partial_\mu F_{\lambda\nu} - \partial_\nu F_{\mu\lambda}), \quad (\text{S204})$$

which vanishes identically when $F_{\mu\nu}$ is built from partial derivatives of a single differentiable vector field A_μ , due to the symmetry of second derivatives:

$$\partial_\lambda \partial_\mu A_\nu = \partial_\mu \partial_\lambda A_\nu. \quad (\text{S205})$$

The identity in Equation (S202)—known as the Bianchi identity in differential geometry—is thus not an assumption but a mathematical consequence of the abelian field's definition. It encodes two central features of classical Maxwell theory: the absence of magnetic monopoles and the integrability of the field strength tensor. Specifically, the equation

$$\partial_{[\lambda} F_{\mu\nu]} = 0 \quad (\text{S206})$$

guarantees that the magnetic field $\vec{B}$ is divergence-free:

$$\nabla \cdot \vec{B} = 0, \quad (\text{S207})$$

which classically implies that there are no isolated magnetic charges—i.e., no magnetic monopoles—in standard electromagnetism.

In the TSRT framework, this result emerges naturally from the geometric structure of abelian trembling. Because the deviation field $\Xi^\mu(\tau)$ generates a symmetric perturbation $\xi_{\mu\nu} = \partial_\mu \Xi_\nu + \partial_\nu \Xi_\mu$, and because the effective potential $A_\mu(x)$ arises from a projected and averaged derivative of this deviation field, the resulting field strength tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ satisfies the curl structure identically. As a result, the Bianchi identity holds at every point in spacetime, and no magnetic monopoles are introduced at the level of trembling-induced electromagnetic dynamics. TSRT explains the absence of magnetic monopoles not as an empirical anomaly, but as a geometric inevitability of the abelian trembling sector. Because abelian deviations commute and remain globally integrable across spacetime, the associated vector potential A_μ can always be locally defined as a differentiable field. Its antisymmetric curl then yields a field strength tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, whose magnetic components are necessarily divergence-free. The absence of magnetic monopoles is thus a direct consequence of the coherence and integrability of the abelian geometric deviations—no ad hoc constraints are required.

If future experiments were to detect monopole-like phenomena, TSRT predicts that they must originate not from the abelian sector, but from non-Abelian or topologically nontrivial trembling configurations. In such scenarios, the deviation fields $\Xi^\mu(\tau)$ would encode curvature structures incompatible with global $U(1)$ integrability, allowing for emergent magnetic sources via topological defects or compact non-integrable bundles. These effects would naturally break the curl-free structure of $F_{\mu\nu}$, permitting magnetic source terms to appear as consequences of the extended trembling geometry. However, within the minimal abelian $U(1)$ sector analyzed here, such configurations do not arise. In the TSRT framework Equation (S197) reflects the underlying geometric coherence of the abelian trembling deviations. Because the deviation field $\Xi^\mu(\tau)$ yields a symmetric perturbation $\xi_{\mu\nu} = \partial_\mu \Xi_\nu + \partial_\nu \Xi_\mu$, and the effective potential A_μ is derived from a projection of $\partial_\mu \Xi^\nu$, the resulting $F_{\mu\nu}$ —as defined in Equation (S203)—is guaranteed to satisfy the curl condition. The antisymmetric structure of $F_{\mu\nu}$ and the commutativity of the underlying abelian trembling sector ensure that $\partial_{[\lambda} F_{\mu\nu]} = 0$ holds at every point in spacetime without additional assumptions.

Thus, the homogeneous Maxwell equation in TSRT emerges not from a postulated gauge symmetry but from the intrinsic geometric structure of the trembling-induced vector potential. These two equations jointly imply that the electromagnetic field is not a fundamental external entity but a coarse-grained feature of abelian trembling geometry.

Furthermore, the motion of a charged particle through an abelian trembling field in TSRT is governed by an effective dynamical equation:

$$m \frac{du^\mu}{d\tau} = q F^\mu{}_\nu(x) u^\nu, \quad (\text{S208})$$

but this relation is not introduced by hand, as shown below.

Equation (S208) emerges directly from the coarse-grained limit of the full trembling geodesic equation derived in Section S4. There, the motion of a test particle under the influence of a trembling-induced deviation field $\Xi^\mu(\tau)$ is governed by a modified geodesic equation of the form:

$$\frac{du^\mu}{d\tau} + \Gamma^\mu_{\rho\sigma}(x) u^\rho u^\sigma = \Delta^\mu(x, \tau), \quad (\text{S209})$$

where $\Gamma^\mu_{\rho\sigma}(x)$ are the Christoffel symbols of the background geometry, and Δ^μ encodes the perturbative influence of the trembling-induced metric fluctuations. In the presence of coherent abelian trembling, these deviations introduce a small, symmetric metric perturbation $\xi_{\mu\nu}(x) = \partial_\mu \Xi_\nu(x) + \partial_\nu \Xi_\mu(x)$, and hence induce an additional contribution to the affine structure via:

$$\delta\Gamma^\mu_{\rho\sigma}(x) = \frac{1}{2} \bar{g}^{\mu\lambda} (\nabla_\rho \xi_{\sigma\lambda} + \nabla_\sigma \xi_{\rho\lambda} - \nabla_\lambda \xi_{\rho\sigma}). \quad (\text{S210})$$

To extract the leading-order effect of these fluctuations on the trajectory of the particle, we take a coarse-grained average over the proper-time fluctuations and project the deviation structure onto an effective background. The perturbation $\xi_{\mu\nu}(x)$ is traced back to a coherent oscillation of the form $\Xi^\mu(\tau) = \epsilon(\tau) u^\mu(\tau)$, with $\epsilon(\tau)$ oscillatory. This leads to an effective vector potential:

$$A_\mu(x) := \langle \partial_\mu \Xi^\nu(x) \rangle g_{\nu\rho} u^\rho, \quad (\text{S211})$$

and correspondingly a field strength tensor:

$$F_{\mu\nu}(x) := \partial_\mu A_\nu(x) - \partial_\nu A_\mu(x). \quad (\text{S212})$$

The effective coarse-grained equation of motion is then obtained by expressing the deviation-induced perturbation Δ^μ in terms of $F^\mu_\nu(x)u^\nu$. The crucial step here is to recognize that the projected acceleration due to the trembling perturbation is transverse to u^μ and proportional to the antisymmetric combination of derivatives arising in F^μ_ν . Thus, the averaged deviation equation becomes:

$$\frac{du^\mu}{d\tau} = \frac{q}{m} F^\mu_\nu(x) u^\nu, \quad (\text{S213})$$

where q arises from the conserved Noether charge associated with the $U(1)$ symmetry of the abelian trembling sector, as derived in Section S14.

This shows that the Lorentz-force-type equation is not assumed but rather deduced from geodesic motion in a fluctuating abelian geometry. The entire structure of classical electromagnetic interaction—including the field, the coupling, and the response—thus emerges from the causal, deterministic dynamics of trembling deviations in TSRT.

Equation (S213) is formally identical to the classical Lorentz force law known from electrodynamics. However, in TSRT, its origin is entirely geometric and deterministic: the right-hand side arises not from a postulated interaction term added to the equations of motion, but from the deviation of geodesic motion due to abelian trembling-induced metric perturbations. In other words, the “force” acting on the particle is a kinematic effect that appears when proper-time evolution is carried out on a background that has been coherently deformed by trembling geometry. To understand this, recall that in general relativity, particles follow geodesics of a curved spacetime. If the spacetime metric undergoes microscopic deterministic fluctuations—such as those introduced by trembling deviations $\Xi^\mu(\tau)$ —then the effective path of the particle deviates from that of a geodesic in the background metric $\bar{g}_{\mu\nu}$. When such fluctuations are coherent and abelian in nature, the resulting deviation structure projects onto an effective vector potential $A_\mu(x)$, and the metric perturbation manifests as a field strength $F_{\mu\nu}$. From this, one can derive the above equation by calculating the coarse-grained acceleration of the particle’s trajectory in the perturbed geometry. The appearance of $qF^\mu_\nu u^\nu$ on the right-hand side thus represents the net effect of traveling through a trembling-induced geometry, where the vector potential has emerged from coherent phase-type deviations.

Therefore, in TSRT, the Lorentz force is not an added empirical law but an emergent property of the particle’s motion through a dynamically fluctuating, yet causally consistent, abelian metric field. This reformulation preserves all essential features of classical electromagnetism—field equations, conserved current, and the force law—yet grounds them in the underlying geometric dynamics of trembling spacetime. The familiar electromagnetic interaction thus becomes a manifestation of spacetime structure itself.

As shown in Equation (S211), the emergent vector potential $A_\mu(x)$ is defined from this deviation structure by coarse-graining the projected derivative of the trembling field:

$$A_\mu(x) := \langle \partial_\mu \Xi^\nu(x) \rangle g_{\nu\rho}(x) u^\rho(x), \quad (\text{S214})$$

where the angular brackets $\langle \cdot \rangle$ denote coarse-graining over small-scale proper-time oscillations. From this effective potential, the abelian field strength tensor is defined by

$$F_{\mu\nu}(x) := \partial_\mu A_\nu(x) - \partial_\nu A_\mu(x). \quad (\text{S215})$$

Taking the divergence of this field strength leads to a dynamical equation:

$$\partial^\mu F_{\mu\nu}(x) = j_\nu(x). \quad (\text{S216})$$

The key insight in TSRT is that the source term $j_\nu(x)$ on the right-hand side does not originate from an externally specified field or matter distribution, as it does in classical electrodynamics. Instead, it arises intrinsically from the collective influence of local trembling deviations—more precisely, from the divergence of the ensemble-averaged acceleration patterns generated by the evolving deviation field $\Xi^\mu(\tau)$.

To see this explicitly, we substitute the definition of $F_{\mu\nu}$ into its divergence:

$$\begin{aligned} j_\nu(x) &= \partial^\mu F_{\mu\nu}(x) \\ &= \partial^\mu (\partial_\mu A_\nu(x) - \partial_\nu A_\mu(x)) \\ &= \partial^\mu \partial_\mu A_\nu(x) - \partial^\mu \partial_\nu A_\mu(x). \end{aligned} \quad (\text{S217})$$

Since $A_\mu(x)$ is constructed from ensemble-averaged trembling deviations, the derivatives in the above expression act on averaged geometric quantities. Therefore, we can write:

$$j_\nu(x) := \langle \partial^\mu \partial_\mu A_\nu(x) - \partial^\mu \partial_\nu A_\mu(x) \rangle, \quad (\text{S218})$$

as quoted previously. This form makes it clear that $j_\nu(x)$ encodes the divergence of curvature gradients generated by macroscopic trembling, and thus reflects a coherent accumulation of microscopic geometric fluctuations.

Importantly, this current does not need to be prescribed. It is not introduced through external matter fields, nor does it require the postulation of point charges or quantized sources. Instead, $j_\nu(x)$ emerges from the internal imbalance of the trembling pattern—specifically, from the non-uniformity in how proper-time deviations $\Xi^\mu(\tau)$ accumulate in different regions of space-time. In physical terms, when a population of trembling trajectories collectively exhibits a preferred direction of proper-time deviation—such as a drift or phase bias relative to the background geodesic structure—this generates a macroscopic flow pattern, which is then interpreted as an effective electric current or charge density. Hence, in TSRT, electromagnetic interaction is not a result of imposing gauge invariance on a quantum field. It instead arises deterministically from the intrinsic geometry of spacetime trembling. The vector potential $A_\mu(x)$, the field strength $F_{\mu\nu}(x)$, and the source term $j_\nu(x)$ all descend from the statistical behavior of proper-time deviation fields, governed by causal, Lorentz-invariant evolution. The conservation of the resulting current,

$$\partial^\nu j_\nu(x) = 0, \quad (\text{S219})$$

is guaranteed by the antisymmetry of $F_{\mu\nu}$ and the commutativity of partial derivatives, fur-

ther confirming that charge conservation in TSRT is a geometric consequence of the trembling structure—not an imposed symmetry.

In this way, TSRT provides a first-principles derivation of electromagnetic source terms and conservation laws, firmly rooting them in spacetime geometry rather than field-theoretic assumptions. It is important to emphasize that the Bianchi identity (S197) is a purely geometric statement: it reflects the antisymmetry and derivation structure of $F_{\mu\nu}$, and holds identically for any connection constructed from a gradient field. Its validity confirms the consistency of the trembling-based origin of A_μ , while Equation (S194) dynamically relates this geometric structure to observable effects such as charge and current densities.

S10.4 Equation of Motion and Lorentz Force

The effective dynamics of a particle subjected to abelian trembling fluctuations are governed not by the imposition of minimal coupling, as in standard gauge theory [58–61], but by the geometric influence of the coarse-grained metric deviation field $\xi_{\mu\nu}(x, \tau)$ along the particle’s worldline. In the presence of coherent phase-type trembling, this influence manifests as a force term that alters the particle’s geodesic path through spacetime. The proper-time evolution of the four-velocity $u^\mu = dx^\mu/d\tau$ of a charged particle with mass m and emergent charge q satisfies the effective equation of motion, as in Equations (S208) and (S213):

$$m \frac{du^\mu}{d\tau} = q F^\mu{}_\nu u^\nu, \quad (\text{S220})$$

which exactly reproduces the Lorentz force law of classical electrodynamics. However, unlike its traditional derivation from the postulate of minimal coupling in a gauge-theoretic Lagrangian, Equation (S220) arises in TSRT from the response of the particle’s path to small but coherent abelian deviations in the local spacetime geometry.

This result follows from the application of the geodesic deviation principle in a trembling background. As shown in Section S9, abelian trembling fluctuations produce symmetric perturbations of the form:

$$\xi_{\mu\nu} = A_\mu A_\nu, \quad (\text{S221})$$

or more generally, can be encoded via their contribution to an effective vector potential A_μ , defined as a coarse-grained projection of $\partial_\mu \Xi^\nu$ along the background trajectory. These deviations produce symmetric metric perturbations, as in Equation (S182), of the form

$$\Xi^\mu(\tau) = \epsilon(\tau) u^\mu(\tau), \quad \epsilon(\tau) = \epsilon_0 \cos(\omega\tau + \phi), \quad (\text{S222})$$

where $\epsilon(\tau)$ encodes the proper-time oscillation amplitude. This ansatz reflects abelian $U(1)$ phase trembling, in which deviations are aligned with the particle’s four-velocity $u^\mu(\tau)$, preserving direction but varying in phase—ensuring coherence under Lorentz transformations and enabling long-range interaction.

When coarse-grained over proper time, these deviations yield an emergent vector potential $A_\mu(x)$, and the resulting curvature is encoded in the field strength tensor. The induced curvature associated with this deviation leads to an acceleration term that is locally equivalent to a

force law, and whose antisymmetric part is captured precisely by the field strength tensor $F_{\mu\nu}$. Therefore, the Lorentz force in TSRT is not a phenomenological input but an emergent geometric consequence of motion through a spacetime whose structure is undergoing abelian trembling. The acceleration of the particle reflects the deviation from inertial motion induced by the geometric torsion of its local frame, projected into proper-time evolution. This is conceptually equivalent to how tidal forces arise in curved spacetimes—but here the curvature is trembling-induced and coherent in phase. Moreover, this geometric derivation ensures that the resulting dynamics respect both local Lorentz covariance and $U(1)$ gauge invariance, since $F_{\mu\nu}$ remains invariant under $A_\mu \rightarrow A_\mu + \partial_\mu \alpha(x)$, and the force law is constructed entirely from tensors under diffeomorphisms.

In summary, Equation (S220) reveals that charged particle motion in TSRT arises as geodesic deviation in a coherently trembling abelian geometry. What appears as a force in conventional theory is reinterpreted as the natural deviation from straight-line motion in a trembling spacetime background, demonstrating the profound geometric unification at the heart of the theory. Thus, what classical electrodynamics describes as a force acting on a charge is, in TSRT, the manifestation of a particle’s inertial response to a coherently trembling spacetime—showing that even Lorentz’s law is a shadow of geometry.

S10.5 Conservation Law and Emergent Charge

In the TSRT framework, the emergence of electric charge is not a fundamental input, but a consequence of symmetry under abelian trembling transformations. Specifically, the phase-type deviation field $\Xi^\mu(\tau)$, when associated with the $U(1)$ sector (as introduced in Section S9 and Equation (S222)), generates a local geometric transformation of the form:

$$\Xi^\mu(\tau) \rightarrow \Xi^\mu(\tau) + \partial^\mu \alpha(\tau), \quad (\text{S223})$$

where $\alpha(\tau)$ is a smooth scalar function encoding local phase shifts in proper time. This transformation leaves the coarse-grained vector potential $A_\mu(x)$, defined in Equation (S214), invariant up to a total derivative:

$$A_\mu(x) \rightarrow A_\mu(x) + \partial_\mu \alpha(x), \quad (\text{S224})$$

ensuring that the field strength tensor

$$F_{\mu\nu} := \partial_\mu A_\nu - \partial_\nu A_\mu \quad (\text{S225})$$

remains unchanged:

$$F_{\mu\nu} \rightarrow F_{\mu\nu}. \quad (\text{S226})$$

Thus, the trembling sector respects local $U(1)$ gauge invariance, as established in the derivation of the inhomogeneous Maxwell equation (Equation (S194)) and the conserved current structure introduced in Section S14.

Because the full trembling geometry is invariant under this continuous internal symmetry, the action functional remains unchanged under infinitesimal $U(1)$ phase transformations. This

implies the existence of a conserved quantity via Noether's theorem [38]:

$$\frac{d}{d\tau}q = 0, \quad (\text{S227})$$

where q is the conserved Noether charge associated with the symmetry. Unlike in standard quantum field theory, where electric charge is introduced as an intrinsic property of fields or particles, in TSRT the conserved quantity q arises directly from the geometric invariance of the trembling deviation field. That is, the concept of charge is not fundamental but emergent: the invariance of the internal trembling phase implies the existence of a current-like structure and its corresponding conserved scalar. This quantity acquires physical interpretation only a posteriori—through its coupling to the emergent vector field A_μ , its role in generating the effective field strength $F_{\mu\nu}$, and its appearance in the Lorentz-like force law derived in Equation (S220). These properties collectively identify q as the electric charge of the particle, emergent from the underlying trembling symmetry.

Importantly, the conservation law in Equation (S227) is geometric in origin and holds universally across all proper-time evolution. It ensures that the internal structure of trembling particles—once established—remains coherent across spacetime, consistent with the observed conservation of electric charge in all known electromagnetic interactions. This redefinition of electric charge as a geometric Noether charge reinforces the central thesis of TSRT: all physical structures, including gauge interactions and conserved quantities, arise from the deterministic trembling of spacetime itself, with symmetry playing the foundational role.

S10.6 Magnetic Dipoles from Coherent Abelian Trembling

In classical electrodynamics, magnetic dipoles arise from closed-loop currents or intrinsic particle spin. These phenomena are often postulated or introduced phenomenologically, relying on either current distributions or intrinsic quantum angular momentum. In contrast, the TSRT framework reinterprets magnetic dipole structures as emergent features of coherent, spatially non-uniform trembling patterns within the abelian deviation field $\Xi^\mu(\tau)$.

Let us consider a localized spacetime region in which the proper-time deviation field exhibits coherent oscillatory behavior aligned with the four-velocity (as in Equation (S182)):

$$\Xi^\mu(\tau) = \epsilon(\tau, x)u^\mu(\tau), \quad \text{with} \quad \epsilon(\tau, x) = \epsilon_0(x) \cos[\omega\tau + \phi(x)], \quad (\text{S228})$$

where $\epsilon_0(x)$ is a slowly varying amplitude field, and $\phi(x)$ encodes the local phase structure of trembling across space. If the spatial gradient of the phase field $\phi(x)$ has a rotational (vortex-like) character, such as $\phi(x) \sim \theta$ in polar coordinates, the associated coarse-grained vector potential $A_\mu(x)$ defined (see also Equation (S185)) as

$$A_\mu(x) := \langle \partial_\mu \Xi^\nu(x) \rangle g_{\nu\rho}(x) u^\rho(x) \quad (\text{S229})$$

acquires a curl with the spatial structure of a magnetic dipole.

Specifically, the magnetic field in classical and relativistic electrodynamics is extracted from the spatial components of the antisymmetric field strength tensor $F_{\mu\nu}$. In the context of TSRT,

where this tensor arises from the coarse-grained projection of proper-time trembling deviations, the magnetic field components B^i are identified as:

$$B^i(x) = \frac{1}{2} \epsilon^{ijk} F_{jk}(x), \quad (\text{S230})$$

where ϵ^{ijk} is the three-dimensional Levi-Civita symbol, and the indices j, k run over the spatial coordinates 1, 2, 3. This expression reflects the fact that the magnetic field is encoded in the spatial curl of the vector potential:

$$B^i(x) = \epsilon^{ijk} \partial_j A_k(x). \quad (\text{S231})$$

This relation follows from the definition of the field strength tensor in the abelian case:

$$F_{\mu\nu}(x) := \partial_\mu A_\nu(x) - \partial_\nu A_\mu(x), \quad (\text{S232})$$

so that, in spatial indices,

$$F_{jk}(x) = \partial_j A_k(x) - \partial_k A_j(x), \quad (\text{S233})$$

and substituting into the antisymmetric contraction with ϵ^{ijk} yields the standard magnetic field expression.

To characterize the macroscopic magnetic structure generated by this trembling-induced field, one introduces the magnetic dipole moment $\vec{m}$. In classical theory, $\vec{m}$ is defined via the spatial distribution of currents. In TSRT, an analogous definition is constructed using the effective coarse-grained current density $j^\mu(x)$, which arises from the divergence of the trembling-induced field strength:

$$j^\mu(x) := \partial^\nu F_{\nu\mu}(x). \quad (\text{S234})$$

This current captures the net geometric flow of trembling deviations across spacetime and includes any local accumulation or circulation induced by spatial asymmetries.

The magnetic dipole moment vector m^i is then defined through the spatial moment of this current, using the standard cross-product integral form:

$$m^i := \frac{1}{2} \int d^3x \epsilon^{ijk} x^j j^k(x). \quad (\text{S235})$$

This expression measures the extent to which effective currents loop or circulate within a finite region of space, producing a nonzero magnetic moment.

The above definition of the magnetic dipole moment, Equation (S235), has thus far been presented in its conventional three-vector form, appropriate for the non-relativistic or laboratory frame description of magnetic systems. However, to ensure compatibility with the relativistic and covariant nature of the TSRT framework, it is essential to reinterpret this quantity in manifestly Lorentz-invariant terms. This is accomplished by embedding the magnetic dipole moment within the antisymmetric angular momentum tensor $M^{\mu\nu}$, which combines both orbital and intrinsic contributions to electromagnetic moments. In the context of trembling spacetime, the

relevant contribution arises from the spatial moment of the conserved current $j^\mu(x)$, defined as:

$$M^{\mu\nu} := \int d^3x (x^\mu j^\nu(x) - x^\nu j^\mu(x)). \quad (\text{S236})$$

This tensor is antisymmetric by construction and transforms covariantly under Lorentz transformations. Its spatial components M^{ij} encode the magnetic moment structure, while its mixed time-space components M^{0i} correspond to electric dipole moments in general configurations.

This formulation confirms that the trembling-induced magnetic dipole moment in TSRT is fully Lorentz-covariant. It arises from the integration of the geometric current $j^\mu(x)$ against spacetime positions, forming the antisymmetric angular momentum tensor $M^{\mu\nu}$. The conventional three-vector magnetic moment $\vec{m}$, introduced in Equation (S235), appears as the spatial part of this covariant structure in the low-velocity limit:

$$m^i = \frac{1}{2} \epsilon^{ijk} M^{jk}. \quad (\text{S237})$$

This demonstrates that the magnetic response of abelian trembling fields inherits the correct transformation properties under Lorentz boosts and rotations, grounded not in coordinate-dependent constructs but in intrinsic spacetime geometry. In TSRT, this magnetic dipole moment is not attributed to circulating electric charges or intrinsic quantum spin. Instead, it reflects coherent spatial structure in the proper-time phase deviations $\epsilon(\tau, x)$. When the phase $\phi(x)$ of the trembling field exhibits angular asymmetries—such as azimuthal winding—spatial derivatives of the coarse-grained potential $A_\mu(x)$ yield a nonzero curl, producing an effective magnetic field. The associated current $j^k(x)$ acquires a circulating component, leading to a measurable magnetic dipole moment.

Thus, TSRT provides a geometric explanation for magnetic dipoles entirely in terms of coherent, abelian, phase-type trembling. No physical current loops or intrinsic spin variables are required. Instead, the dipole arises from directional coherence in trembling-induced deviations, governed by causal proper-time evolution and phase alignment with the four-velocity. This mechanism preserves Lorentz invariance, avoids any violations of causal structure, and ensures compatibility with the broader TSRT framework. The emergent magnetic field is therefore not a fundamental input, but a derived macroscopic effect—arising from spatial asymmetries in the geometry of trembling spacetime. As such, TSRT recovers the full structure of classical electromagnetic multipole fields from first-principles, replacing external source assumptions with intrinsic geometric coherence.

S10.7 Conclusion

Electromagnetism in Trembling Spacetime Relativity Theory (TSRT) emerges as a uniquely determined consequence of the abelian sector of admissible trembling fluctuations. These fluctuations, classified in Section S9, correspond to coherent, non-mixing deviations of the form $\Xi^\mu(\tau) = \epsilon(\tau)u^\mu(\tau)$, which are fully characterized by their invariance under local phase shifts in proper time. This structure enforces a $U(1)$ symmetry—not by postulate or analogy with quantum mechanics, but as the only permissible abelian deformation class that preserves causal

propagation, local Lorentz invariance, and the positivity of proper time.

From this trembling structure, the effective vector potential $A_\mu(x)$ arises through coarse-graining, acquiring the interpretation of a connection under the emergent gauge group. The resulting field strength tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ naturally satisfies Maxwell's equations in the appropriate limit, and the corresponding Lorentz force law governing the dynamics of a charged particle follows from the induced geodesic deviation in the presence of abelian fluctuations. Moreover, electric charge itself is no longer a fundamental input but an emergent Noether charge associated with invariance under internal phase transformations of the trembling field. Its conservation arises geometrically, as a direct consequence of the action's symmetry under abelian internal reparametrizations in proper time. This analysis further reveals that even magnetic dipole moments, traditionally attributed to circulating currents or quantum spin, arise in TSRT from spatially structured proper-time deviations. Coherent abelian trembling with angular phase gradients produces effective currents and fields that reproduce dipole behavior, completing the geometric reinterpretation of classical electromagnetic phenomena.

In totality, the entire structure of classical electromagnetism—including gauge potential, field strength, source current, force law, and charge conservation—is realized within TSRT as a deterministic and causal manifestation of a specific class of metric trembling. The $U(1)$ gauge symmetry is not added to the theory but is uniquely selected by the internal consistency of the trembling spacetime framework. This firmly roots electromagnetism within the geometric dynamics of spacetime itself, eliminating the need for probabilistic quantization or external field assumptions. Thus, TSRT provides a compelling and complete geometric origin for electromagnetic interaction, reconciling the local realism of classical physics with the gauge symmetries of modern field theory, while preserving the deterministic evolution of proper-time-based trembling.

Having derived the structure of emergent abelian interactions, we next examine the more intricate emergence of non-abelian gauge forces from richer trembling patterns. First, we show that chiral non-abelian trembling modes naturally give rise to the weak nuclear force.

S11 Supplementary Material: Emergent Weak Force from Chiral Trembling

S11.1 Chiral Internal Fluctuations and $SU(2)_L \times U(1)_Y$ Symmetry

As rigorously demonstrated in Section S9, admissible classes of trembling-induced metric deviations are exhaustively classified by their geometric embedding and algebraic closure within Lorentzian four-dimensional spacetime. Among these, a unique class of internal non-Abelian fluctuations exists that preserves proper-time causality, respects Lorentz covariance, and remains consistent with compact group structure: the chiral sector governed by $SU(2)_L \times U(1)_Y$.

These fluctuations are not spacetime displacements but rather internal deviations defined within a fibered internal bundle $E \rightarrow \mathcal{M}$, where $\mathcal{M}$ denotes the base spacetime manifold and E encodes spinorial and angular trembling directions. The deviation fields $\Xi^A(\tau)$ associated with this bundle evolve along the particle's worldline in proper time τ , where the internal indices $A, B, \dots$ label directions in the fiber. The distinguishing feature of this sector is its chiral asymmetry. As noted in Section S9, TSRT permits internal trembling fields to undergo coherent, norm-preserving transformations in proper time:

$$\Xi^A(\tau) \mapsto R^A_B(\tau) \Xi^B(\tau), \quad R(\tau) \in SU(2)_L, \quad (\text{S238})$$

where $R(\tau)$ is a local, proper-time-dependent element of the weak isospin group $SU(2)_L$. These rotations act nontrivially only on the left-handed sector $E_L \subset E$, leaving the right-handed sector E_R inert at leading order:

$$E = E_L \oplus E_R, \quad R(\tau) \cdot E_R = E_R. \quad (\text{S239})$$

The spontaneous selection of left-handed coupling arises from the structure of trembling correlations in proper time, which respect the decomposition of spinor representations under the local Lorentz group $SO(3,1) \simeq SL(2, \mathbb{C})$. In four spacetime dimensions, the irreducible representations of the Lorentz group split naturally into left-handed and right-handed Weyl spinors. Trembling-induced internal motions must transform consistently under these spinorial representations to preserve local causal structure. Because the coherent trembling correlations are constrained by causal propagation and proper-time ordering, they selectively enhance correlations among left-handed excitations without imposing an external symmetry-breaking field. Thus, the left-right asymmetry of the weak force emerges geometrically from the causal-trembling framework itself, reflecting the intrinsic structure of spinor decompositions in relativistic spacetime. (See also Section S9 for the foundational classification of trembling-induced representations.)

As a result, the algebra of allowed transformations in this sector is non-Abelian:

$$[T^a, T^b] = i f^{abc} T^c, \quad (\text{S240})$$

where T^a are the generators of $\mathfrak{su}(2)_L$, and f^{abc} are its structure constants. These generators correspond to infinitesimal elements of internal trembling symmetry acting on left-handed internal degrees of freedom. Additionally, the internal trembling fields may undergo global or

local phase oscillations of the form:

$$\Xi^A(\tau) \mapsto e^{i\theta_Y(\tau)} \Xi^A(\tau), \quad (\text{S241})$$

with $\theta_Y(\tau) \in \mathbb{R}$ a proper-time-dependent hypercharge phase. These oscillations commute with all $SU(2)_L$ transformations and define the abelian symmetry group $U(1)_Y$, as also described in the abelian sector (see Section S10).

Together, the chiral and hypercharge degrees of freedom form the direct product:

$$\mathcal{G}_{\text{EW}} = SU(2)_L \times U(1)_Y, \quad (\text{S242})$$

which is the full electroweak gauge structure. Importantly, this structure is not assumed in TSRT—it emerges uniquely from the internal trembling classification constrained by:

1. Lorentz invariance of the metric deformation $\xi_{\mu\nu} = \partial_\mu \Xi_\nu + \partial_\nu \Xi_\mu$,
2. Proper-time causality: $g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} > 0$,
3. Compactness of the internal symmetry group,
4. Chirality consistency with spinor bundles in 4D spacetime,
5. Irreducibility: no subgroup of $SU(2)_L \times U(1)_Y$ gives rise to new conserved charges independent of the known forces.

No other internal trembling symmetry satisfies all five constraints. In particular, groups such as $SU(N > 3)$ or exceptional Lie algebras violate compact embeddability or exceed the allowable dimensionality of internal trembling curvature (see again Section S9).

Thus, the weak interaction arises in TSRT not by fiat, but as the only chirally asymmetric, non-Abelian interaction compatible with trembling-induced geometry in four dimensions. The full electroweak group $SU(2)_L \times U(1)_Y$ follows from the combined rotational and phase structure of internal trembling deviations, establishing a geometric origin for the Standard Model's electroweak symmetry. Because the trembling-induced currents are constructed geometrically and conserve causal structure, the emergent $SU(2)_L$ gauge symmetry is automatically anomaly-free at leading order, consistent with the requirements for a physically realizable weak interaction.

S11.2 Definition of the Weak Gauge Fields from Chiral Deviations

Building on the classification of allowed trembling structures presented in Section S9.4, we now explore the geometric realization of the weak interaction. In particular, we focus on the chiral non-Abelian trembling sector—introduced in Paragraph 3 of that section—which arises from internal, proper-time-dependent fluctuations that are confined to the left-chiral subspace $E_L \subset E$. These fluctuations break left–right symmetry and naturally induce parity-violating gauge interactions, as observed in the weak force. The associated metric perturbation has the form:

$$\xi_{\mu\nu}(\tau) = \sum_a W_{(\mu}^a(\tau) \sigma_{\nu)}^a, \quad (\text{S243})$$

where $W_\mu^a(\tau)$ are non-Abelian deviation fields, σ_ν^a encode the chiral projectors aligned with the weak isospin generators, and $a \in \{1, 2, 3\}$ labels the generators of the $SU(2)_L$ gauge group.

Let $\Xi^A(\tau) \in E_L$ be the internal trembling field defined along the proper-time worldline $x^\mu(\tau)$. Under local $SU(2)_L$ transformations $R^A_B(\tau)$, these fields transform as:

$$\Xi^A(\tau) \mapsto R^A_B(\tau) \Xi^B(\tau), \quad R(\tau) \in SU(2)_L. \quad (\text{S244})$$

Infinitesimally, the transformation reads:

$$\delta \Xi^A(\tau) = \theta^a(\tau) (T^a)^A_B \Xi^B(\tau), \quad (\text{S245})$$

where $T^a \in \mathfrak{su}(2)_L$ are Hermitian generators of the weak isospin algebra, and $\theta^a(\tau) \in \mathbb{R}$ are local rotation parameters.

To project these internal oscillations into macroscopic spacetime structure, we define the emergent weak gauge fields by coarse-graining over proper-time cycles:

$$W_\mu^a(x) := \langle \partial_\mu \Xi^A(x, \tau) \cdot (T^a)_{AB} \cdot \Xi^B(x, \tau) \rangle. \quad (\text{S246})$$

Here, $\langle \cdot \rangle$ denotes proper-time coarse-graining to suppress fast internal oscillations, while preserving macroscopic directional coherence. The contraction over internal indices uses the fiber metric on E , ensuring full covariance under local Lorentz and gauge transformations. This construction mirrors the abelian case discussed in Section S10, but extends it to the non-Abelian regime. Due to the non-commutativity of T^a , the emergent field $W_\mu^a(x)$ reflects the algebraic richness of $SU(2)_L$, encoding the averaged drift of left-handed internal motion across spacetime.

Inclusion of Hypercharge and Completion of the Electroweak Sector. In parallel, abelian phase-type trembling confined to a single internal direction gives rise to the hypercharge field. Let $\theta_Y(\tau)$ be a scalar internal fluctuation associated with $U(1)_Y$ symmetry. Its proper-time derivative yields the emergent hypercharge field:

$$B_\mu(x) := \langle \partial_\mu \theta_Y(\tau) \rangle. \quad (\text{S247})$$

As in the electromagnetic case, $B_\mu(x)$ transforms under local phase shifts and corresponds to a $U(1)_Y$ connection. The full electroweak structure thus consists of the triplet of weak isospin gauge fields $\{W_\mu^a(x)\}_{a=1}^3 \in \mathfrak{su}(2)_L$ and the abelian hypercharge field $B_\mu(x) \in \mathfrak{u}(1)_Y$, together forming the gauge group:

$$\mathcal{G}_{\text{EW}} = SU(2)_L \times U(1)_Y. \quad (\text{S248})$$

These fields transform under their respective local gauge transformations:

$$W_\mu^a(x) \mapsto U(x) W_\mu^a(x) U^{-1}(x) + \frac{i}{g} U(x) \partial_\mu U^{-1}(x), \quad U(x) \in SU(2)_L, \quad (\text{S249})$$

$$B_\mu(x) \mapsto B_\mu(x) + \partial_\mu \alpha(x), \quad \alpha(x) \in \mathbb{R}. \quad (\text{S250})$$

These transformation laws follow directly from the geometric behavior of the deviation fields

$\Xi^A(\tau)$ and $\theta_Y(\tau)$, ensuring compatibility with the underlying causal structure of trembling spacetime.

Summary. In TSRT, the weak isospin and hypercharge gauge fields are not postulated, but emerge deterministically from internal left-chiral and abelian proper-time fluctuations. Their algebraic properties and transformation behavior arise from the coherent structure of the trembling deviations. This yields a first-principles geometric origin of the electroweak gauge group $SU(2)_L \times U(1)_Y$, unifying it with the previously derived electromagnetic case and showing that the full gauge sector of the Standard Model is encoded in the internal symmetry structure of trembling spacetime.

S11.3 Field Strength Tensor and Yang–Mills Structure

As shown in Section S11.2, the weak gauge fields $W_\mu^a(x) \in \mathfrak{su}(2)_L$ emerge from coarse-grained chiral trembling deviations within the internal left-handed bundle E_L . Once these effective gauge fields are defined, their dynamical content must be described by a curvature structure that captures the nontrivial response of the underlying trembling geometry to infinitesimal parallel transport across spacetime. This curvature structure arises from the non-commutativity of local internal chiral rotations, as discussed in the classification of trembling sectors in Section S9. In geometric terms, the failure of successive infinitesimal rotations to commute signals an intrinsic field strength—the hallmark of non-Abelian gauge dynamics.

The field strength tensor $F_{\mu\nu}^a$ associated with the non-Abelian $SU(2)_L$ gauge connection W_μ^a is given by the standard Yang–Mills expression [49]:

$$F_{\mu\nu}^a := \partial_\mu W_\nu^a - \partial_\nu W_\mu^a + g f^{abc} W_\mu^b W_\nu^c, \quad (\text{S251})$$

where g is the effective weak coupling constant, and f^{abc} are the structure constants of the $\mathfrak{su}(2)$ Lie algebra, here given by the Levi-Civita symbol ϵ^{abc} in a suitable basis. This tensor $F_{\mu\nu}^a$ arises from the commutator of the covariant derivatives acting on a left-handed trembling deviation field $\Xi^A(\tau)$, and encodes the degree of curvature induced by transporting the internal deviation across an infinitesimal plaquette in spacetime. In precise terms, for a left-handed deviation $\Xi^A(\tau)$, one has:

$$[D_\mu, D_\nu]\Xi^A = -ig F_{\mu\nu}^a (T^a)^A_B \Xi^B, \quad (\text{S252})$$

where $D_\mu = \partial_\mu - ig T^a W_\mu^a$ is the covariant derivative derived from the trembling connection, as in Equation (54) of Section S9.

The curvature tensor $F_{\mu\nu}^a$ has two key physical interpretations in TSRT. First, it reflects the non-integrability of internal chiral rotations: successive transport around a closed loop in spacetime induces a net trembling misalignment, measurable by $F_{\mu\nu}^a$. Second, it encapsulates self-interactions among the weak gauge fields, a feature that distinguishes non-Abelian forces from the abelian case. For the $U(1)_Y$ hypercharge sector, where internal oscillations are abelian and commute, the corresponding field strength tensor takes the Maxwellian form:

$$B_{\mu\nu} := \partial_\mu B_\nu - \partial_\nu B_\mu, \quad (\text{S253})$$

which describes the integrable phase curvature of the abelian internal deviation $\theta_Y(\tau)$. This construction mirrors the derivation of the electromagnetic field tensor $F_{\mu\nu}$ in Section S10, confirming the general structure of gauge curvature from trembling symmetry.

Together, the tensors $\{F_{\mu\nu}^a, B_{\mu\nu}\}$ represent the full electroweak curvature content induced by the chiral trembling sector. These quantities are not inserted by hand, nor derived from a Lagrangian principle, but follow directly from the deterministic geometric evolution of internal deviations under the core constraints of TSRT. Specifically, the symmetry structures arise from compact Lie groups that admit finite-dimensional representations, ensuring internal closure and unitary evolution. The deviations must also respect local Lorentz invariance and proper-time causality, preserving the relativistic consistency of the underlying geometry. Moreover, they must be geometrically embeddable into four-dimensional trembling spacetime without requiring extra dimensions or hidden fibers. Finally, the internal structures must retain coherence under coarse-grained parallel transport, guaranteeing that their macroscopic effects remain covariant and physically meaningful within the emergent geometry. This geometric derivation provides a first-principles foundation for the Yang–Mills structure of the weak interaction. It establishes that the non-Abelian nature of weak gauge fields—including their curvature, self-coupling, and parity-violating character—are emergent consequences of left-handed internal trembling geometry. No quantization, external gauge symmetry, or field-theoretic assumptions are required: the entire curvature structure follows from the metric-consistent behavior of internal deviation fields within the TSRT framework.

S11.4 Equation of Motion and Chiral Force

The influence of chiral trembling deviations on particle motion is captured in TSRT as a geometric deformation of the proper-time trajectory within an extended configuration space that includes both external spacetime coordinates and internal chiral degrees of freedom. These internal degrees of freedom are geometrically realized via trembling fields defined on the fiber bundle $E_L \subset E$, associated with left-handed weak isospin symmetry.

As established in Section S11.2, proper-time-coherent chiral trembling induces non-Abelian and abelian gauge structures corresponding to the electroweak sector. Specifically, these give rise to coarse-grained gauge connections: a non-Abelian connection $W_\mu^i(x) \in \mathfrak{su}(2)_L$, representing weak isospin interactions, and an abelian connection $B_\mu(x) \in \mathfrak{u}(1)_Y$, associated with hypercharge.

These emergent fields are derived from averaging over internal trembling patterns and therefore reflect the statistical geometry of chiral deviations. To describe the dynamics of particles influenced by these emergent gauge fields, we generalize the standard geodesic equation to the context of a principal fiber bundle. In this generalized setting, the particle’s motion is governed not only by the geometry of the external spacetime manifold $\mathcal{M}$, but also by its evolution through the internal symmetry space E_L . Thus, the full configuration space is the product bundle $\mathcal{M} \times E_L$, and the dynamics must include both spacetime curvature and gauge field interactions. Let $u^\mu := \frac{dx^\mu}{d\tau}$ denote the four-velocity of the particle along its worldline, and let the internal degrees of freedom be represented by a weak isospin doublet $\psi(x)$. The coupling of

ψ to the emergent gauge fields is described by a covariant derivative:

$$D_\mu := \partial_\mu - igT^i W_\mu^i(x) - ig'Y B_\mu(x), \quad (\text{S254})$$

where $T^i \in \mathfrak{su}(2)_L$ are the generators of the weak isospin group, typically realized as $T^i = \frac{1}{2}\sigma^i$ (with σ^i the Pauli matrices), $Y \in \mathbb{R}$ is the hypercharge operator acting on the internal state, and g, g' are the coupling constants associated with $SU(2)_L$ and $U(1)_Y$, respectively.

The force exerted by the gauge fields on the particle is derived from the curvature tensors associated with these connections. The non-Abelian field strength tensor for the weak interaction is given by the standard Yang–Mills form:

$$F_{\mu\nu}^i := \partial_\mu W_\nu^i - \partial_\nu W_\mu^i + g\epsilon^{ijk}W_\mu^j W_\nu^k, \quad (\text{S255})$$

where ϵ^{ijk} are the structure constants of $\mathfrak{su}(2)_L$. Similarly, the abelian field strength tensor corresponding to the hypercharge sector is:

$$B_{\mu\nu} := \partial_\mu B_\nu - \partial_\nu B_\mu. \quad (\text{S256})$$

The total force on the particle is then obtained by computing the covariant Lorentz-like force law in the presence of both field strength tensors. Because the particle carries weak isospin and hypercharge, the acceleration in spacetime is given by:

$$m \frac{du^\mu}{d\tau} = g F^{\mu\nu i}(x) T^i u_\nu + g' B^{\mu\nu}(x) Y u_\nu, \quad (\text{S257})$$

where each term on the right-hand side reflects the projection of the corresponding field strength tensor onto the particle's four-velocity, weighted by the appropriate internal symmetry operator. The left-hand side describes the proper-time derivative of the four-velocity, i.e., the spacetime acceleration.

This equation represents the chiral generalization of the classical Lorentz force law to the electroweak gauge structure in TSRT. Unlike in classical field theory, where the gauge fields are externally introduced, in TSRT they emerge naturally from the geometric coarse-graining of chiral trembling deviations. The coupling constants g and g' encapsulate the effective interaction strengths of the underlying geometric fields. Thus, the equation of motion (S257) is not postulated, but derived from the structure of trembling spacetime. It expresses the influence of emergent gauge fields on particle trajectories, making manifest the deep geometric origin of electroweak interactions within the TSRT framework. This equation is the geometric analogue, in TSRT, of the Lorentz force law—but generalized to a non-Abelian internal bundle structure. Unlike the standard gauge theory derivation, where Equation (S257) is imposed via minimal substitution in the Lagrangian, here it follows from deterministic deviation dynamics and the preservation of proper-time causal structure in the internal trembling configuration space.

The interpretation of Equation (S257) proceeds as follows. The first term on the right-hand side represents the influence of non-Abelian chiral curvature on the trajectory of a left-handed particle. Since the generators T^i act non-trivially only on the left-handed sub-bundle E_L (as established in Section S11), this term inherently violates parity, consistent with the

experimentally observed chiral asymmetry of the weak force. Right-handed fields remain inert under this component of the interaction. The second term captures the effect of the abelian hypercharge curvature. Unlike the chiral term, this contribution is parity-symmetric and applies equally to both left- and right-handed components. However, in TSRT it typically plays a subdominant role relative to the chiral term in explaining electroweak asymmetry. Meanwhile, the left-hand side of the equation retains the classical form of the proper-time derivative of momentum, indicating that the particle trajectory remains geodesic with respect to the total connection induced by trembling. Importantly, this connection now incorporates internal field curvature, rendering the motion inherently gauge-sensitive.

This structure yields several key physical consequences. Parity violation is not introduced via chiral projectors, but rather emerges from the geometric restriction of internal trembling to the left-handed fiber bundle E_L , a constraint rigorously derived in Section S9. Flavor transitions and weak isospin mixing arise naturally from the non-commutative structure of the generators T^i , which permit trembling-induced transitions between internal states. These transitions are the geometric counterpart of neutrino oscillations, manifesting as dynamical rotations among weak isospin doublets. Furthermore, the gauge couplings g and g' are not arbitrary constants, but emerge from the amplitude and frequency spectra of internal trembling modes. They can be interpreted as coarse-grained projection weights derived from the statistical properties of the chiral trembling field. Finally, the gauge forces themselves arise not as imposed interactions, but as deviations from trembling geodesy. Equation (S257) thus represents a second-order geometric consequence of enforcing proper-time consistency in a curved, chiral-trembling background.

This formulation reveals the weak force as a purely geometric interaction resulting from the internal curvature of non-Abelian chiral trembling modes. It shares the same structural origin as the electromagnetic force described in Section S10, yet gains its distinctive parity-violating and flavor-mixing properties through the non-commuting symmetry algebra of $SU(2)_L$. No auxiliary quantum assumptions or Lagrangian-based postulates are required. The TSRT perspective thus yields a unified, causally consistent, and deterministically geometric account of the electroweak interaction as internal fiber curvature, arising solely from symmetry-constrained proper-time trembling.

S11.5 Conserved Quantities and Symmetry Breaking

As established in Section S9, the trembling deviations that define the weak interaction arise from internal chiral oscillations constrained by the compact, non-Abelian symmetry group $SU(2)_L \times U(1)_Y$. This symmetry governs the admissible transformations of the internal deviation field $\Xi^A(\tau) \in E_L$, and remains locally preserved throughout proper-time evolution as long as the trembling amplitudes remain symmetric and unconstrained.

The presence of this full symmetry group implies the existence of conserved currents. As derived in Section S14, invariance of the proper-time action under infinitesimal internal transformations yields two Noether currents:

- The weak isospin current $J_{SU(2)}^\mu$, associated with $SU(2)_L$ transformations,
- The hypercharge current J_Y^μ , arising from $U(1)_Y$ phase oscillations.

Both currents satisfy local conservation laws, expressed respectively as

$$D_\mu J_{SU(2)}^\mu = 0, \quad (\text{S258})$$

$$\partial_\mu J_Y^\mu = 0, \quad (\text{S259})$$

with D_μ the covariant derivative associated with the non-Abelian gauge connection. These conservation laws are not imposed externally but follow from the trembling symmetry class as a direct consequence of the geometric Noether principle in TSRT.

However, as also discussed in Section S9, the internal symmetry group $SU(2)_L \times U(1)_Y$ can be dynamically reduced without requiring any external scalar field. In TSRT, this reduction—analogue to spontaneous symmetry breaking in quantum field theory—arises through saturation and directional reorganization of internal trembling amplitudes. In particular, when the trembling deviations $\Xi^A(\tau)$ acquire proper-time coherence along a preferred direction in internal space, the trembling geometry no longer remains invariant under the full internal symmetry group. This coherence can arise from several geometric mechanisms within the trembling fiber bundle. One possibility is amplitude saturation in a particular chiral direction, whereby the internal deviation field $\Xi^A(\tau)$ preferentially stabilizes along a specific eigenvector of the chiral symmetry generator. Another mechanism involves phase alignment, in which certain trembling components lock into phase relationships that preserve coherence across proper time. Alternatively, a proper-time-consistent reorientation of the trembling basis vectors can dynamically privilege a subspace of the internal fiber, effectively selecting a symmetry-reducing direction. In all cases, the resulting coherence yields a spontaneous breakdown of the full $SU(2)_L \times U(1)_Y$ symmetry to its residual $U(1)_{\text{EM}}$ subgroup.

This geometric process selects a preferred internal trembling direction, thereby breaking the original gauge symmetry:

$$\mathcal{G}_{\text{EW}} = SU(2)_L \times U(1)_Y \quad (\text{S260})$$

down to the unbroken electromagnetic subgroup:

$$\mathcal{G}_{\text{EM}} = U(1)_{\text{EM}}. \quad (\text{S261})$$

The emergent gauge fields reorganize accordingly. The original weak and hypercharge connections mix through a proper-time basis rotation to yield new physical modes:

$$Z_\mu = \cos \theta_W W_\mu^3 - \sin \theta_W B_\mu, \quad (\text{S262})$$

$$A_\mu = \sin \theta_W W_\mu^3 + \cos \theta_W B_\mu, \quad (\text{S263})$$

where the Weinberg angle θ_W is defined geometrically via the coupling ratio:

$$\theta_W = \arctan \left(\frac{g'}{g} \right). \quad (\text{S264})$$

In this new basis, the field A_μ remains massless and mediates the residual $U(1)_{\text{EM}}$ interaction, while the field Z_μ becomes associated with a massive weak neutral current due to directional saturation of internal trembling. Crucially, this mass is not introduced via a Higgs

vacuum expectation value, but results from coherence constraints on the trembling deviation spectrum, which restrict proper-time variance in certain internal directions. This mechanism was first introduced geometrically in Section S9 and now acquires dynamical expression. From this geometric viewpoint, the photon field A_μ emerges as the residual direction in internal trembling space that remains dynamically unconstrained by amplitude or phase alignment, preserving the unbroken $U(1)_{\text{EM}}$ symmetry. In contrast, the broken components of the original gauge structure—such as $W_\mu^\pm$ and Z_μ —acquire effective mass as a direct consequence of the trembling geometry losing full internal invariance. This mass generation does not result from spontaneous symmetry breaking in the quantum-theoretic sense, but instead from deterministic geometric selection: a coherence-driven alignment of trembling amplitudes that renders certain internal directions dynamically inactive. In this way, electroweak symmetry breaking in TSRT is fully geometric, causal, and proper-time-resolved. The net effect is a redefinition of conserved quantities: the original weak isospin and hypercharge currents give way to a new conserved current:

$$J_{\text{EM}}^\mu = J_{SU(2)}^\mu \sin \theta_W + J_Y^\mu \cos \theta_W, \quad (\text{S265})$$

representing the electromagnetic charge current. The associated charge, now interpreted as electric charge, remains conserved due to the unbroken $U(1)_{\text{EM}}$ symmetry. This approach thus yields a full electroweak breaking mechanism within TSRT, without requiring a scalar Higgs field, potential minima, or quantization assumptions. It also aligns with empirical expectations: W and Z bosons emerge as massive, while the photon remains massless, and the observable low-energy interaction structure reduces to electromagnetism. The symmetry reduction is internal, causal, and fully geometric.

S11.6 Conclusion

Within the deterministic and causally covariant framework of Trembling Spacetime Relativity Theory (TSRT), the electroweak interaction emerges not as an imposed quantum gauge symmetry, but as a geometric inevitability arising from internal chiral trembling. As shown in Section S9, the only admissible compact, non-Abelian trembling structure that distinguishes left- and right-handed deviations while remaining consistent with Lorentz covariance and proper-time positivity is governed by the internal symmetry group

$$\mathcal{G}_{\text{EW}} = SU(2)_L \times U(1)_Y. \quad (\text{S266})$$

This internal trembling acts exclusively on the left-handed subbundle $E_L \subset E \rightarrow \mathcal{M}$, producing gauge connections W_μ^i and B_μ via coarse-grained projections of proper-time-dependent deviation dynamics, as formalized in Section S11.2. These gauge fields exhibit non-trivial curvature structures—encoded in the Yang–Mills field strength tensors $F_{\mu\nu}^i$ and $B_{\mu\nu}$ —that arise directly from the non-commuting transformation algebra of internal chiral trembling, as detailed in Section S9.6. The equation of motion governing particles in this internal trembling geometry generalizes the Lorentz force to include chiral gauge couplings. As derived in Section S11.4, the resulting expression

$$m \frac{du^\mu}{d\tau} = g F^{\mu\nu i} T^i u_\nu + g' B^{\mu\nu} Y u_\nu \quad (\text{S267})$$

describes the full electroweak interaction from first principles, showing that chiral coupling, parity violation, and flavor transitions all emerge as kinematic consequences of non-Abelian internal trembling geometry. Moreover, symmetry breaking within TSRT is not achieved via a scalar Higgs mechanism, but results from the deterministic coherence and amplitude saturation of trembling modes in the internal fiber bundle. As shown in Section S11.5, the basis rotation

$$Z_\mu = \cos \theta_W W_\mu^3 - \sin \theta_W B_\mu, \quad (\text{S268})$$

$$A_\mu = \sin \theta_W W_\mu^3 + \cos \theta_W B_\mu \quad (\text{S269})$$

arises from geometric reorientation of the internal trembling structure. This breaks the original gauge group $SU(2)_L \times U(1)_Y$ to the electromagnetic subgroup $U(1)_{\text{EM}}$, yielding the correct massless and massive gauge boson spectrum without introducing extrinsic symmetry-breaking potentials.

In summary, TSRT provides a unified geometric explanation for all essential features of the electroweak interaction. The gauge symmetry $SU(2)_L \times U(1)_Y$ emerges from internal chiral trembling modes constrained by the proper-time dynamics of the deviation field. The associated gauge fields and their non-Abelian curvatures arise from proper-time-coherent geometric deviations within the internal fiber structure. Chiral asymmetry and parity violation are not imposed but result directly from the restriction of trembling dynamics to the left-handed subbundle, reflecting the inherent geometric structure of E_L . Finally, symmetry breaking and the emergence of gauge boson masses are explained by the development of coherent internal alignment—rather than a quantum vacuum condensate—demonstrating that mass generation is a deterministic outcome of trembling-induced geometry. This reformulation reinterprets the electroweak sector of the Standard Model as an emergent, deterministic structure of trembling spacetime, and demonstrates how internal gauge geometry arises naturally from the causal architecture of proper-time evolution. Having established the emergence of the chiral weak interaction from asymmetric trembling, we now turn to the more complex structure of non-abelian color trembling and the strong nuclear force.

S12 Supplementary Material: Emergent Strong Nuclear Force from Non-Abelian Color Trembling

S12.1 Color-Trembling Geometry and $SU(3)_C$ Symmetry

As established in Section S9, the strong nuclear force in TSRT arises from the unique class of internal non-Abelian trembling modes characterized by eight compact internal directions forming a closed Lie algebra. This algebra is isomorphic to $\mathfrak{su}(3)$, the Lie algebra of the special unitary group $SU(3)_C$, known from quantum chromodynamics (QCD) as the gauge group of color charge. Unlike the Abelian sector of electromagnetism (see Section S10) or the chiral non-Abelian structure of the weak force (see Section S11), the color sector is fully internal, parity-symmetric, and topologically richer. In TSRT, this trembling sector is defined within an internal vector bundle $E_C \rightarrow \mathcal{M}$, where $\mathcal{M}$ denotes the four-dimensional spacetime manifold, and E_C encodes the color fiber structure. Internal trembling modes do not shift spacetime positions but instead rotate the internal deviation vectors $\Xi_\mu^a(\tau)$ in color space. These trembling directions are labeled by an index $a = 1, \dots, 8$, corresponding to the eight generators T^a of $\mathfrak{su}(3)$, which satisfy the non-Abelian commutation relations:

$$[T^a, T^b] = i f^{abc} T^c, \quad (\text{S270})$$

where f^{abc} are the totally antisymmetric structure constants of the color algebra.

As introduced in Paragraph 4 of Section S9.4, the strong nuclear force within TSRT arises from a class of non-chiral, non-Abelian trembling patterns involving internal color degrees of freedom. These fluctuations are symmetric under left–right exchange and act purely within the internal fiber structure, giving rise to metric perturbations governed by the symmetry algebra of $SU(3)_C$. Unlike the chiral weak sector, where internal trembling selectively couples to left-handed components, the color sector induces fluctuations that are equally distributed across all components, with no handedness bias. This reflects the observed parity symmetry of the strong interaction. The underlying geometric structure responsible for these effects consists of a set of eight independent deviation vectors $\Xi_\mu^a(\tau)$, indexed by $a = 1, \dots, 8$, each corresponding to one of the eight generators of the Lie algebra $\mathfrak{su}(3)_C$.

Each deviation vector $\Xi_\mu^a(\tau)$ describes an internal oscillation along a particular color direction, evolving with proper time along the particle’s worldline. These color-directed trembling fields induce spacetime curvature via a geometric prescription that mirrors the covariant structure of gauge theory. Specifically, each mode contributes a symmetric perturbation to the spacetime metric:

$$\xi_{\mu\nu}^{(a)} = \partial_\mu \Xi_\nu^a + \partial_\nu \Xi_\mu^a. \quad (\text{S271})$$

This structure ensures that each individual contribution respects the symmetry $\xi_{\mu\nu}^{(a)} = \xi_{\nu\mu}^{(a)}$, making the total metric perturbation consistent with the required properties of a physical spacetime geometry. The full trembling-induced deviation of the metric is obtained by summing

over all eight color directions:

$$\xi_{\mu\nu} = \sum_{a=1}^8 \xi_{\mu\nu}^{(a)}. \quad (\text{S272})$$

This superposition encodes a coherent geometric excitation spanning the entire $\mathfrak{su}(3)_C$ internal space. Unlike quantum chromodynamics (QCD), where gauge fields $G_\mu^a(x)$ are postulated as fundamental quantum fields, the TSRT framework derives this structure deterministically from microscopic trembling of the underlying geometry.

Each $\Xi_\mu^a(\tau)$ thus plays the role of a color-aligned deviation field, evolving deterministically along the particle's worldline and modulating the local spacetime curvature. The superposition of all eight such fields results in a metric perturbation that captures the full non-Abelian structure of $SU(3)_C$, leading to interactions that mirror those of the strong force in the Standard Model.

This geometric formulation bypasses the need for postulated color charge or fundamental gauge bosons. Instead, color arises as a manifestation of the eightfold internal structure of space-time trembling, governed by symmetry constraints that ensure causal evolution ($d\tau/d\lambda > 0$) and the consistency of spacetime geometry. The resulting picture provides a deterministic and fully covariant origin for the strong nuclear force, embedded directly in the fine-grained structure of the trembling spacetime manifold. Because causal coherence constrains large-scale color fluctuations, the trembling-induced gauge fields cannot sustain free color charges over macroscopic separations. This geometric suppression of isolated color excitations naturally leads to confinement, ensuring that only color-neutral bound states can exist at large scales, in agreement with observed hadronic phenomena. The emergence of $SU(3)_C$ as the relevant symmetry group is not an assumption but a derivation: TSRT imposes strict geometric constraints—compactness, closure, proper-time covariance, and compatibility with Lorentz symmetry—that limit the space of admissible non-Abelian internal trembling groups. As shown in the classification analysis of Section S9, $SU(3)$ is the only simple, compact, irreducible Lie group with eight generators that can be embedded into the trembling structure of a four-dimensional Lorentzian spacetime without violating causal propagation or internal fiber coherence. Moreover, the internal symmetry $SU(3)_C$ naturally supports the required structure for color charge conservation and gluonic self-interactions, both of which emerge from the geometric curvature of the associated trembling connection (as will be shown in subsequent subsections). Unlike the weak sector, which exhibits parity violation and chiral asymmetry, the strong sector in TSRT is inherently symmetric under left-right transformations and does not distinguish chirality in its trembling modes.

In summary, TSRT explains the strong nuclear force as the unique and inevitable manifestation of compact, non-Abelian internal metric trembling in an eight-dimensional color fiber space. The full $SU(3)_C$ symmetry, its curvature structure, and its confinement behavior all emerge from deterministic geometric deviations in proper-time, rooted entirely in the causal architecture of trembling spacetime.

S12.2 Emergent Gluon Fields from Coarse-Grained Trembling

The internal trembling fields can be decomposed along the eight generators T^a of the Lie algebra $\mathfrak{su}(3)_C$, corresponding directly to the eight independent gluon fields of the strong interaction.

Each trembling mode along a generator direction mediates the exchange of color charge, in full analogy with the Standard Model description of gluons. Building upon the internal color-trembling geometry defined in Section S12.1, the eight proper-time-dependent deviation vectors $\Xi_\mu^a(\tau) \in E_C$, associated with the generators $T^a \in \mathfrak{su}(3)$, give rise to observable structures in spacetime through their coarse-grained projection onto the particle's worldline. These internal deviations evolve deterministically and coherently under the constraints of TSRT—namely causal propagation, Lorentz covariance, compact symmetry closure, and proper-time consistency.

To extract their macroscopic influence on external dynamics, we define the emergent gluon fields as coarse-grained vector potentials:

$$A_\mu^a(x) := \langle \partial_\mu \Xi_\nu^a(x, \tau) \rangle u^\nu(\tau), \quad (\text{S273})$$

where $\langle \cdot \rangle$ denotes ensemble or proper-time averaging over rapid internal oscillations, and $u^\nu(\tau)$ is the particle's four-velocity along its trembling geodesic. The resulting eight vector fields $A_\mu^a(x)$, each associated with a specific direction in the internal color space, constitute a local connection on a principal $SU(3)_C$ fiber bundle over spacetime. In TSRT, the gluon field $A_\mu^a(x)$ arises from coarse-grained internal color trembling fields $\Xi_\mu^a(\tau)$ that transform under local $SU(3)_C$ rotations. These trembling fields live in an internal color fiber bundle, and at each spacetime point x , the internal fiber admits a local orthonormal basis of generators $T^a \in \mathfrak{su}(3)$, normalized such that

$$\text{Tr}(T^a T^b) = \frac{1}{2} \delta^{ab}. \quad (\text{S274})$$

We define the coarse-grained gluon field by:

$$A_\mu(x) := \langle \Xi_\mu^a(\tau) \rangle T^a, \quad (\text{S275})$$

where the angular brackets $\langle \cdot \rangle$ denote proper-time coarse-graining and the contraction with T^a embeds the result into the Lie algebra $\mathfrak{su}(3)$. Now consider a local gauge transformation

$$U(x) \in SU(3)_C, \quad (\text{S276})$$

which acts on the internal color fiber by rotating the trembling deviations:

$$\Xi_\mu(\tau) \mapsto U(x) \Xi_\mu(\tau) U^{-1}(x), \quad (\text{S277})$$

where $\Xi_\mu(\tau) := \Xi_\mu^a(\tau) T^a$.

The coarse-grained gluon field then transforms as:

$$A_\mu(x) := \langle \Xi_\mu(\tau) \rangle \mapsto \langle U(x) \Xi_\mu(\tau) U^{-1}(x) \rangle \quad (\text{S278})$$

$$= U(x) \langle \Xi_\mu(\tau) \rangle U^{-1}(x) = U(x) A_\mu(x) U^{-1}(x). \quad (\text{S279})$$

However, this describes only the homogeneous part of the gauge transformation. To derive the full inhomogeneous law, we must consider how the covariant derivative transforms. Let the

covariant derivative be defined by:

$$D_\mu := \partial_\mu - igA_\mu(x). \quad (\text{S280})$$

We require that the covariant derivative of a color-carrying matter field $\psi(x)$, transforming as $\psi(x) \mapsto U(x)\psi(x)$, itself transforms covariantly:

$$D_\mu\psi(x) \mapsto U(x)D_\mu\psi(x). \quad (\text{S281})$$

This condition implies:

$$D_\mu\psi \mapsto (\partial_\mu - igA'_\mu)U\psi = (\partial_\mu U)\psi + U\partial_\mu\psi - igA'_\mu U\psi \quad (\text{S282})$$

$$= U(\partial_\mu\psi - igA_\mu\psi) \Rightarrow (\partial_\mu U)\psi + U\partial_\mu\psi - igA'_\mu U\psi = U\partial_\mu\psi - igUA_\mu\psi, \quad (\text{S283})$$

so we equate both sides and rearrange:

$$(\partial_\mu U)\psi - igA'_\mu U\psi = -igUA_\mu\psi. \quad (\text{S284})$$

Multiplying both sides on the right by ψ^{-1} and isolating A'_μ , we find:

$$A'_\mu = UA_\mu U^{-1} + \frac{i}{g}U\partial_\mu U^{-1}. \quad (\text{S285})$$

This is the standard non-Abelian gauge transformation law for the gluon field and is derived here from first principles within TSRT, as a consistency requirement on the transformation of internal trembling deviations and the covariant derivative structure they induce. The derivation parallels the classical Yang–Mills theory [49], but here it emerges naturally from the deterministic trembling structure of spacetime, without postulating a separate quantum field framework. This transformation law is therefore not postulated but follows directly from the internal geometric behavior of the trembling deviations $\Xi_\mu^a(\tau)$ under infinitesimal local color rotations in $SU(3)_C$. In particular, such deviations transform under the adjoint representation of the Lie algebra $\mathfrak{su}(3)$, according to:

$$\Xi_\mu^a(\tau) \mapsto \Xi_\mu^a(\tau) + f^{abc}\theta^b(\tau)\Xi_\mu^c(\tau), \quad (\text{S286})$$

where $\theta^b(\tau)$ are infinitesimal rotation parameters along the proper-time trajectory, and f^{abc} are the structure constants of $\mathfrak{su}(3)$, encoding the non-Abelian nature of internal color space. This transformation mirrors the analogous behavior in the $SU(2)_L$ sector (see Section S11), and underlies the construction of the gluon field as a coarse-grained projection of color-coherent trembling.

As the internal trembling modes evolve, their local reparametrization under $SU(3)_C$ transformations induces the non-Abelian gauge transformation of the emergent gluon field $A_\mu(x) := A_\mu^a(x)T^a$, preserving the geometric structure of the covariant derivative:

$$D_\mu := \partial_\mu - igA_\mu(x), \quad (\text{S287})$$

and ensuring local gauge covariance of the trembling-induced dynamics. The gauge principle, in this formulation, emerges from the symmetry of internal trembling geometry rather than being externally imposed.

Crucially, the gluon fields $A_\mu^a(x)$ are not interpreted as quantum fields, wave functions, or operator-valued distributions, as they are in conventional Quantum Chromodynamics (QCD). Instead, they arise deterministically from the coarse-grained, covariant projection of the internal metric deviation fields $\Xi_\mu^a(\tau)$. Their non-Abelian character originates from the intrinsic geometry of the internal fiber structure, not from quantization. The structure constants f^{abc} arise naturally from the internal curvature algebra of coherent color oscillations, and not from any imposed algebraic constraint.

Because of this geometric origin, the gluon fields inherently encode the non-linear self-interaction terms characteristic of Yang–Mills theories. These interactions emerge not as quantized virtual exchanges, but as macroscopic statistical manifestations of the curved color-bundle geometry. The curvature associated with these non-Abelian connections will be defined precisely in the next subsection. Moreover, this projection-based emergence of gauge fields generalizes the abelian and chiral constructions previously established. The same fundamental mechanism—coarse-graining of deterministic internal deviations—underlies the emergence of the electromagnetic vector potential $A_\mu(x)$ from abelian longitudinal trembling (Section S10), the weak gauge fields $W_\mu^i(x)$ from chiral $SU(2)_L$ deviations (Section S11), and now the strong gluon fields $A_\mu^a(x)$ from non-chiral $SU(3)_C$ oscillations. This unified construction demonstrates that all gauge-like structures in TSRT derive from a common geometric origin: symmetry-preserving proper-time-resolved trembling fields governed by internal bundle dynamics.

Thus, in the TSRT framework, gluons are not fundamental particles in the quantum sense, but emergent curvature-carrying fields resulting from the coherent evolution of internal color geometry. Their transformation law, self-interaction structure, and coupling to matter all follow from deterministic principles of spacetime trembling, ensuring compatibility with causality, Lorentz invariance, and internal gauge symmetry. This geometric reinterpretation not only aligns with the observed structure of strong interactions but also embeds QCD-like dynamics within a broader and deeper deterministic theory of spacetime.

S12.3 Non-Abelian Field Strength Tensor and Curvature

The deterministic color-trembling geometry developed in Sections S9 and S12 leads directly to a coarse-grained curvature structure over spacetime, corresponding to the field strength of the emergent gluon connection. This field strength tensor encapsulates the geometric response of internal color-deviation modes to infinitesimal parallel transport along the particle’s worldline and is structurally analogous to the Yang–Mills curvature of conventional gauge theory. From the emergent vector potentials $A_\mu^a(x)$ defined in Equation (S273), we define the non-Abelian field strength tensor:

$$G_{\mu\nu}^a := \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g f^{abc} A_\mu^b A_\nu^c, \quad (\text{S288})$$

where f^{abc} are the fully antisymmetric structure constants of $SU(3)_C$, and g is the effective strong coupling constant derived from the average amplitude and spectral coherence of color

trembling modes. Each term in Equation (S288) has a geometric interpretation:

- The first two terms $\partial_\mu A_\nu^a - \partial_\nu A_\mu^a$ describe the antisymmetric part of the local change in gluon fields across spacetime, reflecting the net curvature arising from infinitesimal displacements.
- The final term $gf^{abc}A_\mu^b A_\nu^c$ represents intrinsic self-interaction curvature, arising not from externally postulated non-Abelian symmetry, but from the deterministic, non-commutative structure of internal metric trembling. As discussed in Section S11 for the $SU(2)_L$ sector, such non-linearities follow directly from the structure of compact Lie algebras governing coherent internal oscillations.

In this framework, the color field strength $G_{\mu\nu}^a$ is not imposed by demanding invariance under gauge transformations; instead, it emerges naturally from the coarse-grained non-integrability of the internal deviation fields $\Xi_\mu^a(\tau)$. The presence of the structure constant term reflects the fact that parallel transport in internal color space is path-dependent—an intrinsic geometric feature of non-Abelian trembling. Moreover, this field strength tensor simultaneously encodes several intertwined geometric features. First, it captures the curvature induced in external spacetime by projections of internal trembling fluctuations, translating internal deviation dynamics into spacetime curvature effects. Second, it reflects the algebraic torsion and twisting within the internal fiber bundle defined by the $SU(3)_C$ color group, where the non-abelian structure constants govern the orientation dynamics of trembling directions. Third, it encapsulates the self-interactions among distinct color modes evolving over proper time, which are intrinsically encoded in the non-linear, compact structure of the color algebra and manifest geometrically in the non-commutativity of the trembling directions. This unification of curvature, algebraic structure, and trembling dynamics reveals the full complexity of strong interactions within TSRT. In particular, the gluon self-interaction terms in $G_{\mu\nu}^a$ are not added to preserve local invariance, as in quantum field theory, but arise directly from the non-abelian metric geometry of the color fiber. These terms underlie the emergence of non-linear dynamics, color confinement, and asymptotic behavior discussed in subsequent subsections.

Thus, TSRT offers a fully geometric derivation of strong-force curvature, compatible with both the fiber bundle structure of gauge theory and the deterministic, causal trembling evolution of the spacetime manifold. The field strength tensor $G_{\mu\nu}^a$ stands as a geometric observable encoding the local interaction curvature experienced by trembling color charges and bridges the microstructure of internal metric deviation with the macroscopic behavior of color fields.

S12.4 Color Charge, Covariant Coupling, and Interaction Dynamics

In the TSRT framework, the interaction between material excitations and the emergent non-Abelian geometry generated by internal color trembling is governed by a covariant extension of parallel transport. This structure incorporates the fiber geometry of the internal color bundle and naturally encodes how spacetime fluctuations mediate force-like behavior through geometric curvature. As discussed in Sections S9 and S11, this approach generalizes the usual notion of a gauge field to a fully geometric object arising from coarse-grained internal trembling.

To describe how internal color fluctuations interact with localized mass-energy, we consider a matter-carrying excitation—such as a baryon or quark-like localized structure—represented by an effective field $\psi(x)$. In the TSRT context, this field does not denote a quantum-mechanical wavefunction in the probabilistic sense, but rather a geometric object encoding the localized density of proper-time-parametrized particle motion. It lives on a spacetime manifold and carries internal structure consistent with the color fiber bundle associated with $SU(3)_C$. In this sense, $\psi(x)$ represents a particle-like excitation with color charge, evolving deterministically under the influence of spacetime trembling. When the internal structure of such a particle transforms under a representation of the Lie algebra $\mathfrak{su}(3)_C$, the coupling to the trembling-induced geometry is governed by the color-covariant derivative:

$$D_\mu := \partial_\mu - ig T^a A_\mu^a(x), \quad (\text{S289})$$

where $A_\mu^a(x)$ are the eight emergent gauge fields derived by coarse-graining over the proper-time-dependent color deviation vectors $\Xi_\mu^a(\tau)$, as introduced in Equation (S273). The symbols T^a denote the generators of $\mathfrak{su}(3)$, typically realized as Gell-Mann matrices acting on the internal color fiber space, and g is the strong coupling constant, whose magnitude reflects the amplitude and spectral coherence of the underlying trembling modes.

This structure ensures that color-charged matter experiences interaction forces not via postulated fields, but as a natural consequence of its deterministic evolution through the nontrivially curved, trembling geometry of spacetime. The operator D_μ governs the parallel transport of color degrees of freedom and thereby defines the geometric connection on the principal $SU(3)_C$ -bundle over spacetime. Unlike traditional gauge theory—where D_μ is introduced ad hoc to enforce local symmetry—here it emerges intrinsically from the causal propagation of color-indexed internal trembling configurations along a particle’s proper-time trajectory.

In the TSRT framework, the interaction between matter and the emergent non-Abelian color geometry is not introduced axiomatically, but arises as a deterministic consequence of the internal trembling structure of spacetime. Specifically, as shown in Section S12, coarse-grained internal deviations $\Xi_\mu^a(\tau)$ give rise to the gluon gauge fields $A_\mu^a(x) \in \mathfrak{su}(3)_C$, which mediate the strong interaction through the curved geometry of an internal color bundle.

To understand how these gluon fields couple to matter, we begin by considering a fermionic matter field $\psi(x)$ whose internal structure transforms under a representation of the color symmetry group $SU(3)_C$. This field represents a color-carrying excitation in the trembling spacetime and should not be confused with a quantum-mechanical wavefunction: rather, $\psi(x)$ here is a density-supporting geometric object with well-defined transformation properties under the internal symmetry group. The color-covariant derivative acting on $\psi(x)$ is defined by:

$$D_\mu \psi := \partial_\mu \psi - ig T^a A_\mu^a(x) \psi, \quad (\text{S290})$$

where T^a are the generators of $\mathfrak{su}(3)$ in the representation appropriate to ψ , $A_\mu^a(x)$ are the emergent gluon fields, and g is the strong coupling constant, which in TSRT is related to the amplitude and coherence scale of the color-sector trembling.

To derive the full dynamics, we construct the emergent Lagrangian density that governs

the coupling between the matter field and the coarse-grained curvature generated by internal trembling. The full Lagrangian emerges by imposing local covariance of the matter field's dynamics with respect to the color bundle connection, and by accounting for the curvature induced by the non-Abelian gauge potential:

$$\mathcal{L}_{\text{strong}} = -\frac{1}{4}G_{\mu\nu}^a G^{a\mu\nu} + \bar{\psi}(i\gamma^\mu D_\mu - m)\psi. \quad (\text{S291})$$

Here, the first term represents the energy density of the color-curvature tensor:

$$G_{\mu\nu}^a := \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc}A_\mu^b A_\nu^c, \quad (\text{S292})$$

which arises from the non-commutative structure of internal color oscillations (as derived in the next subsection). The second term describes the motion of the matter field ψ through the emergent color geometry, with D_μ ensuring gauge-covariant propagation. This Lagrangian is not postulated, but rather derived from first principles within TSRT. The internal trembling fields $\Xi_\mu^a(\tau)$, through coarse-graining, induce both the gluon potential A_μ^a and the corresponding curvature $G_{\mu\nu}^a$. The coupling term $\bar{\psi}\gamma^\mu T^a A_\mu^a \psi$ results from demanding that the matter field propagate consistently in this emergent fiber geometry. In other words, it arises as the unique lowest-order, Lorentz-invariant, gauge-covariant interaction term between matter and geometry that is compatible with the internal trembling dynamics.

Thus, the interaction structure described by Equation (S291) is a geometric consequence of the TSRT framework. It reflects not an imposed symmetry principle, but a causal and deterministic projection of internal metric fluctuations into external fields and forces. This completes the derivation of the coupling between matter and the strong force geometry in trembling spacetime. This Lagrangian, in Equation (S291), reproduces the dynamical form of QCD but fundamentally differs in origin. Each term arises not from imposed gauge symmetry principles, but from the deterministic structure of trembling spacetime geometry, as derived from proper-time-coherent metric fluctuations. The gauge structure is a derived feature—not a postulate.

The color current corresponding to this interaction is given by:

$$J^{\mu a}(x) := \bar{\psi}(x)\gamma^\mu T^a \psi(x), \quad (\text{S293})$$

which serves as the source of gluonic curvature in the effective field equations. Its conservation is not global but covariant under local color rotations, reflecting the geometry of the internal trembling manifold:

$$D_\mu J^{\mu a} := \partial_\mu J^{\mu a} + gf^{abc}A_\mu^b J^{\mu c} = 0. \quad (\text{S294})$$

This covariant conservation law is a geometric consequence of the TSRT action's invariance under proper-time-dependent internal $SU(3)_C$ transformations. It ensures that color charge remains coherent under deterministic parallel transport and that internal trembling dynamics preserve local consistency and interaction locality.

In summary, TSRT provides a fully geometric origin for color interactions. The emergence of covariant coupling, current conservation, and gluonic curvature all follow from the fibered

structure of the internal trembling bundle and its coarse-grained projection onto spacetime. The standard gauge-theoretic features of the strong force are reinterpreted here as macroscopic consequences of causal, Lorentz-covariant internal deviation geometry, closing the geometric unification of forces under trembling spacetime dynamics.

S12.5 Confinement from Trembling Self-Interaction Geometry

Among the most striking features of the strong nuclear force is the phenomenon of color confinement: individual color-charged particles such as quarks and gluons are never observed in isolation but always appear as part of color-neutral composites. In conventional quantum chromodynamics (QCD), this behavior is attributed to the non-linear structure of the non-Abelian Yang–Mills equations, leading to infrared-divergent potentials and chromodynamic flux tubes. However, these insights depend on non-perturbative and numerical methods and lack a deterministic or geometric explanation. Within the TSRT framework, confinement emerges directly from the self-interaction geometry of internal color trembling, without requiring quantization or vacuum-based gauge field assumptions. As shown in Section S9 and S12.4, the internal deviation fields $\Xi_\mu^a(\tau)$ produce a non-Abelian trembling structure governed by the compact, eight-dimensional color algebra $\mathfrak{su}(3)_C$. These deviations project, through proper-time coarse-graining, onto a set of emergent gluon fields $A_\mu^a(x)$, whose curvature is described by the effective field strength tensor, as in Equation (S292),

$$G_{\mu\nu}^a := \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc}A_\mu^b A_\nu^c. \quad (\text{S295})$$

The final term, $gf^{abc}A_\mu^b A_\nu^c$, reflects the inherent self-interaction among internal trembling modes. Crucially, in TSRT, this term arises not from the requirement of gauge invariance, but from the non-commutative geometry of the internal trembling fiber bundle $E_C \rightarrow \mathcal{M}$. The algebraic structure of $\mathfrak{su}(3)_C$ ensures that any internal trembling deviation induces curvature in multiple coupled directions, with energy cost increasing nonlinearly under attempted spatial separation.

As a particle’s internal trembling state attempts to deviate along a single color direction over extended spatial domains, the associated self-interaction terms in $G_{\mu\nu}^a$ amplify. The internal color curvature cannot remain localized or dilute over long distances without violating proper-time causality or destabilizing the internal trembling structure. This leads to the formation of persistent regions of concentrated internal curvature—analogueous to flux tubes—but interpreted here as trembling curvature filaments. These filaments represent dynamically stabilized regions of internal deviation energy, with energy density proportional to the internal curvature and growing with spatial extension. The attempt to isolate a single color charge (i.e., a trembling configuration with net nonzero deviation in a particular T^a -direction) induces geometrically increasing resistance from the internal fiber structure, leading to effective divergence of trembling energy. This geometric instability forbids the propagation of isolated color charges, enforcing confinement.

Only those trembling configurations that close internally—meaning combinations of deviations whose net trembling direction vanishes in the color algebra $\mathfrak{su}(3)_C$ —can propagate stably

in spacetime. These include mesons, which correspond to pairwise combinations of quark and antiquark-like trembling directions such that their internal deviations cancel each other, and baryons, which arise from triplets of trembling directions whose algebraic sum vanishes according to the trace-free closure condition of $SU(3)_C$, consistent with the group's structure constants. These color-neutral combinations represent configurations that minimize internal trembling curvature while preserving causal propagation and bounded norm evolution of the total deviation field. They thus constitute the only dynamically stable and physically admissible solutions within the strong interaction sector of TSRT.

In this geometric formulation, confinement is not an assumed or computationally inferred phenomenon, as in conventional quantum chromodynamics (QCD), but a necessary consequence of the structure of the theory. It arises from the intrinsic non-Abelian self-coupling of internal trembling directions, from the compact and closed algebraic nature of $\mathfrak{su}(3)_C$, and from the requirement that the proper-time evolution of all trembling modes must respect curvature boundedness and maintain deterministic flow. These constraints together yield confinement as a geometric theorem rather than a quantum postulate. This geometric inevitability provides a causal, non-quantum origin for color confinement. It recasts one of the deepest puzzles of the strong interaction as a necessary outcome of trembling spacetime structure, unified with the mechanisms underlying electromagnetic and weak interactions.

S12.6 Conclusion

In Trembling Spacetime Relativity Theory (TSRT), the strong nuclear interaction emerges as a geometric inevitability from the non-Abelian structure of internal color trembling. Rather than introducing gauge fields and confinement mechanisms by postulate—as is standard in quantum chromodynamics (QCD)—TSRT derives these phenomena deterministically from the compact, fibered geometry of internal deviation fields.

The eight color-aligned trembling modes $\Xi_\mu^a(\tau)$, corresponding to the generators of the compact Lie algebra $\mathfrak{su}(3)_C$, are the unique internal trembling degrees of freedom consistent with the geometric constraints articulated in Section S9: Lorentz covariance, proper-time causality, compact group structure, and algebraic closure. Coarse-graining these internal deviations along the particle worldline yields a set of emergent vector fields $A_\mu^a(x)$, which behave as gluon-like connections defining a non-Abelian gauge structure over spacetime.

The field strength tensor $G_{\mu\nu}^a$, constructed from these connections, includes nonlinear self-interaction terms directly inherited from the structure constants f^{abc} of the color algebra. As detailed in Section S12.4, the interaction of these fields with color-charged matter follows from the covariant derivative acting in the internal trembling bundle, and the associated current conservation law reflects the underlying geometric symmetry of internal $SU(3)_C$ rotations.

Most critically, TSRT offers a deterministic explanation for color confinement. As demonstrated in Section S12, the self-interaction geometry of internal trembling modes leads to an energetic instability whenever color charge attempts to propagate in isolation. Internal curvature concentrates into filament-like trembling structures that grow in energy with spatial extension. Only trembling configurations that are color-neutral—i.e., whose internal deviations algebraically close in $\mathfrak{su}(3)_C$ —can propagate stably, giving rise to mesons and baryons as the

only physically admissible composite states. This geometric confinement mechanism requires no recourse to quantization or vacuum expectation values and is rooted entirely in causal proper-time evolution.

With this construction, the strong force completes the fourfold classification of fundamental interactions within TSRT. Electromagnetism emerges from external abelian phase-type trembling and is associated with the symmetry group $U(1)$. The weak interaction arises from chiral internal non-Abelian trembling, governed by the group $SU(2)_L \times U(1)_Y$. Gravitation, in contrast, results from macroscopic coarse-grained metric trembling that respects diffeomorphism invariance, thereby shaping the emergent spacetime geometry. Finally, the strong interaction originates from internal compact color trembling, whose structure is fully encoded by the non-Abelian Lie algebra $\mathfrak{su}(3)_C$. Each of these four interactions is thus realized as a distinct, symmetry-constrained trembling sector of the underlying deterministic spacetime geometry. Each of these forces arises not from field quantization or externally imposed symmetries, but as a manifestation of permissible geometric deviations of the spacetime metric along proper-time-parametrized particle trajectories. The trembling field $\Xi^\mu(\tau)$, and its internal extensions, encode all interactions as aspects of a unified, causally coherent geometric substrate.

In this light, the strong interaction is reinterpreted as a non-Abelian curvature phenomenon internal to trembling spacetime itself—not as an autonomous gauge theory, but as the inevitable outcome of geometric coherence and compact internal dynamics. The unification achieved here is not merely algebraic or aesthetic—it is causal, geometric, and physically deterministic. In parallel, we show how coarse-grained collective trembling generates gravitational effects consistent with classical general relativity.

S13 Supplementary Material: Emergent Gravity from Coarse-Grained Curvature

S13.1 Macroscopic Deviations and Diffeomorphism Invariance

Among the four fundamental interaction classes derived in TSRT, gravity arises in a categorically distinct geometric regime. Unlike electromagnetism, the weak force, and the strong force—which originate from internal or compact fibered trembling degrees of freedom—gravitational phenomena emerge from large-scale, externally directed metric deviations that accumulate coherently across macroscopic spacetime regions. This sets gravity apart as the only interaction associated with the base manifold $\mathcal{M}$ itself, rather than with internal fiber structures. Importantly, not all trembling fluctuations contribute equally to emergent macroscopic curvature. Only coherent, causally organized trembling patterns, which maintain consistent directional correlations over scales larger than the microscopic resolution length, survive the coarse-graining process to generate observable curvature. Purely random or uncorrelated trembling averages out without producing macroscopic effects, preserving the overall local flatness at the finest scales.

As established in Section S9, the external trembling field $\Xi^\mu(\tau)$ corresponds to proper-time-parametrized deviations of the particle trajectory, defined globally on the manifold and transforming covariantly under spacetime diffeomorphisms. When these deviations are rapidly oscillating and properly bounded (as ensured by the proper-time causality condition discussed in Section S23), their coherent accumulation over many cycles leads to an emergent macroscopic deformation of the background metric:

$$g_{\mu\nu}^{\text{eff}}(x) = \bar{g}_{\mu\nu}(x) + \langle \xi_{\mu\nu}(x, \tau) \rangle, \quad (\text{S296})$$

where $\bar{g}_{\mu\nu}$ is the unperturbed background geometry (e.g., Minkowski), and the symmetric tensor $\xi_{\mu\nu}$ is defined by:

$$\xi_{\mu\nu}(x, \tau) := \partial_\mu \Xi_\nu(x, \tau) + \partial_\nu \Xi_\mu(x, \tau). \quad (\text{S297})$$

The operation $\langle \cdot \rangle$ denotes coarse-graining over high-frequency proper-time oscillations, extracting the slow-varying, long-wavelength curvature contributions of the trembling deviations.

The resulting effective metric $g_{\mu\nu}^{\text{eff}}$ encodes the emergent gravitational geometry. Crucially, it possesses invariance under general coordinate transformations of the form:

$$x^\mu \rightarrow x^{\mu'} = f^\mu(x), \quad (\text{S298})$$

with f^μ an invertible, differentiable map on $\mathcal{M}$. This defines the diffeomorphism group $\text{Diff}(\mathcal{M})$, and its invariance reflects the principle that physics is independent of coordinate labels. In the TSRT framework, this symmetry is not postulated, as in general relativity, but emerges from the covariant structure of the deviation field $\Xi^\mu(\tau)$. Since the deviation is defined on the manifold and transforms as a contravariant vector field under reparametrization, the symmetric combination $\xi_{\mu\nu}$ transforms as a tensor. The effective metric $g_{\mu\nu}^{\text{eff}}$ thereby inherits this tensorial transformation law, ensuring full compatibility with $\text{Diff}(\mathcal{M})$.

Physically, the emergence of diffeomorphism invariance in TSRT guarantees that the ob-

servable curvature, geodesic motion, and gravitational field equations remain invariant under coordinate changes, despite originating from deterministic, microscopic trembling. This reinforces the view that spacetime structure in TSRT is not a static background, but a dynamic, relational geometry shaped by causal deviations. Thus, macroscopic external trembling deviations form the geometric origin of gravitation in TSRT. Through proper-time averaging and causal accumulation, they give rise to a diffeomorphism-invariant metric structure on $\mathcal{M}$, unifying the conceptual foundations of general relativity with the causal geometry of trembling spacetime.

S13.2 Effective Metric and Emergent Geometry

In TSRT, the smooth macroscopic geometry of spacetime is not fundamental but emerges from the deterministic coarse-graining of microscopic, proper-time-resolved trembling deviations. These deviations are encoded by the proper-time-dependent vector field $\Xi^\mu(x, \tau)$, which represents small causal displacements from the background worldline, as first introduced in Section S9. These displacements induce symmetric perturbations in the background metric via:

$$\xi_{\mu\nu}(x, \tau) := \partial_\mu \Xi_\nu(x, \tau) + \partial_\nu \Xi_\mu(x, \tau), \quad (\text{S299})$$

ensuring compatibility with the Lorentzian symmetry structure of spacetime and the preservation of local causality, as formalized in Section S23.

The coarse-grained accumulation of these perturbations over proper time defines an emergent, effective metric field:

$$g_{\mu\nu}^{\text{eff}}(x) := \bar{g}_{\mu\nu}(x) + \langle \xi_{\mu\nu}(x, \tau) \rangle, \quad (\text{S300})$$

where $\bar{g}_{\mu\nu}(x)$ denotes the background (typically Minkowski or maximally symmetric) metric, and the averaging operator $\langle \cdot \rangle$ denotes coarse-graining over rapidly oscillating proper-time modes. This averaging preserves causal propagation and statistical locality, guaranteeing that the effective metric remains smooth and well-defined at macroscopic scales.

The emergent metric $g_{\mu\nu}^{\text{eff}}$ defines the spacetime interval, determines causal structure (light-cones and time-like paths), and forms the basis for geodesic propagation and curvature. The Levi-Civita connection associated with this metric is given by:

$$\Gamma_{\mu\nu}^\rho = \frac{1}{2} g_{\text{eff}}^{\rho\lambda} \left(\partial_\mu g_{\nu\lambda}^{\text{eff}} + \partial_\nu g_{\mu\lambda}^{\text{eff}} - \partial_\lambda g_{\mu\nu}^{\text{eff}} \right), \quad (\text{S301})$$

ensuring that the connection is torsion-free and metric-compatible:

$$\nabla_\lambda g_{\mu\nu}^{\text{eff}} = 0. \quad (\text{S302})$$

Critically, in contrast to general relativity, where the metric and connection are assumed as independent postulates subject to variational principles, TSRT derives both $g_{\mu\nu}^{\text{eff}}$ and $\Gamma_{\mu\nu}^\rho$ as statistical consequences of underlying proper-time deviations. The Levi-Civita connection in Equation (S301) is not a structural axiom but an emergent descriptor of the average motion of particles through a trembling spacetime substrate. This reinterpretation marks a conceptual departure from classical geometry: curvature is not prescribed but generated, not assumed but

derived. The effective connection encodes the cumulative displacement behavior of all trembling paths in a causal, Lorentz-covariant setting, transforming the metric structure of spacetime into a secondary, emergent feature. This bridges the microscopic causal theory of TSRT with the macroscopic structure of general relativistic gravity.

S13.3 Curvature and Einstein Field Equations

Once the coarse-grained effective metric $g_{\mu\nu}^{\text{eff}}(x)$, defined in Equation (S300), is established from proper-time-averaged external deviations $\Xi^\mu(\tau)$, the curvature of spacetime becomes computable in terms of the emergent Levi-Civita connection constructed from this metric, as given in Equation (S301). All curvature quantities retain their standard tensorial definitions from differential geometry, but their interpretation is now deeply rooted in statistical and geometric features of trembling spacetime, rather than postulated spacetime structure. We begin with the emergent Christoffel symbols:

$$\Gamma_{\mu\nu}^\sigma := \frac{1}{2}g_{\text{eff}}^{\sigma\lambda} \left(\partial_\mu g_{\lambda\nu}^{\text{eff}} + \partial_\nu g_{\lambda\mu}^{\text{eff}} - \partial_\lambda g_{\mu\nu}^{\text{eff}} \right). \quad (\text{S303})$$

These connection coefficients define the unique torsion-free, metric-compatible connection associated with the coarse-grained metric $g_{\mu\nu}^{\text{eff}}$.

From this connection, the full curvature structure is determined by the Riemann tensor:

$$R_{\mu\nu\rho}^\sigma := \partial_\mu \Gamma_{\nu\rho}^\sigma - \partial_\nu \Gamma_{\mu\rho}^\sigma + \Gamma_{\mu\rho}^\lambda \Gamma_{\nu\lambda}^\sigma - \Gamma_{\nu\rho}^\lambda \Gamma_{\mu\lambda}^\sigma, \quad (\text{S304})$$

which measures the non-commutativity of parallel transport and captures the intrinsic curvature generated by long-range coherence in trembling patterns. This curvature arises deterministically as a coarse-grained expression of the microscopic deviations $\Xi^\mu(\tau)$.

Contracting the Riemann tensor yields the Ricci tensor and scalar curvature:

$$R_{\mu\nu} := R_{\mu\lambda\nu}^\lambda, \quad (\text{S305})$$

$$R := g_{\text{eff}}^{\mu\nu} R_{\mu\nu}, \quad (\text{S306})$$

which quantify volume distortion and scalar curvature, respectively.

From these quantities, we define the Einstein tensor:

$$G_{\mu\nu} := R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}^{\text{eff}} R, \quad (\text{S307})$$

which satisfies the contracted Bianchi identity:

$$\nabla^\mu G_{\mu\nu} = 0. \quad (\text{S308})$$

This identity is guaranteed by the geometric structure of the emergent metric, which remains a smooth Lorentzian manifold after coarse-graining. No additional assumptions are required beyond those already inherent to the trembling construction.

To determine the dynamics of the emergent metric, we compare the divergence-free Einstein tensor with a conserved source tensor. In standard general relativity, this source is the stress-

energy tensor $T_{\mu\nu}$, whose conservation is required for consistency. In TSRT, the source tensor $T_{\mu\nu}^{\text{eff}}$ arises from both conventional matter and the averaged influence of persistent trembling patterns:

$$T_{\mu\nu}^{\text{eff}} = T_{\mu\nu} + \delta T_{\mu\nu}(\Xi), \quad (\text{S309})$$

where $T_{\mu\nu}$ captures classical matter and field contributions, and $\delta T_{\mu\nu}(\Xi)$ reflects coarse-grained geometric stress-energy sourced by coherent deviations $\Xi^\mu(\tau)$.

Requiring consistency of divergence-free structures on both sides leads naturally to the emergent Einstein field equations:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}^{\text{eff}}R = 8\pi GT_{\mu\nu}^{\text{eff}}, \quad (\text{S310})$$

where the gravitational constant G arises as an effective coupling constant that relates the amplitude and spatial correlation length of the trembling field to large-scale curvature.

Thus, in TSRT, the Einstein field equations emerge not as axioms or principles, but as the necessary large-scale field equations required to describe the curvature sourced by coarse-grained trembling. They are derivable from the dynamical geometry induced by persistent microscopic deviations, ensuring that the resulting theory matches general relativity at large scales while remaining fully embedded in the deterministic trembling structure of spacetime.

Importantly, not all trembling contributions are fully encoded in the coarse-grained metric $g_{\mu\nu}^{\text{eff}}$. Fine-scale, rapidly fluctuating modes contribute to subleading corrections, quantum analogues, or residual stress terms not captured by $T_{\mu\nu}^{\text{eff}}$ alone. These may generate corrections to curvature at very small scales but do not affect the large-scale Einstein equations, which reflect the average geometry determined by statistically coherent trembling. In this way, TSRT provides a unified, first-principles explanation for the emergence of spacetime curvature and its dynamical evolution via Einstein's equations, grounded entirely in deterministic, causal geometric fluctuations.

The validity of Equation (S310) is not imposed externally but derived as a necessary consequence of several intertwined physical and geometric principles. First, the coarse-grained trembling fields $\Xi^\mu(\tau)$ must generate a smooth effective geometry that remains compatible with proper-time causality, ensuring that all particle trajectories maintain positive-definite time evolution. Second, the emergent metric $g_{\mu\nu}^{\text{eff}}$ must be covariantly consistent with the propagation of these trembling fields, meaning it must be metric-compatible under the covariant derivative ∇^{eff} defined by the Levi-Civita connection. Third, the resulting geometry must be invariant under general coordinate transformations, adhering to the full diffeomorphism group $\text{Diff}(\mathcal{M})$, thereby ensuring general covariance. Finally, local energy-momentum conservation must be preserved within the coarse-grained geometry, requiring that $\nabla^\mu T_{\mu\nu}^{\text{eff}} = 0$ holds identically. These conditions collectively ensure that the Einstein field equations emerge as a consistency requirement of trembling-induced curvature, rather than as axiomatic postulates.

From these constraints, the Einstein equations emerge as a necessary closure condition on the deterministic evolution of the trembling-induced curvature. The gravitational field equations are thus recast as emergent large-scale identities derived from proper-time-resolved trembling dynamics. This derivation reinterprets general relativity not as a foundational law of nature, but

as an effective macroscopic limit of a deeper geometric framework. The curvature of spacetime is no longer a primary variable—it is the averaged imprint of the deterministic, Lorentz-invariant trembling of the spacetime fabric, governed by TSRT.

S13.4 Geodesic Motion and the Weak Equivalence Principle

Given the emergent effective metric $g_{\mu\nu}^{\text{eff}}(x)$, defined in Equation (S300), and the associated Levi-Civita connection from Equation (S301), the large-scale motion of a free test particle in TSRT is governed by the coarse-grained geodesic equation:

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\nu\lambda}^\mu \frac{dx^\nu}{d\tau} \frac{dx^\lambda}{d\tau} = 0, \quad (\text{S311})$$

where τ denotes the proper time parameter along the particle's worldline, and $\Gamma_{\nu\lambda}^\mu$ are the Christoffel symbols constructed from the coarse-grained effective metric $g_{\mu\nu}^{\text{eff}}(x)$.

This equation reflects the weak equivalence principle in its emergent form: all freely falling particles follow the same spacetime trajectories, independent of their internal properties, provided the geometric background is the same. Crucially, in TSRT this is not an imposed axiom, but a derived consequence of the averaging procedure applied to the underlying microscopic trembling dynamics. In the regime where the rapid fluctuations in the metric $\xi_{\mu\nu}(\tau)$ vary on timescales or length scales smaller than the resolution of the experiment or observational apparatus, the net effect is encapsulated in a smooth coarse-grained geometry that satisfies the Einstein field equations. The geodesic equation (S311) arises from a variational principle applied to the effective proper-time interval associated with the coarse-grained metric $g_{\mu\nu}^{\text{eff}}(x)$. Consider the action functional for a test particle moving along a worldline $x^\mu(\tau)$, defined by the proper-time integral:

$$S[x^\mu] = \int_{\tau_1}^{\tau_2} \sqrt{g_{\mu\nu}^{\text{eff}}(x)} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} d\tau. \quad (\text{S312})$$

This expression measures the accumulated proper time between two events along a path in the emergent geometry. The goal is to determine the path $x^\mu(\tau)$ that extremizes this quantity, subject to fixed endpoints.

To perform the variation, define a perturbed path:

$$x^\mu(\tau) \rightarrow x^\mu(\tau) + \delta x^\mu(\tau), \quad (\text{S313})$$

and compute the first-order variation of the action. Denoting the velocity vector as $u^\mu := dx^\mu/d\tau$, the variation yields:

$$\begin{aligned} \delta S &= \int_{\tau_1}^{\tau_2} \delta \left(\sqrt{g_{\mu\nu}^{\text{eff}}(x)} u^\mu u^\nu \right) d\tau \\ &= \int_{\tau_1}^{\tau_2} \frac{1}{2} \left(g_{\mu\nu}^{\text{eff}} u^\mu u^\nu \right)^{-1/2} \left(\partial_\rho g_{\mu\nu}^{\text{eff}} \delta x^\rho u^\mu u^\nu + 2g_{\mu\nu}^{\text{eff}} u^\mu \frac{d}{d\tau} \delta x^\nu \right) d\tau. \end{aligned} \quad (\text{S314})$$

Using integration by parts on the second term and discarding boundary terms (since the

endpoints are fixed), we obtain:

$$\delta S = \int_{\tau_1}^{\tau_2} \left[\frac{1}{2} \left(g_{\mu\nu}^{\text{eff}} u^\mu u^\nu \right)^{-1/2} \left(\partial_\rho g_{\mu\nu}^{\text{eff}} u^\mu u^\nu - 2 \frac{d}{d\tau} \left(g_{\rho\nu}^{\text{eff}} u^\nu \right) \right) \right] \delta x^\rho d\tau. \quad (\text{S315})$$

Requiring that $\delta S = 0$ for arbitrary variations δx^ρ implies that the integrand must vanish. Multiplying through by $\sqrt{g_{\mu\nu}^{\text{eff}} u^\mu u^\nu}$, and rearranging terms, one recovers the Euler–Lagrange equation:

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\nu\lambda}^\mu(x) \frac{dx^\nu}{d\tau} \frac{dx^\lambda}{d\tau} = 0, \quad (\text{S316})$$

where $\Gamma_{\nu\lambda}^\mu$ are the Christoffel symbols constructed from the effective metric $g_{\mu\nu}^{\text{eff}}(x)$.

This result matches Equation (S311), confirming that the geodesic equation indeed follows from extremizing the effective proper time. The derivation mirrors the classical route taken in General Relativity, but the underlying physics is distinct in TSRT. Here, the effective metric $g_{\mu\nu}^{\text{eff}}(x)$ itself is not a fundamental input, but an emergent object resulting from the statistical averaging of microscopic trembling deviations $\xi_{\mu\nu}(\tau)$. The spacetime curvature encoded in the Christoffel symbols and in the associated Einstein tensor does not originate from an externally imposed energy–momentum tensor, but from the geometric coarse-graining of deterministic, submicroscopic fluctuations in the spacetime fabric. Thus, in TSRT, the geodesic equation and the gravitational field both emerge together as large-scale consequences of the same trembling dynamics, firmly rooted in causal and variational geometric principles.

However, this emergent geodesic behavior only captures the macroscopic limit. At finer resolution—particularly near microscopic or sub-Planckian scales—the full dynamics of the particle includes nontrivial contributions from the rapid, deterministic fluctuations encoded in $\xi_{\mu\nu}(\tau)$. These residual trembling-induced deviations are described by the full trembling geodesic equation derived in Section S4, which includes nonzero effective acceleration terms arising from the geometric deformation field $\delta\Gamma_{\alpha\beta}^\mu \sim \partial\xi_{\mu\nu}$.

Therefore, Equation (S311) does not replace the microscopic dynamics, but represents their average over proper time and spatial fluctuations, filtered through a coarse-graining process. The microscopic structure continues to exist beneath the smooth effective geometry, contributing to phenomena such as inertia, mass corrections, and quantum-like statistical effects, even though its influence on classical trajectories has been averaged out.

In summary, the geodesic equation in TSRT embodies the weak equivalence principle as an emergent large-scale phenomenon, while fully preserving the underlying microscopic trembling responsible for the generation of both mass and curvature. This dual structure allows TSRT to unify deterministic microphysics with relativistic geometrodynamics in a single coherent framework.

A key implication of this result is the automatic emergence of the weak equivalence principle (WEP). In TSRT, the gravitational field is not a separate physical entity coupled to matter via an external interaction term, but rather the manifestation of a universal geometric background shaped by coarse-grained trembling deviations. Since these metric deviations arise from the external trembling field $\Xi^\mu(\tau)$, which does not couple to internal degrees of freedom such as spin, charge, or color (see Sections S9 and S13), all particles experience the same spacetime geometry. It follows directly from this geometric structure that all freely falling particles—regardless

of their mass, spin, charge, or internal configuration—must follow identical geodesics in the emergent metric $g_{\mu\nu}^{\text{eff}}$. In other words, the universality of free fall is not an empirical postulate but a geometric consequence of the trembling-induced spacetime structure: All freely falling particles, regardless of mass or internal structure, follow identical geodesics in the emergent metric $g_{\mu\nu}^{\text{eff}}$. This statement expresses the weak equivalence principle as an inevitable result of the coarse-grained geometry arising from proper-time trembling in TSRT.

This is precisely the content of the weak equivalence principle, recovered here as a natural and unavoidable consequence of the emergent curvature structure rather than an assumed axiom. The universality of free-fall arises because the effective connection $\Gamma_{\nu\lambda}^{\mu}$ is constructed from the same coarse-grained trembling field for all matter configurations. There is no room in TSRT for particle-dependent connections or metric deformations: the proper-time-trembling of spacetime itself generates a universal background geometry, and particles move along its geodesics according to the same causal variational law. Thus, TSRT not only recovers the classical behavior of freely falling bodies as described by general relativity, but also explains the geometric origin of this behavior within a deeper framework of deterministic spacetime dynamics. The weak equivalence principle becomes an emergent symmetry of trembling spacetime, rooted in the statistical uniformity of macroscopic curvature across all worldlines.

S13.5 Energy-Momentum Tensor and Conservation Laws

In the TSRT framework, the energy-momentum tensor $T_{\mu\nu}$ is not an external source term inserted into the geometry, but an emergent quantity that encodes the total coarse-grained influence of both classical matter and the persistent structure of microscopic trembling deviations. This tensor appears in the emergent Einstein field equations derived in Equation (S310):

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}^{\text{eff}}R = 8\pi G T_{\mu\nu}. \quad (\text{S317})$$

The total energy-momentum tensor splits naturally into two contributions:

$$T_{\mu\nu} = T_{\mu\nu}^{\text{matter}} + T_{\mu\nu}^{\text{tremble}}, \quad (\text{S318})$$

where $T_{\mu\nu}^{\text{matter}}$ represents the conventional stress-energy tensor of particles, fields, and classical sources, while $T_{\mu\nu}^{\text{tremble}}$ arises from the averaged metric deviations induced by microscopic trembling fields $\Xi^{\mu}(\tau)$, as developed in Sections S9 and S13.

Explicitly, the coarse-grained trembling contribution may be interpreted as a vacuum-like geometric stress-energy term:

$$T_{\mu\nu}^{\text{tremble}} := \left\langle \partial_{\mu}\Xi^{\lambda}\partial_{\nu}\Xi_{\lambda} + \frac{1}{2}\bar{g}_{\mu\nu}\bar{g}^{\rho\sigma}\partial_{\rho}\Xi^{\lambda}\partial_{\sigma}\Xi_{\lambda} \right\rangle, \quad (\text{S319})$$

which includes both directional deviations and isotropic tension-like terms. These contributions reflect the inertial and curvature-generating properties of spacetime trembling and persist even in regions devoid of classical matter, making them relevant candidates for gravitational vacuum energy or effective dark energy sources.

A central feature of TSRT is that the total energy-momentum tensor is covariantly conserved:

$$\nabla^\mu T_{\mu\nu} = 0, \quad (\text{S320})$$

where ∇^μ is the covariant derivative associated with the emergent effective metric $g_{\mu\nu}^{\text{eff}}$. This conservation law is not postulated independently but arises from two interlocking geometric principles.

First, in the classical theory of General Relativity, the conservation law follows directly from the diffeomorphism invariance of the Einstein–Hilbert action. Under infinitesimal coordinate transformations, this invariance implies the contracted Bianchi identity:

$$\nabla^\mu \left(R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \right) = 0, \quad (\text{S321})$$

which, when combined with the Einstein field equations $G_{\mu\nu} = \kappa T_{\mu\nu}$, leads to the conservation of energy–momentum:

$$\nabla^\mu T_{\mu\nu} = 0. \quad (\text{S322})$$

This result is well-established in the literature [28, 31, 34] and is a consequence of Noether’s theorem applied to diffeomorphism symmetry [38].

In TSRT, this conservation law continues to hold, but its justification differs in origin. The Einstein tensor $G_{\mu\nu}$ is not introduced via postulation, but rather emerges as the coarse-grained manifestation of fine-scale metric trembling, as described in Section S13. Because the effective large-scale geometry retains the differentiable structure and symmetry properties of a Lorentzian manifold, the emergent metric $g_{\mu\nu}^{\text{eff}}$ inherits diffeomorphism invariance by construction. Consequently, the Bianchi identities remain satisfied by the emergent curvature tensors, and the covariant conservation of the energy–momentum tensor $T_{\mu\nu}$ follows as a derived geometric result. In this way, TSRT recovers the classical conservation law not by assumption, but as an inevitable consequence of the underlying causal and geometric consistency of the trembling-induced effective spacetime.

Thus, the relation

$$\nabla^\mu T_{\mu\nu} = 0 \quad (\text{S323})$$

is fully preserved in TSRT and acquires a new physical interpretation: it expresses not merely the symmetry of a postulated field theory, but the causal coherence of energy flow across regions of spacetime whose geometry arises from averaged trembling motion. This result establishes a profound departure from traditional formulations: in standard general relativity, energy-momentum conservation is typically enforced as a compatibility condition between the matter sector and the geometry. In TSRT, it is an unavoidable consequence of the trembling-induced geometry itself, which is constructed from proper-time coherent fluctuations that preserve local causal structure. The causal flow of geometric deviation ensures that energy and momentum are locally conserved within the emergent curvature field. Furthermore, the trembling contribution $T_{\mu\nu}^{\text{tremble}}$ opens new interpretive directions. It provides a natural, geometric candidate for gravitational vacuum effects and may underlie phenomena attributed to dark energy or modifications to inertial mass. Because the trembling field is inherently Lorentz covariant and causally con-

sistent, its energy-momentum structure respects all physical symmetries of the emergent metric and naturally integrates into cosmological and astrophysical contexts.

In summary, the energy-momentum tensor in TSRT is not an externally defined input but a geometric output of the theory's foundational structure. Its conservation emerges from the same causal and variational principles that govern the formation of curvature, unifying matter dynamics with gravitational geometry under a single deterministic framework rooted in proper-time trembling.

S13.6 Conclusion

In Trembling Spacetime Relativity Theory (TSRT), gravity is not treated as a separate interaction mediated by a quantized field, nor is it introduced axiomatically through postulated curvature equations. Instead, it emerges as the macroscopic manifestation of coherent, causally consistent fluctuations in the external geometry of spacetime itself. These fluctuations, encoded in the deviation field $\Xi^\mu(\tau)$, induce symmetric metric perturbations $\xi_{\mu\nu}(x, \tau)$, which upon coarse-graining yield the effective metric:

$$g_{\mu\nu}^{\text{eff}}(x) = \bar{g}_{\mu\nu}(x) + \langle \xi_{\mu\nu}(x, \tau) \rangle. \quad (\text{S324})$$

From this emergent metric, all elements of gravitational geometry follow deterministically. The Levi-Civita connection, Riemann curvature tensor, and Einstein tensor are derived directly from the coarse-grained structure of the trembling-induced geometry, as shown in Equations (S301)–(S306). The Einstein field equations,

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}^{\text{eff}}R = 8\pi GT_{\mu\nu}, \quad (\text{S325})$$

arise not by assumption, but as a consistency condition imposed by proper-time determinism, diffeomorphism invariance, and curvature closure of the emergent geometry.

The total energy-momentum tensor $T_{\mu\nu}$, comprising both conventional matter contributions $T_{\mu\nu}^{\text{matter}}$ and trembling-induced vacuum-like terms $T_{\mu\nu}^{\text{tremble}}$, satisfies the covariant conservation law:

$$\nabla^\mu T_{\mu\nu} = 0, \quad (\text{S326})$$

as required by the Bianchi identity and geometric self-consistency of the effective manifold.

In this way, gravity in TSRT completes the fourfold classification of fundamental interactions. Each of the known forces corresponds to a distinct, irreducible sector of trembling-induced geometric symmetry. Electromagnetism arises from external abelian trembling and is governed by the symmetry group $U(1)$. The weak interaction emerges from internal chiral non-abelian trembling, associated with the gauge structure $SU(2)_L \times U(1)_Y$. The strong interaction is rooted in internal color trembling and described by the non-abelian compact group $SU(3)_C$. Finally, gravitation results from external coarse-curvature trembling and exhibits full covariance under the diffeomorphism group $\text{Diff}(\mathcal{M})$. These four interaction types represent the complete spectrum of geometrically admissible trembling configurations within the TSRT framework. Unlike the internal gauge forces, which act within compact internal fiber bundles, gravity is distin-

guished by its action on the spacetime base manifold itself, with diffeomorphism invariance as its defining geometric symmetry. This symmetry is not imposed, but arises directly from the covariance of the trembling deviation field $\Xi^\mu(\tau)$, whose proper-time evolution respects the full general coordinate group $\text{Diff}(\mathcal{M})$.

TSRT thus offers a unified, deterministic foundation for all four fundamental forces. Each emerges from a common causal mechanism—spacetime trembling—interpreted through the lens of metric deviation symmetry and proper-time covariance. General relativity is not an independent framework within this view, but rather the large-scale limit of external trembling geometry, reinterpreted as the macroscopic curvature induced by fine-scale, deterministic fluctuations. This completion of the gravitational sector demonstrates that TSRT contains within its geometric core the entire structure of classical spacetime physics, now derived rather than postulated. It provides a causal, quantization-free account of curvature, motion, and field generation, anchoring gravity within the same trembling origin that governs all other interactions. We now derive conservation laws as natural consequences of the geometric symmetries underlying trembling spacetime.

S14 Supplementary Material: Conservation Laws from Geometric Symmetries

One of the central structural achievements of Trembling Spacetime Relativity Theory (TSRT) is that conservation laws are not externally imposed, but arise directly from the geometric symmetries of the deviation field $\Xi^\mu(\tau)$, which governs deterministic microscopic trembling along proper-time worldlines. This stands in marked contrast to classical field theories, where conservation principles follow from symmetry assumptions applied to externally defined Lagrangians. In TSRT, all symmetries and associated conservation laws are rooted in the causal, Lorentz-covariant geometry of trembling deviations. As established in Section S9, each admissible symmetry class—external abelian, internal chiral, internal color, and diffeomorphic—corresponds to a distinct, irreducible trembling mode. Each such mode gives rise to a conserved current through a proper-time version of Noether’s theorem, derived from internal invariance under infinitesimal geometric transformations of the trembling field.

This section consolidates the conservation laws associated with each of the four trembling-induced force sectors, referring explicitly to their geometric origins as developed in Sections S10–S13. Each conserved quantity is shown to follow directly from the specific trembling symmetry—whether a $U(1)$ phase shift, a local $SU(2)_L \times U(1)_Y$ rotation, a color-trembling transformation in $SU(3)_C$, or the general coordinate covariance of macroscopic trembling curvature.

In TSRT, therefore, conservation of electric charge, weak isospin, hypercharge, color, and energy–momentum is not merely preserved by construction, but emerges as a geometric necessity from the structure of trembling spacetime itself.

S14.1 Geometric Invariance and Noether-Type Correspondence

Let $\mathcal{A}[\Xi^\mu(\tau)]$ denote the geometric action functional governing the proper-time evolution of the deviation field $\Xi^\mu(\tau)$ along a particle worldline. As detailed in Section S9, physically admissible classes of trembling correspond to continuous symmetry transformations of $\Xi^\mu(\tau)$, consistent with the constraints of Lorentz invariance, causal propagation, and compact group closure.

Suppose the trembling field undergoes an infinitesimal transformation,

$$\Xi^\mu(\tau) \rightarrow \Xi^\mu(\tau) + \delta\Xi^\mu(\tau), \quad (\text{S327})$$

which leaves the total action invariant:

$$\delta\mathcal{A} = 0. \quad (\text{S328})$$

Such a transformation defines a local geometric symmetry of the deviation dynamics. The connection between continuous symmetries and conserved quantities was first discovered by Emmy Noether [38] in 1918, in the context of classical field theory. Her theorem established that for every differentiable symmetry of the action in a Lagrangian system, there exists a corresponding conserved current. This result profoundly shaped modern theoretical physics, forming the foundation of conservation laws in electromagnetism, general relativity, and quantum field theory. In Trembling Spacetime Relativity Theory (TSRT), a similar correspondence emerges—but it is not rooted in postulated Lagrangians or quantized field operators. Instead, it arises from the in-

trinsic geometry of proper-time trembling. Each class of internal or external geometric deviation preserves a symmetry of the trembling manifold and thus induces a conserved current. These currents reflect geometric quantities whose divergence vanishes under the emergent covariant derivative structure associated with the coarse-grained effective metric $g_{\mu\nu}^{eff}$, introduced in Section S13.

In direct analogy with Noether's theorem—but grounded here in the deterministic structure of geometric fluctuations—each symmetry class gives rise to a current J^μ satisfying:

$$\nabla_\mu J^\mu = 0, \quad (\text{S329})$$

where ∇_μ is the covariant derivative with respect to $g_{\mu\nu}^{eff}$. Unlike conventional Noether currents that depend on field variables in the Lagrangian, TSRT currents originate from symmetry-preserving components of the trembling deviation fields $\Xi^A(\tau)$, projected into spacetime.

We now list the four principal classes of trembling-induced forces identified in TSRT, their associated symmetry groups, and the corresponding conserved currents.

1. Electromagnetism (Abelian Trembling, $U(1)$ Symmetry) In Section S10, the deviation field takes the form $\Xi^\mu(\tau) = \epsilon(\tau)u^\mu(\tau)$, inducing an emergent electromagnetic potential $A_\mu(x)$ via Equation (S185). The conserved electric current arises from the invariance of the trembling field under global phase shifts:

$$J_{\text{EM}}^\mu = qw^\mu, \quad \nabla_\mu J_{\text{EM}}^\mu = 0. \quad (\text{S330})$$

Here, q is the Noether charge associated with the $U(1)$ phase symmetry of the abelian sector.

2. Weak Force (Chiral Non-Abelian Trembling, $SU(2)_L$ Symmetry) In Section S11, the left-handed deviation fields $\Xi^A(\tau) \in E_L$ transform under $SU(2)_L$ and generate emergent gauge fields $W_\mu^a(x)$ through coarse-grained projection (Equation (S246)). The associated weak isospin current takes the form:

$$J_{\text{weak}}^\mu = \bar{\psi}\gamma^\mu T^a \psi, \quad \nabla_\mu J_{\text{weak}}^\mu = 0, \quad (\text{S331})$$

where $T^a \in \mathfrak{su}(2)_L$ are the Pauli matrix generators.

3. Hypercharge (Abelian Phase Trembling, $U(1)_Y$ Symmetry) Also in Section S11, the scalar deviation $\theta_Y(\tau)$ leads to the emergent hypercharge field $B_\mu(x)$ (Equation (S247)). The corresponding conserved current:

$$J_Y^\mu = \bar{\psi}\gamma^\mu Y \psi, \quad \nabla_\mu J_Y^\mu = 0, \quad (\text{S332})$$

where Y is the hypercharge operator acting on the internal state.

4. Strong Force (Color Non-Abelian Trembling, $SU(3)_C$ Symmetry) In Section S12, the deviation fields $\Xi_\mu^a(\tau)$ project to emergent gluon fields $A_\mu^a(x)$. The corresponding color

current appears in the covariant derivative and coupling structure, and is given by:

$$J_{\text{color}}^\mu = \bar{\psi} \gamma^\mu T^a \psi, \quad \nabla_\mu J_{\text{color}}^\mu = 0, \quad (\text{S333})$$

with $T^a \in \mathfrak{su}(3)_C$ acting in the color fiber space.

Consequences Each of these conserved currents arises deterministically in TSRT as a consequence of symmetry-preserving deviations in the internal trembling modes. Their covariant conservation follows from the geometric structure of the coarse-grained manifold and generalizes the traditional Noether framework into a purely geometric setting without the need for postulated action principles. These currents serve as both diagnostic tools and interaction carriers, tightly linking the internal symmetry structure of trembling spacetime to observable conservation laws.

In Section S14.2, we will learn that each admissible symmetry sector introduced in Sections S10–S13 yields such a conservation law, i.e., $U(1)$ phase-type symmetry $\rightarrow$ conserved electric charge, $SU(2)_L \times U(1)_Y$ chiral symmetry $\rightarrow$ conserved weak isospin and hypercharge, $SU(3)_C$ color symmetry $\rightarrow$ conserved color current, and Diffeomorphism invariance $\rightarrow$ conservation of energy–momentum. These are not separate postulates but unified expressions of a deeper geometric principle: that all physical conservation laws emerge from the internal invariance of the deviation field $\Xi^\mu(\tau)$ under causal, proper-time-symmetric trembling in spacetime. This consolidation affirms the foundational role of geometric symmetry in determining the conservation structure of TSRT.

S14.2 Conserved Quantities by Trembling Symmetry Class

As established in Sections S9–S13, each admissible symmetry class of the trembling deviation field $\Xi^\mu(\tau)$ in TSRT corresponds to a specific type of geometric invariance. These include phase-type, chiral, color, and macroscopic diffeomorphic trembling modes, each associated with a compact symmetry group that respects proper-time covariance and causal propagation. From each symmetry group acting on either the external spacetime manifold or an internal fiber bundle, a conserved current emerges through the generalized Noether correspondence described in Section S14. These conserved quantities are not auxiliary or postulated labels (as in standard gauge theory), but intrinsic outcomes of the symmetry structure encoded in the deviation dynamics. The deterministic trembling framework naturally gives rise to conservation laws as consequences of underlying geometric symmetries. These conservation laws are not postulated externally, but emerge from the invariance of the trembling dynamics under specific classes of transformations preserving causal structure.

The key symmetries are:

- **Proper-time translations:** invariance under infinitesimal shifts $\tau \rightarrow \tau + \epsilon$ leads to energy conservation.
- **Local gauge transformations:** invariance of internal trembling fibers under compact group actions (e.g., $U(1)$, $SU(2)_L$, $SU(3)_C$) leads to conserved currents associated with electric charge, weak isospin, and color charge.

- **Diffeomorphism covariance:** invariance under smooth coordinate transformations on the spacetime manifold leads to conservation of energy–momentum.

These symmetries guarantee the existence of corresponding conserved quantities via a generalized Noether theorem adapted to trembling spacetime dynamics. In particular, if the trembling-induced action is invariant under a continuous transformation generated by a vector field $X^\mu(x)$, then the associated current J^μ satisfies, as in Equation (S329):

$$\nabla_\mu J^\mu = 0, \quad (\text{S334})$$

where ∇_μ denotes the covariant derivative with respect to the coarse-grained metric $g_{\mu\nu}^{\text{eff}}$ induced by trembling (see Section S14.1).

Examples include conservation of electric charge from local $U(1)$ symmetry of abelian trembling sectors, conservation of color charge from local $SU(3)_C$ symmetry of color-trembling sectors, and conservation of energy and momentum from spacetime translation invariance embedded in diffeomorphism invariance. The following paragraphs detail the precise form of the conserved quantities arising from each trembling symmetry class, with reference to the corresponding emergent force sector.

$U(1)$ Abelian Trembling and Electric Charge Conservation. As detailed in Sections S9 and S10, the simplest admissible trembling symmetry is a global phase rotation of the deviation field along the particle’s worldline:

$$\Xi^\mu(\tau) \rightarrow e^{i\alpha} \Xi^\mu(\tau), \quad \alpha \in \mathbb{R}. \quad (\text{S335})$$

This defines an external Abelian $U(1)$ symmetry group, under which the trembling action remains invariant. By the generalized geometric Noether principle introduced above, this invariance implies the existence of a conserved scalar current j^μ obeying:

$$\partial_\mu j^\mu = 0. \quad (\text{S336})$$

This current is not imposed externally but emerges from the coarse-grained ensemble of phase-coherent trembling paths. Its associated conserved quantity is identified as electric charge. The corresponding gauge potential A_μ , defined as a proper-time-averaged projection of derivatives of the deviation field (cf. Supplementary Equation (S185)), couples directly to j^μ through the effective minimal coupling structure. Thus, the conservation of electric charge arises as a direct geometric consequence of $U(1)$ -symmetric trembling deviations in TSRT.

$SU(2)_L \times U(1)_Y$ Chiral Trembling and Electroweak Charges. As established in Sections S9 and S11, the electroweak interaction emerges from internal chiral trembling deviations of the field $\Xi^A(\tau)$, which lives in a fiber bundle over spacetime with handed substructure $E = E_L \oplus E_R$. Left-handed components undergo local non-Abelian rotations governed by:

$$\Xi^A(\tau) \rightarrow R^A_B(\tau) \Xi^B(\tau), \quad R(\tau) \in SU(2)_L, \quad (\text{S337})$$

while all components may simultaneously transform under an internal abelian hypercharge phase:

$$\Xi^A(\tau) \rightarrow e^{i\theta_Y(\tau)} \Xi^A(\tau), \quad \theta_Y(\tau) \in \mathbb{R}. \quad (\text{S338})$$

These symmetry transformations leave the trembling action invariant and thus yield conserved Noether currents. Specifically, the left-handed $SU(2)_L$ symmetry produces weak isospin currents $J^{\mu i}$, and the $U(1)_Y$ phase symmetry gives rise to a hypercharge current J_Y^μ , satisfying:

$$D_\mu J^{\mu i} = 0, \quad (\text{S339})$$

$$\partial_\mu J_Y^\mu = 0, \quad (\text{S340})$$

where D_μ denotes the covariant derivative associated with the emergent non-Abelian gauge field W_μ^i . These conservation laws are purely geometric consequences of chiral trembling symmetry and require no external postulates. The intrinsic left-right asymmetry of the internal trembling dynamics ensures that only left-handed states carry non-Abelian charges—providing a deterministic and Lorentz-covariant origin for parity violation in weak interactions.

$SU(3)_C$ Color Trembling and Color Charge Conservation. As detailed in Sections S9 and S12, the strong nuclear force arises from internal non-Abelian trembling deviations $\Xi_\mu^a(\tau)$, where the index $a = 1, \dots, 8$ labels the eight generators of the compact Lie algebra $\mathfrak{su}(3)_C$. These deviation fields transform under the adjoint representation of the color symmetry group:

$$\Xi_\mu^a(\tau) \rightarrow U^{ab}(x(\tau)) \Xi_\mu^b(\tau), \quad U(x) \in SU(3)_C. \quad (\text{S341})$$

Invariance of the trembling action under these local internal rotations leads to the emergence of a conserved color current $J^{\mu a}$, which satisfies the fully covariant conservation equation:

$$D_\mu J^{\mu a} = \partial_\mu J^{\mu a} + g f^{abc} A_\mu^b J^{\mu c} = 0, \quad (\text{S342})$$

where f^{abc} are the antisymmetric structure constants of $\mathfrak{su}(3)_C$, g is the effective coupling determined by the curvature of internal trembling, and A_μ^a denotes the emergent gluon-like gauge fields derived from coarse-grained trembling geometry.

This conservation law is not added by hand—it reflects the internal algebraic structure of color trembling itself. As shown in Section S12, the nonlinear self-interaction of these internal trembling directions produces confinement as a geometric inevitability: color charges remain conserved but cannot be separated beyond the coherence length of the internal trembling manifold.

Diffeomorphism Invariance and Energy–Momentum Conservation. As developed in Section S13, the gravitational sector of TSRT arises from coherent macroscopic averaging of external trembling deviations, resulting in an effective metric tensor $g_{\mu\nu}^{\text{eff}} = \bar{g}_{\mu\nu} + \langle \xi_{\mu\nu} \rangle$. The invariance of the geometric action under infinitesimal coordinate transformations $x^\mu \rightarrow x^\mu + \epsilon^\mu(x)$ leads, by the principle of general covariance, to the covariant conservation of the total

energy–momentum tensor:

$$\nabla^\mu T_{\mu\nu} = 0, \quad (\text{S343})$$

where ∇^μ is the covariant derivative defined by the Levi-Civita connection associated with $g_{\mu\nu}^{\text{eff}}$. The total stress-energy tensor includes both material and trembling-induced geometric contributions:

$$T_{\mu\nu} = T_{\mu\nu}^{\text{matter}} + T_{\mu\nu}^{\text{tremble}}. \quad (\text{S344})$$

This conservation law guarantees that energy and momentum are preserved within the effective curved spacetime geometry and that gravitational curvature evolves consistently with the energetic content of the entire system—including microscopic fluctuations encoded in $T_{\mu\nu}^{\text{tremble}}$.

To summarize, each of the fundamental conservation laws in TSRT—electric charge, weak isospin and hypercharge, color charge, and energy–momentum—emerges as a direct consequence of a geometric symmetry of the trembling deviation field. These symmetries are not imposed externally but follow from the classification in Section S9, which defines all physically admissible trembling sectors. The conserved quantities are therefore unified through a single variational principle and are anchored in the causal, Lorentz-invariant structure of deterministic spacetime trembling.

S14.3 Unified Conservation Mechanism via Trembling Geometry

In Trembling Spacetime Relativity Theory (TSRT), all conservation laws emerge from a unified geometric principle: the invariance of the deviation field $\Xi^\mu(\tau)$ under continuous proper-time parameterized transformations that preserve the action functional $\mathcal{A}[\Xi^\mu]$. As detailed in Section S9, each class of admissible trembling symmetry—whether internal or external, abelian or non-abelian—defines a specific mode of causal deviation in spacetime geometry. These continuous geometric symmetries correspond directly to conserved quantities via a generalized Noether correspondence (see also Section S10 for $U(1)$, Section S11 for $SU(2)_L \times U(1)_Y$, and Section S12 for $SU(3)_C$). This unification offers a geometric reinterpretation of Noether’s theorem that does not depend on externally chosen Lagrangians or imposed field structures. Instead, the conservation laws arise from deterministic internal symmetries of the trembling deviation field $\Xi^\mu(\tau)$, constrained by Lorentz invariance, causal propagation, compactness, and metric embeddability in four-dimensional spacetime.

Specifically:

- Global phase invariance of the abelian trembling field yields conservation of electric charge ($U(1)$, Section S10).
- Local chiral and hypercharge invariance of internal trembling yields weak isospin and hypercharge conservation ($SU(2)_L \times U(1)_Y$, Section S11).
- Local color rotation symmetry of non-Abelian internal trembling yields conservation of color charge ($SU(3)_C$, Section S12).
- Diffeomorphism invariance of macroscopic trembling-induced curvature yields conservation of energy–momentum (Section S13).

All these conservation laws are generated by the same variational structure acting on the same deviation field, applied across different trembling symmetry sectors. No additional fields, quantization assumptions, or gauge postulates are required. The conserved quantities in TSRT are therefore intrinsic geometric invariants of the deviation structure itself—emerging from causal trembling geometry and not from quantized dynamics. This unification affirms the deterministic and covariant foundations of TSRT and highlights that interaction and conservation are not separate principles but consequences of the same causal geometric mechanism.

S14.4 Comparison with Standard Field Theory

In conventional quantum field theory (QFT), conserved quantities originate from the invariance of Lagrangian densities under externally imposed symmetry transformations on quantized fields. Noether’s theorem ensures that each continuous symmetry of the action leads to a conserved current. However, this framework assumes a priori the existence of specific symmetry groups—such as $U(1)$, $SU(2)_L$, and $SU(3)_C$ —which are not derived from deeper geometric or physical principles but are selected to match empirical observations. The Standard Model’s gauge structure is thus phenomenologically justified but not theoretically necessitated.

By contrast, Trembling Spacetime Relativity Theory (TSRT) derives these symmetry groups directly from the geometric structure of deterministic spacetime deviations. As shown in Section S9, only four irreducible symmetry classes are compatible with the foundational TSRT constraints: Lorentz covariance, proper-time positivity, compact Lie structure, and embeddability of metric deviations in four-dimensional spacetime. These symmetry classes correspond precisely to the four known forces and their associated conservation laws, each arising from a distinct trembling geometry:

- $U(1)$ phase symmetry from abelian trembling: electric charge conservation (Section S10),
- $SU(2)_L \times U(1)_Y$ chiral symmetry from internal left-handed trembling: weak isospin and hypercharge conservation (Section S11),
- $SU(3)_C$ non-abelian symmetry from color trembling: color charge conservation (Section S12),
- Diffeomorphism invariance from coarse-grained metric trembling: energy–momentum conservation (Section S13).

These symmetry–conservation pairs are not imposed externally but are intrinsic to the geometry of proper-time-driven deviations $\Xi^\mu(\tau)$. TSRT thus reverses the conceptual order of standard theory: instead of assuming symmetries to derive conservation laws, it derives both from a single geometric variational principle acting on the deviation field. This shift marks a foundational advancement over traditional approaches. TSRT explains not only how conservation laws arise but why only certain symmetries exist at all. It resolves the arbitrariness of gauge group selection in QFT and reveals that the structure of interaction and conservation is not a free choice, but a geometric inevitability in causally consistent trembling spacetime.

S14.5 Conclusion

Trembling Spacetime Relativity Theory (TSRT) provides a unified geometric origin for all known conservation laws by embedding them within the symmetry structure of deterministic spacetime deviations. Rather than introducing conserved quantities through externally imposed symmetries or quantized field prescriptions, TSRT derives them as inevitable consequences of the causal, Lorentz-invariant evolution of the deviation field $\Xi^\mu(\tau)$. Each class of trembling symmetry identified in Section S9—Abelian phase-type, internal chiral, color non-Abelian, and macroscopic diffeomorphic—generates a corresponding conserved current via a generalized, proper-time-based Noether correspondence. These currents represent electric charge, weak isospin and hypercharge, color charge, and energy–momentum, respectively, as demonstrated in the symmetry-specific derivations of Sections S10 through S13. In this framework, conservation laws are not formal constraints or empirical postulates but intrinsic features of a unified causal geometry. The deviation field $\Xi^\mu(\tau)$ encodes both the internal coherence of microscopic space-time trembling and the global propagation of conserved quantities through curved geometry. As a result, symmetry, interaction, and conservation are not independent structures—they co-emerge from the same geometric principle.

This geometric unification strengthens the explanatory power of TSRT, replacing the modular assumptions of conventional field theory with a single deterministic framework grounded in proper-time covariance. In TSRT, all observed conservation laws — electric charge, color charge, energy, momentum, and angular momentum — emerge inevitably from the internal symmetries of trembling spacetime. Conservation is not an auxiliary principle imposed onto physical laws, but a structural necessity embedded in the causal architecture of the universe itself. The trembling structure also determines the range of fundamental forces. We now analyze how this emerges from the underlying geometric framework.

S15 Supplementary Material: Emergent Range of Fundamental Forces in TSRT

The spatial range of a physical force is traditionally explained in terms of the mass of its mediating bosons. In conventional quantum field theory, massless exchange particles like the photon and graviton yield long-range forces (electromagnetism and gravity), while massive mediators such as the $W^\pm$, Z^0 , and confined gluons produce short-range weak and strong interactions. However, this framework presupposes quantization and invokes additional constructs such as spontaneous symmetry breaking and the Higgs mechanism to explain why some bosons are massive and others not.

Trembling Spacetime Relativity Theory (TSRT) replaces this paradigm with a deterministic, geometric account rooted in the causal coherence of proper-time-dependent metric deviations. Each fundamental force emerges from a distinct symmetry class of trembling deviation fields $\Xi(\tau)$, and the effective spatial range of each force is determined not by mediator mass but by how long the geometric correlations in these deviations persist across spacetime.

The key diagnostic tool is the proper-time correlation function:

$$C^\mu_\nu(x, x') := \langle \Xi^\mu(x) \Xi_\nu(x') \rangle, \quad (\text{S345})$$

which quantifies the coherence between deviations at spacetime points x and x' . The correlation length ℓ_c is defined by the decay scale of this function and determines how far a trembling-induced interaction can propagate before decoherence renders it ineffective. Importantly, this correlation length ℓ_c is not assumed or fit to experiment; rather, it emerges from the internal group-theoretic, topological, and causal properties of each trembling sector. For example, the $U(1)$ symmetry of abelian electromagnetism allows for global phase coherence and long-range propagation, while the non-abelian torsion and internal curvature of the $SU(2)_L$ and $SU(3)_C$ sectors induce rapid decoherence and short interaction range.

This section proceeds in two parts. First, we examine how the correlation function arises from the coarse-grained statistics of proper-time fluctuations and how it is connected to spectral coherence and causal propagation. Second, we classify the interaction ranges of all four fundamental forces—electromagnetism, weak interaction, strong interaction, and gravity—based on their trembling symmetry class. This analysis demonstrates that the observed hierarchy of force ranges is a structural prediction of TSRT, not a phenomenological assumption.

S15.1 Trembling Coherence and Emergent Interaction Range

In the TSRT framework, the spatial range of any fundamental interaction is not imposed from external considerations, nor derived from particle masses, but emerges intrinsically from the causal coherence properties of the proper-time-dependent trembling fields that give rise to the interaction. This coherence is quantitatively captured by the spacetime correlation function of the deviation field, which measures the extent to which fluctuations at distinct spacetime points

remain phase-aligned. We define the proper-time correlation tensor:

$$C_{\nu}^{\mu}(x, x') := \langle \Xi^{\mu}(x) \Xi_{\nu}(x') \rangle, \quad (\text{S346})$$

where $\Xi^{\mu}(x)$ denotes the trembling-induced deviation vector field at point x , and the angular brackets $\langle \cdot \rangle$ represent a coarse-graining operation—either statistical or ensemble-averaged—over unresolved internal frequencies and phases. This two-point function is conceptually analogous to correlation functions in statistical mechanics, wave propagation theory, and stochastic field theory, but in TSRT, it arises from deterministic geometric fluctuations in spacetime itself.

The decay of this correlation function with spacetime separation $|x - x'|$ defines a characteristic coherence length ℓ_c , which serves as the effective interaction range:

$$C_{\nu}^{\mu}(x, x') \approx 0 \quad \text{for} \quad |x - x'| \gg \ell_c. \quad (\text{S347})$$

This scale ℓ_c does not depend on field quantization or virtual boson exchange, but is governed by the internal geometry, symmetry group structure, and spectral properties of the underlying deviation field. Associated with this coherence length is a characteristic temporal scale τ_c , linked via the dominant trembling frequency $\omega_c \sim 1/\tau_c$. In principle, ω_c can be derived from first principles: for instance, in some cases it may be connected to causal curvature bounds or minimal time resolution scales, as discussed elsewhere in TSRT (e.g., in relation to Planck-like energy limits). In highly curved or topologically twisted internal bundles, the natural frequencies of trembling become large, and coherence rapidly decays. Different trembling sectors—each classified by a specific internal symmetry group—lead to qualitatively different behaviors for the correlation function. Abelian sectors with trivial curvature permit global phase coherence and long-range correlation. In contrast, non-Abelian sectors with torsion, curvature, or compact internal topology introduce rapid decoherence, curvature-driven scrambling, and effective interaction suppression over distance.

In what follows, we systematically examine each of the four fundamental interactions—electromagnetic, weak, strong, and gravitational—and show how their observed range hierarchy emerges naturally from the decay behavior of their trembling correlation functions. This provides a first-principles geometric explanation of force range, with no need for boson mass assumptions or quantization postulates.

S15.2 Electromagnetic Interaction ($U(1)$ Abelian)

In TSRT, the electromagnetic force emerges from a sector of spacetime trembling characterized by abelian phase-like deviations. These deviations are encoded in a proper-time-dependent vector field aligned with the particle's four-velocity:

$$\Xi^{\mu}(\tau) = \epsilon(\tau) u^{\mu}(\tau), \quad (\text{S348})$$

where $\epsilon(\tau)$ is a scalar proper-time oscillation amplitude, and $u^{\mu}(\tau)$ is the future-directed unit four-velocity vector along the particle's worldline, normalized as $u^{\mu} u_{\mu} = -1$.

A typical form of this oscillation is given by:

$$\epsilon(\tau) = \epsilon_0 \cos(\omega\tau + \phi), \quad (\text{S349})$$

where ϵ_0 is the amplitude, ω is the characteristic frequency of trembling, and ϕ is a proper-time phase. This form reflects the simplest harmonic excitation of the proper-time degrees of freedom and leads to purely longitudinal trembling—i.e., deviations that do not rotate or twist in the tangent bundle but rather remain aligned with the particle’s instantaneous direction of motion.

Symmetry Structure and Coherence These deviations form a representation of the abelian $U(1)$ group, whose Lie algebra is 1-dimensional and commutative. As such, the phase-like structure of $\epsilon(\tau)$ is globally integrable and free from topological obstructions or torsional instabilities. The lack of structure constants (i.e., the vanishing of all f^{abc}) implies that proper-time oscillations in different regions do not interfere destructively through non-abelian mixing. Consequently, the deviations maintain coherence over arbitrarily large intervals in proper time, provided no external interactions (e.g., medium-induced decoherence) interfere. This makes the electromagnetic trembling sector maximally coherent among all four force-generating classes.

Correlation Function and Emergent Range The proper-time coherence can be formally captured by defining a two-point correlation tensor:

$$C^\mu_\nu(x, x') := \langle \Xi^\mu(x) \Xi_\nu(x') \rangle, \quad (\text{S350})$$

where the averaging $\langle \cdot \rangle$ is performed over unresolved or unobservable internal degrees of freedom—such as unmeasured phase offsets or fast internal cycles.

Substituting the abelian trembling form, we obtain:

$$C^\mu_\nu(x, x') = \langle \epsilon(\tau(x)) u^\mu(x) \cdot \epsilon(\tau(x')) u_\nu(x') \rangle \quad (\text{S351})$$

$$= \langle \epsilon(\tau(x)) \epsilon(\tau(x')) \rangle u^\mu(x) u_\nu(x'), \quad (\text{S352})$$

assuming $u^\mu(x) \approx u^\mu(x')$ for nearby points, or in the slowly varying regime, we can approximate:

$$C^\mu_\nu(x, x') \approx u^\mu u_\nu \langle \epsilon(\tau) \epsilon(\tau') \rangle. \quad (\text{S353})$$

Using the harmonic form of $\epsilon(\tau)$, we compute the scalar correlation function:

$$\langle \epsilon(\tau) \epsilon(\tau') \rangle = \epsilon_0^2 \langle \cos(\omega\tau + \phi) \cos(\omega\tau' + \phi) \rangle \quad (\text{S354})$$

$$= \epsilon_0^2 \left\langle \frac{1}{2} [\cos(\omega(\tau - \tau')) + \cos(\omega(\tau + \tau') + 2\phi)] \right\rangle. \quad (\text{S355})$$

Averaging over uniformly distributed phase $\phi \in [0, 2\pi)$, the second term vanishes:

$$\langle \epsilon(\tau) \epsilon(\tau') \rangle = \frac{\epsilon_0^2}{2} \cos(\omega(\tau - \tau')). \quad (\text{S356})$$

Thus, the full correlation tensor becomes:

$$C^\mu_\nu(x, x') \approx \frac{\epsilon_0^2}{2} u^\mu u_\nu \cos(\omega(\tau - \tau')). \quad (\text{S357})$$

This correlation function oscillates but does not decay exponentially, and hence retains significant magnitude even at large proper-time separations. Unlike non-abelian correlations (which decay due to internal phase decoherence), this function reflects persistent phase coherence. As a result, the coherence length ℓ_c is formally infinite, limited only by external factors (e.g., scattering, thermal noise, or boundary conditions).

Interpretation and Force Range The correlation length ℓ_c determines the effective range over which trembling modes remain phase-locked and thus generate interaction. For abelian $U(1)$ trembling:

$$\ell_c^{\text{EM}} \rightarrow \infty \quad (\text{in vacuum, with no external decoherence}). \quad (\text{S358})$$

This result is deeply rooted in the geometric structure of the underlying symmetry, namely, $U(1)$ is topologically S^1 with zero curvature and trivial torsion, no group-induced phase cancellation occurs, and coarse-graining does not disrupt long-range phase alignment. Therefore, in TSRT, the infinite range of electromagnetism arises as a deterministic and geometric consequence of the abelian structure of the trembling sector—not as an empirical postulate. This coherence structure mirrors the classical field behavior of Maxwell theory, but it emerges here from the geometry of proper-time fluctuations in trembling spacetime.

S15.3 Weak Interaction ($SU(2)_L$ Chiral Non-Abelian)

In TSRT, the weak interaction arises from chiral, non-abelian internal trembling modes confined to the left-handed subbundle $E_L \subset E$. The deviation fields associated with this sector take the form:

$$\Xi^A(\tau) \in E_L, \quad \Xi^A(\tau) \mapsto R^A_B(\tau) \Xi^B(\tau), \quad R(\tau) \in SU(2)_L, \quad (\text{S359})$$

where $A, B = 1, 2$ label internal isospin directions, and $R(\tau)$ denotes a proper-time-dependent local rotation in the internal symmetry group $SU(2)_L$. These fields transform under the fundamental (doublet) representation of the $SU(2)_L$ group, which has three generators $T^a = \frac{1}{2}\sigma^a$, with σ^a the Pauli matrices and structure constants $f^{abc} = \epsilon^{abc}$.

Non-Abelian Structure and Phase Decoherence Unlike the abelian $U(1)$ sector, the $SU(2)_L$ group is non-commutative and compact, and it possesses a non-trivial internal curvature. This curvature is captured by the Lie bracket:

$$[T^a, T^b] = if^{abc}T^c, \quad (\text{S360})$$

which implies that any combination of rotations in different group directions leads to internal phase mixing. As a consequence, chiral deviations that initially start aligned in one direction tend to rotate and scramble through the internal fiber as proper time progresses.

This effect is not just algebraic but geometric: it reflects the torsional and curvature-rich structure of the fiber bundle $E_L \rightarrow \mathcal{M}$, where $\mathcal{M}$ is the base spacetime manifold. These geometrically induced internal rotations cause rapid phase decoherence, limiting the range over which any given deviation field remains correlated with itself.

Proper-Time Correlation Function Let us define the two-point proper-time correlation function for internal chiral deviations, which reside in the left-handed subbundle associated with the weak isospin group:

$$C^{AB}(x, x') := \langle \Xi^A(x, \tau) \Xi^B(x', \tau') \rangle. \quad (\text{S361})$$

Here, the indices $A, B \in \{1, 2\}$ label the internal weak isospin directions, and $\Xi^A(x, \tau)$ represents the trembling-induced deviation field at spacetime point x , evaluated along the proper-time τ . This function measures the statistical coherence between internal deviations at two spacetime points and is crucial in determining how the weak interaction propagates. In the TSRT framework, these trembling deviations are inherently oscillatory in proper time, with rapid local phase fluctuations due to the non-abelian nature of the underlying $SU(2)_L$ symmetry. To study macroscopic effects, we perform a coarse-graining over proper-time cycles, averaging out high-frequency components. Under this averaging, the dominant surviving contribution comes from symmetric, phase-aligned deviations, i.e., terms where the internal indices match. This results in a diagonal structure in isospin space is

$$C^{AB}(x, x') \approx \delta^{AB} C(x, x'). \quad (\text{S362})$$

To analyze the spatial extent over which chiral deviations $\Xi^A(\tau) \in E_L$ remain causally coherent, we define the proper-time-averaged two-point correlation function, given in Equation (S361), where the brackets denote coarse-graining over fast internal trembling cycles. The chiral deviations transform under $SU(2)_L$ and evolve according to non-commuting generators. As a result, their evolution produces phase dispersion due to the curvature and torsion of the chiral fiber bundle.

Now, in a globally flat or abelian space, correlation functions can maintain long-range structure, as coherent phase alignment persists. However, in the non-abelian chiral setting, each internal deviation rotates through different group directions as it propagates, and these rotations do not commute. The cumulative effect is a randomization of phase across different spacetime points due to non-abelian parallel transport and internal curvature. The loss of phase coherence is thus not assumed—it is a dynamical consequence of the symmetry group’s geometry. Mathematically, this scrambling leads to an exponential suppression of correlations, much like what occurs in disordered spin systems or curved fiber bundles in gauge theory. Therefore, we model the decay of $C^{AB}(x, x')$ by the leading-order term consistent with Gaussian phase dispersion:

$$C^{AB}(x, x') \sim \delta^{AB} e^{-|x-x'|/\ell_c^{\text{weak}}}, \quad (\text{S363})$$

where ℓ_c^{weak} is the coherence length dynamically determined by the internal curvature scale of the emergent $SU(2)_L$ connection.

This form follows from the non-abelian, non-commutative structure of the chiral gauge

group, the absence of globally flat, integrable transport on the fiber bundle, the statistical decoherence of phase-aligned trembling vectors under coarse-graining, and analogy with known exponential decay of correlations in non-abelian gauge fields over curved or torsional bundles. In this way, TSRT yields a physical mechanism for interaction range based on deterministic trembling dynamics rather than imposed boson mass. The exponential decay reflects the fact that in a non-Abelian setting, phase alignment is rapidly lost due to internal rotations and torsion, unlike the abelian electromagnetic case where commutativity ensures global phase integrability.

Geometric Bound on Coherence Length In TSRT, the coherence length ℓ_c^{weak} is not treated as an arbitrary input. Instead, it emerges naturally from the internal geometry and curvature of the chiral trembling sector. More precisely, the spatial rate of decoherence is governed by the effective field strength of the emergent weak gauge fields $W_\mu^a(x)$, derived from coarse-graining the deviation fields over proper time. This field strength is defined via the standard Yang–Mills expression:

$$F_{\mu\nu}^a(x) = \partial_\mu W_\nu^a - \partial_\nu W_\mu^a + g \epsilon^{abc} W_\mu^b W_\nu^c, \quad (\text{S364})$$

where g is the weak coupling constant and ϵ^{abc} are the structure constants of the $SU(2)_L$ Lie algebra.

To extract a single scalar quantity that reflects the overall curvature strength, we compute the norm of the field strength tensor:

$$\|F_{\mu\nu}\|^2 := F_{\mu\nu}^a F^{a\mu\nu}, \quad (\text{S365})$$

where repeated indices are summed according to the Einstein summation convention. This scalar curvature invariant captures the average magnitude of non-Abelian rotation and torsion in the fiber bundle.

Specifically, the coherence length ℓ_c^{weak} is determined by the local curvature of the coarse-grained non-Abelian connection associated with $SU(2)_L$ symmetry. This curvature is described by the emergent field strength tensor:

$$F_{\mu\nu}^a(x) = \partial_\mu W_\nu^a - \partial_\nu W_\mu^a + g \epsilon^{abc} W_\mu^b W_\nu^c, \quad (\text{S366})$$

where $W_\mu^a(x)$ are the coarse-grained gauge fields derived from chiral trembling deviations, g is the weak coupling constant, and ϵ^{abc} are the structure constants of $SU(2)_L$. To assess the net impact of curvature on coherence loss, we compute the norm in Equation (S365) of the field strength; this scalar quantity encodes the average magnitude of the non-Abelian curvature and directly reflects the extent to which internal phase coherence is disrupted across spacetime. In analogy with the behavior of wave packets on curved manifolds and with correlation decay in spin systems with gauge interactions, stronger curvature leads to more rapid scrambling of internal states. Therefore, the correlation length ℓ_c^{weak} is geometrically suppressed by curvature via:

$$\ell_c^{\text{weak}} \sim \frac{1}{\sqrt{\|F_{\mu\nu}\|}}. \quad (\text{S367})$$

This inverse-square-root relation is not heuristic—it results from the fact that the effective dispersion of internal trembling directions is governed by parallel transport on a non-commutative fiber. As curvature increases, parallel transport around infinitesimal loops deviates more strongly from identity, leading to rapid decoherence of the trembling vector fields. The coherence length thus shrinks in direct response to the internal geometric complexity of the $SU(2)_L$ bundle.

In summary, ℓ_c^{weak} is a derived quantity—rooted in the symmetry group, connection curvature, and fiber torsion of the trembling structure—and provides a deterministic geometric explanation for the limited range of the weak force. Thus, the short range of the weak interaction in TSRT arises not from a boson mass, but from the intrinsic geometric instability of the chiral $SU(2)_L$ trembling sector. The coherence length ℓ_c^{weak} becomes a derived quantity determined by the curvature geometry of spacetime deviations.

Interpretation and Empirical Connection The rapid falloff of the correlation function (Equation (S363)) implies that chiral trembling deviations do not extend far in space before losing coherence. In empirical terms, the weak interaction is known to have a range of order $\sim 10^{-18}$ m. TSRT interprets this not as a parameter to be inserted by hand, but as a direct consequence of the internal non-abelian geometry of $SU(2)_L$ trembling. Importantly, even without using the mass of the $W^\pm$ and Z^0 bosons, TSRT explains the short range of the weak interaction from purely geometric considerations as follows. Non-zero structure constants $f^{abc} = \epsilon^{abc}$ introduce internal curvature, compact group topology leads to torsion and phase dispersion, and chiral asymmetry (left-handedness only) introduces further coherence-breaking. Thus, the weak force in TSRT has short range not because of particle mass insertion, but because of deterministic decoherence driven by internal fiber curvature.

S15.4 Strong Interaction ($SU(3)_C$ Non-Chiral Non-Abelian)

In the TSRT framework, the strong force emerges from internal trembling deviations $\Xi_\mu^a(\tau)$, where the index $a = 1, \dots, 8$ labels directions in the internal color space governed by the symmetry group $SU(3)_C$. These deviations are non-chiral (i.e., they do not distinguish between left- and right-handed sectors) and transform under the adjoint representation of the color gauge group:

$$\Xi_\mu^a(\tau) \mapsto \Xi_\mu^a(\tau) + f^{abc} \theta^b(\tau) \Xi_\mu^c(\tau), \quad (\text{S368})$$

where f^{abc} are the structure constants of $\mathfrak{su}(3)_C$, encoding the non-abelian Lie algebra commutation relations:

$$[T^a, T^b] = i f^{abc} T^c. \quad (\text{S369})$$

Internal Geometry and Fiber Curvature The color fiber bundle associated with $SU(3)_C$ is both non-abelian and topologically nontrivial. Unlike the $U(1)$ and $SU(2)_L$ cases, the $SU(3)$ group has a higher-dimensional internal symmetry space (8 generators), admits more intricate curvature due to its larger set of non-zero f^{abc} , and supports topologically stable non-integrable fiber bundles with nontrivial holonomy. As a result, color trembling fields evolve according to

a rich torsional geometry, and the local phases of these internal oscillations rapidly become uncorrelated.

Two-Point Correlation Function and Coherence Decay In the TSRT framework, the strong interaction arises from internal metric deviations $\Xi_\mu^a(x, \tau)$ associated with color degrees of freedom. These deviations live in the adjoint representation of the $SU(3)_C$ gauge group, with index $a = 1, \dots, 8$ labeling directions in color space. The internal dynamics of these deviations are governed by the non-abelian transformation law, given in Equation (S368), where f^{abc} are the structure constants of $\mathfrak{su}(3)_C$, encoding the intrinsic curvature and torsional properties of the internal symmetry bundle. The complex non-abelian structure of $SU(3)_C$ results in rapid local phase rotation and scrambling among internal directions, especially due to the compactness and nontrivial holonomy of the associated fiber bundle. These properties cause the internal trembling fields to decohere swiftly across spacetime. To quantify the interaction range, we define the coarse-grained two-point correlation tensor:

$$C_{\mu\nu}^{ab}(x, x') := \left\langle \Xi_\mu^a(x, \tau) \Xi_\nu^b(x', \tau') \right\rangle. \quad (\text{S370})$$

Because of the high internal dimensionality and the strong coupling among modes, the off-diagonal contributions average out under proper-time integration. Thus, the leading contribution to the correlation tensor is diagonal in both color and spacetime indices:

$$C_{\mu\nu}^{ab}(x, x') \sim \delta^{ab} \delta_{\mu\nu} e^{-|x-x'|/\ell_c^{\text{strong}}}, \quad (\text{S371})$$

where ℓ_c^{strong} is the coherence length over which color trembling retains phase memory. The exponential decay reflects a statistical loss of internal alignment, driven by the non-integrable curvature and non-commutativity of $SU(3)_C$. This form is not assumed arbitrarily. It follows from the randomizing effect of non-commuting internal transformations, leading to Gaussian phase diffusion, the stochastic suppression of off-diagonal terms in high-dimensional Lie algebras under proper-time coarse-graining, and the statistical independence of color directions at spacetime points separated by more than ℓ_c^{strong} .

Geometric Bound on Coherence Length As in the weak sector, the coherence length in the strong interaction is not a postulated parameter but a derived geometric quantity. The emergent coarse-grained gluon fields $A_\mu^a(x)$, which encode the collective motion of the internal trembling fields, define a non-Abelian curvature tensor given by:

$$G_{\mu\nu}^a(x) := \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g_s f^{abc} A_\mu^b A_\nu^c, \quad (\text{S372})$$

where f^{abc} are the structure constants of $SU(3)_C$ and g_s is the strong coupling constant. These terms capture the torsional and rotational structure of the internal fiber bundle generated by trembling.

To estimate the coherence length, we consider the scalar norm of this curvature tensor:

$$\|G_{\mu\nu}\|^2 := G_{\mu\nu}^a G^{a\mu\nu}. \quad (\text{S373})$$

The coherence length is then bounded by the inverse square root of the curvature magnitude:

$$\ell_c^{\text{strong}} \sim \frac{1}{\sqrt{\|G_{\mu\nu}\|}}. \quad (\text{S374})$$

This relation arises naturally from geometric considerations. Stronger curvature introduces more rapid internal rotation and deformation of the local fiber structure, preventing the persistence of aligned color directions across distance. As a result, any trembling coherence collapses rapidly, and colored excitations lose their internal phase correlation beyond femtometer scales.

Interpretation: Geometric Confinement This geometric mechanism leads to a natural confinement effect: when $|x - x'| > \ell_c^{\text{strong}}$, color-carrying deviations Ξ_μ^a at x and x' become statistically decorrelated. As such, any composite excitation carrying net color charge cannot propagate coherently across spacetime. This naturally explains the absence of free quarks and gluons, without invoking quantized gauge fields or potential energy wells. The confinement radius ℓ_c^{strong} becomes the intrinsic upper limit for color propagation. This effect mirrors empirical confinement phenomena but arises here deterministically from the topological and geometric structure of trembling spacetime.

Comparison with the Weak Sector Compared to the $SU(2)_L$ weak sector, the $SU(3)_C$ color symmetry introduces more directions, higher curvature, and a more compact fiber structure. Consequently,

$$\ell_c^{\text{strong}} \ll \ell_c^{\text{weak}}, \quad (\text{S375})$$

reflecting the much shorter range of the strong interaction, not because of a larger coupling constant per se, but because of the geometric structure of its internal coherence.

Summary The short range of the strong interaction in TSRT is a deterministic outcome of high-dimensional, non-abelian fiber curvature. It emerges from the geometry of internal trembling fields and their rapidly decaying coherence. The derived correlation structure and coherence length bound yield a geometric explanation for confinement and support the observed short-range behavior of QCD without invoking quantization or postulated field excitations.

S15.5 Gravitational Interaction (External Deviations and Coarse-Averaged Curvature)

In TSRT, gravity is not introduced as an independent interaction mediated by a quantized field. Instead, it arises from long-wavelength, externally directed trembling deviations of the particle worldline:

$$\Xi^\mu(\tau) := x^\mu(\tau) - \bar{x}^\mu(\tau), \quad (\text{S376})$$

where $x^\mu(\tau)$ is the true trembling trajectory and $\bar{x}^\mu(\tau)$ is the corresponding classical path in flat spacetime. These deviations are projected into spacetime and coarse-grained over proper time to yield an effective metric correction:

$$g_{\mu\nu}^{\text{eff}}(x) := \eta_{\mu\nu} + \langle \partial_\mu \Xi_\nu(\tau) + \partial_\nu \Xi_\mu(\tau) \rangle, \quad (\text{S377})$$

as established in Section S13.

External vs. Internal Trembling Unlike the internal fiber-based trembling modes responsible for gauge interactions (electromagnetic, weak, and strong), gravitational deviations are not associated with compact internal symmetry groups, do not rotate under internal gauge transformations, and are globally defined as vector fields in spacetime. This absence of internal torsion, curvature, or phase degeneracy implies that there is no structural mechanism that scrambles the direction or phase of the external deviations.

Correlation Tensor and Global Coherence In TSRT, therefore, gravitation emerges not from compact internal trembling modes but from persistent deviations in the external spacetime geometry. These deviations are described by the vectorial trembling field in Equation (S376), which is aligned with the four-velocity and evolves under proper-time without being embedded in a nontrivial internal symmetry group. To quantify the coherence of such deviations, we define the proper-time coarse-grained two-point correlation tensor:

$$C^\mu_\nu(x, x') := \langle \Xi^\mu(x, \tau) \Xi_\nu(x', \tau') \rangle, \quad (\text{S378})$$

where the brackets $\langle \cdot \rangle$ denote averaging over unresolved proper-time phases. Since these deviations are purely external and free from internal torsion or compact group-induced scrambling, the correlation function remains robust over arbitrarily large separations:

$$C^\mu_\nu(x, x') \sim \delta^\mu_\nu, \quad \text{for } |x - x'| \gg L_{\text{local}}, \quad (\text{S379})$$

where L_{local} represents the coarse-graining scale over which local mass-energy configurations are resolved. Unlike in the gauge sectors, the absence of a compact internal symmetry or non-Abelian curvature prevents phase scrambling. This makes the correlation tensor effectively non-decaying, resulting in global coherence across macroscopic—and even cosmological—scales.

Coherence Length and Interaction Range Because no internal structure disrupts the propagation of $\Xi^\mu(\tau)$, the gravitational coherence length is not merely large—it diverges:

$$\ell_c^{\text{gravity}} \rightarrow \infty. \quad (\text{S380})$$

This implies that the external trembling deviations remain phase-coherent across arbitrarily large distances, enabling curvature to accumulate cumulatively over long spacetime paths. Consequently, gravity manifests as a truly long-range interaction, consistent with observational data. This conclusion is not an empirical insertion but follows from three geometric facts, i.e., the trembling field $\Xi^\mu(\tau)$ is vectorial and external, there is no compact internal fiber to induce torsion or phase dispersion, and the proper-time evolution respects global Lorentz covariance and causal propagation.

Emergence of the Gravitational Field The gravitational interaction emerges as the curvature of the coarse-grained effective metric $g_{\mu\nu}^{\text{eff}}$, constructed from coherent superpositions of

$\Xi^\mu(\tau)$. The relevant curvature structures include the Riemann tensor:

$$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\lambda_{\nu\sigma} \Gamma^\rho_{\mu\lambda} - \Gamma^\lambda_{\mu\sigma} \Gamma^\rho_{\nu\lambda}, \quad (\text{S381})$$

and the associated Einstein tensor:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu}^{\text{eff}} R, \quad (\text{S382})$$

where $\Gamma^\lambda_{\mu\nu}$ are the Levi-Civita connection symbols derived from the coarse-grained metric. These geometric quantities are not postulated but arise naturally from the persistent, coherent oscillations of external trembling fields. The causal structure of spacetime, the evolution of matter, and the formation of curvature all follow from the deterministic, Lorentz-invariant trembling of spacetime itself. This provides a geometric origin for gravitation without the need for massless gravitons or imposed curvature axioms.

Contrast with Gauge Interactions This infinite range contrasts sharply with the short correlation lengths associated with non-abelian internal group curvature in $SU(3)_C$, chiral torsion in $SU(2)_L$, and even abelian phase decoherence in thermally screened plasmas for $U(1)$. In TSRT, the reason gravity is long-range is structural: there is no internal scrambling mechanism to destroy the phase alignment of spacetime-directed trembling. This explanation circumvents the need for field quantization or graviton exchange.

Summary The gravitational interaction in TSRT emerges from coherent, large-scale external trembling deviations. These deviations maintain their phase coherence indefinitely because they lie entirely in spacetime, they are not influenced by compact internal symmetry groups, and they lack internal curvature and torsion. This leads naturally to an infinite-range interaction mediated by a deterministic, coarse-grained metric field. Thus, gravity in TSRT is not only emergent but also structurally long-range—a direct consequence of the unscreened and untwisted nature of spacetime deviations.

S15.6 Comparative Scaling and Interpretation

While the precise numerical values of the coherence lengths ℓ_c for each interaction may ultimately be calibrated against experimental measurements, the Trembling Spacetime Relativity Theory (TSRT) provides a rigorous, geometric rationale for their relative ordering. In TSRT, these coherence lengths are not inserted arbitrarily or assumed from phenomenology; rather, they emerge from the structural properties of the deviation fields $\Xi(\tau)$ and the underlying internal symmetries that govern their evolution. The ordering of interaction ranges is thus a derived consequence of the following geometric principles, leading to a scaling:

$$\ell_c^{\text{strong}} \ll \ell_c^{\text{weak}} \ll \ell_c^{\text{EM}} \approx \ell_c^{\text{grav}}. \quad (\text{S383})$$

Each step in this hierarchy reflects the intrinsic coherence behavior of the corresponding trembling class:

Electromagnetism ($U(1)$ Abelian):

Phase-type deviations $\Xi^\mu(\tau) = \epsilon(\tau)u^\mu(\tau)$ belong to a commutative, globally integrable group with trivial curvature. No internal torsion or topological obstruction disrupts coherence, allowing phase alignment to persist over macroscopic—and even cosmological—scales. The correlation tensor $C^\mu_\nu(x, x')$ decays slowly or not at all. Hence, $\ell_c^{\text{EM}} \rightarrow \infty$ in the absence of medium effects.

Weak Interaction ($SU(2)_L$ Chiral Non-Abelian):

Chiral trembling fields $\Xi^A(\tau) \in E_L$ evolve within a non-abelian, fiber-curved structure. The presence of non-commuting generators $T^a \in \mathfrak{su}(2)_L$, internal torsion, and left-handed asymmetry introduces rapid local phase rotation and destructive interference. These effects reduce the coherence length ℓ_c^{weak} , as expressed in the exponential decay of correlation functions. While the exact value is not imposed, TSRT predicts that decoherence occurs well within subnuclear scales, consistent with the known short range of the weak force.

Strong Interaction ($SU(3)_C$ Non-Chiral Non-Abelian):

Color-trembling fields $\Xi_\mu^a(\tau)$ in the 8-dimensional internal fiber space are subjected to intense phase scrambling due to the non-integrability of the color bundle and the non-abelian nature of $SU(3)_C$. The structure constants f^{abc} induce rapid torsion and multi-dimensional interference effects. The correlation tensor decays on femtometer scales or shorter, yielding $\ell_c^{\text{strong}} \ll \ell_c^{\text{weak}}$. Confinement of color charge arises geometrically as a loss of coherent signal over distances larger than this intrinsic cutoff.

Gravity (External Vectorial Deviation):

Gravitational effects emerge from external trembling $\Xi^\mu(\tau)$, which are globally defined in spacetime and are not coupled to any internal, rotating symmetry group. Consequently, there are no phase decoherence mechanisms or internal torsional disruptions. The associated correlation tensor remains approximately constant across large spacetime separations. Hence, $\ell_c^{\text{grav}} \rightarrow \infty$, structurally mirroring the behavior of the electromagnetic sector, but arising from an entirely distinct (non-fibered) geometric mechanism.

Unified Interpretation. The hierarchy of coherence lengths in TSRT therefore reflects the commutativity or non-commutativity of the underlying symmetry group (e.g., $U(1)$ vs. $SU(3)_C$), the dimensionality and curvature of the internal fiber space, the presence or absence of torsion, chirality, or compactness in the deviation dynamics, and the distinction between internal vs. external trembling modes. This results in a theoretical framework where the range of a fundamental interaction is not imposed as a field-theoretic parameter (such as mass or coupling suppression), but emerges as a geometric measure of phase coherence length:

$$\ell_c \sim (\text{determinant of phase-integrability over deviation group manifold})^{-1/2}. \quad (\text{S384})$$

In this sense, TSRT offers a genuinely first-principles derivation of the known qualitative hierarchy of the four fundamental forces:

Color Confinement $\ll$ Weak Decay $\ll$ Electromagnetic Propagation $\sim$ Gravitational Curvature.

This ordering is robust and requires no quantization, no postulated mediators, and no imposed Lagrangians. It flows entirely from the symmetry class and geometric coherence of trembling deviations.

S15.7 No Need for Massive Bosons

As established in Sections S10–S13, TSRT provides a fully geometric account of interaction range without invoking massive mediators, Higgs couplings, or spontaneous symmetry breaking. In this framework, no particle acquires range-limiting mass from field-theoretic mechanisms. Instead, the spatial extent of each interaction arises from the internal coherence and causal propagation properties of its corresponding trembling mode. More specifically, each trembling sector induces a proper-time deviation field $\Xi(\tau)$, whose spatial correlation function $C(x, x')$ determines the interaction's effective range. The correlation length ℓ_c is not inserted phenomenologically, but derived from the internal symmetry structure, coherence decay rate, and causal integrability of the deviation field.

Electromagnetic and gravitational fields emerge from abelian or external deviations with trivial internal torsion, leading to $\ell_c \rightarrow \infty$. No photon or graviton mass is required to explain long-range propagation.

Weak interactions arise from $SU(2)_L$ chiral trembling, which induces rapid phase decoherence and fiber torsion. The resulting finite-range interaction emerges naturally from the curvature and rotation properties of the internal trembling manifold (see Section S11).

Strong interactions stem from color-trembling in the $SU(3)_C$ sector. Here, compact fiber curvature, high-dimensional non-abelian structure, and internal interference all contribute to a coherence cutoff below the femtometer scale—without requiring gluon confinement postulates or quantum chromodynamics (see Section S12).

In all cases, the force range is thus controlled by the coherence and symmetry properties of trembling spacetime itself. This offers a deterministic, Lorentz-invariant alternative to quantum field theory's reliance on mass terms and symmetry breaking. Interaction range in TSRT is not a free parameter but a derived feature of the internal geometric dynamics of proper-time fluctuations.

S15.8 Summary and Conclusion: Force Range as a Geometric Manifestation

In Trembling Spacetime Relativity Theory (TSRT), the spatial range of each fundamental interaction is a direct consequence of the causal, spectral, and symmetry properties of its underlying trembling deviation field. This coherence is quantified through the correlation tensor $C(x, x') = \langle \Xi(x)\Xi(x') \rangle$, which controls the effective propagation scale ℓ_c .

Electromagnetism arises from globally coherent abelian deviations aligned with particle four-velocity. These produce long-range correlations with $\ell_c^{\text{EM}} \rightarrow \infty$, without requiring massless photon assumptions (see Section S10).

Weak interactions result from torsion-rich, chiral, non-abelian trembling in $SU(2)_L$, where decoherence and local curvature break long-range alignment. TSRT predicts $\ell_c^{\text{weak}} \ll \ell_c^{\text{EM}}$, consistent with short-range forces (see Section S11).

Strong interactions emerge from $SU(3)_C$ color trembling with maximal non-abelian interference. The resulting $\ell_c^{\text{strong}} \ll \ell_c^{\text{weak}}$ accounts for confinement and nuclear scale suppression (see Section S12).

Gravity originates from large-scale external deviations coarse-grained into the background metric. These fluctuations retain global coherence, yielding $\ell_c^{\text{grav}} \rightarrow \infty$, and reproducing universal long-range curvature effects without postulating graviton exchange (see Section S13).

These results show that TSRT does not rely on field-theoretic assumptions like spontaneous symmetry breaking, quantization, or mediator masses to explain the observed force ranges. Instead, it derives these ranges from first-principles geometric analysis of trembling coherence in proper time. The symmetry group, curvature, and causal integrability of each deviation field fully determine the resulting interaction scale. This geometric understanding of interaction range not only reproduces the observed hierarchy among the four known forces, but elevates the concept of force range from a phenomenological input to a structural prediction. In TSRT, coherence governs reach: long-range behavior follows from integrable, untwisted trembling; short-range behavior reflects curvature-driven decoherence.

In the next section, we show that only four distinct fundamental interactions arise as irreducible symmetry classes of trembling spacetime.

S16 Supplementary Material: Uniqueness of the Four Fundamental Interactions in TSRT

S16.1 Introduction: Why Uniqueness Matters

In TSRT, the four known fundamental interactions emerge as the only geometrically and causally consistent classes of trembling modes permitted by the structure of spacetime. Their uniqueness follows from strict constraints on metric deviations: causality, Lorentz covariance, compact symmetry closure, and proper-time coherence. No other symmetry classes or interaction types can arise without violating these fundamental requirements. Throughout the history of physics, a central aspiration has been not merely to describe the laws of nature, but to uncover the underlying principles that render these laws inevitable. The deepest theoretical frameworks are those that explain why the structures of the physical world could not have been otherwise.

In the Standard Model of particle physics, the gauge groups $U(1) \times SU(2) \times SU(3)$ governing the electromagnetic, weak, and strong interactions are selected based on phenomenological success. Although mathematically elegant, these groups are externally imposed rather than internally derived. No intrinsic principle within the Standard Model forbids the existence of other gauge symmetries or additional forces, leaving the model with an acknowledged degree of arbitrariness.

Trembling Spacetime Relativity Theory (TSRT) seeks to overcome this limitation. It does not merely aim to unify the known forces but to demonstrate that no additional fundamental forces are possible once the principles of causal consistency, geometric coherence, and trembling structure are imposed. The theory demands closure: the permitted types of interactions must be uniquely determined by the underlying geometry and causality of spacetime itself. Proving uniqueness is thus a crucial milestone. It transforms TSRT from a framework that accommodates existing forces into one that necessitates them — and forbids the appearance of others. In doing so, it distinguishes itself sharply from quantum field theory, string theory, and related paradigms where additional gauge groups, new forces, or speculative symmetries can be introduced without violating basic structural assumptions.

In this section, we complete the proof initiated in Section S9. Building upon the classification of trembling-induced metric fluctuations into four distinct, causally permissible classes, we demonstrate that these four classes exhaust all geometrically and physically admissible structures. No fifth fundamental force, no hidden sector, and no exotic symmetry remains consistent with the trembling spacetime framework. This result establishes TSRT as a theory that achieves not only unification but inevitability — a profound step toward a truly fundamental understanding of the physical universe.

S16.2 Summary of the Classification Results

The classification developed in Section S9 revealed that, under the foundational constraints of Trembling Spacetime Relativity Theory (TSRT)—namely strict causality through the condition $d\tau/d\lambda > 0$, Lorentz covariance, and geometric coherence of trembling structures—only four distinct classes of permissible metric fluctuations exist.

First, trace-like isotropic trembling, corresponding to scalar perturbations of the metric, emerges naturally and is associated with gravitational interaction. This trembling mode, developed further in Section S13, reflects uniform rescalings of spacetime intervals and leads, through coarse-graining, to effective large-scale curvature.

Second, abelian vector-like trembling, involving directional but commutative deformations respecting an internal $U(1)$ symmetry, gives rise to electromagnetism. As derived in Section S10, this trembling structure reproduces the gauge freedoms and field dynamics of Maxwellian electrodynamics.

Third, chiral non-abelian trembling, characterized by left–right asymmetric geometric distortions consistent with an $SU(2)$ internal structure, underpins the weak nuclear force. The emergence of chirality, mass generation, and parity violation are all natural consequences of this trembling class, elaborated in Section S11.

Fourth, non-chiral non-abelian color trembling, representing fully isotropic internal fluctuations aligned with $SU(3)_C$ symmetry, accounts for the strong nuclear force. This class, explored in Section S12, captures the essential features of color confinement and asymptotic freedom observed in quantum chromodynamics.

Each trembling class satisfies all necessary physical conditions: causal positivity, Lorentz-invariant consistency, and coherent phase-space structure. Importantly, the classification is exhaustive: no other independent trembling class exists. Hypothetical alternatives—such as antisymmetric perturbations, higher-spin structures, or mixed-parity fluctuations—either violate causal monotonicity, destabilize the metric, or fail to sustain coherent phase-space evolution. Thus, TSRT achieves a one-to-one mapping between trembling-induced geometric structures and the four known fundamental forces, establishing a uniquely determined foundation for the physical interactions observed in nature.

S16.3 Why Additional Fundamental Forces Cannot Exist

Having classified all permissible trembling structures in Section S9, we now rigorously establish why no additional fundamental forces can arise within Trembling Spacetime Relativity Theory (TSRT).

Exhaustion of Trembling Classes. The exhaustive decomposition of the trembling-induced metric deviations $\xi_{\mu\nu}(\tau)$ into their irreducible components—specifically into trace-like scalar fluctuations, abelian vector-like deformations, and non-abelian tensorial structures—leaves no available degrees of freedom for further independent trembling modes that are consistent with Lorentz invariance and the strict causality constraint $d\tau/d\lambda > 0$. Any hypothetical attempt to introduce additional trembling structures would inevitably violate these core physical requirements. Antisymmetric perturbations of the metric, which would involve $\xi_{\mu\nu} = -\xi_{\nu\mu}$, are immediately ruled out, as they contradict the fundamental symmetry of the metric tensor and destabilize the proper-time structure. Higher-spin trembling modes, involving contractions beyond bilinear combinations of the four-velocity, necessarily break Lorentz symmetry or induce negative contributions to the proper-time increment, violating causal monotonicity. Furthermore, exotic group structures that extend beyond the $U(1) \times SU(2) \times SU(3)$ framework would

either demand additional spacetime dimensions—contrary to TSRT’s strict four-dimensional construction—or lead to unphysical behavior such as negative or imaginary proper-time excursions.

Thus, from a geometric, causal, and group-theoretic perspective, the classification of trembling structures within TSRT is complete and exclusive: no further fundamental forces can arise without sacrificing internal consistency, causal structure, or geometric admissibility.

Group-Theoretic Closure. The gauge structures that naturally emerge from the trembling classes—namely $U(1)$, $SU(2)$, and $SU(3)$ —exhaust the set of compact, simply connected Lie groups that can be associated with internal symmetries under local Lorentz-invariant trembling, without invoking higher-dimensional extensions or compromising causal coherence. Each group plays a distinct and necessary role: $U(1)$ arises as the unique non-trivial Abelian gauge group consistent with a single vector degree of freedom, $SU(2)$ appears as the minimal non-Abelian compact group capable of supporting chirality, a feature essential for the structure of the weak interaction, and $SU(3)$ naturally accommodates the non-chiral internal symmetries associated with the strong force.

This complete correspondence between trembling classes and minimal symmetry groups ensures that no additional gauge structures are produced within TSRT. Larger or more exotic symmetry groups could only arise through artificial extensions of the trembling framework, such as postulating higher spacetime dimensions or introducing extraneous internal degrees of freedom—approaches that TSRT explicitly forbids on the grounds of internal consistency, causal stability, and geometric necessity.

Causal Consistency and Proper-Time Monotonicity. The strict causality constraint

$$\frac{d\tau}{d\lambda} > 0 \tag{S385}$$

serves as a decisive filter that eliminates any trembling patterns whose cumulative fluctuations could cause local reversal, stagnation, or irregular progression of proper time. This requirement dynamically imposes strong limitations on the permissible amplitude and spectral structure of trembling fields, ensuring that fluctuations remain bounded and causally stable at all curvature scales. Furthermore, it restricts the correlation patterns among different fluctuation modes, forbidding configurations that would destabilize the causal structure.

Consequently, the emergence of any additional fundamental force would necessitate a new, independent trembling class. Such a class, however, would inevitably violate one or more foundational requirements—either disrupting group-theoretic closure, breaking Lorentz invariance, or undermining the proper-time monotonicity central to TSRT. Thus, the causal constraint acts as both a safeguard and a constructive principle, reinforcing the uniqueness of the four fundamental interactions.

Conclusion. The four known fundamental interactions—gravity, electromagnetism, the weak force, and the strong force—comprehensively exhaust all causally and geometrically permissible trembling structures within Trembling Spacetime Relativity Theory (TSRT). No fifth funda-

mental force, no exotic gauge extension, and no speculative interaction sector can arise without violating the foundational principles of Lorentz invariance, causal stability, or the proper-time positivity condition.

Thus, the uniqueness of the four forces emerges not as an external assumption, but as a rigorous and unavoidable consequence of the internal logic of TSRT, firmly rooted in the causal, geometric, and symmetry structure of trembling spacetime itself.

S16.4 Internal Consistency and Mutual Compatibility

Having established that exactly four fundamental trembling classes are permitted within Trembling Spacetime Relativity Theory (TSRT), it is essential to demonstrate not only their individual admissibility but also their mutual compatibility. Specifically, the trembling modes corresponding to gravitational, electromagnetic, weak, and strong interactions must be able to coexist within the same spacetime fabric without inducing internal contradictions, violations of causal structure, or conflicts with the proper-time evolution constraint.

This mutual compatibility is a critical requirement for the completeness and internal coherence of TSRT as a unified physical theory.

Orthogonality of Trembling Structures. The decomposition of metric fluctuations $\xi_{\mu\nu}(\tau)$ into irreducible components ensures that the trembling modes associated with different forces are geometrically orthogonal. Each class corresponds to a distinct transformation property under Lorentz symmetry and affects independent aspects of the spacetime fabric: The trace-like component describes isotropic scaling fluctuations that modulate the overall spacetime volume without introducing anisotropy, underpinning gravitational phenomena. The Abelian vector-like component, linked to $U(1)$ gauge freedom, generates phase shifts along worldlines while preserving spacetime isotropy, corresponding to electromagnetic interactions. The chiral non-Abelian trembling modes transform according to the $SU(2)$ group and introduce handedness-sensitive distortions essential for capturing weak interaction behavior. Finally, the non-chiral non-Abelian trembling modes, structured by $SU(3)$ color symmetry, encode internal transformations responsible for strong nuclear interactions. Because these trembling components originate from distinct geometric sectors, their superposition does not generate cross-terms that could compromise Lorentz invariance or the strict positivity of proper time evolution. Each trembling mode operates independently at the fundamental level, ensuring that their combined effects on spacetime remain mutually compatible and causally coherent.

Causal Structure Preservation. Each fundamental trembling class, considered individually, satisfies the strict causal constraint

$$\frac{d\tau}{d\lambda} > 0, \tag{S386}$$

ensuring forward progression of proper time along physical worldlines. Crucially, the orthogonality of the trembling structures implies that their combined effects remain causally coherent. The causal geometry imposes bounded amplitude conditions on each trembling mode, preventing any local reversal or stagnation of proper time evolution.

Specifically, trace-like gravitational trembling uniformly rescales proper-time intervals across all directions without introducing anisotropic distortions, thereby preserving monotonicity. Abelian vector-like trembling associated with electromagnetic effects manifests as transverse phase modulations along worldlines, which respect the forward causal structure without altering spacetime separations in a way that could violate positivity. Chiral and non-chiral non-Abelian trembling modes, responsible for weak and strong interactions respectively, exhibit internal group symmetries that constrain fluctuations to maintain timelike evolution when averaged over sufficiently small spacetime neighborhoods.

Thus, the mutual compatibility of all trembling classes is ensured both geometrically and causally. Their interplay does not threaten the foundational requirement of forward proper-time evolution, but rather reinforces the coherence of spacetime dynamics within the framework of TSRT.

Gauge Structure Consistency. The gauge groups $U(1)$, $SU(2)$, and $SU(3)$ that emerge from the distinct trembling sectors are mutually consistent and geometrically natural. The Abelian group $U(1)$ commutes trivially with both $SU(2)$ and $SU(3)$, ensuring that electromagnetic fluctuations coexist without interfering with the non-Abelian structures associated with the weak and strong forces. Meanwhile, $SU(2)$ and $SU(3)$ act on entirely different internal degrees of freedom: weak isospin and color charge, respectively. This separation guarantees that the dynamics of weak and strong interactions do not conflict at the fundamental trembling level. Importantly, the coexistence of these gauge structures does not necessitate any artificial cross-coupling or imposed symmetry mixing. Electroweak unification, for example, arises dynamically through spontaneous symmetry breaking mechanisms associated with the chiral trembling sector, rather than being introduced externally or enforced through overlapping trembling patterns. The product structure $U(1) \times SU(2) \times SU(3)$, familiar from the Standard Model, thus appears here as a natural consequence of the causal geometry of trembling spacetime, not as a phenomenologically motivated postulate.

This seamless emergence of compatible gauge structures underscores the internal coherence and elegance of TSRT's framework.

No Interference in Fundamental Spectra. A further aspect of internal consistency in TSRT is the distinctness of the physical spectra associated with each trembling class. Because each trembling structure modulates different geometric features of the spacetime metric and its phase structure, their emergent excitations correspond to fundamentally separate particle modes. The trace-like trembling gives rise to gravitons, encoding perturbations in isotropic spacetime volume. The Abelian vector-like trembling generates photons, corresponding to phase-preserving oscillations tied to electromagnetic interactions. The chiral non-Abelian trembling leads to weak bosons, mediating parity-violating processes, while the non-chiral non-Abelian trembling produces gluons, responsible for color charge interactions.

Crucially, these emergent spectra remain orthogonal and non-interfering even when superposed, ensuring that each force retains its distinct identity at the level of fundamental excitations. No unintended mixing or coupling arises between the spectra, preserving the clear separability of gravitational, electromagnetic, weak, and strong interactions within the trem-

bling spacetime framework. This spectral orthogonality is a direct reflection of the causal and geometric foundations of TSRT and reinforces its predictive consistency.

Conclusion. The four trembling classes identified within TSRT not only exhaust the permissible geometric structures consistent with causal and Lorentzian constraints, but also coexist without internal contradiction or destabilization of the spacetime fabric. Their orthogonality, causal stability, and gauge coherence ensure that gravity, electromagnetism, the weak force, and the strong force emerge as distinct yet mutually compatible aspects of trembling geometry. Thus, TSRT provides a coherent, unified, and causally complete geometric foundation for all four fundamental interactions, validating both the internal logical consistency and the physical viability of the theory.

S16.5 The End of the Standard Model's Open-Ended Symmetries

A persistent conceptual difficulty within the Standard Model of particle physics lies in the arbitrariness of its gauge group structure. Although the combination $U(1) \times SU(2) \times SU(3)$ accurately describes electromagnetic, weak, and strong interactions, it is not derived from a deeper necessity. Instead, these symmetries are postulated to fit empirical observations, without an internal mechanism that explains why precisely these groups—and no others—govern fundamental interactions.

Furthermore, the Standard Model permits speculative extensions without intrinsic constraint. Additional $U(1)$ sectors, enlarged $SU(N)$ groups, grand unification models such as $SU(5)$ or $SO(10)$, and higher-dimensional symmetries proposed in string theories all exemplify the open-ended flexibility of the framework. While this flexibility enables diverse model-building efforts, it also highlights a fundamental conceptual incompleteness: the Standard Model offers no principle that forbids the proliferation of gauge groups or the emergence of new fundamental forces.

How TSRT Closes the Gauge Structure. In contrast to the Standard Model, TSRT offers a geometric and causal foundation that inherently limits the number and type of allowable fundamental forces. The classification of trembling structures ensures that only four distinct types of geometric deviations are admissible, each corresponding to a distinct emergent interaction.

The simplest trembling mode corresponds to large-scale, trace-like metric deviations, giving rise to emergent curvature and gravitational phenomena. Trace-like trembling induces scalar curvature distortions while preserving causal coherence, resulting in macroscopic spacetime curvature without introducing additional internal degrees of freedom.

The second admissible mode consists of coherent phase-like oscillations, leading to the emergence of an abelian $U(1)$ gauge symmetry. Abelian vector-like trembling generates electromagnetic interactions, associated naturally with $U(1)$ gauge invariance, where phase redundancy arises from the microscopic structure of proper-time fluctuations.

The third mode involves internal chiral rotations, constrained by the Lorentzian spinor structure, and leads naturally to an $SU(2)_L$ symmetry. Chiral non-Abelian trembling produces

the weak force, governed by an emergent $SU(2)_L$ gauge structure, and reflects the spontaneous geometric selection of left-handed coupling.

Finally, compact internal oscillations along eight color directions generate an $SU(3)_C$ symmetry associated with the strong interaction. Non-chiral, non-Abelian trembling modes lead to strong nuclear interactions, characterized by a coarse-grained emergent $SU(3)_C$ color gauge field, with confinement behavior naturally arising from causal coherence constraints. No additional independent trembling class satisfies the combined demands of Lorentz invariance, strict forward evolution of proper time ($d\tau/d\lambda > 0$), compact symmetry closure, and geometric decomposability under local spacetime structure. As a result, TSRT naturally and necessarily closes the gauge group structure, achieving a completeness that the Standard Model merely postulates without deeper justification.

Absence of Arbitrary Postulates. A defining feature of TSRT is the elimination of arbitrary assumptions. Within this framework, no gauge group is introduced independently of the underlying trembling geometry, no internal symmetries are imposed without direct geometric necessity, and no speculative unification—such as enlarged symmetry groups or hidden sectors—is permitted beyond what the structure of trembling spacetime naturally enforces. The complete set of fundamental interactions arises inevitably from the irreducible decomposition of permissible metric fluctuations. In TSRT, the architecture of nature is not shaped by external postulates or model-building freedom; it is an intrinsic consequence of causality and the trembling structure of spacetime itself.

Predictive and Structural Economy. Because the four fundamental forces emerge as inevitable consequences of trembling spacetime geometry, TSRT attains a level of predictive power far surpassing that of the Standard Model. Specifically, it demonstrates why the observed fundamental forces must exist, rather than being postulated; it establishes why there are exactly four—and no more—fundamental interactions; and it accounts for why the associated gauge symmetries possess precisely their observed structures. This intrinsic predictive power exemplifies the principle of structural economy: the explanation of a maximal range of phenomena from a minimal set of underlying assumptions. TSRT thereby fulfills one of the highest standards expected of a fundamental physical theory.

Conclusion. The Standard Model remains one of the most successful phenomenological frameworks in physics; however, it leaves the origin and necessity of its gauge symmetries fundamentally unexplained. In contrast, TSRT completes the conceptual structure by demonstrating that gauge symmetries are not arbitrary constructs, but are inevitable outcomes of the causal trembling of spacetime geometry itself. This natural closure, rooted in geometric and causal necessity rather than empirical postulation, marks a profound philosophical and scientific advancement over all previous approaches to fundamental physics.

S16.6 The Fundamental Simplicity of TSRT

One of the most striking features of Trembling Spacetime Relativity Theory (TSRT) is its fundamental simplicity. While earlier frameworks—ranging from the Standard Model to string theory—required increasingly elaborate assumptions, extensions, and free parameters to account for physical phenomena, TSRT achieves natural closure with minimal initial input.

Minimal Foundations. TSRT rests on two core principles. The first is the deterministic trembling of spacetime: the metric exhibits ever-present, fine-grained fluctuations that encode the unresolved structure of spacetime in a strictly deterministic, non-probabilistic manner. The second principle is the strict forward evolution of proper time: the requirement that $d\tau/d\lambda > 0$ must hold universally, ensuring both local and global causality and forbidding any backward or stalled progression of proper time. From these two foundational premises—without invoking additional assumptions—the full range of observed physical phenomena emerges naturally. The existence and form of uncertainty principles, the emergence of gauge fields, the unification of the four fundamental forces, the derivation of black hole thermodynamics and holography, and the underlying causal structure of relativistic spacetime all arise as necessary consequences of this trembling geometric framework.

Natural Unification without Forced Structures. Unlike grand unified theories (GUTs), string theories, or loop quantum gravity approaches, TSRT does not attempt unification by artificially enlarging the symmetry group or postulating hidden dimensions. Instead, the four fundamental forces arise naturally from the decomposition of the permitted trembling structures of spacetime itself. Each trembling class corresponds directly to a specific interaction, carrying well-defined causal and geometric properties that are intrinsic to the structure of spacetime. No superfluous constructs, such as extra spatial dimensions, hidden sectors, or supersymmetry, are required to explain the observed phenomena. The unification in TSRT is thus not imposed by external model-building choices but is a natural and inevitable outcome of the geometric and causal structure of trembling spacetime.

A Theory Without Free Floating Parameters. In TSRT, fundamental quantities and structures are not postulated arbitrarily but emerge naturally from the underlying geometric and causal principles. Planck’s constant $\hbar$ is not introduced as an external input; it arises from the causal cutoff structure inherent to trembling spacetime. Gauge invariance is not assumed a priori but follows from the covariance properties and local symmetry requirements of the trembling field. The number and type of fundamental forces are not guessed or imposed; they are uniquely dictated by the causal evolution of proper time and the irreducible decomposition of the spacetime metric. This elimination of free-floating parameters marks a major advance in theoretical economy and coherence.

Philosophical Significance. Throughout the history of physics, major theoretical advances have been characterized by greater simplicity, coherence, and unification. From Newton’s laws to Einstein’s general relativity, progress was achieved by replacing complex, ad hoc frameworks

with simpler, deeper principles. TSRT continues this tradition by replacing the multitude of assumptions and independent postulates found in quantum field theory and particle physics with a single coherent geometric foundation. It restores causality and determinism without diminishing the richness of quantum phenomena, and it shows that uncertainty, forces, and curvature are not distinct mysteries to be patched together, but rather inevitable manifestations of a trembling spacetime geometry.

Conclusion. TSRT achieves in a single unified structure what decades of quantum gravity research, string theory, and beyond-the-Standard-Model physics have sought with increasingly elaborate constructs: a causal, geometric, deterministic, and comprehensive explanation of nature. Its simplicity is not a limitation; it is a profound triumph of conceptual clarity and physical understanding. With this conceptual closure achieved, the path is now clear to derive each of the four fundamental forces explicitly from their associated trembling structures in the sections that follow.

S16.7 Summary and Implications

The systematic classification of metric fluctuations within Trembling Spacetime Relativity Theory (TSRT) has revealed a profound and far-reaching result: only four fundamental trembling classes are consistent with the causal structure of spacetime, and each class maps naturally and uniquely onto one of the four known fundamental forces.

Summary of Core Results. Trace-like isotropic trembling leads naturally to emergent gravitational interactions, as developed explicitly in Section S13. Abelian vector-like trembling gives rise to electromagnetism through its intrinsic $U(1)$ gauge structure, as derived in Section S10. Chiral non-Abelian trembling generates the weak nuclear force by producing handedness-dependent $SU(2)_L$ gauge fields, as demonstrated in Section S11. Non-chiral non-Abelian color trembling produces the strong nuclear force through emergent $SU(3)_C$ color symmetries, as explored in Section S12. The causal requirement $d\tau/d\lambda > 0$ plays a central role throughout: it not only bounds the trembling amplitudes but also restricts the permissible geometric structures, ensuring that no further independent forces can arise without violating either causality or proper-time monotonicity.

Implications for Fundamental Physics. The results achieved here elevate TSRT far beyond a mere descriptive framework. TSRT achieves true unification: the four fundamental forces are not imposed by hand but emerge as inevitable consequences of the causal trembling of spacetime. There is no arbitrary choice of symmetry groups, no ad hoc gauge structures inserted by external fiat; instead, the allowed geometries dictate the very existence and form of the interactions. Moreover, TSRT naturally predicts that no new independent fundamental forces exist beyond the four known today, eliminating the conceptual arbitrariness that plagues extensions of the Standard Model.

Next, we summarize the axiomatic structure that underlies the classification of interactions in TSRT.

S17 Supplementary Material: Axiomatic Summary of Interaction Classification in TSRT

This appendix summarizes, in axiomatic and logically sequential form, how the four known fundamental forces emerge uniquely from the geometric structure of trembling spacetime in TSRT. It is intended as a concise pedagogical reference for readers seeking a step-by-step overview of the argument that spans the main text and supplementary materials.

S17.1 Axioms: Geometric Foundations of TSRT

Axiom 1: Trembling Spacetime Geometry. Spacetime coordinates deviate microscopically along proper-time trajectories due to deterministic geometric fluctuations:

$$x^\mu(\tau) = \bar{x}^\mu(\tau) + \Xi^\mu(\tau), \quad \xi_{\mu\nu} := \partial_\mu \Xi_\nu + \partial_\nu \Xi_\mu. \quad (\text{S387})$$

These deviations induce metric fluctuations around a smooth background $\bar{g}_{\mu\nu}$, giving rise to an effective geometry governed by:

$$g_{\mu\nu}(x, \tau) = \bar{g}_{\mu\nu}(x) + \xi_{\mu\nu}(x, \tau). \quad (\text{S388})$$

Axiom 2: Physical Consistency Constraints. Trembling deviations $\Xi^\mu(\tau)$ are only admissible if they satisfy Lorentz covariance (with $\xi_{\mu\nu}$ transforming as a symmetric tensor), preserve causal structure by ensuring positive proper time ($d\tau^2 = g_{\mu\nu}dx^\mu dx^\nu > 0$), evolve deterministically with bounded norm over τ , and admit only compact symmetry groups supporting finite-dimensional unitary representations.

Axiom 3: Geometric Embedding Constraints. All trembling symmetry structures must be realizable purely within the four-dimensional Lorentzian spacetime manifold, without invoking additional spatial dimensions or unphysical internal fibers.

S17.2 Logical Derivation: Irreducible Force Classes from Trembling

Step 1: Deviation-Induced Geometry. Trembling deviations modify spacetime trajectories via $x^\mu(\tau) = \bar{x}^\mu(\tau) + \Xi^\mu(\tau)$, inducing a metric fluctuation $\xi_{\mu\nu} = \partial_\mu \Xi_\nu + \partial_\nu \Xi_\mu$.

Step 2: Imposing Physical Constraints. Lorentz covariance ensures relativistic consistency. Proper-time positivity enforces causal propagation at every point. Deterministic norm evolution restricts the allowed fluctuation amplitudes. The compactness of the symmetry group guarantees unitary, finite-dimensional realizations, and all structures must be embeddable within the causal geometry of four-dimensional spacetime.

Step 3: Classification of Allowed Trembling Symmetries. Exactly four trembling structures satisfy these constraints. Coarse-grained trace-like trembling, where macroscopic averages

of $\Xi^\mu(\tau)$ modify the spacetime curvature, leads to gravitational interactions governed by diffeomorphism invariance. Abelian vector-like trembling, corresponding to phase-like deviations along the worldline, gives rise to the electromagnetic $U(1)$ gauge symmetry. Chiral non-Abelian trembling structures, selectively coupled to left-handed sectors, produce the $SU(2)_L$ symmetry associated with the weak interaction. Non-chiral non-Abelian trembling, involving internal color degrees of freedom, generates the $SU(3)_C$ symmetry responsible for the strong force.

Step 4: Proof of Exhaustiveness and Uniqueness. Low-rank compact groups such as $U(1)$, $SU(2)$, and $SU(3)$ are fully realized by the identified trembling classes. Larger gauge groups, such as $SU(4)$ or $SO(10)$, would require extensions of spacetime dimensionality or break causal coherence, while non-compact groups fail to yield finite-norm, unitary, causally consistent evolution. Subgroups and reducible representations provide no new independent trembling structures. Therefore, the classification is exhaustive.

S17.3 Conclusion

To conclude the classification of fundamental interactions in TSRT, we now summarize the underlying geometric and causal axioms that uniquely determine the existence and structure of the four admissible trembling sectors.

Axiom 1: Proper-Time Covariance. All trembling fields $\Xi^\mu(\tau)$ must be defined along worldlines parametrized by proper time, preserving invariance under local diffeomorphisms and ensuring causal monotonicity ($d\tau/d\lambda > 0$).

Axiom 2: Lorentz Covariance. The trembling deviations must transform consistently under local Lorentz transformations, maintaining compatibility with the spinor decomposition of fields in four-dimensional Minkowski spacetime.

Axiom 3: Compactness and Closure. Internal trembling degrees of freedom must form compact Lie groups under coarse-grained averaging, ensuring boundedness of excitation amplitudes and mathematical closure of trembling-induced symmetry operations.

Axiom 4: Causal Preservation. Trembling-induced metric deviations must respect causal structure at all scales, preventing backward-in-time propagation or breakdown of local light cones.

Axiom 5: Minimal Geometric Decomposability. The trembling structure must be decomposable into irreducible geometric classes, each preserving the minimal necessary structure to generate independent interaction types without redundancy.

Under these constraints, TSRT admits exactly four fundamental trembling sectors:

- Trace-like metric trembling, leading to macroscopic curvature and gravity.
- Phase-like trembling, leading to $U(1)$ gauge structure and electromagnetism.
- Chiral internal trembling, leading to $SU(2)_L$ gauge structure and the weak interaction.

- Non-chiral compact internal trembling, leading to $SU(3)_C$ gauge structure and the strong interaction.

No other independent trembling class satisfies all five axioms simultaneously. Thus, TSRT predicts not only the existence but also the uniqueness and completeness of the four observed fundamental forces, grounding their emergence in deterministic, causal, and geometric principles. Finally, we return to the microscopic trembling dynamics to show how quantum mechanical behavior, specifically the Schrödinger and Dirac equations, emerges statistically from the underlying causal structure.

S18 Supplementary Material: Deterministic Causal Emergence of Quantum Dynamics: Schrödinger and Dirac Equations from TSRT

This section demonstrates how the standard quantum wave equations—the non-relativistic Schrödinger equation and the relativistic Dirac equation—arise as emergent statistical constructs from the deterministic framework of Trembling Spacetime Relativity Theory (TSRT). In contrast to quantum mechanics, which begins with wavefunctions and probabilistic axioms, TSRT treats particles as fundamentally real, persistent entities that traverse trembling trajectories in a geometrically dynamic spacetime. No probabilistic superposition, quantization postulates, or Planck constant are assumed at the outset. Instead, the apparent quantum behavior emerges from coarse-graining the unresolved phases of microscopic deterministic motion. We analyze this emergence in three regimes. First, in the non-relativistic limit, the trembling of particle paths introduces a geometric correction to the classical Hamilton–Jacobi equation, resulting in a modified dynamical law that becomes mathematically equivalent to the Schrödinger equation when expressed in complex form. Second, in the special relativistic regime, internal angular trembling degrees of freedom give rise to an effective spinor field whose evolution satisfies a Dirac-like equation, derived from the Lorentz-covariant geometry of internal motion. Finally, we extend the construction to curved spacetime, showing that the general-relativistic Dirac equation also arises from the same trembling geometric dynamics when internal fluctuations are propagated through a spacetime with nontrivial curvature.

In all three cases, the familiar quantum wave equations are shown to reflect not fundamental indeterminism, but coarse-grained descriptions of deeply deterministic geometric processes. The statistical interpretation of quantum mechanics—traditionally enforced by the Copenhagen principle—is here recovered as a derived consequence of the unresolved fine-scale structure of trembling spacetime.

S18.1 Trembling-Induced Statistical Fields

Let a particle of inertial mass m move in flat Minkowski spacetime, with proper time parameter τ . In the TSRT framework, the trajectory of the particle is not strictly inertial but undergoes a fine-scale trembling due to the structure of spacetime itself. Denote the full trajectory, as introduced in Equation (S389), by:

$$x^\mu(\tau) = x_{\text{class}}^\mu(\tau) + \Xi^\mu(\tau), \quad (\text{S389})$$

where $x_{\text{class}}^\mu(\tau)$ is the classical path in the absence of trembling, and $\Xi^\mu(\tau)$ is a deterministic but irregular function describing the spacetime trembling. The trembling is non-random and intrinsic to spacetime, but may vary in microscopic phase across particles.

We now consider a coarse-grained probabilistic description of a single particle following deterministic trembling trajectories. Although the underlying dynamics are strictly deterministic, incomplete information about the initial trembling phase leads to effective statistical uncertainty. This motivates a probabilistic density function $\rho(\vec{x}, t)$, which encodes the likelihood of

finding the particle at position $\vec{x}$ and time t . Even when describing a single particle, this function reflects uncertainty in its trembling-influenced trajectory due to unresolved initial phase information. Associated with this ensemble is a mean velocity field $\vec{v}(\vec{x}, t)$, representing the associated average local velocity of particles at each point. We introduce a macroscopic action field $S(\vec{x}, t)$, which encapsulates the cumulative effect of the underlying trembling dynamics on the mean motion, and is defined such that the velocity field satisfies the relation $\vec{v} = \nabla S/m$. Because TSRT is deterministic and conservative, the ensemble obeys the standard continuity equation [62, 63]:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0. \quad (\text{S390})$$

We now examine the dynamics of S . If trembling were absent, the mean particle motion would follow classical mechanics, and S would satisfy the standard Hamilton–Jacobi equation for a classical non-relativistic particle in a potential $V(\vec{x})$. In this case, the dynamics of a classical non-relativistic particle in a potential $V(\vec{x})$ can be described by the Hamilton–Jacobi equation (HJE), which is a fundamental reformulation of classical mechanics [64, 65].

Starting from the standard HJE for Hamilton’s principal function $S(\vec{x}, t)$,

$$\frac{\partial S}{\partial t} + H\left(\vec{x}, \frac{\partial S}{\partial \vec{x}}, t\right) = 0, \quad (\text{S391})$$

where $H(\vec{x}, \vec{p}, t)$ is the Hamiltonian of the system. For a particle of mass m moving in a potential $V(\vec{x})$, the Hamiltonian takes the form

$$H(\vec{x}, \vec{p}) = \frac{\vec{p}^2}{2m} + V(\vec{x}). \quad (\text{S392})$$

Substituting the canonical momentum $\vec{p} = \nabla S$ (as derived from the generating function properties of S [64, 66]), we obtain

$$H(\vec{x}, \nabla S) = \frac{(\nabla S)^2}{2m} + V(\vec{x}). \quad (\text{S393})$$

Thus, the HJE of Equation (S391) reduces to the following form for a classical particle in a potential:

$$\frac{\partial S}{\partial t} + \frac{(\nabla S)^2}{2m} + V(\vec{x}) = 0. \quad (\text{S394})$$

Equation (S394) governs the evolution of $S(\vec{x}, t)$, which encodes the system’s dynamics through its relationship with the particle’s momentum and energy [65, 67].

However, the presence of microscopic trembling modifies the kinetic content of the ensemble. Although each individual path remains deterministic, the irregular structure of $\Xi^\mu(\tau)$, defined in Equation (S389), introduces additional fine-scale motion—akin to a sub-path kinetic energy—that contributes to the total energy budget of the ensemble. This contribution manifests as a scalar correction term, denoted Q_{TSRT} . This expression represents the expected energy of a single particle whose position is probabilistically distributed according to $\rho(\vec{x}, t)$. The energy contributions are weighted by the likelihood of the particle’s presence at each point. To derive Q_{TSRT} from first principles rather than by analogy, we treat it as a geometric en-

ergy correction rooted in the fine-scale trembling structure. Our aim is not to reproduce the Schrödinger equation by fiat, but to uncover its mathematical equivalence from the physical trembling geometry.

To identify the form of Q_{TSRT} , we analyze the ensemble-averaged energy:

$$\langle E \rangle = \int d^3x \rho(\vec{x}, t) \left[\frac{1}{2} m \vec{v}^2 + V(\vec{x}) + \varepsilon_{\text{tremble}}(\vec{x}, t) \right], \quad (\text{S395})$$

where $\varepsilon_{\text{tremble}}$ is the local energy contribution from the trembling motion, averaged over the microscopic phase. Since the trembling arises from deterministic geometry, this energy must be a function of local properties of the density field ρ and its derivatives.

To determine the form of the trembling energy contribution $\varepsilon_{\text{tremble}}$, we assume it arises from the local inhomogeneity of the spatial distribution $\rho(\vec{x}, t)$. Since the trembling is geometric and deterministic, it cannot depend on any stochastic postulate, but may depend on local spatial gradients of the density. The function $\varepsilon_{\text{tremble}}$ must vanish in the case of uniform density, implying that it must involve spatial derivatives of ρ . Additionally, rotational symmetry restricts $\varepsilon_{\text{tremble}}$ to depend only on scalar combinations of gradients. Dimensional analysis and symmetry considerations strongly constrain the possible form of the trembling-induced energy correction. Since the trembling arises from fine-scale geometric motion superimposed on the classical trajectory, its energy contribution must appear as an effective scalar term added to the ensemble-averaged energy density.

We begin by recalling that the coarse-grained statistical description of a trembling ensemble leads to a probability density field $\rho(\vec{x}, t)$, representing the likelihood of finding the particle at position $\vec{x}$ at time t , due to unresolved initial phases of trembling motion. Associated with this field is a mean velocity field $\vec{v}(\vec{x}, t)$, derived from an effective macroscopic action $S(\vec{x}, t)$ via the classical relation:

$$\vec{v}(\vec{x}, t) = \frac{\nabla S(\vec{x}, t)}{m}. \quad (\text{S396})$$

Even though each particle follows a deterministic path with a well-defined trembling component $\Xi^\mu(\tau)$, our ignorance of the precise trembling phase leads to the emergence of an ensemble-level mean-field energy density, expressible, as introduced in Equation (S395), as:

$$\langle E \rangle = \int d^3x \rho(\vec{x}, t) \left[\frac{1}{2} m \vec{v}^2(\vec{x}, t) + V(\vec{x}) + \varepsilon_{\text{tremble}}(\vec{x}, t) \right], \quad (\text{S397})$$

where $\varepsilon_{\text{tremble}}$ represents the local geometric energy density correction due to unresolved trembling, and must be a function of ρ and its spatial derivatives.

Thus, to determine the form of $\varepsilon_{\text{tremble}}$, we impose the following constraints:

1. **Scalar nature:** The correction must be a scalar quantity so that it is independent of the coordinate system.
2. **Rotational symmetry:** It should remain invariant under spatial rotations, implying dependence only on scalar combinations of derivatives of ρ .
3. **Vanishing in uniform regions:** If $\rho(\vec{x}, t)$ is spatially constant (i.e., trembling is evenly distributed), there should be no net correction. This rules out dependence on ρ alone and suggests that only gradients of ρ contribute.

4. Dimensional consistency: The correction must have dimensions of energy per unit mass:

$$[\varepsilon_{\text{tremble}}] = \frac{[\text{length}]^2}{[\text{time}]^2}. \quad (\text{S398})$$

The simplest such scalar, constructed from the Laplacian of $\sqrt{\rho}$, captures the spatial curvature of the amplitude and yields:

$$\varepsilon_{\text{tremble}}(\vec{x}, t) = -\frac{\mathcal{T}^2}{2m} \frac{\nabla^2 \sqrt{\rho(\vec{x}, t)}}{\sqrt{\rho(\vec{x}, t)}}, \quad (\text{S399})$$

where $\mathcal{T}$ is the characteristic geometric action scale emerging from the fine-scale structure of proper-time-resolved metric trembling. This same parameter appears in the trembling-induced quantum potential Q_{TSRT} defined in Equation (S400), and governs the phase dynamics via the Hamilton–Jacobi-like equation (S412).

Unlike Planck’s constant $\hbar$, which is introduced axiomatically in conventional quantum theory, the TSRT action scale $\mathcal{T}$ arises deterministically from the causal and Lorentz-invariant correlation structure of the proper-time fluctuations $\Xi^\mu(\tau)$. It quantifies the minimal unit of coherent spacetime deviation and provides the dimensional bridge between internal geometric oscillations and emergent statistical dynamics. This correction is often referred to as a geometric quantum potential, but here it arises from fully deterministic trembling dynamics rather than any postulate of quantization. We define:

$$Q_{\text{TSRT}} := -\frac{\mathcal{T}^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}}. \quad (\text{S400})$$

To see how this term modifies the classical Hamilton–Jacobi equation, we recall that in the absence of trembling, the action field $S(\vec{x}, t)$ evolves according to the standard classical relation, Equation (S394),

$$\frac{\partial S}{\partial t} + \frac{(\nabla S)^2}{2m} + V(\vec{x}) = 0. \quad (\text{S401})$$

However, in the presence of geometric trembling, the total effective energy density gains an additional contribution from Q_{TSRT} , modifying the Hamilton–Jacobi equation to:

$$\frac{\partial S}{\partial t} + \frac{(\nabla S)^2}{2m} + V(\vec{x}) + Q_{\text{TSRT}} = 0. \quad (\text{S402})$$

This corrected Hamilton–Jacobi equation encapsulates the impact of trembling fluctuations on ensemble dynamics. It is no longer purely classical, but now contains a geometric energy correction that depends on the spatial distribution of ρ . Notably, this term is non-zero even for a single particle, due to the coarse-grained projection of unresolved trembling phase, and yet retains a fully deterministic origin.

So, in TSRT, Q_{TSRT} arises from the internal causal structure of spacetime and does not imply probabilistic postulates. It reflects the redistributed kinetic energy induced by geometric inhomogeneities in trembling amplitude, and its appearance is essential for recovering the emergent quantum-like dynamics without invoking any quantum principles. This corrected Equation (S402) is not introduced as an ad hoc modification but follows directly from averag-

ing over the deterministic trembling paths of the ensemble. Each individual particle undergoes fine-scale motion encoded by $\Xi^\mu(\tau)$, whose influence on energy is small but persistent.

When we consider the effective description of a single particle whose trembling phase is unresolved, the coarse-grained energy contribution manifests as a smooth field Q_{TSRT} in Equation (S402), depending on the density profile. Thus, the appearance of this term is a natural consequence of coarse-graining geometric fluctuations at sub-resolution scales—it reflects a redistribution of geometric energy, not a probabilistic interpretation. To unify the continuity Equation (S390) and the corrected Hamilton–Jacobi Equation (S402) into a single evolution equation, we now introduce a complex-valued function $\psi(\vec{x}, t)$ defined as:

$$\psi(\vec{x}, t) := \sqrt{\rho(\vec{x}, t)} e^{iS(\vec{x}, t)/\mathcal{T}}. \quad (\text{S403})$$

This function combines the ensemble probability density $\rho(\vec{x}, t)$ and the action field $S(\vec{x}, t)$ into a single expression. The amplitude $|\psi| = \sqrt{\rho}$ encodes the spatial density of particle presence, while the phase $\arg(\psi) = S/\mathcal{T}$ encodes the local momentum and energy structure through its derivatives. In other words, the fine-scale deterministic trembling of particle worldlines leads, under coarse-graining, to an effective statistical description at macroscopic scales, $\psi(\vec{x}, t)$ denoting a complex-valued field encoding the averaged trembling-induced amplitude for the particle to occupy position $\vec{x}$ at time t . This field emerges as a natural statistical summary of unresolved microfluctuations. Crucially, ψ is not a fundamental wavefunction in the traditional sense; rather, it encodes the coarse-grained structure of causal spacetime trembling, and thus inherits its properties from deterministic underlying dynamics.

S18.2 Non-Relativistic Limit: Toward the Schrödinger Equation

The evolution of ψ is constrained by the causal structure of trembling worldlines. The proper-time monotonicity and local Lorentz covariance imply that the field must evolve through a first-order differential equation in time, while spatial derivatives must enter at second order, preserving rotational invariance and locality. At leading order in the coarse-graining expansion, the evolution equation for ψ takes the form:

$$i\mathcal{T} \frac{\partial \psi}{\partial t} = -\frac{\mathcal{T}^2}{2m} \nabla^2 \psi + V(\vec{x})\psi, \quad (\text{S404})$$

where $\mathcal{T}$ is the characteristic action scale determined by the minimal causal resolution of trembling spacetime, and $V(\vec{x})$ represents an effective external potential. It is developed, step-by-step, in what follows.

Note that, identifying $\mathcal{T}$ with Planck’s constant $\hbar$ up to geometric normalization factors, Equation (S404) recovers the familiar time-dependent Schrödinger equation.

Thus, quantum mechanics arises not from postulated probability amplitudes, but from deterministic coarse-grained causal dynamics of trembling spacetime. The complex structure of the emergent field ψ arises naturally from the asymmetry between forward and backward local fluctuations along proper-time increments. Because proper time flows monotonically forward, the coarse-grained evolution of trembling exhibits an intrinsic causal directionality, favoring a complex first-order time evolution equation rather than a second-order one. This geometric

asymmetry selects a Schrödinger-type structure rather than a purely classical wave equation.

We now derive the evolution Equation (S404) satisfied by ψ by computing both the time derivative and the spatial Laplacian explicitly.

First, we determine the time derivative. Using the chain rule:

$$\frac{\partial \psi}{\partial t} = \frac{\partial}{\partial t} \left(\sqrt{\rho} e^{iS/\mathcal{T}} \right) \quad (\text{S405})$$

$$= \left(\frac{1}{2\sqrt{\rho}} \frac{\partial \rho}{\partial t} + \frac{i}{\mathcal{T}} \frac{\partial S}{\partial t} \right) \psi. \quad (\text{S406})$$

Second, we determine the spatial Laplacian. Applying the Laplacian to ψ , we again use the product rule:

$$\nabla^2 \psi = \nabla^2 \left(\sqrt{\rho} e^{iS/\mathcal{T}} \right) \quad (\text{S407})$$

$$= e^{iS/\mathcal{T}} \left[\nabla^2 \sqrt{\rho} + \frac{2i}{\mathcal{T}} \nabla \sqrt{\rho} \cdot \nabla S + \frac{i}{\mathcal{T}} \sqrt{\rho} \nabla^2 S - \frac{1}{\mathcal{T}^2} \sqrt{\rho} (\nabla S)^2 \right]. \quad (\text{S408})$$

Third, we multiply the time derivative by $i\mathcal{T}$ to obtain

$$i\mathcal{T} \frac{\partial \psi}{\partial t} = \left(\frac{i\mathcal{T}}{2\sqrt{\rho}} \frac{\partial \rho}{\partial t} - \frac{\partial S}{\partial t} \right) \psi. \quad (\text{S409})$$

To substitute for $\frac{\partial \rho}{\partial t}$ and $\frac{\partial S}{\partial t}$, we now use the continuity Equation (S390) and the corrected Hamilton–Jacobi Equation (S402):

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \left(\rho \frac{\nabla S}{m} \right) = 0, \quad (\text{S410})$$

$$\Rightarrow \frac{\partial \rho}{\partial t} = -\nabla \cdot \left(\rho \frac{\nabla S}{m} \right), \quad (\text{S411})$$

and from Equation (S402), we obtain

$$\frac{\partial S}{\partial t} = -\frac{(\nabla S)^2}{2m} - V(\vec{x}) - Q_{\text{TSRT}}. \quad (\text{S412})$$

Substituting Equation (S411) and Equation (S412) into the expression for $i\mathcal{T} \partial \psi / \partial t$ in Equation (S409), and using the definition of Equation (S400), i.e.,

$$Q_{\text{TSRT}} = -\frac{\mathcal{T}^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}}, \quad (\text{S413})$$

we obtain:

$$i\mathcal{T} \frac{\partial \psi}{\partial t} = \left(\frac{i\mathcal{T}}{2\sqrt{\rho}} \left[-\nabla \cdot \left(\rho \frac{\nabla S}{m} \right) \right] + \left[\frac{(\nabla S)^2}{2m} + V + Q_{\text{TSRT}} \right] \right) \psi. \quad (\text{S414})$$

Meanwhile, the Laplacian $\nabla^2 \psi$ can be computed explicitly. Using the definition

$$\psi(\vec{x}, t) = R(\vec{x}, t) e^{iS(\vec{x}, t)/\mathcal{T}}, \quad (\text{S415})$$

where R and S are real functions, we apply the Laplacian operator by the product rule:

$$\nabla^2 \psi = \left(\nabla^2 R - \frac{1}{\mathcal{T}^2} R (\nabla S)^2 + \frac{i}{\mathcal{T}} (2 \nabla R \cdot \nabla S + R \nabla^2 S) \right) e^{iS/\mathcal{T}}. \quad (\text{S416})$$

Similarly, the time derivative of ψ reads

$$\frac{\partial \psi}{\partial t} = \left(\frac{\partial R}{\partial t} + \frac{i}{\mathcal{T}} R \frac{\partial S}{\partial t} \right) e^{iS/\mathcal{T}}. \quad (\text{S417})$$

In the non-relativistic limit relevant to the emergent Schrödinger dynamics, we focus on slowly varying amplitudes, meaning that terms involving spatial derivatives of R are smaller compared to those involving R and S . Therefore, second derivatives of R can be neglected compared to terms involving derivatives of S . Substituting Equation (S416) and Equation (S417) into the generalized dynamics of Equation (S414), and recalling that in this regime the dominant kinetic contribution comes from the gradient of S , we obtain:

$$i\mathcal{T} \left(\frac{\partial R}{\partial t} + \frac{i}{\mathcal{T}} R \frac{\partial S}{\partial t} \right) = -\frac{\mathcal{T}^2}{2m} \left(\nabla^2 R - \frac{1}{\mathcal{T}^2} R (\nabla S)^2 + \frac{i}{\mathcal{T}} (2 \nabla R \cdot \nabla S + R \nabla^2 S) \right) + V(\vec{x})R. \quad (\text{S418})$$

Multiplying through by $e^{iS/\mathcal{T}}$ is consistent throughout, and the imaginary and real parts must separately vanish to preserve the complex structure of the equation. Grouping the dominant terms, the imaginary part yields:

$$\mathcal{T} \frac{\partial R}{\partial t} + \frac{\mathcal{T}}{m} \nabla R \cdot \nabla S + \frac{\mathcal{T}}{2m} R \nabla^2 S = 0, \quad (\text{S419})$$

which governs the conservation of probability density in this emergent dynamics. To derive the continuity equation from this, we recall that $R^2 = |\psi|^2$ represents the probability density. Using the identity $\frac{\partial R^2}{\partial t} = 2R \frac{\partial R}{\partial t}$, and multiplying Equation (S419) by $2R$, we obtain:

$$\frac{\partial R^2}{\partial t} + \frac{2}{m} R \nabla R \cdot \nabla S + \frac{1}{m} R^2 \nabla^2 S = 0. \quad (\text{S420})$$

Now observe that:

$$\nabla \cdot (R^2 \nabla S) = 2R \nabla R \cdot \nabla S + R^2 \nabla^2 S. \quad (\text{S421})$$

Substituting this into the equation above yields:

$$\frac{\partial R^2}{\partial t} + \frac{1}{m} \nabla \cdot (R^2 \nabla S) = 0, \quad (\text{S422})$$

which is the continuity-like form governing the conserved flow of trembling-induced probability density.

The real part yields:

$$-R \frac{\partial S}{\partial t} = -\frac{(\nabla S)^2}{2m} R + V(\vec{x})R, \quad (\text{S423})$$

which simplifies to

$$\frac{\partial S}{\partial t} + \frac{(\nabla S)^2}{2m} + V(\vec{x}) = 0, \quad (\text{S424})$$

resembling a Hamilton-Jacobi type equation for S .

The effective coarse-grained description of trembling spacetime dynamics reveals a fundamental asymmetry between forward and backward local variations along worldlines, rooted in the causal ordering of proper time increments. This asymmetry naturally selects a complex structure for the emergent statistical description, with the imaginary unit i encoding the oriented evolution along causal trajectories. As a result, the low-energy limit of trembling dynamics is governed by a first-order evolution equation in proper time, which takes the form of a Schrödinger equation. Recombining Equation (S422) and Equation (S424) into a single complex equation for ψ , and using the identities in Equation (S416) and Equation (S417) and assuming slowly varying amplitude R , so that higher-order derivatives of R can be neglected, the coarse-grained complex dynamics reduce to:

$$i\mathcal{T} \frac{\partial \psi}{\partial t} = -\frac{\mathcal{T}^2}{2m} \nabla^2 \psi + V(\vec{x})\psi. \quad (\text{S425})$$

This equation is identical in form to the conventional time-dependent Schrödinger equation, provided we identify $\mathcal{T}$ with $\hbar$, the quantum of action. However, in TSRT, this equation does not arise from a quantization procedure or operator formalism. Instead, it follows deterministically from the coarse-grained behavior of trembling deviations in proper time. Thus, the appearance of quantum dynamics in the non-relativistic limit is a direct result of the geometric averaging of spacetime fluctuations. The Schrödinger equation is not a fundamental postulate but an emergent equation arising from Lorentz-invariant causal geometry under coarse-grained trembling dynamics.

Equation (S425) is formally identical in structure to the conventional time-dependent Schrödinger equation:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V(\vec{x})\psi, \quad (\text{S426})$$

with the key difference that $\mathcal{T}$, not $\hbar$, plays the role of the action scale. In TSRT, this is not a postulate but a geometric consequence of the fine-scale deterministic trembling. The complex-valued field ψ is not a fundamental wavefunction—it is a statistical encoding of unresolved trembling-induced fluctuations.

The fundamental action scale $\mathcal{T}$ governing trembling amplitudes defines the normalization of the emergent field. Identifying $\mathcal{T}$ with Planck's constant $\hbar$ (up to universal geometric factors) aligns the trembling-induced statistical dynamics precisely with the conventional Schrödinger equation. Thus, the appearance of $\hbar$ is not a fundamental mystery: it reflects the minimal causal resolution scale of trembling spacetime. In all situations where $\mathcal{T} = \hbar$, standard quantum mechanics emerges as the exact statistical shadow of TSRT. But crucially, the formalism reveals that quantum dynamics need not be postulated—they can be derived from a deeper, causal trembling geometry of spacetime.

Causal Origin of the Complex Structure. The emergence of the imaginary unit i in the Schrödinger equation is not an arbitrary mathematical choice, but a direct consequence of the causal structure of trembling spacetime. Proper-time coarse-graining over the fine-scale deterministic trembling introduces an intrinsic asymmetry between forward and backward proper-time increments. This asymmetry encodes an arrow of causal evolution, leading naturally to a first-order time derivative in the statistical dynamics. In the mathematical formalism, the complex structure introduced by i captures the orientation of this causal flow in phase space.

Thus, the complex-valued field $\psi(\vec{x}, t)$ arises as a natural coarse-grained descriptor of proper-time-ordered trembling dynamics, with the imaginary unit representing the causal embedding of fluctuations in time evolution. Quantum mechanics, in this view, is not fundamentally probabilistic, but geometrically causal: the statistical complex amplitudes reflect the organized microscopic trembling constrained by the causal structure of spacetime.

S18.3 Beyond the Smooth-Amplitude Approximation: Refined Schrödinger Equation from TSRT

In the preceding subsection, the non-relativistic Schrödinger equation was derived by assuming that the spatial amplitude variations of the field ψ were smooth, enabling the neglect of second derivatives of the amplitude in comparison with phase-gradient terms. However, this approximation does not always hold—particularly in contexts where the spatial probability density exhibits rapid modulation, abrupt localization, or significant curvature. In such cases, TSRT provides a refined evolution equation that incorporates all terms from the full proper-time-coarse-grained dynamics.

We begin from the generalized equation derived in Equation (S414), which expresses the emergent coarse-grained evolution of ψ as:

$$i\mathcal{T}\frac{\partial\psi}{\partial t} = \left(\frac{i\mathcal{T}}{2\sqrt{\rho}} \left[-\nabla \cdot \left(\rho \frac{\nabla S}{m} \right) \right] + \left[\frac{(\nabla S)^2}{2m} + V(\vec{x}) + Q_{\text{TSRT}} \right] \right) \psi, \quad (\text{S427})$$

where $\psi = \sqrt{\rho}e^{iS/\mathcal{T}}$, and $Q_{\text{TSRT}} := -\frac{\mathcal{T}^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}}$ is the TSRT quantum potential. Substituting the full definitions into Equation (S427) and eliminating ρ and S in favor of ψ , we arrive at an exact expression for the refined Schrödinger equation. For simplicity, we work in one spatial dimension and restore all amplitude-derivative terms.

Let $\psi(x, t) = \sqrt{\rho(x, t)}e^{iS(x, t)/\mathcal{T}}$. Then,

$$\frac{\partial\psi}{\partial t} = \left(\frac{1}{2\sqrt{\rho}} \frac{\partial\rho}{\partial t} + \frac{i}{\mathcal{T}} \frac{\partial S}{\partial t} \right) \psi, \quad (\text{S428})$$

$$\partial_x \psi = \left(\frac{1}{2\sqrt{\rho}} \partial_x \rho + \frac{i}{\mathcal{T}} \partial_x S \right) \psi, \quad (\text{S429})$$

$$\partial_x^2 \psi = \left[\frac{1}{2\sqrt{\rho}} \partial_x^2 \rho - \frac{1}{4\rho^{3/2}} (\partial_x \rho)^2 + \frac{i}{\mathcal{T}} \left(\partial_x^2 S + \frac{1}{\rho} \partial_x \rho \partial_x S \right) - \frac{1}{\mathcal{T}^2} (\partial_x S)^2 \right] \psi. \quad (\text{S430})$$

Inserting these expressions into the general form of the TSRT dynamics, we obtain the

refined evolution equation for ψ :

$$i\mathcal{T}\frac{\partial\psi}{\partial t} = \left[-\frac{\mathcal{T}^2}{2m} \left(\frac{1}{2\sqrt{\rho}} \partial_x^2 \rho - \frac{1}{4\rho^{3/2}} (\partial_x \rho)^2 \right) + V(x) + \frac{1}{2m} (\partial_x S)^2 - \frac{i\mathcal{T}}{2\sqrt{\rho}} \left(\frac{\rho \partial_x^2 S}{m} + \frac{\partial_x \rho \partial_x S}{m} \right) \right] \psi. \quad (\text{S431})$$

To express the refined TSRT dynamics entirely in terms of ψ , we use the following relations:

$$\frac{\partial_x \rho}{\rho} = 2 \operatorname{Re} \left(\frac{\partial_x \psi}{\psi} \right), \quad \frac{\partial_x^2 \rho}{\rho} = 2 \operatorname{Re} \left(\frac{\partial_x^2 \psi}{\psi} \right) + 2 \left[\operatorname{Re} \left(\frac{\partial_x \psi}{\psi} \right) \right]^2, \quad \frac{\partial_x S}{\mathcal{T}} = \operatorname{Im} \left(\frac{\partial_x \psi}{\psi} \right),$$

along with

$$\frac{\partial_x^2 S}{\mathcal{T}} = \operatorname{Im} \left(\frac{\partial_x^2 \psi}{\psi} \right) - 2 \operatorname{Im} \left(\frac{\partial_x \psi}{\psi} \right) \operatorname{Re} \left(\frac{\partial_x \psi}{\psi} \right).$$

Inserting these identities into Equation (S431) and performing algebraic simplification, we obtain:

$$\begin{aligned} i\mathcal{T}\frac{\partial\psi}{\partial t} = & -\frac{\mathcal{T}^2}{2m} \left[\operatorname{Re} \left(\frac{\partial_x^2 \psi}{\psi} \right) + \left(\operatorname{Re} \left(\frac{\partial_x \psi}{\psi} \right) \right)^2 \right] \psi \\ & + V(x)\psi + \frac{\mathcal{T}^2}{2m} \left(\operatorname{Im} \left(\frac{\partial_x \psi}{\psi} \right) \right)^2 \psi \\ & - \frac{i\mathcal{T}^2}{2m} \left[\operatorname{Im} \left(\frac{\partial_x^2 \psi}{\psi} \right) + 2 \operatorname{Im} \left(\frac{\partial_x \psi}{\psi} \right) \operatorname{Re} \left(\frac{\partial_x \psi}{\psi} \right) \right] \psi. \end{aligned} \quad (\text{S432})$$

This final form is the refined TSRT evolution equation valid beyond the smooth-amplitude approximation. All terms are expressed directly in terms of ψ and its first and second spatial derivatives, reflecting both amplitude and phase curvature effects.

To better understand the impact of this refinement, let us now decompose Equation (S432) into two parts:

$$i\mathcal{T}\frac{\partial\psi}{\partial t} = \left(-\frac{\mathcal{T}^2}{2m} \nabla^2 + V(x) \right) \psi + \Delta_{\text{TSRT}}[\psi], \quad (\text{S433})$$

where the first term is precisely the conventional Schrödinger evolution and the second term $\Delta_{\text{TSRT}}[\psi]$ captures the deviations introduced by retaining amplitude curvature effects:

$$\begin{aligned} \Delta_{\text{TSRT}}[\psi] := & -\frac{\mathcal{T}^2}{2m} \left[\operatorname{Re} \left(\frac{\partial_x^2 \psi}{\psi} \right) + \left(\operatorname{Re} \left(\frac{\partial_x \psi}{\psi} \right) \right)^2 - \frac{\partial_x^2 \psi}{\psi} \right] \psi \\ & + \frac{\mathcal{T}^2}{2m} \left[\left(\operatorname{Im} \left(\frac{\partial_x \psi}{\psi} \right) \right)^2 - \operatorname{Im} \left(\frac{\partial_x^2 \psi}{\psi} \right) \right] i\psi. \end{aligned} \quad (\text{S434})$$

The total correction $\Delta_{\text{TSRT}}[\psi]$ thus contains both real and imaginary contributions. These terms encode the effects of amplitude gradients and curvature which are neglected in the standard quantum mechanical treatment.

Why These Terms Disappear in the Smooth-Amplitude Limit. If the amplitude $|\psi| = \sqrt{\rho}$ varies slowly over space, then $\operatorname{Re}(\partial_x \psi / \psi)$ becomes small, and $\partial_x^2 \psi / \psi \approx \operatorname{Re}(\partial_x^2 \psi / \psi)$. Similarly, if the phase $S(x)$ varies slowly or linearly, then $\operatorname{Im}(\partial_x^2 \psi / \psi) \approx 0$, and the imaginary

correction vanishes. In this limit, all correction terms are suppressed, and the conventional Schrödinger equation is recovered to excellent approximation.

Interpretation of the Corrections. The real part of $\Delta_{\text{TSRT}}[\psi]$ represents additional potential-like effects arising from spatial variations in the amplitude of ψ , akin to a generalized curvature-induced quantum potential. The imaginary part introduces non-unitary effects tied to sharp gradients in the phase and amplitude, corresponding to local breakdown of coherent flow or increased decoherence in rapidly oscillating zones.

Implications for Conventional Quantum Mechanics. In standard quantum mechanics, these corrections are absent due to the assumption that ψ is a smooth, analytic wavefunction. However, in regions with steep probability gradients—such as at nodes, boundaries, or in engineered quantum devices with abrupt potentials—these terms could become significant. The TSRT refinement thus offers a framework to go beyond the Schrödinger equation and capture subtle geometric effects invisible to canonical quantization.

Moreover, since the corrections are deterministic and emerge from coarse-grained causal geometry, they provide a route to systematically study when and how quantum mechanics might require modification in extreme regimes without introducing stochasticity or hidden variables.

Refinement Generalized to Three Spatial Dimensions To generalize this refinement to three spatial dimensions, we write:

$$i\mathcal{T}\frac{\partial\psi}{\partial t} = \left(-\frac{\mathcal{T}^2}{2m}\nabla^2 + V(\vec{x})\right)\psi + \Delta_{\text{TSRT}}[\psi], \quad (\text{S435})$$

with the correction term:

$$\begin{aligned} \Delta_{\text{TSRT}}[\psi] := & -\frac{\mathcal{T}^2}{2m} \left[\text{Re}\left(\frac{\nabla^2\psi}{\psi}\right) + \left\|\text{Re}\left(\frac{\nabla\psi}{\psi}\right)\right\|^2 - \frac{\nabla^2\psi}{\psi} \right] \psi \\ & + \frac{\mathcal{T}^2}{2m} \left[\left\|\text{Im}\left(\frac{\nabla\psi}{\psi}\right)\right\|^2 - \text{Im}\left(\frac{\nabla^2\psi}{\psi}\right) \right] i\psi. \end{aligned} \quad (\text{S436})$$

Here, $\nabla\psi$ and $\nabla^2\psi$ denote the gradient and Laplacian of ψ over $\mathbb{R}^3$, and $\|\cdot\|$ is the Euclidean norm. The correction $\Delta_{\text{TSRT}}[\psi]$ splits again into real and imaginary components. The real part introduces curvature-driven corrections to the local energy landscape, while the imaginary part captures dissipative or decohering tendencies when amplitude or phase structure becomes highly non-uniform. In the smooth-amplitude limit, where $\text{Re}(\nabla\psi/\psi)$ and amplitude curvature terms are small, the correction terms vanish. The resulting dynamics then reduce precisely to the standard Schrödinger equation. Thus, this formulation preserves standard quantum mechanics in low-curvature, slowly varying regimes, while naturally extending it into new domains. This yields a fully deterministic, extended Schrödinger-type equation, valid even in regimes where the probability amplitude varies rapidly in space. It reduces to the conventional Schrödinger equation when higher-order amplitude terms are negligible, but remains valid and predictive in strongly curved or high-gradient density regimes—such as near nodes, edges, sharply localized

regions, or in engineered nanostructures with steep potentials.

Interpretation. This refined TSRT evolution equation reveals that quantum mechanics, as normally formulated, approximates the deeper causal trembling dynamics only when the amplitude structure of ψ is sufficiently smooth. The corrections captured by $\Delta_{\text{TSRT}}[\psi]$ are not ad hoc additions but follow necessarily from retaining previously neglected amplitude derivatives in the geometric projection. Their form suggests that measurable deviations from conventional quantum predictions may arise in situations where the probability density changes abruptly—such as near interference nodes, measurement boundaries, or sub-wavelength confinement zones. These refinements are entirely deterministic and causal, and provide a geometric route toward identifying where standard quantum theory may require correction. Such differences may be testable in high-precision, high-gradient quantum experiments, potentially offering direct empirical access to the causal structure and curvature of trembling spacetime. TSRT thus provides a falsifiable refinement of quantum mechanics—grounded not in quantization axioms, but in the Lorentz-invariant, curvature-sensitive geometry of proper-time dynamics.

S18.4 Special Relativistic Limit: Toward the Dirac Equation

In TSRT, the trembling of spacetime is not restricted to spatial displacement alone but can also include internal rotational structures, especially relevant for relativistic particles. Just as the probabilistic behavior derived from coarse-graining trembling motion in the non-relativistic regime gave rise to the Schrödinger Equation (S426), here we explore how relativistic trembling and internal angular degrees of freedom give rise to the Dirac equation.

Let $x^\mu(\tau)$ be the worldline of a particle undergoing deterministic trembling in Minkowski spacetime, parametrized by proper time τ . The trembling deviations from the classical geodesic path are again described by a fine-scale structure, as introduced in Equation (S389):

$$x^\mu(\tau) = x_{\text{class}}^\mu(\tau) + \Xi^\mu(\tau), \quad (\text{S437})$$

where $\Xi^\mu(\tau)$ represents the deterministic trembling in spacetime, but now potentially entangled with internal rotational structure.

We now introduce a second component of the motion: an internal degree of freedom $\chi^\alpha(\tau)$ associated with local angular motion or internal symmetry, which is also subject to trembling. This multicomponent quantity $\chi^\alpha(\tau)$ evolves deterministically, governed by internal trembling equations that couple to spacetime through the local frame defined by $x^\mu(\tau)$. We assume the dynamics of χ^α preserve Lorentz invariance, and that they are governed by a geometric rule that transforms consistently under local frame changes. In analogy to the non-relativistic case, we now consider a probabilistic description of such trembling dynamics, where a single particle's fine-scale phase offset is treated statistically to construct a probabilistic field over spacetime of Ξ^μ and χ^α but governed by the same deterministic trembling laws. The internal degrees of freedom, when coarse-grained across the ensemble, produce an effective field $\Psi(x)$ defined over spacetime. This field is not a fundamental wavefunction, but a statistical encoding of how the internal structure of trembling paths projects onto macroscopic observations. Because

the internal trembling is deterministically coupled to the spacetime trajectory, and because the coarse-grained probabilistic description incorporates a distribution of microscopic phase offsets over the particle's trembling history, we now seek an effective evolution equation for the statistical field $\Psi(x)$ that encodes this internal structure. This field does not represent an ontological wavefunction but rather a compressed geometric descriptor of the internal angular motion observed in a single particle, averaged over unresolvable microscopic phase fluctuations. The form of the governing equation for $\Psi(x)$ must be tightly constrained by the following physical and mathematical requirements:

- (1) Lorentz Covariance: Since the internal trembling is embedded within the Minkowski structure of spacetime, any field $\Psi(x)$ describing its evolution must transform appropriately under Lorentz transformations. This ensures that the same trembling dynamics appear consistent to all inertial observers and that the internal structure does not introduce preferred reference frames. The evolution equation must be formulated using tensors or spinors that respect the Lorentz symmetry of special relativity.
- (2) First-Order Derivatives: The trembling behavior involves local oscillations in proper time, and thus its macroscopic statistical description should obey a first-order differential equation in spacetime derivatives. A second-order equation, such as the Klein–Gordon equation [68, 69], cannot capture the coherence of phase propagation associated with geometric internal rotation. A first-order structure guarantees causal propagation and avoids unphysical solutions with negative norm or superluminal phase velocity.
- (3) Probability Current Conservation: Since the trembling is conservative at the microscopic level, its coarse-grained description must conserve the associated probability current. This implies that the equation must yield a conserved four-current $j^\mu = \bar{\Psi}\gamma^\mu\Psi$ whose divergence vanishes:

$$\partial_\mu j^\mu = 0. \quad (\text{S438})$$

This requirement ensures that probabilistic density—although emergent—obeys a continuity equation analogous to that in standard quantum theory.

- (4) Geometric Encoding of Mass: The inertial mass of the particle must fundamentally arise from the trembling dynamics themselves, not from a postulated external parameter, although, for simplicity, we often assume the mass to be given in problems solved in TSRT where mass emergence does not play a key-role. A detailed study on mass emergence is currently undertaken. Hence, the governing equation must include the mass term as a coupling between the internal geometry (represented by Ψ) and the underlying spacetime structure.

We now ask: what is the most general linear partial differential equation for a field $\Psi(x)$, defined over Minkowski spacetime, that satisfies all of the above constraints? Mathematically, this leads to the requirement that the evolution equation must be first-order in spacetime derivatives, Lorentz covariant, linear in the field Ψ , and must include a mass term. The general form of such an equation is:

$$\Gamma^\mu \partial_\mu \Psi(x) = -\frac{imc}{\mathcal{T}} \Psi(x), \quad (\text{S439})$$

where Γ^μ are matrices acting on the components of Ψ , and $\mathcal{T}$ is the TSRT action scale introduced

in Equation (S399) in the derivation of the Schrödinger equation. This constant plays the role of a geometric unit of action arising from the internal trembling amplitude and frequency content, and it naturally replaces Planck's constant $\hbar$ in this deterministic framework (with $\hbar$ emerging from the trembling cutoff as established in Section S19). The role of the matrices Γ^μ is to mediate the coupling between the spacetime derivatives and the internal degrees of freedom encoded in $\Psi(x)$. These matrices must be chosen such that the full equation is Lorentz covariant and yields a conserved current. This requirement restricts Γ^μ to obey a Clifford algebra:

$$\{\Gamma^\mu, \Gamma^\nu\} = 2\eta^{\mu\nu}\mathbb{I}, \quad (\text{S440})$$

where $\eta^{\mu\nu}$ is the Minkowski metric and $\mathbb{I}$ is the identity matrix. Equation (S440) is to ensure Lorentz covariance and a conserved current [5, 70]. This structure traces back to Clifford's geometric algebras [71], but its physical necessity was first demonstrated by Dirac in his relativistic electron theory. The only matrices satisfying these conditions and leading to a conserved current are the Dirac gamma matrices γ^μ , which form the foundation of spinor theory in relativistic quantum mechanics. This is detailed further in this section.

Thus, the only permissible choice is $\Gamma^\mu = \gamma^\mu$, and the evolution equation becomes:

$$i\mathcal{T}\gamma^\mu\partial_\mu\Psi(x) = mc\Psi(x). \quad (\text{S441})$$

This equation is formally identical to the Dirac equation, except that $\mathcal{T}$, not $\hbar$, governs the relationship between internal angular frequency and mass. Therefore, in all situations where $\mathcal{T} = \hbar$, this equation is identical to Dirac's. This result is not an assumption but a deduction from the structure of trembling spacetime. It shows that relativistic spinor fields—and all their associated properties—can be derived as effective probabilistic descriptors of deterministic, trembling, single-particle trajectories.

This equation satisfies all of the initial requirements:

- It is Lorentz covariant by construction, since γ^μ transforms as a four-vector and Ψ as a spinor.
- It is first-order in derivatives, ensuring causal signal propagation.
- It yields a conserved current $j^\mu = \bar{\Psi}\gamma^\mu\Psi$, with $\bar{\Psi} = \Psi^\dagger\gamma^0$.
- The mass term appears naturally as a frequency scale set by the internal geometric oscillations of the particle.

In summary, Equation (S441) is the unique linear, relativistically invariant, first-order evolution equation for a probabilistic field encoding the internal structure of a geometrically trembling particle. It emerges from the mathematical and physical demands of TSRT, rather than being postulated as in quantum mechanics. The Dirac formalism is thus revealed not as a foundational mystery, but as a statistical surface of a deeper geometric process. The matrices Γ^μ introduced in Equation (S439) are not arbitrary placeholders, but must reflect the intrinsic algebraic structure, stated in Equation (S440), of the underlying trembling degrees of freedom. To understand their nature, we must examine the mathematical structure that governs how internal geometric fluctuations—particularly those associated with local angular motion—transform under spacetime symmetries.

Recall that in TSRT, the internal degree of freedom $\chi^\alpha(\tau)$, introduced to describe angular trembling, evolves according to deterministic geometric rules coupled to the local spacetime frame. These rules must preserve Lorentz invariance, meaning that χ^α must transform under a well-defined representation of the Lorentz group, and that any coupling between χ^α and spacetime derivatives must respect this symmetry. This condition places strict algebraic constraints on the allowable matrix structures Γ^μ used in the dynamical equation for $\Psi(x)$. To be precise, consider the requirement that the evolution equation for $\Psi(x)$,

$$\Gamma^\mu \partial_\mu \Psi(x) = -\frac{imc}{\mathcal{T}} \Psi(x), \quad (\text{S442})$$

must be Lorentz covariant. Under a Lorentz transformation Λ^μ_ν , the spacetime coordinate transforms as $x^\mu \rightarrow x'^\mu = \Lambda^\mu_\nu x^\nu$, and the derivative transforms accordingly as $\partial_\mu \rightarrow \Lambda^\nu_\mu \partial_\nu$. Simultaneously, the field Ψ must transform under some spinor representation $S(\Lambda)$ of the Lorentz group:

$$\Psi(x) \rightarrow \Psi'(x') = S(\Lambda) \Psi(x). \quad (\text{S443})$$

For the full equation to remain invariant under this combined transformation, the matrices Γ^μ must satisfy:

$$S^{-1}(\Lambda) \Gamma^\mu S(\Lambda) = \Lambda^\mu_\nu \Gamma^\nu. \quad (\text{S444})$$

This transformation law implies that the set of matrices Γ^μ transforms like a four-vector under Lorentz transformations, ensuring that the contraction $\Gamma^\mu \partial_\mu$ behaves as a Lorentz scalar operator acting on spinors. This is a hallmark of the Dirac equation and immediately suggests a deeper algebraic structure underpinning Γ^μ .

To determine this structure explicitly, we demand that the operator $\Gamma^\mu \partial_\mu$ generate a relativistically invariant quadratic form when squared. In other words, the square of the operator must yield the invariant d'Alembertian operator plus a mass term. This imposes the algebraic constraint:

$$(\Gamma^\mu \partial_\mu)^2 = \Gamma^\mu \Gamma^\nu \partial_\mu \partial_\nu = \eta^{\mu\nu} \partial_\mu \partial_\nu \mathbb{I}, \quad (\text{S445})$$

which is only possible if the matrices Γ^μ satisfy the anticommutation relation:

$$\{\Gamma^\mu, \Gamma^\nu\} := \Gamma^\mu \Gamma^\nu + \Gamma^\nu \Gamma^\mu = 2\eta^{\mu\nu} \mathbb{I}, \quad (\text{S446})$$

where $\eta^{\mu\nu} = \text{diag}(-1, +1, +1, +1)$ is the Minkowski metric in standard coordinates, and $\mathbb{I}$ is the identity matrix acting on the internal degrees of freedom. This defining relation, Equation (S446), identifies the Γ^μ as generators of a Clifford algebra over Minkowski spacetime [5, 70, 71]. In four dimensions, the unique (up to similarity transformations) finite-dimensional irreducible representation of this Clifford algebra is given by the Dirac gamma matrices γ^μ . These matrices act on four-component spinors and satisfy:

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} \mathbb{I}, \quad (\text{S447})$$

precisely matching the condition in Equation (S446), and Equation (S440). Therefore, we are mathematically compelled to identify $\Gamma^\mu = \gamma^\mu$ to satisfy all the physical and geometric

requirements imposed by TSRT. This identification is unique up to unitary equivalence: in four-dimensional Minkowski spacetime, the Clifford algebra generated by the anticommutation relations in Equation (S446) admits a unique (up to isomorphism) irreducible representation on four-component complex spinors. Thus, the standard Dirac matrices γ^μ are not just a convenient choice—they are the only admissible algebraic structure capable of encoding the trembling-induced coupling between spacetime and internal degrees of freedom in a Lorentz-covariant manner.

Substituting this result into the general evolution Equation (S442), we obtain:

$$i\mathcal{T}\gamma^\mu\partial_\mu\Psi(x) = mc\Psi(x), \quad (\text{S448})$$

which is formally equivalent to the standard Dirac equation in relativistic quantum mechanics, with one crucial difference: the constant $\mathcal{T}$, which emerges from the intrinsic trembling structure of spacetime in TSRT, plays the role of the fundamental unit of action instead of Planck's constant $\hbar$. This result has profound implications. It shows that the Dirac equation—long regarded as one of the fundamental postulates of relativistic quantum theory—arises naturally and unavoidably as the coarse-grained statistical description of a single particle's internal trembling in spacetime. The spinorial structure, the linear coupling to spacetime derivatives, the presence of mass as a geometric frequency, and the conserved current structure all emerge from the deeper deterministic dynamics of TSRT. What appears as quantum behavior in standard theory is revealed, in TSRT, to be a manifestation of underlying geometric order.

In this view, the gamma matrices γ^μ are no longer mysterious algebraic artifacts. They encode how the internal phase structure of trembling particles couples to the spacetime manifold, giving rise to the rich phenomenology of spin, relativistic propagation, and antiparticle symmetry—without invoking any probabilistic axioms. This emergent Dirac Equation (S448), derived from the internal deterministic structure of trembling spacetime, retains all the key physical features of its quantum counterpart. We now elaborate on three foundational aspects—its relativistic consistency, the appearance of antiparticle states, and its reduction to non-relativistic spin dynamics—all of which arise naturally in the TSRT framework.

Relativistic Consistency and Recovery of the Klein–Gordon Equation. To confirm that the emergent Dirac equation is fully compatible with special relativity, we verify that it implies the standard relativistic dispersion relation. Starting from the Dirac equation:

$$i\mathcal{T}\gamma^\mu\partial_\mu\Psi(x) = mc\Psi(x), \quad (\text{S449})$$

we apply the operator $(i\mathcal{T}\gamma^\nu\partial_\nu + mc)$ to both sides:

$$(i\mathcal{T}\gamma^\nu\partial_\nu + mc)(i\mathcal{T}\gamma^\mu\partial_\mu\Psi) = (i\mathcal{T}\gamma^\nu\partial_\nu)(i\mathcal{T}\gamma^\mu\partial_\mu\Psi) + mc(i\mathcal{T}\gamma^\mu\partial_\mu\Psi). \quad (\text{S450})$$

Using the Clifford algebra identity $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$, we compute:

$$(i\mathcal{T})^2\gamma^\nu\gamma^\mu\partial_\nu\partial_\mu = -\mathcal{T}^2\eta^{\mu\nu}\partial_\mu\partial_\nu = -\mathcal{T}^2\Box, \quad (\text{S451})$$

where $\square := \eta^{\mu\nu} \partial_\mu \partial_\nu$ is the d'Alembert operator in flat spacetime. The mass term contributes:

$$mc(i\mathcal{T}\gamma^\mu \partial_\mu \Psi) = mc \cdot mc\Psi = m^2 c^2 \Psi. \quad (\text{S452})$$

Combining both contributions—the trembling spacetime-induced dynamics and the mass term—we obtain the second-order differential equation

$$\left(\square + \frac{m^2 c^2}{\mathcal{T}^2}\right) \Psi(x) = 0, \quad (\text{S453})$$

where $\square = \partial^\mu \partial_\mu$ is the standard d'Alembert operator in flat spacetime, and $\Psi(x)$ denotes the spinorial field resulting from the trembling geometry.

To analyze the physical implications of Equation (S453), we consider plane-wave solutions of the form

$$\Psi(x) = u(p) e^{-ip_\mu x^\mu / \mathcal{T}}, \quad (\text{S454})$$

where $u(p)$ is a constant spinor amplitude and p_μ is the four-momentum.

Substituting the ansatz (S454) into Equation (S453) yields

$$\left(-\frac{p^\mu p_\mu}{\mathcal{T}^2} + \frac{m^2 c^2}{\mathcal{T}^2}\right) u(p) e^{-ip_\mu x^\mu / \mathcal{T}} = 0, \quad (\text{S455})$$

which simplifies, upon canceling the common factor $u(p) e^{-ip_\mu x^\mu / \mathcal{T}} \neq 0$, to

$$p^\mu p_\mu = m^2 c^2. \quad (\text{S456})$$

In explicit components, writing $p^\mu = (E/c, \vec{p})$ with E the energy and $\vec{p}$ the spatial momentum, the invariant reads

$$\left(\frac{E}{c}\right)^2 - |\vec{p}|^2 = m^2 c^2, \quad (\text{S457})$$

which rearranges to the familiar relativistic energy-momentum relation

$$E^2 = p^2 c^2 + m^2 c^4. \quad (\text{S458})$$

Thus, the Klein–Gordon equation (S453) derived from trembling spacetime dynamics directly enforces the correct relativistic dispersion relation (S458). This confirms that the emergent Dirac-type structure in TSRT preserves the principles of special relativity at the level of observable dynamics, ensuring that no deviation from the standard relativistic energy-momentum behavior occurs.

Negative-Frequency Modes and Antiparticles. A distinctive feature of the Dirac equation is its support for negative-frequency (or equivalently, negative-energy) solutions. In conventional quantum theory, these solutions were historically problematic, leading Dirac to introduce the concept of antiparticles via the hole theory to resolve the issue of unbounded Hamiltonians. In TSRT, however, the origin of these negative-frequency modes is fully geometric and deterministic, arising from the internal trembling structure of spacetime itself. Importantly, it does

not involve any violation of causality or reversal of physical proper time evolution.

The key point is that, although the global progression of proper time τ remains strictly causal and monotonically increasing, the internal phase dynamics associated with the particle's trembling can exhibit effective phase configurations that correspond to a local reversal of internal orientation. Specifically, in the geometric decomposition of trembling modes, certain internal oscillatory patterns can evolve with a phase that, when coarse-grained, appears as a negative-frequency solution relative to the global time parameter.

However, this phase reversal is purely internal to the trembling dynamics and does not imply that the particle's worldline or external motion violates causality or flows backward in time. Proper time τ and coordinate time t both remain globally forward-directed. What reverses is the internal phase winding within the trembling metric deviations, not the macroscopic temporal evolution. Mathematically, when the effective Dirac field $\Psi(x)$ is constructed from these internal geometric modes, the phase-sensitive structure admits both positive- and negative-frequency components. Upon coarse-graining, these manifest as the two types of solutions of the Dirac equation: one corresponding to particles and the other to antiparticles.

Thus, in TSRT, antiparticles do not arise from a quantized field-theoretic reinterpretation but emerge naturally from the symmetry structure of deterministic trembling configurations. They represent distinct internal phase orientations of the same trembling spacetime system. This interpretation restores physical clarity to the concept of antimatter: antiparticles are not fundamentally different entities, but rather alternate deterministic modes of the same underlying trembling geometry, revealed statistically through ensemble projection. Charge conjugation and time-reversal symmetries are embedded directly within the geometric properties of space-time trembling itself, without requiring negative-energy particles or unphysical assumptions. Crucially, because the underlying proper time τ always increases and causal structure is preserved at all stages, there is no contradiction with the strict causality postulate of TSRT. To make this structure explicit, consider plane-wave solutions of the emergent Dirac equation in TSRT of the form

$$\Psi(x) = u(p)e^{-ip_\mu x^\mu/\mathcal{T}}, \quad (\text{S459})$$

where $u(p)$ is a constant spinor amplitude associated with momentum $p^\mu = (E/c, \vec{p})$. Substituting into the Dirac equation, one obtains the algebraic condition

$$(\gamma^\mu p_\mu - mc)u(p) = 0, \quad (\text{S460})$$

which is satisfied when p^μ fulfills the mass-shell condition $p^\mu p_\mu = m^2 c^2$. However, because the internal trembling modes allow for phase-reversed internal configurations, a second class of solutions arises naturally by considering

$$\Psi(x) = v(p)e^{+ip_\mu x^\mu/\mathcal{T}}, \quad (\text{S461})$$

where $v(p)$ is another constant spinor amplitude. Substituting into the Dirac equation yields

$$(\gamma^\mu p_\mu + mc)v(p) = 0, \quad (\text{S462})$$

which again requires $p^\mu p_\mu = m^2 c^2$, preserving the correct relativistic dispersion relation.

The presence of both types of solutions is a direct consequence of the phase structure of the underlying trembling spacetime geometry. The positive-frequency solutions correspond to standard particle modes, while the negative-frequency solutions correspond to antiparticle modes. Importantly, in TSRT, both $u(p)$ and $v(p)$ describe different internal phase configurations of the same deterministic trembling system, rather than distinct quantum field excitations. No negative total energies occur, and the proper time τ remains globally forward-directed throughout. Thus, the mathematical structure of the Dirac equation emerging from TSRT automatically accommodates both particles and antiparticles as deterministic outcomes of the geometric trembling dynamics.

Non-Relativistic Limit and Spin Dynamics. In the low-energy (non-relativistic) limit, where the trembling frequency is much smaller than $mc^2/\mathcal{T}$, the Dirac equation simplifies to an effective two-component equation describing spin dynamics. This reduction can be carried out via a Foldy–Wouthuysen transformation, which systematically decouples positive- and negative-frequency modes and expands the dynamics in powers of $1/c$ [72]. To illustrate the structure, we decompose the four-component Dirac spinor Ψ into two two-component spinors,

$$\Psi = \begin{pmatrix} \phi \\ \chi \end{pmatrix}, \quad (\text{S463})$$

where ϕ represents the large (upper) components and χ represents the small (lower) components. The Dirac equation can then be written as a coupled system:

$$i\mathcal{T} \frac{\partial \phi}{\partial t} = (mc^2 + V(\vec{x})) \phi + c \vec{\sigma} \cdot (-i\mathcal{T}\nabla - \vec{A}) \chi, \quad (\text{S464})$$

$$i\mathcal{T} \frac{\partial \chi}{\partial t} = (-mc^2 + V(\vec{x})) \chi + c \vec{\sigma} \cdot (-i\mathcal{T}\nabla - \vec{A}) \phi. \quad (\text{S465})$$

In the non-relativistic regime, the rest energy mc^2 dominates, so χ is much smaller than ϕ . Therefore, to leading order, we can solve Equation (S465) for χ as

$$\chi \approx \frac{1}{2mc} \vec{\sigma} \cdot (-i\mathcal{T}\nabla - \vec{A}) \phi. \quad (\text{S466})$$

Substituting this approximation into Equation (S464), we obtain an effective equation for ϕ :

$$i\mathcal{T} \frac{\partial \phi}{\partial t} = (mc^2 + V(\vec{x})) \phi + c \vec{\sigma} \cdot (-i\mathcal{T}\nabla - \vec{A}) \left(\frac{1}{2mc} \vec{\sigma} \cdot (-i\mathcal{T}\nabla - \vec{A}) \phi \right) \quad (\text{S467})$$

$$= (mc^2 + V(\vec{x})) \phi + \frac{1}{2m} \left(\vec{\sigma} \cdot (-i\mathcal{T}\nabla - \vec{A}) \right)^2 \phi. \quad (\text{S468})$$

Expanding the squared term using the Pauli matrix identity

$$(\vec{\sigma} \cdot \vec{a}) (\vec{\sigma} \cdot \vec{b}) = \vec{a} \cdot \vec{b} + i \vec{\sigma} \cdot (\vec{a} \times \vec{b}), \quad (\text{S469})$$

where $\vec{a} = -i\mathcal{T}\nabla - \vec{A}$ and $\vec{b} = -i\mathcal{T}\nabla - \vec{A}$, we obtain:

$$\left(\vec{\sigma} \cdot (-i\mathcal{T}\nabla - \vec{A})\right)^2 = \left(-i\mathcal{T}\nabla - \vec{A}\right)^2 + i\vec{\sigma} \cdot \left((-i\mathcal{T}\nabla) \times (-\vec{A}) + (-\vec{A}) \times (-i\mathcal{T}\nabla)\right). \quad (\text{S470})$$

The cross terms simplify to

$$i\vec{\sigma} \cdot (2i\mathcal{T}\nabla \times \vec{A}) = -2\mathcal{T}\vec{\sigma} \cdot (\nabla \times \vec{A}). \quad (\text{S471})$$

Thus, the effective equation becomes

$$i\mathcal{T}\frac{\partial\phi}{\partial t} = \left[mc^2 + V(\vec{x}) + \frac{1}{2m} \left(-i\mathcal{T}\nabla - \vec{A}\right)^2 - \frac{\mathcal{T}}{m} \vec{\sigma} \cdot \vec{B} \right] \phi, \quad (\text{S472})$$

where $\vec{B} = \nabla \times \vec{A}$.

Finally, subtracting the constant rest mass energy mc^2 (which merely shifts the energy reference) yields the Pauli-type equation:

$$i\mathcal{T}\frac{\partial\phi}{\partial t} = \left[\frac{1}{2m} \left(-i\mathcal{T}\nabla - \vec{A}\right)^2 + V(\vec{x}) - \frac{\mathcal{T}}{2m} \vec{\sigma} \cdot \vec{B} \right] \phi. \quad (\text{S473})$$

Here, $\vec{A}$ plays the role of an emergent gauge field arising from the coupling of internal trembling modes to external geometry, $\vec{B} = \nabla \times \vec{A}$ acts as the corresponding geometric analogue of a magnetic field, and $\vec{\sigma}$ are the Pauli matrices representing internal spin degrees of freedom.

Crucially, in TSRT, the spin coupling term $\vec{\sigma} \cdot \vec{B}$ does not arise from an abstract quantum mechanical postulate or an externally imposed g-factor. It emerges directly from the geometric structure of internal rotational trembling—that is, from the response of the particle’s internal trembling field to local spacetime curvature and twisting.

These results jointly demonstrate that the Dirac formalism—including its relativistic dispersion, intrinsic spin structure, and antiparticle symmetry—emerges from deterministic trembling geometry, not from quantization axioms. TSRT thus provides a deeper, causal foundation for the formal structures of relativistic quantum theory. The statistical emergence of complex amplitudes, spinorial transformation laws, and wave equations is a consequence of how trembling spacetime organizes itself geometrically at microscopic scales. Rather than being foundational, the quantum formalism is a shadow cast by a deeper deterministic structure—one whose causal integrity and relativistic consistency are built into the very geometry of trembling spacetime. In this geometric framework, the field $\Psi(x)$ does not represent a fundamental wavefunction in the conventional quantum mechanical sense. Instead, it is a statistical construct: a coarse-grained projection of the underlying deterministic internal dynamics of a single particle as it undergoes trembling motion in spacetime. This trembling includes not only translational deviations from classical geodesics but also rotational fluctuations associated with internal degrees of freedom. The spinorial nature of $\Psi(x)$ is not imposed a priori. Rather, it emerges naturally from the algebraic and geometric structure of these internal fluctuations. The requirement of Lorentz covariance imposes strict constraints on how the internal degrees of freedom may transform. When these constraints are analyzed—particularly under the assumption that the internal trembling variables form a closed algebra consistent with the spacetime Clifford structure—it

becomes inevitable that $\Psi(x)$ must transform as a spinor. Thus, what in quantum mechanics is regarded as a mysterious intrinsic property of matter—spin—arises here as a necessary outcome of the relativistic covariance of internal trembling geometry. Similarly, the parameter $\mathcal{T}$, which governs both the phase evolution of $\Psi(x)$ and the magnitude of trembling-induced corrections to classical dynamics, is not an arbitrary constant. It is a derived quantity that encodes the amplitude and frequency content of the trembling fluctuations in TSRT. If nature selects trembling amplitudes and frequencies such that $\mathcal{T} \equiv \hbar$, then quantum mechanics arises as an exact effective theory from TSRT. However, if future high-precision experiments detect deviations from standard quantum predictions at extremely fine scales, this could signal a measurable difference between $\mathcal{T}$ and $\hbar$, offering a pathway for direct experimental validation or falsification of the trembling spacetime hypothesis. But crucially, this equivalence is not axiomatic—it is a phenomenological consequence of how spacetime trembles, not a foundational principle.

In summary, the Dirac equation in TSRT arises not from quantization rules, operator methods, or wave-particle duality, but as an effective description of the coarse-grained internal geometry of a single, deterministically trembling particle. The entire formalism of relativistic quantum theory—complete with spin, antiparticles, Lorentz covariance, and probability amplitudes—emerges as the macroscopic statistical shadow of finer-scale geometric dynamics in trembling spacetime.

Why First-Order Time Derivatives Are Inevitable. In the Dirac sector, the first-order time evolution is a structural necessity. Since proper time must increase monotonically along worldlines, the coarse-grained statistical description must preserve this one-way ordering. A second-order time derivative (as appears in the Klein-Gordon equation) would allow solutions with negative-frequency components corresponding to backward-in-time propagation, conflicting with causal consistency. The trembling-induced Dirac equation ensures that the statistical field $\Psi(x)$ evolves through a first-order time derivative, maintaining causal integrity at the coarse-grained level. Thus, even the structure of relativistic quantum spinors is rooted in the deterministic, causal framework of trembling spacetime.

S18.5 From Special to General Relativistic Dirac Dynamics in TSRT

To describe particles with intrinsic spin, the trembling dynamics must respect the local spinor decomposition of Lorentz symmetry. The proper-time-resolved internal structure of worldlines leads, upon coarse-graining, to a multicomponent field $\Psi(x)$ satisfying a generalized Dirac Equation (S448) of the form:

$$\left(i\gamma^\mu\partial_\mu - \frac{m}{\mathcal{T}}\right)\Psi(x) = 0, \quad (\text{S474})$$

where γ^μ are the standard gamma matrices associated with the local Minkowski metric. This equation captures the trembling-induced correlation structure necessary to encode spin- $\frac{1}{2}$ degrees of freedom while preserving causal evolution along worldlines. Once again, quantum spinor dynamics emerge not from postulated quantum behavior, but as a statistical reflection of fine-scale, causally ordered trembling.

In the previous subsection, the emergence of the Dirac equation was derived under the assumption of a flat Minkowski background. However, TSRT applies equally well to curved

spacetimes, where curvature arises as a macroscopic consequence of large-scale trembling induced by matter. In this context, we now extend the derivation of the Dirac equation to accommodate general relativistic effects within the trembling spacetime framework. In general relativity, the motion of matter in a curved manifold is governed by the Einstein field equations, which relate energy-momentum to spacetime curvature [21]. In TSRT, curvature still appears as a macroscopic spacetime structure described by a smooth metric $g_{\mu\nu}(x)$, much like in general relativity. However, this curvature is not regarded as fundamental. Instead, it is interpreted as a large-scale, coarse-grained manifestation of deeper metric fluctuations—namely, the trembling of spacetime driven by matter and energy content at finer scales.

From this perspective, the full geometric background experienced by a particle in TSRT consists of two interwoven components: a smooth background curvature, comparable in form to the metric of general relativity, and a fine-scale trembling component that encodes microscopic, deterministic deviations from classical geodesic motion. The particle’s motion thus takes place within a spacetime that is both curved and trembling, with the background curvature shaping the large-scale structure of the manifold, and the trembling fluctuations governing the sub-resolution dynamics. Accordingly, the metric $g_{\mu\nu}(x)$ is treated as an effective description of the underlying trembling ensemble, capturing the average curvature induced by matter distributions. But on top of this, individual particles follow deterministic trembling paths, which must be modeled by extending the geodesic and spinor equations to include local inertial frames and spin transport across the curved background.

The Dirac Equation (S448) in flat spacetime, as derived in the previous section, emerges in TSRT not as a postulate, but as the statistical encoding of a single particle’s internal angular trembling when viewed through a coarse-grained geometric lens. The spinor field $\Psi(x)$ describes the projected evolution of internal degrees of freedom—specifically, deterministic angular oscillations—embedded within a trembling spacetime background. Crucially, this formulation rests on proper time evolution and local Lorentz invariance, not on quantum principles.

To extend this construction to curved spacetime, we remain entirely within the TSRT framework and ask: How must internal trembling be encoded when the particle moves through a region where spacetime curvature is no longer negligible? The answer begins with the recognition that in general relativity, the presence of mass–energy modifies the spacetime geometry itself. In TSRT, macroscopic curvature is not introduced as a fundamental field independent of trembling geometry. Rather, it emerges as the large-scale average of underlying metric fluctuations induced by matter and energy. However, as mentioned above, this does not imply that curvature and trembling are inseparable or indistinguishable in practice. On the contrary, it is both meaningful and operationally useful to decompose the full spacetime geometry into two components: a smooth background metric $\bar{g}_{\mu\nu}(x)$, representing the coarse-grained curvature analogous to general relativity, and a superimposed trembling component $\delta g_{\mu\nu}(x, \tau)$, encoding local, proper-time-dependent fluctuations, as we did in Equation (S28). Unlike conventional perturbative treatments of general relativity, where $\delta g_{\mu\nu}$ typically represents weak gravitational waves or matter-induced curvature, here it arises intrinsically from microscopic, causally constrained trembling of spacetime itself. It encodes local fluctuations that preserve Lorentz invariance on average but induce stochastic deviations at fine scales. Thus, while the background curvature

and trembling both originate from the same unified geometric dynamics, they operate at different scales and fulfill different roles: the background guides large-scale geodesics and defines the macroscopic structure of spacetime, whereas the trembling governs fine-scale particle dynamics and internal structure. This layered description allows TSRT to recover the predictive power of general relativity at large scales while providing a deeper, deterministic account of quantum-like effects at small scales.

A particle undergoing internal trembling in a curved spacetime cannot rely on globally defined inertial coordinates as in flat Minkowski geometry. Instead, its internal degrees of freedom must be referred to a locally inertial frame at each point along the worldline [50, 73, 74]. This is achieved through the introduction of a tetrad field (also called a vierbein), denoted $e_a^\mu(x)$, which serves as a local orthonormal basis at each spacetime point x . The tetrad relates the curved spacetime coordinates (with Greek indices $\mu, \nu, \dots$) to a locally flat Minkowski frame (with Latin indices $a, b, \dots$). The relationship is defined through the identity

$$g_{\mu\nu}(x) = e_\mu^a(x) e_\nu^b(x) \eta_{ab}, \quad (\text{S475})$$

where $g_{\mu\nu}(x)$ is the curved spacetime metric and $\eta_{ab} = \text{diag}(-1, +1, +1, +1)$ is the flat Minkowski metric. This formalism was first introduced by Weyl in 1928 as a geometric tool to unify gravity and electromagnetism [75], and was later rigorously developed into modern tetrad gravity by Cartan [76] and others.

Conversely, one can reconstruct the local Minkowski metric from the inverse tetrads,

$$\eta_{ab} = e_a^\mu(x) e_b^\nu(x) g_{\mu\nu}(x). \quad (\text{S476})$$

The tetrads $e_a^\mu(x)$ and their inverses $e_\mu^a(x)$ satisfy the orthonormality conditions,

$$e_a^\mu(x) e_\mu^b(x) = \delta_a^b, \quad (\text{S477})$$

$$e_\mu^a(x) e_b^\mu(x) = \delta_b^a. \quad (\text{S478})$$

In this construction, a vector V^μ in the curved coordinate basis is mapped to the local inertial frame by contraction with the tetrad:

$$V^a(x) = e_\mu^a(x) V^\mu(x), \quad (\text{S479})$$

and conversely, vectors in the local frame can be pushed forward to the curved spacetime frame,

$$V^\mu(x) = e_a^\mu(x) V^a(x). \quad (\text{S480})$$

This mapping allows the internal trembling variables—defined in local inertial coordinates as components $\chi^a(\tau)$ —to be consistently interpreted across the curved manifold. The evolution of these variables with respect to proper time must now include the effects of curvature, which enter via the spatial and temporal variation of the tetrad fields and the associated spin connection. Hence, in the TSRT framework, the tetrad $e_a^\mu(x)$ is not merely a mathematical device but a physical structure that links the locally defined internal trembling of a particle to the globally

curved geometry of spacetime. It plays a dual role. Geometrically it is translating between curved and flat frames; Dynamically, it is enabling the correct transport and coupling of spinor (trembling-internal) degrees of freedom across curved backgrounds. The trembling internal variables $\chi^\alpha(\tau)$ now evolve with respect to this local frame, not the global coordinate basis. The deterministic trembling evolution still occurs in proper time τ , but the local spacetime geometry now modulates the internal dynamics. This requires that any coarse-grained statistical field $\Psi(x)$ encoding this internal structure must evolve according to local inertial coordinates and be transported consistently across the curved geometry.

From this geometric viewpoint, the transition from the flat-spacetime Dirac equation to its curved-spacetime counterpart in TSRT arises as follows. Proper time remains the dynamical parameter in TSRT, even in curved spacetime. Each particle follows a deterministic path through a metric that now includes both trembling and curvature contributions. The internal trembling degrees of freedom are no longer embedded in a fixed Minkowski background. They must be referenced against the local orthonormal tetrad, which varies from point to point due to curvature. The trembling equations thus acquire explicit dependence on the local tetrad structure. The statistical field $\Psi(x)$, which encodes the averaged internal motion across an ensemble of trembling phases, must now be a spinor field defined on the curved manifold, transforming under the local Lorentz group defined by the tetrads. The transport of internal geometric structure across spacetime—specifically the directional rotation and phase of internal trembling—is governed by parallel transport in curved spacetime, which requires a spin connection $\Omega_\mu(x)$ to maintain Lorentz covariance.

The final form of the evolution equation is not imposed, but determined by the requirement that the coarse-grained statistical field $\Psi(x)$ respects proper time evolution, internal Lorentz covariance, and the geometric consistency of trembling across curved space.

To derive the evolution equation for a trembling particle in a curved spacetime within the TSRT framework, we must first account for the fact that spinors are defined only in flat spacetime. Since curved manifolds do not possess a global Minkowski structure, we rely on the tetrad formalism to introduce local inertial frames at every point in the manifold. Recall from Equation (S475) that the tetrad field $e_a^\mu(x)$ maps local inertial frame indices a to curved spacetime indices μ , and ensures that the local Minkowski metric η_{ab} and the global curved metric $g_{\mu\nu}(x)$ are related consistently.

Now, consider the flat-spacetime Dirac Equation (S448) that emerged in TSRT as a coarse-grained description of deterministic internal trembling:

$$i\mathcal{T}\gamma^\mu\partial_\mu\Psi(x) = mc\Psi(x). \quad (\text{S481})$$

In curved spacetime, this equation must be generalized in such a way that: it respects the local Lorentz symmetry of the internal trembling degrees of freedom, that it allows the spinor field $\Psi(x)$ to be transported covariantly over the curved manifold, and that it remains valid under general coordinate transformations.

The first modification is to replace the constant gamma matrices γ^μ , defined in flat coordinates, with local gamma matrices γ^a defined in each local inertial frame. These matrices satisfy

the Clifford algebra [6, 45, 59, 60, 77–79],

$$\{\gamma^a, \gamma^b\} = 2\eta^{ab}\mathbb{I}. \quad (\text{S482})$$

To translate these matrices back into curved spacetime coordinates, we use the tetrad field:

$$\gamma^\mu(x) := e^\mu_a(x)\gamma^a, \quad (\text{S483})$$

so that contractions like $\gamma^\mu\partial_\mu$ become locally meaningful.

The second modification is to upgrade the partial derivative ∂_μ to a covariant derivative that is appropriate for spinor fields. This is essential because spinors do not transform as tensors under general coordinate transformations; they require a special structure called the spin connection.

The covariant derivative for spinors in curved spacetime is defined as

$$D_\mu\Psi(x) := (\partial_\mu + \Omega_\mu(x))\Psi(x), \quad (\text{S484})$$

where $\Omega_\mu(x)$ is the spin connection, a crucial object that ensures the derivative transforms covariantly under both local Lorentz transformations and general coordinate changes.

The spin connection $\Omega_\mu(x)$ encodes how the local inertial frame rotates as one moves along the manifold. It is given by

$$\Omega_\mu(x) = \frac{1}{4}\omega_\mu^{ab}(x)\gamma_a\gamma_b, \quad (\text{S485})$$

where $\omega_\mu^{ab}(x)$ are the spin connection coefficients (or Ricci rotation coefficients), which depend on the tetrad field $e^\mu_a(x)$ and its derivatives. These coefficients arise from the requirement that the covariant derivative of the tetrad vanishes ($D_\mu e^\alpha_\nu = 0$), ensuring consistency between the curved spacetime and local Lorentz frames. Explicitly, they are computed as

$$\omega_\mu^{ab}(x) = e^a_\nu \left(\partial_\mu e^{b\nu} + \Gamma^\nu_{\mu\sigma} e^{b\sigma} \right) - e^b_\nu \left(\partial_\mu e^{a\nu} + \Gamma^\nu_{\mu\sigma} e^{a\sigma} \right), \quad (\text{S486})$$

where $\Gamma^\nu_{\mu\sigma}$ are the Christoffel symbols of the Levi-Civita connection.

This formalism was pioneered by Weyl and Cartan in the 1920s [76, 80] and later refined by physicists like Fock and Ivanenko to describe fermions in general relativity [81]. Modern treatments, such as those in [34, 60], rigorously derive these results from the principle of local gauge invariance under the Lorentz group.

In Equation (S486), $\Gamma^\nu_{\mu\sigma}$ is the Christoffel symbol constructed from the spacetime metric $g_{\mu\nu}(x)$, ensuring compatibility with the Levi-Civita connection (torsion-free and metric-preserving). Substituting these ingredients into the flat-space Dirac equation and expressing everything in terms of local geometric structures, we arrive at the fully covariant generalization within TSRT,

$$i\mathcal{T}\gamma^a e^\mu_a(x) (\partial_\mu + \Omega_\mu(x))\Psi(x) = mc\Psi(x). \quad (\text{S487})$$

This equation reduces to the flat-space TSRT Dirac equation when $e^\mu_a(x) \rightarrow \delta^\mu_a$ and $\Omega_\mu(x) \rightarrow 0$, it respects the local Lorentz invariance of the trembling-induced spinor degrees of freedom, and it ensures the correct parallel transport and phase evolution of $\Psi(x)$ in a curved trembling

background.

Thus, this equation emerges not by imposing minimal coupling or quantum quantization rules, but by requiring that the deterministic trembling dynamics of a single particle—defined with respect to local inertial frames—remain coherent and statistically interpretable as the particle propagates through a geometrically curved spacetime. This derivation is structurally identical to that used in quantum field theory in curved spacetime [45, 74], but conceptually distinct: here the entire structure emerges from deterministic internal motion projected statistically, rather than from postulated quantum principles.

In other words, Equation (S487) is mathematically equivalent to the covariant Dirac equation in general relativity [45, 74], but its physical origin is radically different. In quantum field theory, this equation is introduced via minimal coupling, imposed to maintain Lorentz invariance and covariance under general coordinate transformations. In TSRT, by contrast, it arises from coarse-graining the deterministic internal trembling of a single particle propagating through curved spacetime. The statistical field $\Psi(x)$ does not represent a wavefunction in a Hilbert space but a geometric envelope encoding how unresolved trembling phases behave on average in a locally curved setting. The curvature affects the transport and local behavior of trembling, but not the underlying determinism. It is only when coarse-grained over unresolved internal phase fluctuations that the motion appears probabilistic. Thus, the TSRT derivation of Equation (S487) shows that even in curved spacetime, quantum-like spinor dynamics arise as the statistical projection of deterministic internal geometry. This view replaces quantization as a primitive assumption and reveals spinor field evolution in general relativity as a secondary, emergent structure. Hence, although Equation (S487) is familiar in the context of quantum field theory in curved space [28, 45, 73, 74], its derivation in TSRT is novel. It is not introduced to ensure compatibility between general relativity and quantum mechanics, but instead follows from the deeper geometry of trembling spacetime itself. The coupling to curvature through the tetrad and spin connection is not an external imposition, but a natural requirement for preserving the coherence of internal trembling degrees of freedom across a curved manifold.

In this sense, TSRT offers not just a reinterpretation, but a reduction: quantum spinor dynamics in curved spacetime are no longer foundational but emergent from a geometric substructure. The mystery of quantization is resolved by the determinism of trembling motion, and the elegance of the Dirac equation is retained—not as a principle, but as a consequence. Otherwise said, the form of Equation (S487) is mathematically known from quantum field theory in curved spacetime (see e.g. [45, 74]), where it is typically introduced as a generalization of the flat-spacetime Dirac equation to incorporate gravitational effects. However, in those approaches, the equation is postulated based on symmetry and covariance requirements, and the underlying physical interpretation remains within the quantum field theoretic formalism.

By contrast, in the TSRT framework, this equation emerges as a statistical consequence of deterministic trembling dynamics, where both the curvature of spacetime and the internal spinor degrees of freedom are manifestations of the same geometric trembling substrate. The onset is not probabilistic but geometrically deterministic: the particle follows a trembling trajectory through curved spacetime, and the spinor field $\Psi(x)$ represents a coarse-grained statistical descriptor of the ensemble of such trembling paths, averaged over unresolvable internal phase

fluctuations.

Importantly, the curvature of spacetime—typically induced by mass-energy according to Einstein’s field equations—modifies the trembling background by shaping the local tetrad fields $e_a^\mu(x)$. This, in turn, alters the local propagation and phase rotation of the coarse-grained spinor $\Psi(x)$ via the spin connection $\Omega_\mu(x)$. In TSRT, this coupling is not imposed by analogy with minimal coupling procedures; it arises naturally because the trembling structure is embedded in the full geometric background, including both microscopic fluctuations and macroscopic curvature. Thus, the Dirac equation in curved spacetime within TSRT plays a dual role: it describes the local statistical dynamics of internal trembling degrees of freedom, and it incorporates curvature through geometric consistency with the underlying trembling spacetime. It bridges the microscopic scale (trembling proper time evolution) with the macroscopic gravitational field, without appealing to a wavefunction ontology or quantum postulates.

From this perspective, TSRT offers a physically grounded reinterpretation of the known covariant Dirac equation in curved spacetime. The equation is not merely a covariant extension for symmetry’s sake—it becomes the emergent projection of deeper deterministic laws, interpreted statistically. The appearance of general relativistic corrections to spinor dynamics is no longer a secondary quantization step, but rather a geometric consequence of the same trembling framework that gives rise to quantum structure in the flat limit. This reinterpretation also reshapes the conceptual interface between general relativity and quantum mechanics. It suggests that their apparent tension is not fundamental, but results from trying to unify two incomplete effective theories. In TSRT, both gravitational curvature and quantum spin arise from a unified geometric substrate—the trembling of spacetime itself. The extension of the Dirac equation to curved spacetime is well established within the framework of quantum field theory in curved backgrounds [28, 45, 73, 74] and can be compared with the derivation in TSRT outlined here. In those approaches in the literature, the Dirac field is treated as a fundamental quantum object, and its dynamics in curved geometry are formulated using tetrads, spin connections, and covariant derivatives. However, such formulations remain embedded in the probabilistic and operator-based formalism of quantum theory.

In contrast, the present derivation within TSRT is conceptually distinct: it does not assume quantization, nor does it treat the spinor field as fundamental. Instead, the Dirac-like dynamics arise from the coarse-grained description of deterministic trembling motion — including internal angular fluctuations — propagating in a curved spacetime whose geometry itself trembles. While the resulting equation coincides structurally with the curved-spacetime Dirac equation used in quantum field theory, its interpretation and origin are radically different. The mass term, spin structure, and conserved current are not imposed by symmetry or quantization but are derived from geometric and causal properties of the trembling motion. Thus, TSRT offers a deterministic geometric foundation for spinor dynamics in curved spacetime, providing a new physical explanation for mathematical structures that in conventional theory are introduced axiomatically.

S18.6 Conclusion

This supplementary section has demonstrated that both the Schrödinger and Dirac equations emerge as effective statistical consequences of deterministic trembling dynamics within the framework of Trembling Spacetime Relativity Theory (TSRT). In this framework, particles follow real, deterministic paths through a dynamically fluctuating geometric background. The observed quantum behavior of particles arises not from indeterminacy or wave duality, but from a coarse-grained description of unresolved trembling structure at sub-resolution scales. In the non-relativistic regime, we showed that spatial trembling contributes a geometric correction term to the classical Hamilton–Jacobi dynamics. This yields a modified ensemble-level equation that, when re-expressed in complex form, becomes structurally identical to the Schrödinger equation. Importantly, no quantum postulates are assumed: the emergence of the wavefunction, phase dynamics, and effective Planck-scale behavior is entirely rooted in the geometry of trembling trajectories. The analysis in Section S18.3 shows that the Schrödinger equation derived from trembling spacetime is not merely reproduced in its standard form but can be extended when amplitude gradients become non-negligible. By retaining curvature-sensitive terms in the coarse-grained dynamics, TSRT yields a refined evolution equation that captures corrections in regions of strong spatial variation of probability density. This refined formulation reduces to the conventional Schrödinger equation in the smooth-amplitude limit but opens the possibility of probing deviations in high-gradient quantum systems. It reflects the deeper geometric origin of quantum behavior in TSRT and suggests potential experimental signatures beyond the reach of canonical quantum mechanics. In the special relativistic regime, internal angular trembling degrees of freedom—obeying Lorentz-covariant geometric rules—project, upon coarse-graining, into a spinor field whose evolution is governed by a Dirac-type equation. The gamma matrix algebra, mass coupling, and antiparticle structure are not introduced by fiat but arise naturally from the algebraic and geometric properties of the trembling modes. The parameter $\mathcal{T}$, playing the role of an action scale, appears as a derived quantity from the internal trembling frequency and amplitude, rather than a fundamental constant like $\hbar$.

Crucially, we extended the derivation into the general relativistic regime. When trembling trajectories evolve through a curved spacetime background, the internal spinor field is shown to couple to curvature through a covariant derivative with spin connection, yielding the generally covariant Dirac equation. This extension is not postulated but follows necessarily from the requirement that trembling degrees of freedom evolve covariantly under parallel transport along a dynamically perturbed worldline. Thus, TSRT not only recovers but explains the appearance of the general relativistic Dirac equation from deterministic principles—a result previously absent from standard quantum derivations. In all regimes, the apparent quantum behavior emerges as a statistical projection of deeper causal geometry. TSRT provides a fully deterministic, geometrically grounded origin for the formal structures of both non-relativistic and relativistic quantum theory. The trembling of spacetime unifies classical mechanics, special and general relativity, and quantum behavior into a single coherent framework. The elegance of wave equations, spinor fields, and probabilistic dynamics is recast as a lens through which we observe the microscopic geometric fluctuations of spacetime itself.

In this light, TSRT fulfills the vision shared by Einstein, de Broglie, and Schrödinger—that

quantum theory is not an axiomatic structure, but an emergent phenomenon rooted in a deeper geometric substrate. Unlike hidden-variable models or pilot-wave theories, TSRT requires no reinterpretation of measurement or collapse: it replaces quantum axioms with causal geometry. It reconciles determinism with probabilistic appearance, local realism with spin and antimatter, and classical structure with quantum laws—all as shadows of the same trembling spacetime. Having classified the fundamental types of trembling modes and their associated interactions, we now turn to the microscopic causal structure underlying all trembling dynamics. We show that the requirement of causal resolution at the finest scales naturally gives rise to a fundamental action unit, corresponding to Planck’s constant. This emergence is not postulated, but derived as a necessary feature of the spacetime geometry itself.

S19 Supplementary Material: Planck's Constant as Minimal Causal Action and Trembling Cutoff in Spacetime

The emergence of a minimal action scale $\mathcal{S}_c = \hbar$ in TSRT is a direct consequence of the requirement that microscopic trembling fluctuations maintain causal distinguishability between neighboring worldlines. Without a lower bound on the effective phase-space volume, infinitesimally close trajectories would merge or collapse, violating proper-time monotonicity and the causal ordering of events. Thus, Planck's constant $\hbar$ does not arise as an external parameter, but as the minimal geometric unit $\mathcal{S}_c$ required to preserve causal separation at the finest resolvable scales of spacetime structure.

In the TSRT framework, trembling trajectories deform spacetime in a deterministic but fluctuating manner. This section demonstrates that when trembling is bounded by a physically motivated cutoff—imposed to preserve causal and temporal coherence—the energy per geometric mode becomes fixed, leading to a natural emergence of Planck's constant. Rather than being postulated, $\hbar$ appears as a consequence of the geometric limit on how densely spacetime can tremble. This derivation reveals a structural connection between $\hbar$ and the gravitational constant G , encoded in the curvature spectrum of trembling spacetime.

S19.1 Trembling-Induced Curvature and Energy Density

Let spacetime trembling be characterized locally by the geometric field $\Xi^\mu(\tau)$, defined in Equation (S389), which encodes small deviations from a classical geodesic. These deviations induce perturbations in the metric tensor. In general, the trembling metric takes the form:

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + \xi_{\mu\nu}, \quad (\text{S488})$$

as introduced in Equation (S28), where $\bar{g}_{\mu\nu}$ is the background geometry and $\xi_{\mu\nu}$ the trembling deviation field.

In a local inertial frame where $\bar{g}_{\mu\nu} = \eta_{\mu\nu}$, this reduces to:

$$g_{\mu\nu}(x) = \eta_{\mu\nu} + \delta g_{\mu\nu}(x), \quad (\text{S489})$$

with $\delta g_{\mu\nu} \equiv \xi_{\mu\nu}$. These metric perturbations give rise to local curvature.

In general relativity, the Ricci scalar R arises from contractions of the Riemann tensor, which in turn depends on second derivatives of the metric. In TSRT, the trembling field $\xi_{\mu\nu}(x)$ induces small, localized deviations from the background metric $\bar{g}_{\mu\nu}$, and the resulting curvature scalar is governed by the second-order structure of these fluctuations. To leading order, for a spatially localized trembling mode with characteristic amplitude ϵ and angular frequency ω , dimensional and analytic estimates, as done in further Section S19.6, yield:

$$R \approx \frac{\alpha \epsilon^2 \omega^2}{c^2}, \quad (\text{S490})$$

where α is a dimensionless numerical coefficient determined by the exact tensorial structure of the oscillating deviation field $\xi_{\mu\nu}$. A precise equation is derived below and results in Equa-

tion (S509). This relation captures the fact that curvature increases quadratically with amplitude and inversely with the square of the spatial wavelength $\lambda \sim c/\omega$. Although the coefficient α may vary between modes, the dependence on ϵ and ω reflects the universal geometric scaling of trembling-induced curvature.

To relate the trembling-induced curvature to energy content, we begin from the trace of Einstein's field equations. Recall that the standard Einstein field equations in the $(-+++)$ metric signature read:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \quad (\text{S491})$$

where $G_{\mu\nu}$ is the Einstein tensor, Λ is the cosmological constant (neglected here for local curvature induced by trembling), and $T_{\mu\nu}$ is the energy-momentum tensor.

The Einstein tensor itself is defined as

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R, \quad (\text{S492})$$

where $R_{\mu\nu}$ is the Ricci tensor and R is the Ricci scalar.

Taking the trace of the Einstein equations means contracting both sides with $g^{\mu\nu}$. Using the properties of the metric, we compute:

$$g^{\mu\nu} G_{\mu\nu} = g^{\mu\nu} R_{\mu\nu} - \frac{1}{2} g^{\mu\nu} g_{\mu\nu} R \quad (\text{S493})$$

$$= R - \frac{1}{2} (4) R \quad (\text{S494})$$

$$= R - 2R \quad (\text{S495})$$

$$= -R. \quad (\text{S496})$$

Thus, the trace of the left-hand side of Equation (S491) simplifies to $-R$.

Similarly, tracing the right-hand side of Equation (S491) gives

$$g^{\mu\nu} \left(\frac{8\pi G}{c^4} T_{\mu\nu} \right) = \frac{8\pi G}{c^4} g^{\mu\nu} T_{\mu\nu} = \frac{8\pi G}{c^4} T, \quad (\text{S497})$$

where we define

$$T = T^\mu{}_\mu = g^{\mu\nu} T_{\mu\nu} \quad (\text{S498})$$

as the trace of the energy-momentum tensor.

Thus, the traced Einstein equation becomes

$$-R = \frac{8\pi G}{c^4} T, \quad (\text{S499})$$

or equivalently

$$R = -\frac{8\pi G}{c^4} T. \quad (\text{S500})$$

Now, to interpret T , we consider the structure of the energy-momentum tensor $T_{\mu\nu}$ in a local inertial frame, where spacetime is approximately flat and gravitational fields are negligible. In this frame, $T_{\mu\nu}$ has dominant component T_{00} , representing the energy density, and smaller spatial components associated with stresses and pressures. Under the assumption of negligible

pressure and spatial stresses, the energy–momentum tensor reduces effectively to

$$T_{\mu\nu} \approx \rho c^2 \delta_{\mu 0} \delta_{\nu 0}, \quad (\text{S501})$$

where ρ is the mass density and c^2 accounts for energy density according to mass-energy equivalence.

The trace T then simplifies as

$$T = g^{\mu\nu} T_{\mu\nu} \quad (\text{S502})$$

$$\approx g^{00} T_{00} \quad (\text{S503})$$

$$= (-1) \times \rho c^2 \quad (\text{S504})$$

$$= -\rho c^2, \quad (\text{S505})$$

using $g^{00} = -1$ in the $(-+++)$ signature.

Substituting this into Equation (S500), we obtain:

$$R = -\frac{8\pi G}{c^4} (-\rho c^2) \quad (\text{S506})$$

$$= \frac{8\pi G}{c^2} \rho. \quad (\text{S507})$$

This result expresses the Ricci scalar R directly as an exact function of the local energy density ρ under the low-pressure, weak-stress limit, fully consistent with general relativity and directly applicable to the trembling-induced curvature considered in TSRT.

Solving for ρ , the local energy density can then be written as

$$\mathcal{E}_{\text{tremble}} = \rho c^2 = \frac{c^4}{8\pi G} R, \quad (\text{S508})$$

as rigorously discussed in Carroll [34] and adapted here to the context of geometric deviations induced by trembling. Importantly, in TSRT, the Ricci scalar R does not arise from conventional matter sources but from the inherent fluctuations in the metric structure itself. By combining the curvature estimate from Equation (S490) with the energy–curvature relation from Equation (S508), we obtain the trembling-induced energy density. To leading order, and assuming a localized oscillating deviation field $\xi_{\mu\nu}$ with amplitude ϵ and angular frequency ω , the induced Ricci scalar takes the form:

$$R = \alpha \frac{\epsilon^2 \omega^2}{c^2}, \quad (\text{S509})$$

where α is a dimensionless coefficient reflecting the geometric structure of the trembling mode. Substituting into the Einstein-based energy density Equation (S508),

$$\mathcal{E}_{\text{tremble}} = \frac{c^4}{8\pi G} R, \quad (\text{S510})$$

yields:

$$\mathcal{E}_{\text{tremble}} = \frac{\alpha c^2}{8\pi G} \epsilon^2 \omega^2. \quad (\text{S511})$$

This expression follows directly from first principles: the Einstein field equations, evaluated for a trembling-induced curvature, combined with the geometric structure of the deviation field. It is not a phenomenological assumption, but a physical consequence of the causal geometry encoded in TSRT. To compute the total energy associated with one localized trembling mode, we consider a spatial extent $\lambda = c/\omega$, corresponding to a spatial volume:

$$V = \lambda^3 = \left(\frac{c}{\omega}\right)^3. \quad (\text{S512})$$

The total geometric energy per trembling mode is then:

$$E_{\text{tremble}} = \mathcal{E}_{\text{tremble}} \cdot V = \frac{\alpha c^2}{8\pi G} \epsilon^2 \omega^2 \cdot \left(\frac{c}{\omega}\right)^3 = \frac{\alpha \epsilon^2 c^5}{8\pi G \omega}. \quad (\text{S513})$$

Equation (S513) is the exact energy per trembling mode at frequency ω , grounded entirely in the covariant geometry of TSRT and consistent with Einstein's general relativity. It shows that internal metric fluctuations act as energetic, causal sources of curvature, and that their energy content is precisely determined by amplitude, frequency, and the underlying spacetime response. The coefficient α in Equation (S490) reflects the geometric structure of the trembling mode and may in general depend on its polarization, spatial symmetry, and localization profile. However, for localized, isotropic trembling configurations—as expected from coarse-grained spacetime fluctuations constrained by causal coherence—it is known from general relativity and effective curvature averaging techniques (e.g., [34, 82, 83]) that the scalar curvature scales precisely with $R = \epsilon^2 \omega^2 / c^2$. This corresponds to $\alpha = 1$. We therefore adopt the convention that $\alpha = 1$ under causal, isotropic trembling conditions, and henceforth use:

$$R = \frac{\epsilon^2 \omega^2}{c^2}, \quad \mathcal{E}_{\text{tremble}} = \frac{\epsilon^2 c^2 \omega^2}{8\pi G}, \quad E_{\text{tremble}} = \frac{\epsilon^2 c^5}{8\pi G \omega}. \quad (\text{S514})$$

S19.2 The Cutoff Principle in TSRT

In TSRT, experienced proper time must remain real and positive: $d\tau^2 > 0$. Trembling cannot be arbitrarily violent without distorting the causal structure. This imposes a fundamental upper limit on both the frequency and amplitude of trembling. This limit defines a cutoff scale in the fabric of spacetime itself.

Let this cutoff correspond to a maximal curvature fluctuation consistent with causal coherence. We denote: In TSRT, there exists a maximal trembling frequency ω_c , representing the highest allowable oscillation rate consistent with causal structure, and a maximal trembling amplitude ϵ_c , denoting the largest geometrically admissible deviation without violating proper-time positivity. The two are interconnected. These are not adjustable parameters—they are structural properties of trembling spacetime, akin to the speed of light limiting velocities studied in Sections S23 and S24. Once the cutoff is reached, no further trembling information can be encoded geometrically without leading to negative proper time or nonphysical curvature.

S19.3 Planck's Constant from Cutoff Action per Mode

In trembling spacetime, each mode of oscillation contributes a localized packet of energy over a finite duration. The fastest allowed trembling mode is bounded by a fundamental frequency ω_c and amplitude ϵ_c , beyond which causality or curvature constraints forbid sharper structure. For such a mode, the energy content, based on Equation (S514), is

$$E_{\text{tremble}} = \frac{\epsilon_c^2 c^5}{8\pi G \omega_c}, \quad (\text{S515})$$

and the temporal duration of a single oscillation is:

$$T = \frac{2\pi}{\omega_c}. \quad (\text{S516})$$

We now compute the action associated with this mode — defined as the product of energy and duration — to obtain:

$$\mathcal{S}_c = E_{\text{tremble}} \cdot T = \left(\frac{\epsilon_c^2 c^5}{8\pi G \omega_c} \right) \cdot \left(\frac{2\pi}{\omega_c} \right) = \frac{\epsilon_c^2 c^5}{4G \omega_c^2}. \quad (\text{S517})$$

This quantity, $\mathcal{S}_c$, has units of action and is invariant under coordinate transformations. It represents the minimal action content per localized trembling excitation at the cutoff scale. In TSRT, the existence of a trembling cutoff is a direct consequence of the fundamental causality condition: the local proper time must always evolve forward, and no segment of a trembling path may correspond to a reversal of time direction. This requirement imposes a strict constraint on the allowable modes of spacetime fluctuations. Specifically, to maintain forward evolution of proper time, only full oscillations—complete periods—can be meaningfully associated with physical structures. As will be shown in Section S19.4, partial oscillations would imply local backward time evolution within a period, violating causality at the most fundamental level. Consequently, the trembling field cannot support wavelengths shorter than a critical cutoff length, nor periods shorter than a critical cutoff time.

In this framework, the cutoff quantities $2\pi/\omega_c$ (minimal allowed proper time duration) and ϵ_c (minimal allowed amplitude of trembling) emerge naturally. Physical trembling paths are necessarily coarse-grained over at least one full period of oscillation, ensuring that all internal fluctuations preserve a globally forward evolution of proper time without any reversal. Thus, the cutoff reflects a geometric threshold intrinsic to the causal structure of trembling spacetime: no energetic or geometric structure smaller than these limits can be physically resolved or meaningfully defined. Because this action scale arises as the universal bound on localized trembling modes, it plays an identical role to Planck's constant in conventional quantum mechanics — not by assumption, but by geometric necessity. We therefore define

$$\hbar \equiv \mathcal{S}_c = \frac{\epsilon_c^2 c^5}{4G \omega_c^2}. \quad (\text{S518})$$

and show further that it leads to a relation first postulated by Planck, in Section S19.4. Equation (S518) derives Planck's constant in terms of the gravitational coupling G , the speed of light

c , and the TSRT cutoff parameters ϵ_c and ω_c . Inverting the relation yields:

$$\epsilon_c = \sqrt{\frac{4G\hbar\omega_c^2}{c^5}}, \quad (\text{S519})$$

showing that the amplitude of permissible spacetime trembling is constrained by the requirement that the action per cutoff mode match $\hbar$, because, otherwise, causality would be violated.

Thus, Planck's constant is not a fundamental input to TSRT, but rather an emergent scale fixed by the maximal trembling geometry that spacetime can support without violating causality or geometric consistency. This reframes $\hbar$ as a boundary-action per mode — a geometric consequence of the causal cutoff, not a quantum axiom.

S19.4 Derivation of the Planck–Einstein Relation from Causal Geometry

The Planck–Einstein relation, $E = \hbar\omega$, traditionally enters physics as an empirical postulate motivated by black body radiation and the photoelectric effect. In TSRT, however, this relation arises as an inevitable consequence of the causal and geometric structure of trembling spacetime.

In this section, we derive $E = \hbar\omega$ directly from first principles: the requirement that trembling-induced action occur only in complete causal cycles, the emergence of a minimal invariant action scale $\hbar$, and the spectral consequences of improper trembling truncation. Quantization of energy is thus shown to follow not from statistical assumptions or operator algebra, but from the preservation of causal evolution under proper-time-parametrized metric fluctuations.

Having derived $\hbar$ as the action associated with the most intense geometrically allowed trembling mode—the cutoff configuration defined by (ϵ_c, ω_c) —we now examine how this leads to a universal quantization principle across all trembling frequencies ω . In TSRT, the action $\mathcal{S}$ of a cyclic trembling mode is given in Equation S517, namely

$$\mathcal{S} = E \cdot T, \quad (\text{S520})$$

where E is the energy and $T = 2\pi/\omega$ is the oscillation period. This relation holds for all localized modes and is manifestly Lorentz-invariant, as both energy and time transform reciprocally under boosts. This follows from the transformation properties of energy and time under Lorentz boosts; namely, energy transforms as the timelike component of the four-momentum $p^\mu = (E/c, \mathbf{p})$, while time transforms as the timelike component of the position four-vector $x^\mu = (ct, \mathbf{x})$. Hence, under a boost with velocity v along the x -axis, these quantities transform reciprocally as

$$E' = \gamma(E - vp_x), \quad T' = \gamma\left(T - \frac{vx}{c^2}\right), \quad (\text{S521})$$

where $\gamma = (1 - v^2/c^2)^{-1/2}$. For a system at rest ($\mathbf{p} = 0$, $\mathbf{x} = 0$), this reduces to $E'T' = ET$, demonstrating the invariance of the product [6, 37, 57].

Since both E and T are defined with respect to proper time intervals along the trembling path, this expression for $\mathcal{S}$ is frame-independent and geometrically meaningful. It holds for all localized trembling modes, irrespective of their energy or duration, as long as causality is preserved. The special trembling mode at the causal cutoff—where the amplitude and frequency

reach their maximal values consistent with $d\tau^2 > 0$ —yields an action, shown in Equation (S517),

$$\mathcal{S}_c = \frac{\epsilon_c^2 c^5}{4G\omega_c^2}. \quad (\text{S522})$$

TSRT identifies this value, for instance via dimensional analysis in Section S19.6, as the fundamental action quantity, provided in Equation (S519), being

$$\hbar \equiv \mathcal{S}_c. \quad (\text{S523})$$

Importantly, since $\mathcal{S}_c$ is a scalar derived from spacetime curvature and proper-time constraints, it is invariant under Lorentz transformations and cannot be reduced further without violating causal geometry. Thus, $\hbar$ represents the minimal Lorentz-invariant unit of trembling action that preserves causal coherence in spacetime.

Indeed, suppose a trembling configuration were to produce an action smaller than $\mathcal{S}_c$. This would require a mode of either higher frequency $\omega > \omega_c$ or larger curvature amplitude $\epsilon > \epsilon_c$, forbidden per definitionem. However, as defined in Section S19.3, any such configuration necessarily leads to negative or imaginary proper time—violating the causal structure of spacetime. Therefore, trembling configurations yielding $\mathcal{S} < \mathcal{S}_c$ are not physically realizable. Because $\mathcal{S}_c$ is a Lorentz-invariant scalar, and no physically allowed trembling configuration can produce an action below this value without violating causality, it follows that $\mathcal{S}_c = \hbar$ is the minimal Lorentz-invariant action consistent with the causal structure of spacetime. This identification of $\hbar$ as the minimal allowable action per trembling mode is therefore not arbitrary. It is enforced by the deepest geometric structure of TSRT and admits no further reduction without invalidating the theory’s foundational constraint on proper-time realism.

As a result of the causal cutoff structure in trembling spacetime, any physically allowed trembling configuration must consist of a whole number of full oscillation periods. This requirement follows directly from the necessity of maintaining global forward evolution of proper time τ along every segment of the trembling path.

In TSRT, the proper time τ is not a static background parameter but is determined dynamically through the local spacetime metric, fluctuating together with the trembling structure. The relation between τ and the coordinate functions $x^\mu(\tau)$, including the coordinate time $t(\tau)$, is mediated by the local metric deviations. Therefore, the fundamental causality condition established in Equation (S119) demands that τ must advance monotonically relative to the coordinate paths, prohibiting any local reversal or decrease:

$$\frac{d\tau}{d\lambda} > 0, \quad (\text{S524})$$

where λ is any auxiliary affine parameter used to describe the trembling paths.

Because trembling configurations must consist of complete causal oscillations without spectral contamination, the geometric action accumulated over each full period must likewise be quantized into invariant units, setting the foundation for the emergence of Planck’s constant as a minimal action quantum. To understand why incomplete trembling cycles are forbidden within TSRT, consider a local segment of proper-time oscillation truncated before completing

a full period:

$$\delta t(\tau) = \begin{cases} \sin(\omega\tau), & 0 \leq \tau \leq \tau_0, \\ 0, & \text{otherwise,} \end{cases} \quad (\text{S525})$$

with $\tau_0 < T = 2\pi/\omega$ corresponding to an incomplete cycle.

The Fourier transform of this truncated signal is:

$$\tilde{\delta t}(\nu) = \int_0^{\tau_0} \sin(\omega\tau) e^{i2\pi\nu\tau} d\tau, \quad (\text{S526})$$

which leads to spectral components distributed across a wide range of frequencies:

$$\tilde{\delta t}(\nu) = \frac{1}{2i} \left(\frac{e^{i(\omega+2\pi\nu)\tau_0} - 1}{\omega + 2\pi\nu} - \frac{e^{i(-\omega+2\pi\nu)\tau_0} - 1}{-\omega + 2\pi\nu} \right). \quad (\text{S527})$$

This shows that truncation introduces high-frequency harmonics above the fundamental ω , in violation of the trembling frequency cutoff ω_c defined in Equation (8). Additionally, low-frequency sidebands $\nu \ll \omega/2\pi$ also emerge, producing large-wavelength modulations.

In the TSRT framework, the local metric is given by:

$$g_{\mu\nu}(\tau) = \eta_{\mu\nu} + \xi_{\mu\nu}(\tau), \quad (\text{S528})$$

where $\eta_{\mu\nu}$ is the flat background and $\xi_{\mu\nu}(\tau)$ encodes proper-time fluctuations.

When the oscillation is incomplete, $\xi_{\mu\nu}(\tau)$ contains spurious low-frequency terms:

$$\xi_{\mu\nu}(\tau) \sim A_0 \cos(2\pi\nu_0\tau) + \sum_n A_n \cos(2\pi\nu_n\tau), \quad (\text{S529})$$

with $\nu_0 \ll \omega/2\pi$. These long-wavelength modes affect the causal proper-time increment:

$$d\tau^2 = (\eta_{\mu\nu} + \xi_{\mu\nu}(\tau)) \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} d\lambda^2. \quad (\text{S530})$$

This expands to:

$$d\tau^2 = d\tau_0^2 + \delta(d\tau^2), \quad (\text{S531})$$

$$d\tau_0^2 = \eta_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} d\lambda^2, \quad (\text{S532})$$

$$\delta(d\tau^2) = \xi_{\mu\nu}(\tau) \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} d\lambda^2. \quad (\text{S533})$$

If the long-wavelength perturbation $\delta(d\tau^2)$ is negative and sufficiently large, then $d\tau^2$ may become negative, rendering $d\tau$ imaginary, which violates the strict causal progression constraint:

$$\frac{d\tau}{d\lambda} > 0. \quad (\text{S534})$$

Moreover, inclusion of high-frequency components beyond ω_c breaks the TSRT frequency bound derived from curvature constraints, making such truncations doubly forbidden: they introduce both causality-violating low-frequency distortions and unphysical high-frequency ex-

citations.

Therefore, TSRT forbids partial trembling periods not merely due to analytic discontinuities, but because they violate both the geometric frequency cutoff and the forward causal propagation of proper time. Full-period oscillations are the only allowable modes consistent with TSRT's foundational causal geometry. Thus, the total trembling-induced action $\mathcal{S}$ associated with any physical mode must be an integer multiple of a minimal action quantum $\hbar$, corresponding to one full causal oscillation. We therefore write:

$$\mathcal{S}_n = n\hbar, \quad \text{with } n \in \mathbb{N}, \quad (\text{S535})$$

where n counts the number of full trembling periods, and $\mathbb{N}$ denotes the set of positive integers. Combining this quantized action structure of Equation (S535), with the general relation between action, energy, and time for oscillatory modes,

$$\mathcal{S} = E \cdot T, \quad (\text{S536})$$

where T is the oscillation period and $\omega = 2\pi/T$ is the corresponding angular frequency, we obtain

$$E = \frac{\mathcal{S}_n}{T} = \frac{n\mathcal{S}_c}{T} = \frac{n\hbar}{T} = n\mathcal{S}_c\omega = n\hbar\omega. \quad (\text{S537})$$

This is the full form of Planck's postulate, originally proposed in 1901 [84]:

$$E = n\hbar\omega, \quad n \in \mathbb{N}, \quad (\text{S538})$$

under the assumption of Equation (S523). In this view, quantization arises not from operator algebra, curve fitting to experimental data, or statistical rules, but from the causal and geometric constraints on spacetime trembling itself. The discreteness of energy is a consequence of the requirement that trembling-induced action must occur in integer multiples of a fundamental invariant action unit $\hbar$, reflecting the necessity of completing full oscillation cycles under causal proper-time evolution.

Allowing fractional values of n would correspond to partial trembling cycles, which would violate local causality, break Lorentz covariance, and destabilize the emergent spacetime geometry. Therefore, only positive integer multiples are permitted, and $n \in \mathbb{N}$ is strictly enforced.

TSRT thus provides a geometric derivation of Planck's postulate [84]: not only is $E = \hbar\omega$ recovered for the minimal cutoff mode, but the universal structure $E = n\hbar\omega$ emerges naturally as a structural rule governing all trembling excitations permitted by the causal topology of spacetime. Unlike in conventional quantum theory, where $E = n\hbar\omega$ is imposed ad hoc to match black-body radiation and photoelectric observations, TSRT shows that this energy quantization is not assumed but rather enforced by the geometric and causal structure of reality itself. TSRT thus provides a geometric and causal foundation for Planck's postulate [84]. The relation $E = \hbar\omega$ is not imposed externally, nor introduced to fit empirical data. Instead, it emerges as the lowest-energy instance of a more general quantization rule:

$$E = n\mathcal{S}_c\omega = n\hbar\omega, \quad n \in \mathbb{N}, \quad (\text{S539})$$

governing all trembling excitations permitted by the causal structure of spacetime.

This quantization is enforced by the requirement that every physically admissible trembling mode must correspond to an integer number of full causal oscillations. Partial or truncated cycles would introduce low-frequency modulations that violate proper-time monotonicity and destabilize local causal ordering. As a result, energy quantization in TSRT is not optional but structurally enforced: a direct manifestation of the causal coherence of trembling spacetime. Unlike conventional quantum theory, where $E = n\hbar\omega$ is postulated or derived from operator spectra, TSRT reveals this law as a consequence of deeper geometric necessity. Trembling is not merely a kinematic perturbation—it is the engine of quantization itself.

S19.5 The Classical Limit as Coarse-Grained Trembling Geometry

In the TSRT framework, classical physics does not arise from taking $\hbar \rightarrow 0$, as this would contradict the geometric and causal foundation of Planck’s constant, which emerges from the trembling structure of spacetime itself. Instead, the classical regime corresponds to a situation where the total action scale characterizing the system is much larger than the minimal trembling action scale $\mathcal{T} \sim \hbar$, while $\mathcal{T}$ remains finite and universal. This regime is defined by the inequality:

$$\frac{\mathcal{T}}{\mathcal{S}_{\text{sys}}} \ll 1, \quad (\text{S540})$$

where $\mathcal{S}_{\text{sys}}$ denotes the characteristic total action of the system — either accumulated over its dynamical evolution (for long-lived trajectories) or integrated over its spatial or compositional extent (for large, multi-particle, or high-mass systems). In either case, the cumulative effect of fine-scale trembling is averaged out, and the system’s trajectory becomes effectively smooth.

It is important to emphasize that $\mathcal{T}$ represents the minimal unit of causal action allowed by the trembling structure of spacetime, but this does not imply that the total system action must be quantized strictly as $\mathcal{S}_n = n\mathcal{T}$ for some integer n . Rather, the relation $\mathcal{S}_n = n\mathcal{T}$ may arise approximately in specific geometrically coherent modes or harmonic configurations, but is not a general postulate. TSRT accommodates more general configurations where the total action $\mathcal{S}_{\text{sys}}$ exceeds $\mathcal{T}$ without requiring exact integer multiples. The classical limit therefore does not rely on taking $\mathcal{T} \rightarrow 0$ or $n \rightarrow \infty$ individually, but only on the scale separation $\mathcal{T}/\mathcal{S}_{\text{sys}} \ll 1$. This can arise from long lifetimes, extended spatial domains, or composite structures with many contributing degrees of freedom. Classical physics thus emerges as a coarse-grained geometric approximation, valid when the trembling structure is finely layered, spectrally dense, and statistically smoothed across macroscopic resolution scales. In this large-action limit, the discrete structure of proper-time trembling averages out under coarse-graining, and the emergent evolution of the system follows smooth, classical trajectories. The system’s dynamics is governed by an effective variational principle:

$$\delta\mathcal{S}_{\text{eff}}[x^\mu] = 0, \quad (\text{S541})$$

where $\mathcal{S}_{\text{eff}}$ is the coarse-grained action resulting from integrating over many full trembling cycles. This leads to the familiar classical equations of motion: Newton’s laws, the Lorentz force law, or geodesic motion in curved spacetime, depending on the interaction sector.

Thus, classical mechanics emerges as the deterministic, smooth limit of TSRT when the

system's energy and action exceed the geometric trembling scale by many orders of magnitude. Trembling still occurs in this regime, but its effects are averaged out and do not manifest at observable scales. This replaces the conventional quantum-to-classical transition (e.g., via $\hbar \rightarrow 0$) with a geometric criterion based on action scale separation. Importantly, this classical limit retains full consistency with the curvature-regulated uncertainty structure developed earlier. Even in the large- n regime, the phase-space volume per degree of freedom remains bounded from below by $\hbar$, preserving the causal coherence of the underlying spacetime. The emergence of classical determinism is therefore a collective effect of many trembling modes acting in causal unison—not a violation of quantum discreteness, but its macroscopic manifestation. Having shown that classical physics emerges as the coarse-grained limit of quantized trembling dynamics, we now examine how the fundamental constants that appear in this classical regime—such as G , $\hbar$, and c —arise from the internal dimensional structure of trembling spacetime itself.

S19.6 Consistency with the Planck Scale and Dimensional Analysis

In conventional physics, the Planck scale arises by dimensional combination of three constants: Newton's gravitational constant G , Planck's constant $\hbar$, and the speed of light c . These are often taken as fundamental, defining units of length, time, mass, and energy where quantum gravitational effects become significant. In TSRT, however, these constants are not externally inserted—they emerge from the internal structure of trembling spacetime itself.

As shown in previous sections, $\hbar$ arises from the minimal action per complete causal oscillation, c emerges as the maximal signal speed enforced by the null condition on trembling propagation, and G characterizes the coupling strength between coarse-grained metric curvature and effective energy-momentum. Using dimensional analysis, we recover the usual Planck units as consistency checks on the scale hierarchy implied by TSRT. The Planck length ℓ_P , time t_P , and mass m_P are given by:

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}, \quad (\text{S542})$$

$$t_P = \sqrt{\frac{\hbar G}{c^5}}, \quad (\text{S543})$$

$$m_P = \sqrt{\frac{\hbar c}{G}}. \quad (\text{S544})$$

These scales define the minimal length and time resolutions, and the maximal mass-energy scale, compatible with both trembling-induced uncertainty and gravitational backreaction.

Role of the Planck Scale in TSRT. In TSRT, these Planck units do not mark the breakdown of physical theory, but rather the crossover point where trembling deviations saturate causal constraints. At scales larger than ℓ_P , geometric deviations remain coherent and subluminal; at scales near or below ℓ_P , causal coherence requires that proper time remain monotonic, bounding the amplitude and frequency of allowable trembling.

The preceding results show that Planck's constant $\hbar$ arises in TSRT as the action quantity associated with the most intense geometrically admissible trembling mode—defined by the

causal cutoff on oscillation frequency and amplitude. This cutoff scale is determined internally by TSRT's requirement that proper time remain real and positive: the trembling metric must never produce backward or imaginary temporal intervals. To evaluate whether this internally derived cutoff aligns with known physical limits, we compare it with the conventional Planck length ℓ_P and Planck frequency ω_P , both of which emerge in dimensional analysis when combining the gravitational constant G , the speed of light c , and Planck's constant $\hbar$. Recall the TSRT-derived expression for Planck's constant in Equation (S518),

$$\hbar = \frac{\epsilon_c^2 c^5}{4G\omega_c^2}, \quad (\text{S545})$$

where ϵ_c and ω_c are the maximal trembling amplitude and frequency, defined by the causal structure of spacetime. Solving for ϵ_c yields:

$$\epsilon_c = \sqrt{\frac{4G\hbar\omega_c^2}{c^5}}. \quad (\text{S546})$$

Let us now examine whether the values of ω_c and ϵ_c implied by this relation naturally correspond to known physical scales. Suppose that the internal geometric cutoff frequency ω_c , derived from the demand that no trembling mode violates the causal condition $d\tau^2 > 0$, reaches the value:

$$\omega_c = \sqrt{\frac{c^5}{\hbar G}}, \quad (\text{S547})$$

which is the frequency scale at which the causal curvature effects of trembling approach unity in Planck units. Substituting into the above gives:

$$\epsilon_c = \sqrt{\frac{4G\hbar}{c^5} \cdot \frac{c^5}{\hbar G}} = \sqrt{4} = 2, \quad (\text{S548})$$

which yields:

$$\epsilon_c = 2\ell_P, \quad (\text{S549})$$

where $\ell_P = \sqrt{\hbar G/c^3}$ is the Planck length.

This confirms that the causal cutoff derived from TSRT leads to trembling amplitudes and frequencies lying naturally at the Planck scale, without requiring this scale to be imposed externally. The agreement is a consistency check—not an assumption—and reinforces the interpretation of Planck's constant as the minimal action per mode consistent with trembling-induced curvature at the causal boundary. To further verify consistency, we examine the dimensional structure of the TSRT expression for $\hbar$:

$$\hbar = \frac{\epsilon_c^2 c^5}{4G\omega_c^2}. \quad (\text{S550})$$

The units of $\epsilon_c^2 c^5$ are:

$$[\epsilon_c^2 c^5] = \text{L}^2 \cdot (\text{L} \cdot \text{T}^{-1})^5 = \text{L}^7 \cdot \text{T}^{-5},$$

while the units of $G\omega_c^2$ are:

$$[G\omega_c^2] = \text{L}^3 \cdot \text{M}^{-1} \cdot \text{T}^{-6},$$

resulting in:

$$[\hbar] = \frac{L^7 \cdot T^{-5}}{L^3 \cdot M^{-1} \cdot T^{-6}} = M \cdot L^4 \cdot T = M \cdot L^2 \cdot T^{-1},$$

which matches the correct dimensions of action.

In other words, TSRT not only predicts Planck's constant from geometric principles rooted in causal trembling structure, but also demonstrates that its value and physical scaling are consistent with the Planck regime. This consistency strengthens the identification of $\hbar$ as an emergent action quantity, defined by spacetime's structural limitations. The appearance of Planck units is not imposed externally—they arise naturally when the causal geometry of trembling saturates its maximal permissible curvature. This anchors quantum discreteness in the deterministic, covariant trembling of spacetime.

No Arbitrary Units or UV Cutoffs. Unlike effective field theory, TSRT does not impose arbitrary ultraviolet cutoffs. Instead, the Planck scale appears dynamically through the interplay of Lorentz invariance, proper-time causality, and curvature-induced geometric suppression. The minimal action scale $\hbar$, once linked to cutoff trembling configurations, combines with G and c to define the boundary between continuous classical evolution and discrete causal geometry.

Thus, dimensional analysis serves here not as a method of conjecture but as a consistency check on the causal quantization structure already embedded in the trembling framework. The Planck units emerge not as inputs but as structural byproducts of the theory's internal geometry.

S19.7 Physical Meaning of Planck's Constant in TSRT

In TSRT, Planck's constant $\hbar$ emerges not as a quantum mechanical postulate, but as a causal geometric invariant dictated by the structure of trembling spacetime. It represents the minimal action quantity $\mathcal{S}_c$ associated with a single localized trembling mode at the causal cutoff scale—the highest-frequency, smallest-amplitude oscillation spacetime can sustain without violating proper-time realism. Beyond this threshold, trembling would disrupt the positivity of $d\tau^2$, rendering such configurations physically inadmissible. The cutoff scale is therefore not imposed but arises from TSRT's internal requirement that all geometric evolution be causally coherent and Lorentz-invariant. Planck's constant emerges as the critical action content per mode that spacetime permits under these constraints. In this view, $\hbar$ is a structural feature of the metric itself, intrinsically tied to the gravitational constant G , the speed of light c , and the limit geometry of trembling spacetime.

This interpretation offers a deeper unification: quantum discreteness and gravitational curvature are not disparate phenomena, but different aspects of the same causal geometric substrate. The celebrated relation $E = \hbar\omega$ is not an imposed quantization rule—it is the inevitable consequence of a universe whose metric degrees of freedom are bounded by causal geometry. Planck's constant thus ceases to be a bridge between classical and quantum regimes; it becomes a measurable imprint of spacetime's refusal to tremble beyond what causal structure allows. In TSRT, Planck's constant $\hbar$ arises unavoidably as the minimal causal action required to preserve distinguishability among neighboring worldlines and maintain the coherence of spacetime ge-

ometry. It is not an arbitrary constant to be measured and inserted into physical laws, but an inevitable feature of the geometry and dynamics of trembling spacetime. Every quantum phenomenon — uncertainty, discreteness, interference — ultimately traces back to this microscopic causal resolution scale.

Having identified the minimal causal action scale, we now turn to the double-bounded uncertainty structure that trembling spacetime imposes on all physical trajectories, and will retrieve Heisenberg's uncertainty.

S20 Supplementary Material: Derivation of Double-Bounded Uncertainty from Causal Structure

In standard quantum mechanics [85], Heisenberg's uncertainty principle [3] establishes that the product of position and momentum uncertainties is fundamentally bounded from below. It arises from the canonical commutation relation $[\hat{x}, \hat{p}] = i\hbar$, with the precise inequality first derived by Kennard [86] and generalized by Weyl [75]:

$$\Delta x \Delta p \geq \frac{\hbar}{2}. \quad (\text{S551})$$

Traditionally, this relation is interpreted as an intrinsic feature of quantum systems, rooted in operator algebra and probabilistic frameworks. In TSRT, however, the uncertainty principle emerges from purely geometric and causal constraints imposed by trembling spacetime. Moreover, it acquires a richer structure: not only is there a minimal uncertainty product, but causal coherence also enforces an upper bound on phase-space resolution, regulated by local spacetime curvature.

This section derives the full double-bounded uncertainty relation from first principles, revealing how the causal structure of trembling spacetime fundamentally limits both the minimal and maximal spread of physical trajectories.

S20.1 Causal Constraints on Trembling Worldlines

In TSRT, the fine-scale structure of spacetime is characterized by deterministic trembling deviations $\Xi^\mu(\tau)$ superposed upon smooth worldlines $\bar{x}^\mu(\tau)$. These deviations are not arbitrary: they are constrained by the fundamental requirement of causal coherence. The worldline of a particle is thus expressed as in Equation (2),

$$x^\mu(\tau) = \bar{x}^\mu(\tau) + \Xi^\mu(\tau), \quad (\text{S552})$$

where $\bar{x}^\mu(\tau)$ is the coarse-grained trajectory and $\Xi^\mu(\tau)$ represents the causal, Lorentz-invariant deviation field arising from trembling. The proper-time increment along any physical path must remain strictly positive to preserve the arrow of time, as in Equation (S534),

$$\frac{d\tau}{d\lambda} > 0, \quad (\text{S553})$$

where λ is an auxiliary affine parameter along the worldline. Locally, the perturbed spacetime interval satisfies

$$d\tau^2 = g_{\mu\nu} dx^\mu dx^\nu, \quad (\text{S554})$$

and causal structure demands that $d\tau^2 \geq 0$ for all infinitesimal displacements. Trembling deviations $\Xi^\mu(\tau)$ must therefore remain within the light cone at every point, forbidding spacelike excursions that would allow superluminal propagation. These causal conditions impose strict geometric limits on the amplitude, frequency, and correlation properties of the trembling field. In particular, they define a minimal and maximal admissible spread in phase space, which under

coarse-graining manifests as uncertainty relations between position and momentum.

Furthermore, the local deviation field $\Xi^\mu(\tau)$ must remain within the light cone at each point to preserve local Lorentz structure. These constraints impose a minimum and maximum limit on how rapidly the trajectory can deviate in phase space (x^i, p_i) . Rather than relying on operator commutation relations, TSRT attributes uncertainty to deterministic, microscopic fluctuations in the geometry of spacetime. These are modeled by writing the metric, as introduced in (S28), as:

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + \xi_{\mu\nu}, \quad (\text{S555})$$

where $\bar{g}_{\mu\nu}$ is the smooth background metric, typically taken to be Minkowskian in flat regions, and $\xi_{\mu\nu}$ encodes localized geometric perturbations arising from spacetime trembling. These fluctuations affect geodesic motion, perturbing the trajectory of a particle away from the idealized path it would follow in $\bar{g}_{\mu\nu}$ alone. The deviations in the particle's observable position and momentum arise as effective statistical quantities that reflect this trembling.

Let the position be perturbed according to:

$$x \rightarrow x + \xi_x(\tau), \quad (\text{S556})$$

and the momentum as:

$$p \rightarrow p + \xi_p(\tau), \quad (\text{S557})$$

where $\xi_x(\tau)$ and $\xi_p(\tau)$ are time-dependent deviations along the particle's proper-time parameterized worldline. These fluctuations are not stochastic but arise deterministically from the underlying geometry. The resulting spreads in position and momentum, when coarse-grained over unresolved internal phases of trembling, give rise to the emergent uncertainty relation in TSRT.

S20.2 Geometric Covariance Functional and Phase-Space Spread

To quantify the effective spread induced by trembling deviations in phase space, we define the geometric covariance functional $g(\xi)$. Locally, the trembling field produces deviations in position and momentum:

$$\xi_x = \Xi^i(\tau), \quad (\text{S558})$$

$$\xi_p = m \dot{\Xi}_i(\tau), \quad (\text{S559})$$

where m is the particle mass and the dot denotes derivative with respect to proper time τ . The covariance functional is defined as the coarse-grained expectation value:

$$g(\xi) := 2\langle \xi_x \xi_p \rangle = \left\langle \Xi^i(\tau) m \dot{\Xi}_i(\tau) \right\rangle, \quad (\text{S560})$$

which characterizes the effective phase-space area occupied by a trembling trajectory at a given resolution scale.

In TSRT, the statistical spread of position and momentum fluctuations is directly linked to

this geometric covariance. Under coarse-graining, the effective uncertainty relation becomes:

$$\Delta x \Delta p \sim g(\xi), \quad (\text{S561})$$

where Δx and Δp denote the root-mean-square deviations in position and momentum, respectively. Define $\Delta x = \sqrt{\langle \xi_x^2 \rangle}$ and $\Delta p = \sqrt{\langle \xi_p^2 \rangle}$, where the averages are taken over unresolved internal phases in the trembling trajectory. These represent effective spreads in position and momentum, determined by geometric correlations in the trembling field. Now consider the Cauchy–Schwarz inequality [87–89] for any two square-integrable real functions $f(\tau)$ and $q(\tau)$:

$$\left(\int f(\tau)^2 d\tau \right) \left(\int q(\tau)^2 d\tau \right) \geq \left(\int f(\tau) q(\tau) d\tau \right)^2. \quad (\text{S562})$$

Applying this to $f(\tau) = \xi_x(\tau)$ and $q(\tau) = \xi_p(\tau)$, we obtain:

$$\langle \xi_x^2 \rangle \langle \xi_p^2 \rangle \geq \langle \xi_x \xi_p \rangle^2. \quad (\text{S563})$$

The geometric covariance functional is used, as defined in Equation (S560). Then, by direct substitution,

$$\Delta x \Delta p = \sqrt{\langle \xi_x^2 \rangle} \cdot \sqrt{\langle \xi_p^2 \rangle} \geq \frac{1}{2} |g(\xi)|. \quad (\text{S564})$$

This yields the TSRT uncertainty relation:

$$\Delta x \Delta p \geq \frac{1}{2} |g(\xi)|, \quad (\text{S565})$$

which mirrors the structural form of Equation (S551), but now with a right-hand side that originates from deterministic geometric correlations rather than operator algebra. The quantity $g(\xi)$ embodies the causal, curvature-sensitive signature of trembling spacetime and serves as the cornerstone for the TSRT interpretation of uncertainty.

Thus, the trembling structure of spacetime encodes a fundamental phase-space correlation that governs the emergent uncertainty relations—rooted not in probabilistic postulates, but in the causal and geometric properties of microscopic spacetime dynamics.

S20.3 Lower Bound from Minimal Causal Resolution

In the TSRT framework, a fundamental lower bound on uncertainty arises not from wave-particle duality, but from the causal structure of trembling spacetime itself. Specifically, the trembling deviations $\Xi^\mu(\tau)$ must possess a nonzero minimal amplitude in order to ensure resolvable geometric separation between neighboring worldlines and maintain strictly positive proper-time evolution. If the amplitude of deviations were permitted to vanish entirely, the differential increments in proper time $d\tau$ would become arbitrarily small, potentially violating the causal constraint

$$\frac{d\tau}{d\lambda} > 0, \quad (\text{S566})$$

which is necessary to preserve forward evolution and prohibit retrocausal propagation.

Moreover, the trembling-induced spread in spacetime displacements and momentum devia-

tions defines the minimal phase-space volume associated with a localized particle. If this volume were allowed to collapse to zero, it would contradict the metric-based geometric separation required for the existence of a particle-like worldline. Thus, TSRT imposes a minimal spread in phase space—encoded in the amplitude and frequency structure of $\Xi^\mu(\tau)$ —which ensures that proper time remains well-defined, causal ordering is preserved, and no physically realizable configuration collapses below the causal resolution scale. This leads to a nonzero lower bound on the product of spatial and momentum uncertainties, derived rigorously in the following analysis. The proper-time coherence condition requires that no physical trajectory permits degenerate evolution. This leads to the fundamental lower bound which will be derived below and which is also known as Heisenberg’s inequality:

$$\Delta x \Delta p \geq \frac{\hbar}{2}. \quad (\text{S567})$$

Here, $\hbar$ arises naturally as the minimal action per causal oscillation, ensuring the maintenance of distinguishable causal histories at the finest geometric scale.

In TSRT, this inequality is not an imposed quantum axiom; it emerges as a geometric necessity for preserving causal structure under fine-scale trembling. In particular, it reflects the minimal phase-space area that a physical worldline must occupy to prevent the collapse of proper-time ordering. The key insight from TSRT derivations in Section S18, is that all trembling-induced statistical corrections ultimately trace back to the action scale $\mathcal{T}$, which encodes the amplitude and frequency content of the deterministic trembling trajectory $\Xi^\mu(\tau)$. In the non-relativistic case, this trembling action enters the Schrödinger equation as in Equation (S400), i.e.,

$$Q_{\text{TSRT}} = -\frac{\mathcal{T}^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}}. \quad (\text{S568})$$

Analogously, the covariance functional $g(\xi)$, defined in Equation (S560), must reflect the characteristic scale of the fluctuations in phase space induced by trembling spacetime. Since both position and momentum deviations arise from the same underlying geometric field $\Xi^\mu(\tau)$, their mutual covariance inherits its scaling properties from the amplitude and spectral structure of these deviations. To express this rigorously, we reintroduce two central quantities from TSRT: The action scale $\mathcal{T}$: introduced in Equation (S400), this is the minimal geometric action per full oscillation cycle of trembling. It emerges from the coarse-grained statistical dynamics of $\Xi^\mu(\tau)$, and governs the transition from deterministic to statistical behavior—appearing, for example, in the emergent Schrödinger equation (S425).

The cutoff action $\mathcal{S}_c$: defined in Equation (S518), this reflects the maximum allowed causal action per localized trembling mode, constrained by curvature bounds and the gravitational coupling G . In flat spacetime, where curvature is minimal, we have $\mathcal{T} \sim \mathcal{S}_c$.

In this section, we are concerned with the local correlation of fluctuations, and hence it is $\mathcal{T}$, the coarse-grained effective action per mode, that directly governs the covariance scale between ξ_x and ξ_p . We therefore express the covariance functional $g(\xi)$ as:

$$g(\xi) = \alpha \mathcal{T}, \quad (\text{S569})$$

where α is a dimensionless coefficient encoding the normalized spectral shape of trembling modes and the internal correlation structure of the deviation field. This is not a postulate but a result of dimensional consistency, causal symmetry, and the shared origin of spatial and momentum deviations from the same deterministic trembling dynamics. The function $g(\xi)$ measures the local statistical correlation between phase-space variables induced by geometric trembling, and in the limit $\xi \rightarrow 0$, causal smoothness demands $g(\xi) \rightarrow \alpha \mathcal{T}$. To match the established standard quantum uncertainty in flat spacetime, confirmed by numerous experiments, the following value identification must hold, despite the different framework and meaning in TSRT compared to quantum mechanics:

$$\alpha \mathcal{T} = \hbar \quad \Rightarrow \quad g(\xi) = \hbar. \quad (\text{S570})$$

In this flat regime, where gravitational curvature vanishes and trembling is minimal, the geometric structure imposes no additional distortion. Hence, the normalization constant takes the canonical value $\alpha = 1$, and we obtain:

$$\mathcal{T} = \hbar. \quad (\text{S571})$$

This reinforces a key conclusion: the uncertainty principle in standard quantum mechanics is not fundamental but emerges as a limiting case of the deeper geometric correlation structure of trembling spacetime. It reflects the minimal causal resolution permitted by spacetime geometry, not a probabilistic axiom.

In curved or strongly fluctuating geometries, where local coherence and spectral shape deviate from the flat limit, $\alpha \neq 1$, and the covariance scale $g(\xi)$ may depart from $\hbar$. This opens the door to curvature-dependent generalizations of uncertainty, as further explored in later sections.

Thus, TSRT recovers the standard lower bound of the uncertainty principle as a deterministic consequence of causal geometry, while also predicting the conditions under which this bound can be modified.

S20.4 Upper Bound from Curvature-Induced Contraction

While a minimal phase-space spread is required to ensure causal distinguishability (as established in the previous subsection), trembling deviations cannot grow arbitrarily large. At sufficiently high energies or in strongly curved spacetime regions, the geometric fluctuations that underlie trembling must remain constrained to preserve causal coherence and maintain a monotonic progression of proper time. Excessive trembling would lead to local backward propagation or complex-valued proper-time increments, violating the foundational causal structure of TSRT. To understand the source of this constraint, recall from the main article Equation (4) that metric fluctuations induced by trembling are expressed as:

$$\xi_{\mu\nu} = \partial_\mu \Xi_\nu + \partial_\nu \Xi_\mu. \quad (\text{S572})$$

The perturbed spacetime line element becomes:

$$d\tau^2 = g_{\mu\nu}(x) dx^\mu dx^\nu = (\bar{g}_{\mu\nu} + \xi_{\mu\nu}) dx^\mu dx^\nu, \quad (\text{S573})$$

where $\bar{g}_{\mu\nu}$ is the background metric and $\xi_{\mu\nu}$ arises from trembling. The requirement that $d\tau^2 > 0$ for all physical paths ensures that proper time remains real and strictly forward-oriented (cf. Equation (S119)).

As the curvature of spacetime increases, the condition $d\tau^2 > 0$ —which enforces forward progression of proper time—becomes increasingly difficult to satisfy. The local geometry tightens its tolerance for fluctuations, leaving less room for deviation without violating causality. In TSRT, the fine-scale trembling of a particle’s worldline is encoded in the deviation field $\Xi^\mu(\tau)$, which perturbs the background metric via the relation (see Equation (4) in the main article):

$$\xi_{\mu\nu} = \partial_\mu \Xi_\nu + \partial_\nu \Xi_\mu. \quad (\text{S574})$$

These metric perturbations enter the perturbed line element

$$d\tau^2 = (\bar{g}_{\mu\nu} + \xi_{\mu\nu}) dx^\mu dx^\nu, \quad (\text{S575})$$

and must preserve the condition $d\tau^2 > 0$ for all physically admissible displacements dx^μ . This requirement means that the trembling field $\Xi^\mu(\tau)$ cannot become arbitrarily large. If its amplitude grows too much, the induced perturbations $\xi_{\mu\nu}$ can overwhelm the background metric $\bar{g}_{\mu\nu}$, potentially flipping the sign of the line element and rendering proper time imaginary—an explicit violation of causality. To prevent this, TSRT imposes a curvature-sensitive upper bound on the magnitude of the trembling displacement:

$$|\Xi^\mu(\tau)|^2 < \frac{1}{\kappa_{\text{local}}}, \quad (\text{S576})$$

where κ_{local} is a positive-definite scalar curvature invariant that measures how rapidly geodesics diverge or converge in the local region of spacetime. A canonical choice for this quantity is the Kretschmann scalar:

$$\kappa_{\text{local}} := \sqrt{R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma}}. \quad (\text{S577})$$

Intuitively, this bound plays a role similar to a maximum safe oscillation range in a vibrating system. In flat spacetime, where curvature vanishes ($\kappa_{\text{local}} = 0$), the trembling amplitude can in principle be large (though still bounded by global constraints such as those tied to the Planck scale). But in regions of high curvature—such as near a neutron star or black hole—the geometry becomes more “rigid,” and even small trembling excursions risk destabilizing the proper-time structure. The tighter the curvature, the smaller the admissible trembling.

Thus, Equation (S576) ensures that the trembling-induced perturbations remain small enough for the modified spacetime interval to stay causally consistent. It encodes the geometric analogue of a stability criterion: fluctuations must be confined within the causal “elasticity” of the surrounding spacetime. Other scalar invariants (e.g., R , $R_{\mu\nu} R^{\mu\nu}$) may be used depending on the geometric setting. The choice must reflect the dominant local curvature behavior.

Now recall from Section S20.2 the covariance functional $g(\xi)$ introduced in Equation (S560), which quantifies the phase-space correlation due to trembling:

$$g(\xi) := 2\langle \xi_x \xi_p \rangle = m \langle \Xi^i(\tau) \dot{\Xi}_i(\tau) \rangle. \quad (\text{S578})$$

Increased trembling amplitudes $|\Xi^\mu(\tau)|$ raise this covariance, thereby enhancing $g(\xi)$ and enlarging the associated uncertainty product $\Delta x \Delta p$. However, the curvature-induced bound (S576) constrains this enhancement.

Therefore, the geometric covariance functional $g(\xi)$, which governs the effective phase-space spread induced by trembling, must obey an upper bound that reflects the curvature-limited amplitude of $\Xi^\mu(\tau)$. Specifically, since excessive deviation would violate the causal constraint $d\tau^2 > 0$, there exists a maximal covariance consistent with forward-evolving proper time:

$$g(\xi) \leq g_{\max} \sim \frac{mc}{\sqrt{\kappa_{\text{local}}}}. \quad (\text{S579})$$

Here, the right-hand side quantifies the largest allowable phase-space area a particle of mass m can occupy without destabilizing local causal structure. The origin of the factor mc is not ad hoc—it reflects the relativistic momentum scale associated with a particle of rest mass m , given by the relation $p = mc$ in the ultra-relativistic regime where trembling becomes geometrically significant. This momentum scale sets the baseline for the maximal phase-space extension. Meanwhile, the quantity $1/\sqrt{\kappa_{\text{local}}}$ captures the minimal spacetime scale over which geodesics can deviate without causal breakdown. It emerges from the trembling amplitude bound

$$|\Xi^\mu(\tau)|^2 < \frac{1}{\kappa_{\text{local}}}, \quad (\text{S580})$$

which ensures that metric perturbations $\xi_{\mu\nu} \sim \partial_\mu \Xi_\nu$ remain sufficiently small to preserve the sign and positivity of the proper-time interval.

Combining these two insights—(1) that the maximal spatial deviation scales as $\Delta x_{\max} \sim 1/\sqrt{\kappa_{\text{local}}}$, and (2) that the corresponding maximal momentum scale is $\Delta p_{\max} \sim mc$ —yields the product:

$$\Delta x \Delta p \sim \frac{mc}{\sqrt{\kappa_{\text{local}}}}. \quad (\text{S581})$$

This expression represents the geometric ceiling on the uncertainty product, above which the trembling-induced deformations of the metric would exceed causal limits.

Thus, the factor mc arises naturally as the geometric-momentum scale associated with the particle’s inertial mass in trembling spacetime. It ties together the local causal geometry with the particle’s capacity to imprint itself in phase space. The entire bound follows from deterministic constraints—not quantum axioms—and reveals how TSRT unifies inertia, geometry, and uncertainty under a single causal framework. Consequently, as will be further studied in Section S20.5, the uncertainty product obeys a double-bounded structure,

$$\frac{\hbar}{2} \leq \Delta x \Delta p \leq \frac{1}{2} \beta \frac{mc}{\sqrt{\kappa_{\text{local}}}}, \quad (\text{S582})$$

where β is a geometric factor incorporating the spectral properties of the trembling field and the correlation structure.

S20.5 Complete Double-Bounded Uncertainty Relation

At first glance, it may seem paradoxical that a single causal requirement—the condition that proper time must advance monotonically, $d\tau^2 > 0$ —could simultaneously impose both a lower and an upper bound on the phase-space product $\Delta x \Delta p$. However, this is not contradictory: these bounds emerge from opposite asymptotic regimes of the same fundamental constraint.

The lower bound arises from the requirement that trembling deviations $\Xi^\mu(\tau)$ cannot vanish entirely. If they did, neighboring worldlines would become indistinguishable, reducing proper-time increments $d\tau$ to zero and collapsing the causal structure. To prevent such degeneration, spacetime must possess a nonzero minimal trembling amplitude that preserves geometric separability and guarantees forward-directed time evolution. This leads to a lower bound on $\Delta x \Delta p$, set by the irreducible action scale $\mathcal{T} \sim \hbar$.

The upper bound, in contrast, originates from limiting how large the trembling amplitude can grow in regions of high curvature. Excessive trembling would induce metric perturbations $\xi_{\mu\nu}$ capable of overwhelming the background structure $\bar{g}_{\mu\nu}$, flipping the sign of $d\tau^2$ and violating the causal order. To avoid this, the trembling amplitude must remain below a curvature-dependent threshold, which in turn restricts the maximum allowable phase-space spread.

Thus, both bounds are consequences of a single geometric principle—the preservation of forward causal flow—but operate at opposite extremes. The lower bound ensures distinguishability; the upper bound preserves coherence. Their coexistence defines a causally admissible window in phase space, bounded below by the minimal resolution scale and above by the maximum deformation tolerable without causal breakdown.

From the previous subsections, this leads directly to the double-bounded uncertainty relation:

$$\frac{1}{2}\hbar \leq \Delta x \Delta p \leq \frac{1}{2}\beta \frac{mc}{\sqrt{\kappa_{\text{local}}}}, \quad (\text{S583})$$

where κ_{local} is a scalar curvature invariant (e.g., the Kretschmann scalar), and β is a dimensionless geometric coefficient encapsulating the internal structure of the trembling field—its anisotropies, spectral correlations, and curvature coupling. In regions of vanishing curvature, the upper bound diverges, and the inequality reduces to the standard Heisenberg uncertainty relation:

$$\Delta x \Delta p \geq \frac{1}{2}\hbar. \quad (\text{S584})$$

But in curved spacetime, causal coherence restricts the permissible trembling amplitude, contracting the allowed phase-space volume and producing an upper bound as well. To express this more precisely, the geometric covariance functional $g(\xi)$, introduced in Equation (S560), is constrained as

$$\hbar \leq g(\xi) \leq g_{\text{max}} := \beta \frac{mc}{\sqrt{\kappa_{\text{local}}}}, \quad (\text{S585})$$

leading directly to the double-bounded form:

$$\frac{1}{2}\hbar \leq \Delta x \Delta p \leq \frac{1}{2}\beta \frac{mc}{\sqrt{\kappa_{\text{local}}}}. \quad (\text{S586})$$

The lower limit reflects the irreducible action content per causal oscillation. The upper

limit encodes the geometric tolerance of spacetime to trembling-induced deformations, which diminishes in regions of high curvature. Together, they define a causally and geometrically regulated domain for physically admissible uncertainty—grounded not in probabilistic axioms, but in the deterministic structure of spacetime itself.

This result profoundly reframes the interpretation of uncertainty: it is no longer a fixed postulate, but a derived feature of trembling geometry. Heisenberg’s inequality is revealed as the low-curvature limit of a broader, curvature-sensitive causal window. As curvature grows, this window narrows from above, signaling the transition from quantum to trans-quantum regimes within a unified geometric theory. Accordingly, the double-bounded structure of Equation (S586) marks a conceptual turning point. It elevates uncertainty from a quantum principle to a manifestation of causal geometry, and shows that Planck’s constant $\hbar$ itself emerges not as an imposed scale, but as the minimal invariant action compatible with local causal resolution.

In the next subsection, we demonstrate how the conventional Heisenberg relation arises as the special case of this framework when $\kappa_{\text{local}} \rightarrow 0$, recovering standard quantum mechanics as the asymptotic flat-spacetime limit of trembling dynamics.

S20.6 Flat Spacetime Limit and Recovery of Heisenberg Uncertainty

In the limit of vanishing spacetime curvature, $\kappa_{\text{local}} \rightarrow 0$, the TSRT uncertainty relation simplifies dramatically. The upper bound diverges, and only the lower bound remains finite:

$$\Delta x \Delta p \geq \frac{1}{2} \hbar. \quad (\text{S587})$$

This recovers the standard Heisenberg uncertainty relation of quantum mechanics, where the uncertainty product is bounded from below by $\hbar/2$, but can grow arbitrarily large depending on the state. In this flat-spacetime regime, trembling-induced deviations are minimally constrained by curvature, and causal coherence imposes only a minimal geometric resolution scale. There is no upper bound; the uncertainty product can grow arbitrarily large, limited only by the quantum state, just as in conventional quantum mechanics.

Thus, TSRT reproduces quantum uncertainty as the asymptotically flat limit of a more general causal-geometric framework. The Planck-scale action $\hbar$ emerges here not as a postulate, but as the minimal phase-space area compatible with distinguishable causal evolution in the absence of curvature-induced constraints. This picture changes significantly in curved spacetime. As κ_{local} increases, the upper bound on the uncertainty product becomes finite and contracts. One might mistakenly interpret this contraction as a gain in precision—but it is the opposite. Causal coherence in curved regions imposes stricter constraints on permissible trembling amplitudes. Neighboring worldlines must remain tightly synchronized to preserve forward evolution, leaving less room for phase-space variability. In this regime, uncertainty does not loosen—it crystallizes. The “blur” familiar from quantum mechanics becomes a rigid window, not of observational fuzziness, but of geometric necessity. Spacetime itself constrains how distinct a particle’s position and momentum can become.

The phase-space volume a particle can occupy becomes tightly bounded, not due to experimental limitations, but because spacetime itself refuses to encode more detail. The lower bound

persists or grows mildly, while the upper bound descends—narrowing the corridor within which physical motion remains causally viable. This deterministic suppression of allowable spread leads to stronger correlations between position and momentum. It does not yield finer resolution, but a tighter constraint on how sharply these variables can be distinguished from one another.

To illustrate this, consider a visual analogy: imagine photographing a moving object with a camera whose pixels are embedded in a deformable fabric. In flat regions, the pixel grid is stable—you can capture a wide range of motion. But near a massive object, the fabric stretches and curves. The pixel density warps, and the fabric itself restricts the fidelity of your image. The blur is no longer due to movement, but to geometric distortion. The structure of spacetime—not your tools—limits what can be resolved.

In TSRT, this limitation is encoded quantitatively through the phase-space covariance functional $g(\xi)$. In flat spacetime,

$$g(\xi) = \hbar, \quad (\text{S588})$$

whereas in curved regions, the upper limit becomes:

$$g_{\max} \sim \frac{mc}{\sqrt{\kappa_{\text{local}}}}. \quad (\text{S589})$$

This bound reflects the largest allowable phase-space spread consistent with the preservation of $d\tau^2 > 0$. More precisely, using the TSRT action scale $\mathcal{T}$, this constraint reads:

$$g_{\max} = \mathcal{T} \cdot \alpha_{\max} \quad \text{with} \quad \alpha_{\max} \sim \sqrt{\frac{\kappa_{\text{flat}}}{\kappa_{\text{local}}}}. \quad (\text{S590})$$

Finally, relating trembling velocity to curvature via:

$$\left| \frac{d\Xi^\mu}{d\tau} \right|^2 \lesssim \kappa_{\text{local}}, \quad (\text{S591})$$

and noting that $\Delta p \sim m d\Xi^\mu/d\tau$, while $\Delta x \sim \Xi^\mu \sim 1/\sqrt{\kappa_{\text{local}}}$, we obtain:

$$\Delta x \Delta p \lesssim \frac{mc}{\sqrt{\kappa_{\text{local}}}}. \quad (\text{S592})$$

This recovers the upper limit in terms of the curvature-regulated geometric capacity of spacetime to sustain distinguishable motion:

$$g(\xi) \leq \beta \frac{mc}{\sqrt{\kappa_{\text{local}}}}, \quad (\text{S593})$$

where β encodes the spectral properties and directional anisotropies of trembling.

In summary, the uncertainty principle in TSRT emerges from causal geometry, not probabilistic rules. In low-curvature settings, the upper bound vanishes, yielding the familiar Heisenberg form. In high curvature, the causal architecture of spacetime tightens both ends of the uncertainty window. This reinterprets uncertainty as a deterministic feature of spacetime's trembling fabric—one that limits not what we can measure, but what nature permits to exist distinctly.

Thus, the uncertainty principle becomes not a statement about what we fail to measure,

but a statement about what nature refuses to separate.

S20.7 A New Regime of Physics

In standard quantum mechanics, the Heisenberg uncertainty principle [3] imposes a lower bound on the product of the uncertainties in position and momentum:

$$\Delta x \Delta p \geq \frac{\hbar}{2}. \quad (\text{S594})$$

This relation is widely regarded as a universal feature of quantum systems, grounded in operator algebra and the probabilistic interpretation of observables. It is assumed to hold across all energy scales and gravitational environments. Trembling Spacetime Relativity Theory (TSRT) challenges this assumption by showing that uncertainty is not a fixed axiom, but an emergent consequence of causal spacetime geometry. In this framework, geometric trembling induces deterministic fluctuations that yield both a lower and an upper bound on the product $\Delta x \Delta p$, as derived in Equation (S583):

$$\frac{1}{2}\hbar \leq \Delta x \Delta p \leq \frac{1}{2}\beta \frac{mc}{\sqrt{\kappa_{\text{local}}}}. \quad (\text{S595})$$

Here, κ_{local} is a positive-definite curvature scalar, m is the particle's rest mass, and β is a dimensionless coefficient capturing the spectral and directional structure of the trembling field.

This dual-bound relation arises directly from the geometric requirement that proper time remain real and forward-directed. The lower bound ensures that worldlines remain causally distinguishable, while the upper bound reflects the suppression of large trembling excursions in regions of strong curvature. Neither bound relies on measurement theory or operator formalism—they follow purely from causal coherence. In practical terms, increasing curvature tightens the phase-space window. It constrains both position and momentum deviations, enforcing stronger correlations between them. The allowable product $\Delta x \Delta p$ shrinks, not due to increased measurement precision, but because spacetime geometry physically forbids excessive spread. Even in asymptotically flat spacetime ($\kappa_{\text{local}} \rightarrow 0$), trembling is not eliminated. It encodes a minimal action scale $\mathcal{T}$, which reproduces the Heisenberg bound as a limiting case. Thus, Planck's constant $\hbar$ emerges as the geometric footprint of the causal structure in the flat-spacetime limit, rather than a postulated fundamental constant.

Most strikingly, TSRT predicts a critical curvature κ_{crit} at which the lower and upper bounds coincide:

$$\frac{1}{2}\hbar = \Delta x \Delta p = \frac{1}{2}\beta \frac{mc}{\sqrt{\kappa_{\text{crit}}}}, \quad (\text{S596})$$

from which it follows that

$$\kappa_{\text{crit}} = \left(\frac{\beta mc}{\hbar} \right)^2. \quad (\text{S597})$$

This defines a sharp geometric transition between two regimes:

- $\kappa_{\text{local}} > \kappa_{\text{crit}}$: Strong curvature closes the uncertainty window to the canonical Heisenberg limit. Quantum mechanics is recovered as a curvature-saturated limit of TSRT.
- $\kappa_{\text{local}} < \kappa_{\text{crit}}$: Weak curvature expands the upper bound. The particle explores a wider phase-

space domain, revealing causal structures beyond the scope of standard quantum theory.

Importantly, even in exactly flat Minkowski spacetime ($\kappa_{\text{local}} = 0$), the minimal bound $\hbar/2$ persists. What changes is not the existence of uncertainty, but the size of the causal window permitted by the spacetime geometry.

This result reframes Planck’s constant: not as a ceiling on uncertainty, but as the base of a dynamically regulated window. As curvature increases, the causal architecture contracts this window, enforcing a maximal phase-space density consistent with forward-evolving proper time. The minimal phase-space volume arises from the requirement that trembling worldlines do not collapse into causal degeneracy. This constraint anchors a universal action scale, emergent from geometry rather than quantum postulates, and naturally consistent with the observed value of $\hbar$. TSRT thus completes and extends Heisenberg’s principle. It shows that quantum uncertainty is not the endpoint of our understanding, but the low-curvature shadow of a more fundamental causal law. In this new regime of physics, uncertainty is not about limitations of observation—it is about the geometric structure of existence itself.

S20.8 Beyond the Critical Threshold: Can the Bounds Cross?

The double-bounded uncertainty principle in TSRT enforces both a minimum and a maximum allowable phase-space product $\Delta x \Delta p$. At the critical curvature value κ_{crit} , these two bounds coincide, and the uncertainty product becomes sharply defined. But what happens if the local curvature drops below this threshold—i.e., $\kappa_{\text{local}} < \kappa_{\text{crit}}$? Would the upper bound on uncertainty then fall below the causal lower bound $\hbar/2$, leading to a contradiction? TSRT prohibits such a scenario. The lower bound originates from the requirement that proper time remains strictly increasing along every physical worldline. If the upper bound were to fall below this threshold, it would imply that no physically admissible trembling configuration could satisfy causal evolution. This would violate the condition $d\tau^2 > 0$, and hence cannot occur. The resolution lies in the self-regulating nature of trembling spacetime. As κ_{local} approaches κ_{crit} from above, any further suppression of curvature—such as might arise in an artificially flattened or vacuum-diluted region—is compensated by an amplification of internal trembling modes. This increase in geometric fluctuation restores the minimum necessary covariance $g(\xi) \geq \hbar$, and with it, the proper-time coherence of the underlying spacetime structure.

Thus, TSRT enforces a *dynamical floor* on curvature: $\kappa_{\text{local}} \geq \kappa_{\text{crit}}$. This is not a quantum inheritance, but a prediction of TSRT’s causal geometry. Even in the absence of external fields, trembling injects residual structure into spacetime that prevents it from becoming exactly flat. In this sense, κ_{crit} represents not only a mathematical intersection of uncertainty bounds but also a physically enforced lower limit on curvature compatible with causal consistency. This behavior reflects a deeper mechanism: trembling and curvature form a feedback loop. If curvature is suppressed below κ_{crit} , the resulting breakdown in causal resolution triggers a geometric response—namely, the amplification of internal metric fluctuations—to stabilize the structure at or above the critical scale. This “geometric backreaction” acts as a form of intrinsic pressure, dynamically maintaining the causal window required for uncertainty bounds to hold.

In summary, TSRT does not merely impose uncertainty as a structural feature—it actively

preserves it. The bounds cannot cross because the causal architecture of spacetime dynamically forbids configurations that would violate the requirement of forward-evolving proper time. Geometry, trembling, and causality co-conspire to enforce a fundamental window of indeterminacy, securing uncertainty not as a rule of measurement but as a geometric necessity of existence.

S20.9 Physical Interpretation and Implications

In TSRT, the uncertainty principle is not an axiom—it is a geometric theorem. In standard quantum mechanics, Heisenberg’s inequality

$$\Delta x \Delta p \geq \frac{\hbar}{2} \quad (\text{S598})$$

is derived from operator non-commutativity and is interpreted as a fundamental, context-independent bound reflecting intrinsic quantum randomness. This lower bound is traditionally viewed as a universal feature of all physical systems. TSRT, by contrast, reveals that this relation has a deeper origin. Trembling spacetime deviations—modeled by the geometric field $\Xi^\mu(\tau)$ —introduce deterministic perturbations to a particle’s trajectory. These fine-scale modulations are causally coherent yet unresolved at observational scales. When coarse-grained, they appear as statistical spreads in position and momentum, giving rise to uncertainty as an emergent feature of sub-geometric structure. Crucially, trembling is not just a useful model—it is a geometric necessity. As shown in Supplementary Section S21, the very requirement that proper time evolve monotonically, $d\tau^2 > 0$, implies that spacetime must possess internal trembling. A perfectly still geometry would violate causal distinguishability and permit worldlines to collapse into degeneracy. Hence, trembling is inseparable from the fabric of causality.

As derived in Equation (S583), the resulting uncertainty relation in TSRT takes the form:

$$\frac{1}{2}\hbar \leq \Delta x \Delta p \leq \frac{1}{2}\beta \frac{mc}{\sqrt{\kappa_{\text{local}}}}, \quad (\text{S599})$$

where κ_{local} denotes the scalar curvature of the local spacetime region, and β captures the geometric and spectral characteristics of trembling. Both bounds emerge from the same causal principle. If trembling correlations are too weak, the lower bound is violated and proper time loses positivity. If deviations are too large, the upper bound is exceeded, destabilizing causal coherence. Spacetime geometry thus enforces a finite corridor of permissible phase-space spread.

In low-curvature regions, trembling amplitudes can be broad and loosely correlated. The effective geometric covariance then approaches $g(\xi) \approx \hbar$, and the familiar Heisenberg uncertainty principle is recovered as the limiting behavior. No upper bound is operative.

In strongly curved environments, however, the causal structure becomes increasingly rigid. To preserve timelike evolution, trembling amplitudes must contract, enhancing correlation between position and momentum fluctuations. The uncertainty product becomes more tightly bounded—not because of increased measurement precision, but because geometric freedom itself diminishes. What appears to be stronger constraint is in fact a loss of representational capacity in curved spacetime.

This reinterpretation of uncertainty in TSRT carries several profound implications:

First, it restores determinism to the foundations of physics. Particles follow well-defined trembling worldlines. The apparent indeterminacy of quantum measurements is not fundamental but arises from sub-geometric fluctuations too fine to resolve. No stochastic interpretation or wavefunction collapse is needed.

Second, it completes the uncertainty principle by bounding it not only from below but also from above. Uncertainty becomes a finite, curvature-dependent domain of admissible phase-space spread, rather than an open-ended blur.

Third, this framework predicts new experimental signatures. In ultra-strong curvature environments—such as near black hole horizons or during high-energy collisions—deviations from the Heisenberg lower bound may be observable. Such deviations would signal the presence of the geometric ceiling imposed by trembling curvature.

Finally, TSRT resolves long-standing interpretational problems in quantum theory. It eliminates the need for nonlocality, wavefunction collapse, or many-worlds branching. Planck's constant $\hbar$ emerges not as an input, but as a derived consequence of geometric covariance. It is the minimal action required for causal resolution—not a fundamental constant of nature, but a curvature-regulated threshold.

In summary, uncertainty in TSRT is not a constraint on knowledge—it is a constraint on existence. The bounds on $\Delta x \Delta p$ are not epistemic but ontological. In flat spacetime, the Heisenberg floor emerges naturally; in curved spacetime, a ceiling is added. Together, they define a causal window dictated by the geometry of trembling spacetime. This reformulation departs fundamentally from prior frameworks. While quantum field theory in curved spacetime—especially through the work of Hawking [10], Unruh [17], Birrell and Davies [45], and Penrose [90]—has explored the effects of gravity on quantum processes, none of these approaches derive or predict a modification of Heisenberg's inequality. Even in extreme gravitational fields, uncertainty has traditionally been assumed invariant. TSRT overturns this assumption. It reveals that uncertainty itself is emergent. It arises from the interplay of curvature, causal structure, and geometric trembling—not from quantum indeterminacy. Therefore, TSRT does not merely reinterpret the uncertainty principle—it completes it. It embeds it within the geometric and causal foundation of physical law, transforming uncertainty from an abstract limitation into a tangible consequence of spacetime's deepest structure.

To complete this foundation, we now demonstrate why trembling must exist in any causally consistent spacetime, and prove that causal ordering demands it as a geometric inevitability.

S21 Supplementary Material: Causal Necessity of Spacetime Trembling

Trembling spacetime is not an optional feature in TSRT—it is a structural requirement enforced by causal coherence. If the fine-scale geometry were exactly smooth, with no fluctuations in worldline trajectories, then neighboring paths could collapse, proper time could fail to progress consistently, and microscopic causal order would become ill-defined.

This section establishes that trembling must occur in any spacetime that is deterministic, Lorentz-invariant, and causally coherent. The necessity of trembling arises not from empirical assumption, but from geometric and causal first principles. The double-bounded uncertainty relation derived in Section S20 plays a central role.

S21.1 Logical Setup: Phase-Space Bounds in TSRT

The necessity of trembling spacetime in TSRT follows directly from three interlinked causal-geometric conditions:

Proper-Time Monotonicity: Proper time must advance strictly along every physical worldline, ensuring the preservation of forward-directed causality.

Worldline Separability: Neighboring particle trajectories must remain geometrically distinguishable across finite intervals to prevent causal collapse.

Local Causal Admissibility: Any metric deviation from the background must preserve timelike or null intervals, forbidding spacelike jumps in the effective geometry.

Together, these conditions require a bounded phase-space spread for any physically admissible trajectory. In TSRT, this leads to the double-bounded uncertainty relation:

$$\frac{1}{2}\hbar \leq \Delta x \Delta p \leq \frac{1}{2}\beta \frac{mc}{\sqrt{\kappa_{\text{local}}}}, \quad (\text{S600})$$

where $\hbar$ represents the minimal action scale, arising from irreducible cutoff trembling; m is the particle's mass; c is the invariant speed of light; κ_{local} is a scalar measure of the local spacetime curvature; β is a dimensionless geometric factor encoding the spectral structure and anisotropy of the trembling field.

The lower bound enforces minimal causal resolution: without nonzero trembling, proper time could stagnate or reverse, violating causal monotonicity. This constraint is directly linked to the condition:

$$\frac{d\tau}{d\lambda} > 0, \quad (\text{S601})$$

as established in Section S7.

The upper bound arises from the opposite limit: if trembling amplitudes become too large—particularly in regions of high curvature—then deviations $\Xi^\mu(\tau)$ can distort the effective geometry to the point of causality breakdown. The upper limit thus reflects the maximal trembling amplitude consistent with local geometric coherence.

S21.2 Hypothetical Absence of Trembling

Consider a hypothetical scenario in which spacetime is perfectly smooth and static at all scales—devoid of any intrinsic trembling or fine-scale geometric fluctuation. In such a still spacetime, no minimal deviation field $\Xi^\mu(\tau)$ would exist to enforce irreducible phase-space spread. Consequently, the lower bound $\Delta x \Delta p \geq \hbar/2$ would lack a geometric origin and could not arise from first principles. Without trembling, the uncertainty relation would lose its intrinsic causal floor. Only the curvature-dependent upper bound would remain, and in low-curvature regimes, this would diverge—leaving the phase-space volume unconstrained from below.

S21.3 Violation of Causal Structure

In the absence of trembling, regions of extremely low curvature ($\kappa_{\text{local}} \rightarrow 0$) admit no dynamically enforced lower bound on the uncertainty product. The upper bound,

$$\Delta x \Delta p \leq \frac{1}{2} \beta \frac{mc}{\sqrt{\kappa_{\text{local}}}}, \quad (\text{S602})$$

diverges in this limit, permitting $\Delta x \Delta p \rightarrow 0$.

Such unconstrained localization enables arbitrarily sharp specification of both position and momentum. This leads to phase-space configurations in which infinitesimal deviations can produce backward or ill-defined increments of proper time—directly violating the monotonicity condition

$$\frac{d\tau}{d\lambda} > 0, \quad (\text{S603})$$

as defined in Equation (S601). Without a trembling-induced minimal resolution scale, the local causal order becomes unstable. Fine-grained geometric coherence breaks down, allowing the emergence of acausal or non-integrable worldline behavior—especially in flat or near-flat spacetime domains.

S21.4 Necessity of Trembling for Causal Integrity

The absence of trembling would fatally undermine the monotonic progression of proper time, destabilizing local and global causal structure. In this scenario, distinguishable events along neighboring worldlines could collapse into a single geometric locus, violating the minimal resolution required for causal separation. To avoid such degeneracy, spacetime must possess a built-in fluctuation scale—an irreducible trembling amplitude—that guarantees finite separation between causal paths. This minimal structure enforces the lower bound

$$\Delta x \Delta p \geq \frac{1}{2} \hbar, \quad (\text{S604})$$

even in flat or weakly curved regions, where the upper bound diverges.

Trembling thus provides the geometric enforcement of causal consistency: it dynamically stabilizes proper-time evolution, prevents acausal jumps, and preserves Lorentz-invariant separability among worldlines. Without it, spacetime would lose its capacity to support a coherent causal order. Therefore, trembling is not a postulated explanation of uncertainty—it is a derived

structural necessity. It is the only viable mechanism by which spacetime can remain causally admissible at all curvature scales.

S21.5 Clarification on Logical Consistency

To preempt any concerns about circular reasoning, we emphasize that the argument for trembling spacetime is strictly deductive and non-circular. The existence of trembling is not assumed—it is derived. We begin with a single independent premise: that proper time must evolve monotonically along all physically admissible worldlines. This is a foundational requirement of causal structure, unrelated to any assumption about geometric fluctuations. We then examine the consequences of eliminating trembling. In doing so, we find that causal coherence collapses in the low-curvature limit: proper-time monotonicity fails, and distinguishable trajectories become geometrically unstable. This leads to a rigorous conclusion: trembling spacetime is the only structural feature capable of enforcing causal integrity across all curvature regimes. It is not postulated—it is required. The result is a logically complete and non-circular derivation of trembling as a causal necessity.

S21.6 Conclusion

The causal structure of TSRT shows that trembling spacetime is not optional—it is inevitable. Its existence is not inferred from observation or postulated as a mechanism, but derived from first principles. Any spacetime that preserves causal coherence and proper-time monotonicity must exhibit irreducible geometric fluctuations.

A perfectly still spacetime is incompatible with causal evolution. Without trembling, neighboring worldlines would collapse, proper time could stagnate or reverse, and microscopic events would lose distinguishability. Causality would fail not because of dynamics, but because the geometry would lack the minimal resolution needed to define time itself. Trembling therefore serves as a structural safeguard. It stabilizes causal order, enforces a nonzero uncertainty floor, and ensures that events remain separable within a curvature-dependent phase-space window. These fluctuations are not perturbative—they are the geometric backbone of a physically meaningful spacetime. Still spacetime is not just implausible—it is unphysical. Trembling is the only way to prevent degeneracy of trajectories, breakdown of time ordering, and collapse of information-carrying capacity.

Accordingly, trembling spacetime is a necessary feature of any deterministic, causally ordered universe. In what follows, we extend this causal framework to explore how curvature further constrains the information that spacetime can support.

S22 Supplementary Material: Curvature, Uncertainty, and the Limits of Information

Trembling spacetime in TSRT does more than determine particle dynamics—it governs how much information a region of spacetime can physically support. In this framework, the uncertainty relation is bounded both from below and above by local curvature, and this phase-space structure imposes a fundamental limit on the number of distinguishable physical states. Information becomes a geometric quantity, regulated by the causal architecture of spacetime.

S22.1 Causal Geometry as an Information Regulator

The double-bounded uncertainty relation derived in Section S20 implies that information localization is not arbitrary. The resolvable phase-space area—and hence the number of distinguishable configurations—is dynamically constrained by the curvature-regulated structure of trembling spacetime.

Modern approaches to quantum gravity—particularly those involving black hole thermodynamics and the holographic principle—have long suggested a deep connection between geometry and information. A central insight is that entropy, as a measure of information, scales with surface area rather than volume. The Bekenstein–Hawking entropy formula [10, 11] expresses this relationship for black holes:

$$S_{\text{BH}} = \frac{k_B c^3}{4G\hbar} A, \quad (\text{S605})$$

where A is the event horizon area, k_B is Boltzmann’s constant, and the other constants define the Planck scale. This formula suggests a geometric ceiling on entropy: the maximal information content of a gravitational system scales with its boundary.

This led to the holographic principle, first proposed by ’t Hooft [13] and developed by Susskind [14], which asserts that all physical information within a volume can, in principle, be encoded on its boundary surface. However, these approaches remain rooted in standard quantum field theory, where the Heisenberg uncertainty relation

$$\Delta x \Delta p \geq \frac{\hbar}{2} \quad (\text{S606})$$

is treated as a context-independent axiom—even in extreme regimes of curvature.

This assumption reveals a hidden tension. If strong curvature can distort causal structure and limit coherence, should phase-space resolution remain unaffected? Can the number of physically realizable states truly remain fixed as spacetime itself deforms? Trembling Spacetime Relativity Theory (TSRT) answers in the negative. It reframes the uncertainty principle not as a rigid quantum postulate, but as a geometric constraint that tightens under curvature. The uncertainty bounds arise from the structure of spacetime itself, via deterministic but fluctuating metric deviations. As curvature increases, the upper limit on $\Delta x \Delta p$ contracts, shrinking the accessible phase-space volume and reducing the number of distinguishable physical states.

In this view, information is not imposed externally—it emerges as a property of causal geometry. Entropy bounds, long treated as thermodynamic inputs, become geometric consequences

of a curvature-regulated phase-space structure.

S22.2 Trembling, Phase-Space Area, and State Distinguishability

As established in Section S20, the deviation field $\Xi^\mu(\tau)$ generates statistical covariance in phase space:

$$g(\xi) := \langle \Xi^i(\tau) m \dot{\Xi}_i(\tau) \rangle \sim \Delta x \Delta p. \quad (\text{S607})$$

This covariance determines the effective phase-space area per distinguishable physical state. In TSRT, the lower bound

$$\Delta x \Delta p \geq \frac{\hbar}{2} \quad (\text{S608})$$

sets a minimal unit of distinguishability—a geometric scale below which distinct causal trajectories cannot be resolved.

Crucially, TSRT does not treat this uncertainty as a fixed axiom. Instead, it emerges as part of a dynamically bounded, curvature-dependent relation:

$$\frac{1}{2}\hbar \leq \Delta x \Delta p \leq \frac{1}{2}\beta \frac{mc}{\sqrt{\kappa_{\text{local}}}}, \quad (\text{S609})$$

where κ_{local} is a scalar measure of local curvature (such as the Kretschmann scalar), m is the particle’s rest mass, and β is a dimensionless coefficient encoding the spectral structure of trembling.

The lower bound ensures causal distinguishability, while the upper bound arises from curvature constraints that limit the allowable deviation amplitude in phase space. As curvature increases, the upper limit contracts, narrowing the set of admissible phase-space fluctuations.

This structure implies that uncertainty is not merely epistemic or operational—it is a direct expression of spacetime geometry. In low-curvature regions, large trembling amplitudes permit wide excursions in phase space, supporting a large number of distinguishable microstates. But in highly curved regimes, $\Xi^\mu(\tau)$ is geometrically constrained by causal coherence, suppressing both position and momentum deviations and thus reducing the volume of accessible phase space. This has immediate implications for black hole thermodynamics. The Bekenstein–Hawking entropy formula (Equation (S605)) suggests that black hole entropy scales with horizon area rather than volume. TSRT provides a geometric rationale for this: in regions of extreme curvature, the phase-space volume available to trembling trajectories contracts to the point where internal resolution is suppressed. The horizon area thereby marks not only a thermodynamic limit, but a causal cutoff beyond which distinguishability vanishes.

TSRT thus shifts the conceptual foundation of entropy bounds from thermodynamic analogy to geometric necessity. The contraction of trembling-induced phase-space volume is what enforces the entropy limit. Where the holographic principle postulates that information must be encoded on a boundary surface, TSRT derives this as a consequence of causal suppression: increased curvature forces internal fluctuations to compress, compelling information to reside in the geometrically admissible modes.

S22.3 Curvature-Imposed Suppression of Information

In curved regions of spacetime, the upper bound on the uncertainty product becomes finite due to causal coherence constraints:

$$\Delta x \Delta p \leq \frac{1}{2} \beta \frac{mc}{\sqrt{\kappa_{\text{local}}}}. \quad (\text{S610})$$

This constraint sets a maximal number of distinguishable causal paths N_{max} that can exist within a region of characteristic size L :

$$N_{\text{max}} \sim \frac{L}{\Delta x} \sim \sqrt{\kappa_{\text{local}}} \cdot L. \quad (\text{S611})$$

As curvature increases, the smallest resolvable position uncertainty Δx must decrease, thereby reducing the number of distinguishable worldlines and contracting the available informational degrees of freedom.

In traditional black hole thermodynamics, entropy bounds are derived from semiclassical reasoning. Bekenstein proposed that the entropy S of any bounded physical system cannot exceed a value proportional to its boundary area, based on thought experiments involving the generalized second law of thermodynamics [11]. Hawking later connected this bound to surface gravity and event horizon geometry, arriving at the entropy–area law [10]:

$$S_{\text{BH}} = \frac{k_B c^3}{4G\hbar} A. \quad (\text{S612})$$

These results gave rise to the holographic principle [13, 14], which asserts that the maximal number of degrees of freedom in a spatial volume is determined not by its volume, but by the surface area enclosing it. However, these approaches rest on two key assumptions: (i) that the Hilbert space structure of quantum theory is universal and independent of background geometry, and (ii) that the Heisenberg uncertainty principle

$$\Delta x \Delta p \geq \frac{\hbar}{2} \quad (\text{S613})$$

applies equally in all gravitational regimes.

TSRT departs from both assumptions. It derives entropy bounds not from external thermodynamic arguments, but from the internal structure of spacetime geometry. In this theory, the metric undergoes deterministic fine-scale fluctuations $\xi_{\mu\nu}$, governed by the deviation field $\Xi^\mu(\tau)$. To preserve causal integrity—specifically, to maintain real and forward-pointing proper time increments $d\tau^2 = g_{\mu\nu} dx^\mu dx^\nu > 0$ —the amplitude and frequency of these fluctuations must be constrained. This causal constraint gives rise to a double-bounded uncertainty relation:

$$\frac{1}{2} \hbar \leq \Delta x \Delta p \leq \frac{1}{2} \beta \frac{mc}{\sqrt{\kappa_{\text{local}}}}. \quad (\text{S614})$$

As κ_{local} increases, the upper bound contracts, directly limiting the phase-space volume and thus the number of distinguishable microstates. In TSRT, entropy bounds are not heuristic estimates—they are geometric consequences. A region of sharply curved spacetime cannot support an arbitrarily large number of physically resolvable states, because the metric geometry forbids

the requisite trembling amplitudes needed to differentiate them. In this light, the entropy–area law is not imposed by analogy with thermodynamics or imported from semiclassical quantum field theory. It is a consequence of the causal admissibility of metric fluctuations. As local curvature increases, trembling amplitudes must contract, suppressing internal resolution and reducing the phase-space volume available to physical systems.

TSRT therefore reframes the entropy bound as a curvature-regulated geometric limit. Rather than being an externally imposed holographic constraint, it emerges from the causal structure of trembling spacetime itself. A derivation of this result in the context of black hole horizons is given in Section S22.4.2.

S22.4 Information as a Geometric Quantity

The central insight of TSRT is that the maximum amount of physically extractable information from a spacetime region is not arbitrary—it is geometrically bounded. Trembling spacetime imposes a structural limit on distinguishability, linking informational resolution directly to the curvature-dependent causal coherence of neighboring worldlines. As local curvature increases, the allowable phase-space volume contracts, reducing the number of distinguishable trajectories. This leads to a profound consequence: the capacity to encode information is not an overlay on spacetime—it is a direct consequence of its geometric structure. Regions of stronger curvature admit fewer distinguishable causal histories, not due to measurement limitations, but because the geometry itself prohibits finer phase-space differentiation.

In this framework, information is not a statistical abstraction or external bookkeeping device—it is a manifestation of causal geometry. The trembling amplitude sets the minimal unit of phase-space resolution, while curvature bounds its maximum extent. The stronger the curvature, the tighter the causal structure, and the fewer the degrees of freedom that can coexist within a coherent region. Trembling spacetime thus acts as a dynamic information regulator. It binds the number of admissible physical states to the underlying curvature of the manifold, naturally connecting the concepts of curvature, uncertainty, and information into a single geometric framework.

S22.4.1 Holography as a Consequence of Geometric Compression

The holographic principle, originally proposed by 't Hooft [13] and extended by Susskind [14], asserts that the information content of a spacetime region is encoded not in its volume, but on its boundary surface. This principle suggests a fundamental upper limit on the number of degrees of freedom within any spatial domain—one that scales with area rather than volume. It has found deep theoretical support in string theory and the AdS/CFT correspondence [16], where a higher-dimensional bulk theory is exactly dual to a lower-dimensional boundary conformal field theory.

In these frameworks, the dimensional reduction of information is treated as a foundational postulate, arising from the algebraic structure of unitary quantum field theory and conformal symmetry. The uncertainty principle, operator algebra, and Hilbert space structure remain fixed, without offering an intrinsic geometric mechanism for the emergence of this boundary-bound compression.

Trembling Spacetime Relativity Theory (TSRT) provides a complementary and deeper explanation. It does not oppose the holographic principle—it grounds it. In TSRT, the number of physically distinguishable configurations within a region is not arbitrarily imposed, but determined by the internal curvature of spacetime. As local curvature κ_{local} increases, the amplitude of metric trembling must decrease to preserve causal coherence. This enforces a tighter upper bound on the phase-space uncertainty product, as given in Equation (S600):

$$\frac{1}{2}\hbar \leq \Delta x \Delta p \leq \frac{1}{2}\beta \frac{mc}{\sqrt{\kappa_{\text{local}}}}. \quad (\text{S615})$$

This contraction of phase-space area directly reduces the number of microstates that can be physically resolved. As the allowable volume in phase space shrinks, the system is forced to encode information in more compressed forms—conforming not to arbitrary boundary prescriptions, but to the internal geometric constraints of the spacetime region. TSRT thus explains the holographic effect as a geometric necessity. The apparent dimensional reduction is not a postulated feature of quantum gravity—it arises because the internal resolution capacity of curved regions becomes insufficient to support unconstrained correlations. Information must therefore be encoded in lower-dimensional, surface-like structures, consistent with causal admissibility.

Moreover, TSRT does not rely on unitarity or Hilbert space evolution. It replaces quantum mechanical information conservation with a geometric criterion: information exists only if it can be causally supported by trembling-compatible metric fluctuations. In highly curved regions, certain correlations are not destroyed—they become inadmissible. This exclusion is not due to measurement, collapse, or evaporation, but to the geometry itself: proper time must remain forward-directed, and this condition suppresses the phase-space freedom needed to represent particular correlations. As curvature increases near gravitational horizons, the number of distinguishable internal configurations shrinks until only surface-level resolution remains. This transition marks the geometric origin of the holographic principle in TSRT: in regimes of extreme curvature, information content becomes intrinsically two-dimensional, with entropy scaling in proportion to surface area rather than volume. Thus, TSRT does not merely accommodate holography—it explains it. The dimensional compression of information arises naturally from the causal structure of trembling spacetime, where curvature regulates the admissibility of correlations and enforces geometric coherence.

S22.4.2 Black Hole Entropy from TSRT

Within the TSRT framework, the Bekenstein–Hawking entropy law arises not from statistical assumptions, but from geometric necessity. The entropy–area relation follows directly from the causal limits imposed by trembling-induced resolution near the event horizon. At the horizon, curvature reaches its maximal finite value, and the number of causally distinguishable configurations saturates.

A central result in quantum gravity is the Bekenstein–Hawking formula, previously introduced in Equation (S605), which states that the entropy of a black hole is proportional to the area A of its event horizon:

$$S_{\text{BH}} = \frac{k_B c^3}{4G\hbar} A. \quad (\text{S616})$$

While this formula has been derived through semiclassical methods involving quantum field theory near the horizon and gravitational path integrals, the origin of the universal coefficient $1/4$ remains theoretically opaque. Most derivations assume the Heisenberg uncertainty principle as background-independent and exact.

TSRT offers a geometric derivation rooted in curvature-dependent phase-space resolution. The TSRT uncertainty product is bounded both from below and above, as given by Equation (S600):

$$\frac{1}{2}\hbar \leq \Delta x \Delta p \leq \frac{1}{2}\beta \frac{mc}{\sqrt{\kappa_{\text{local}}}}, \quad (\text{S617})$$

where κ_{local} is the local curvature scalar, m is the particle mass, c the speed of light, and β a dimensionless constant encoding spectral properties of trembling.

The smallest resolvable phase-space cell per degree of freedom is set by the lower bound $\delta x \delta p \sim \hbar/2$. The upper bound determines the maximum phase-space area over which trembling correlations can remain causally coherent. Near a black hole horizon, where curvature approaches its maximum allowed value before singular breakdown, this becomes:

$$\Delta x \Delta p \sim \frac{1}{2}\beta \frac{mc}{\sqrt{\kappa_{\text{horizon}}}}. \quad (\text{S618})$$

To estimate the number of distinguishable configurations, we compare the maximal causal phase-space area to the minimal irreducible cell size:

$$N_{\text{micro}} \sim \frac{\Delta x \Delta p}{\hbar/2} \sim \frac{\beta mc r_s^2}{\hbar}, \quad (\text{S619})$$

where we used the approximation $\kappa_{\text{horizon}} \sim 1/r_s^4$, based on the Schwarzschild Kretschmann scalar:

$$K = R_{\mu\nu\alpha\beta} R^{\mu\nu\alpha\beta} \sim \frac{48G^2 M^2}{c^4 r_s^6} \sim \frac{3}{r_s^4}, \quad (\text{S620})$$

and $r_s = 2GM/c^2$ is the Schwarzschild radius. Substituting this into the upper bound yields:

$$\Delta x \Delta p \sim \frac{1}{2}\beta mc r_s^2. \quad (\text{S621})$$

To compute the total number of accessible microstates over the entire horizon, we sum over all independent degrees of freedom across the surface area:

$$A = 4\pi r_s^2, \quad \ell_P^2 = \frac{G\hbar}{c^3}, \quad (\text{S622})$$

so that

$$N_{\text{total}} \sim \frac{A}{4\ell_P^2}. \quad (\text{S623})$$

Here, the factor $\frac{1}{4}$ reflects the projection of trembling-limited resolution onto the two-dimensional horizon, and emerges naturally from geometric constraints—not by statistical counting or Hilbert space quantization.

The corresponding entropy follows from Boltzmann’s relation:

$$S = k_B \log N_{\text{total}} \sim \frac{k_B c^3}{4G\hbar} A. \quad (\text{S624})$$

Thus, TSRT reproduces the Bekenstein–Hawking entropy law, with the area scaling and prefactor emerging from first-principles limits on causal resolution in trembling spacetime.

The interpretation is significant. Entropy does not count microstates in an abstract Hilbert space, but quantifies the number of causally distinguishable configurations that trembling geometry can support under curvature constraints. The shrinking of admissible phase-space area in high-curvature regions ensures that only a finite number of states remain physically realizable. Hence, the entropy–area law becomes a direct consequence of geometric admissibility, not a thermodynamic postulate. The holographic character of black hole entropy is thus recovered as a projection of trembling-induced resolution onto a two-dimensional causal surface. In TSRT, black hole entropy is not a mystery of quantum gravity—it is a geometric manifestation of causal limits. Entropy measures not ignorance, but the maximal number of distinct correlations consistent with forward-propagating proper time under maximal curvature.

S22.4.3 TSRT vs. Unitarity-Based Frameworks

In quantum mechanics and quantum field theory, unitarity ensures that the evolution of isolated systems preserves total information. This principle underlies much of modern quantum gravity, including the formulation of holographic dualities and the widespread belief that black hole evaporation must conserve information. Hawking’s original derivation of black hole entropy [10] appeared to violate unitarity, giving rise to the black hole information paradox—a problem later addressed through string theory and AdS/CFT duality [16, 91–93].

TSRT, by contrast, does not invoke unitarity at all. It is fully deterministic, but not unitary in the quantum sense. Apparent information loss in TSRT does not result from probabilistic collapse or from inaccessible branches of a universal wavefunction. Instead, it arises from a geometric cutoff on phase-space resolution: in regions of high curvature, certain correlations simply cannot exist, because they would violate the causal structure of trembling spacetime. This leads to a key insight: although TSRT does not rely on unitarity, it still reproduces the canonical entropy–area relation of black hole thermodynamics. Both frameworks are governed by a limiting principle—but where standard theories impose unitarity plus thermodynamic consistency, TSRT enforces causal admissibility of metric fluctuations. Both yield the same quantitative result,

$$S_{\text{BH}} = \frac{k_B c^3}{4G\hbar} A, \quad (\text{S625})$$

but from entirely different premises. In TSRT, the factor $\frac{1}{4}$ emerges from the projection of causally distinguishable trembling configurations onto the black hole horizon surface. No thermodynamic argument is required.

Thus, TSRT does not contradict unitary models—it subsumes them under a deeper geometric framework. What conventional theories treat as preserved quantum information, TSRT reinterprets as geometrically admissible correlation structure. And in regions of extreme cur-

vature, where those correlations are forbidden by causal constraints, the information is not lost—it is simply not defined. In summary, TSRT reaches the same entropy–area result as Hawking, Bekenstein, and holographic theories, but without postulating unitarity or invoking Hilbert space evolution. The entropy of a black hole—and the deeper notion of what constitutes physical information—arises not from quantum statistical reasoning, but from the curvature-dependent structure of spacetime itself.

Evaporation and Information Recovery. In standard quantum field theory, black hole evaporation via Hawking radiation is essential to reconciling unitarity with entropy loss [10]. Quantum fluctuations near the event horizon are thought to generate particle–antiparticle pairs, one of which escapes to infinity while the other falls inward. This results in a gradual reduction of the black hole’s mass, analogous to the thermodynamic evaporation of a liquid surface.

However, this picture introduces a deep paradox. If the outgoing Hawking radiation is purely thermal, it carries no imprint of the internal state of the black hole, apparently violating unitarity—the principle that evolution must preserve information. To resolve this, the holographic principle was proposed [13, 14], positing that all information absorbed by a black hole remains encoded on its event horizon. In this view, the missing information is eventually recovered through subtle correlations in the outgoing radiation [15]. AdS/CFT duality [16] further supports this view, demonstrating that black hole evaporation in certain spacetimes is dual to a unitary boundary theory.

TSRT requires no such mechanism. It replaces quantum evaporation with geometric evolution. In TSRT, black hole entropy evolves not through probabilistic particle emission but through deterministic changes in the curvature-dependent structure of trembling spacetime. Information is never lost in TSRT—it is causally suppressed. In regions of extreme curvature, such as near the horizon, the causal structure restricts the number of distinguishable phase-space configurations. As the black hole loses mass (e.g., through external interactions or dynamical processes), the local curvature decreases and these causal constraints relax.

This relaxation enlarges the space of geometrically admissible correlations. The reduction in entropy reflects not radiated information, but the restoration of phase-space resolution. Information reenters the accessible domain of spacetime not by emission, but by causal unfolding. Thus, black hole “evaporation” in TSRT is not a thermodynamic or quantum process—it is the dynamic reopening of causal structure. The term geometric unfolding refers to the reemergence of previously compressed information as trembling constraints ease with decreasing curvature. In this view, the black hole information paradox is resolved without invoking particle pair production, wavefunction collapse, unitarity, or entanglement transfer. TSRT explains the evolution of black hole entropy as a causal geometric phenomenon: information is never destroyed, only temporarily compressed by the curvature of spacetime.

Postulate Economy and Theoretical Elegance. TSRT achieves with minimal assumptions what quantum gravity frameworks reach only through layered hypotheses. Traditional approaches to black hole thermodynamics rely on a patchwork of ingredients—vacuum fluctuations, unitarity, thermal pair production, and entropy analogies. By contrast, TSRT begins from just two principles: (1) spacetime trembles deterministically, and (2) proper time must remain

real and forward-directed. From these, the entire structure of uncertainty, entropy, and black hole thermodynamics is derived. The Bekenstein–Hawking entropy law follows from curvature-regulated phase-space resolution. No quantum field theory is needed; no Hilbert space structure is assumed.

TSRT thus offers a more parsimonious and geometrically grounded resolution to the fundamental problems of black hole physics. Information is not conserved by fiat—it is filtered by curvature. Entropy bounds are not postulated—they emerge from causal admissibility. In this way, TSRT provides a coherent and self-contained alternative to the standard quantum paradigm.

S22.5 Conclusion

Trembling Spacetime Relativity Theory (TSRT) offers a fundamental shift in our understanding of uncertainty, entropy, and information. Unlike quantum field theory or holographic duality frameworks, TSRT does not rely on unitarity or Hilbert space assumptions. Instead, it grounds these concepts in the deterministic, curvature-dependent structure of trembling geometry. This geometric foundation provides a natural and intuitive resolution to long-standing paradoxes such as black hole evaporation. Information is neither duplicated nor lost—it is geometrically compressed, filtered, and later restructured by the causal coherence of spacetime. In this picture, entropy arises not from statistical ignorance, but from a physical limit on distinguishability imposed by local curvature. At the core of this framework is the curvature-bounded uncertainty relation:

$$\frac{1}{2}\hbar \leq \Delta x \Delta p \leq \frac{1}{2}\beta \frac{mc}{\sqrt{\kappa_{\text{local}}}}, \quad (\text{S626})$$

which replaces the standard, lower-only Heisenberg bound. The lower limit prevents acausal collapse of proper time, while the upper limit contracts phase-space resolution in regions of high curvature. These bounds are not epistemic—they are geometric constraints that preserve causal structure.

Within this framework, unitarity is no longer fundamental. Information is not guaranteed to evolve reversibly; instead, it becomes admissible or inadmissible depending on the causal constraints induced by curvature. In low-curvature regions, the full phase-space structure recovers the standard Heisenberg regime. But near black hole horizons or in the early universe, the set of causally permitted correlations shrinks, excluding fine-grained microstates and enforcing entropy bounds as geometric consequences. This provides a direct derivation of the Bekenstein–Hawking entropy law:

$$S_{\text{BH}} = \frac{k_B c^3}{4G\hbar} A, \quad (\text{S627})$$

not from thermodynamic analogy or quantum field theory, but from the projection of curvature-limited phase-space volume onto the event horizon surface. The factor $\frac{1}{4}$ emerges naturally from the causal cutoff geometry, not by assumption. TSRT thereby completes the uncertainty principle. It reinterprets it not as a measurement artifact, but as a curvature-dependent restriction on which correlations spacetime can support. Trembling geometry determines not only how uncertainty manifests, but how information is allowed to exist.

Importantly, TSRT does not contradict the frameworks of Bekenstein, Hawking, 't Hooft,

or Maldacena. It extends them by uncovering the geometric substratum that gives rise to their entropy bounds and holographic behavior. Where previous models impose the holographic principle as an axiom, TSRT derives it as a consequence of causal trembling. Unitarity, in this light, is not a foundational law but a low-curvature approximation. In extreme geometries, causal structure itself forbids certain correlations—thus breaking unitarity not by loss of information, but by preventing its causal realization. TSRT offers a more economical and causally coherent alternative to unitarity-based quantum gravity. It explains black hole thermodynamics and information constraints from a single geometric principle, fulfilling the ideal of theoretical parsimony: explaining the most with the least. Thus, information is not an abstract entity overlaid on spacetime—it is an emergent property of causal geometry itself. Trembling spacetime dynamically binds information capacity to local curvature: the stronger the curvature, the fewer the distinguishable causal histories. This realization unifies the physics of geometry, uncertainty, and entropy into a single deterministic causal framework.

We now turn to a special case of this causal structure: the behavior of particles with vanishing proper time, such as photons.

S23 Supplementary Material: The Special Case of Particles with Vanishing Proper Time

In TSRT, physical worldlines are parametrized by proper time τ , and all trembling-induced deviations evolve along such curves while preserving causal order. However, a special class of trajectories arises when the proper time between infinitesimally adjacent events vanishes: $d\tau = 0$. These trajectories do not admit any internal temporal separation as experienced by the particle, and they lie on the boundary of the light cone in spacetime geometry. From a geometric standpoint, such paths are null: they are neither timelike nor spacelike, and they define the limiting case of causal propagation in trembling spacetime. Importantly, TSRT does not assume a priori that these trajectories correspond to photons or any other known massless particles. Instead, it treats $d\tau = 0$ as a purely mathematical and geometric condition, arising from the metric structure of spacetime and independent of any specific particle identity. Only a posteriori, after analyzing the implications of trembling along such null worldlines, does TSRT identify that these conditions naturally lead to the existence of particles without rest mass—consistent with the physical behavior of photons and other massless excitations in the Standard Model.

The goal of this section is therefore twofold: first, to analyze the structure and constraints of trajectories with vanishing proper time purely from the internal logic of TSRT; and second, to show how these constraints imply masslessness, directional coherence, and propagation at the maximal causal speed c , without requiring any external postulates from classical relativity or quantum field theory. This limiting case plays a foundational role in establishing the universality of the speed of light, the causal coherence of trembling dynamics, and the geometric origin of massless particles within the TSRT framework.

S23.1 Vanishing Proper Time and Null Worldlines

When $d\tau = 0$, the worldline of a particle becomes null, and its spacetime interval satisfies:

$$d\tau^2 = g_{\mu\nu}dx^\mu dx^\nu = 0. \quad (\text{S628})$$

This condition must be preserved under trembling. That is, trembling-induced deviations $\Xi^\mu(\tau)$ must remain confined to what is, in traditional physics, called the light cone. Any transverse perturbation would generate spacelike segments, violating causality. Thus, the trembling structure of massless particles is constrained: only longitudinal (on-cone) fluctuations are permitted. This explains why massless particles such as photons do not exhibit transverse spatial trembling—they propagate along the causal frontier of spacetime. It is important to clarify that this condition does not, by itself, encode the value of the speed of light. Instead, it merely states that for a given class of particles—such as photons—the proper time interval $d\tau^2$ vanishes. This follows directly from the definition of proper time in terms of the spacetime metric:

$$d\tau^2 = g_{\mu\nu}dx^\mu dx^\nu. \quad (\text{S629})$$

Thus, when the contraction of the displacement vector dx^μ with the metric tensor yields zero, we have:

$$d\tau^2 = 0, \quad (\text{S630})$$

because the path through spacetime lies entirely along, what is ex-post-facto called 'the light cone', where the interval is null. This defines a special class of trajectories that are neither timelike nor spacelike, and correspond to particles that do not experience the passage of proper time along their worldline. This alone does not dictate the maximum velocity, nor does it presuppose that the speed of these particles must be the speed of light.

To avoid circular reasoning, we first derive that there exists a particular limiting speed $v_{\max}$, above which the proper time expression would become negative, leading to an unphysical imaginary time. Only after this derivation do we rely on experimental results from relativity (e.g., the Michelson-Morley experiment [20, 94]) and a theoretical proof in section S24 to establish that $v_{\max}$ corresponds to the speed of light c . This ensures that our reasoning remains logically sound and independent of assumptions from classical relativity.

Introducing the three-velocity v , we express

$$v = \left| \frac{dx^i}{dt} \right|. \quad (\text{S631})$$

We now consider what happens if v were to exceed a certain limiting value, say $v_{\max}$. To analyze the causal structure associated with maximal-speed propagation, we start from the general Minkowski line element for a particle moving in flat spacetime:

$$d\tau^2 = dt^2 - \frac{1}{v_{\max}^2} dx^i dx^i, \quad (\text{S632})$$

where we have introduced a generalized maximal speed $v_{\max}$, which plays the same dimensional role as the speed of light c (c emerges from the causal structure of trembling spacetime as derived in Section S24) but is left undefined at this stage. This form is motivated by restoring units explicitly in the spatial term. Rewriting this expression, we find:

$$d\tau^2 = \frac{1}{v_{\max}^2} (v_{\max}^2 dt^2 - dx^i dx^i). \quad (\text{S633})$$

This expression conveniently isolates the role of $v_{\max}$ in bounding the causal interval. We will now analyze what happens when the spatial displacement per unit time exceeds $v_{\max}$.

Now suppose a particle travels at exactly $v = v_{\max}$. By definition of velocity,

$$dx^i dx^i = v_{\max}^2 dt^2, \quad (\text{S634})$$

which upon substitution into the expression for $d\tau^2$ gives:

$$d\tau^2 = \frac{1}{v_{\max}^2} (v_{\max}^2 dt^2 - v_{\max}^2 dt^2) = 0. \quad (\text{S635})$$

This confirms that for motion at the maximal allowable speed, the proper time interval vanishes: $d\tau = 0$. This defines the boundary between timelike and spacelike propagation in the TSRT

framework.

However, if $v > v_{\max}$, then

$$d\tau^2 = \frac{1}{v_{\max}^2}(v_{\max}^2 dt^2 - dx^i dx^i) < 0. \quad (\text{S636})$$

Since $d\tau^2$ must be non-negative for real proper time, we conclude that exceeding $v_{\max}$ would lead to an unphysical result, namely an imaginary proper time. Therefore, $v_{\max}$ is an upper bound on speed. The reasoning we followed here also implies that if in the timeless experience of a photon, it would be postulated that its world is not smooth, but rough, that this roughness automatically trembles for particles moving at a lower speed. From experimental results in special relativity—notably the Michelson–Morley experiment [95–99]—it is established that the upper speed limit for signal propagation corresponds to the speed of light. Independently, in Section S24, we derive from first principles that the maximal causal speed in trembling spacetime is dynamically constrained to be c . Thus, TSRT naturally identifies $v_{\max} = c$, confirming that the speed of light is the universal limit for causal propagation.

S23.2 Proof That Particles with $d\tau^2 = 0$ Cannot Have Rest Mass

We now prove from first principles that any particle propagating along a worldline with vanishing proper time, $d\tau^2 = 0$, must necessarily have zero rest mass. This result follows directly from the geometric structure of spacetime and the causal definition of proper time in TSRT, without relying on any axioms from special relativity.

To derive the relationship between the four-momentum and the rest mass of a particle, we begin by recalling that the motion of a particle through spacetime is described by its worldline $x^\mu(\tau)$, parametrized by proper time τ . The four-velocity u^μ is defined as the tangent vector to this worldline:

$$u^\mu := \frac{dx^\mu}{d\tau}. \quad (\text{S637})$$

This vector represents the infinitesimal displacement of the particle per unit of proper time. It is a geometric object that lives in the tangent space of the spacetime manifold and transforms as a contravariant vector under coordinate transformations.

In special and general relativity, the four-momentum p^μ is then defined by:

$$p^\mu := m u^\mu = m \frac{dx^\mu}{d\tau}, \quad (\text{S638})$$

where m is the rest mass of the particle. This definition ensures that in the rest frame of the particle, the spatial components of p^μ vanish and the time component equals the rest energy $E = m$, in natural units.

To determine the invariant norm of the four-momentum, we compute the contraction of p^μ with its covariant counterpart p_μ . This contraction must be performed using the metric tensor $g_{\mu\nu}$, which defines the scalar product in curved or flat spacetime:

$$p^\mu p_\mu := g_{\mu\nu} p^\mu p^\nu. \quad (\text{S639})$$

This is the general relativistic analog of computing the squared length of a vector, and it yields a scalar quantity that is invariant under coordinate transformations.

Substituting the definition of $p^\mu = m dx^\mu/d\tau$ into this scalar product gives:

$$p^\mu p_\mu = g_{\mu\nu} \left(m \frac{dx^\mu}{d\tau} \right) \left(m \frac{dx^\nu}{d\tau} \right) \quad (\text{S640})$$

$$= m^2 g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}. \quad (\text{S641})$$

Now, by the definition of proper time in terms of the spacetime metric, we have:

$$d\tau^2 = g_{\mu\nu} dx^\mu dx^\nu, \quad (\text{S642})$$

so dividing both sides by $d\tau^2$ yields:

$$g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 1, \quad (\text{S643})$$

which holds for timelike worldlines, where $d\tau^2 > 0$.

Substituting this into the expression for the norm of the four-momentum, we obtain the so-called mass-shell condition:

$$p^\mu p_\mu = m^2 g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = m^2. \quad (\text{S644})$$

This identity—valid in natural units, where $c=1$, tells us that the squared norm of the four-momentum vector is equal to the square of the rest mass, regardless of the particle's velocity or coordinate frame, as long as the motion is timelike. The metric $g_{\mu\nu}$ ensures that this contraction respects the geometric structure of spacetime, including its signature and curvature. It is a central tool in general relativity for converting contravariant components into scalars and for defining physical invariants. Therefore, the link between $p^\mu p_\mu$ and m^2 is mediated by the metric $g_{\mu\nu}$, which encodes how distances and times are measured in the underlying spacetime geometry. This result is not an assumption but follows directly from the geometric structure of relativistic theories—and is preserved and generalized in TSRT.

Now consider a particle whose worldline satisfies:

$$d\tau^2 = g_{\mu\nu} dx^\mu dx^\nu = 0. \quad (\text{S645})$$

This defines a null trajectory: the particle moves along the light cone, and no proper time elapses along its worldline. Therefore, the parametrization by τ is no longer valid, and the expression $dx^\mu/d\tau$ becomes undefined or divergent. Consequently, the definition of four-velocity and momentum using τ fails for null particles. However, we may still parametrize the trajectory using an arbitrary affine parameter λ , such that:

$$dx^\mu = \frac{dx^\mu}{d\lambda} d\lambda. \quad (\text{S646})$$

Then the null condition becomes:

$$g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = 0. \quad (\text{S647})$$

Let us now define a candidate four-momentum vector for such a trajectory as:

$$p^\mu := m \frac{dx^\mu}{d\lambda}. \quad (\text{S648})$$

This definition formally resembles the massive case but allows us to proceed algebraically. Substituting into the norm:

$$p^\mu p_\mu = m^2 g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = m^2 \cdot 0 = 0. \quad (\text{S649})$$

Therefore,

$$p^\mu p_\mu = 0 \quad \Rightarrow \quad m = 0. \quad (\text{S650})$$

In the TSRT framework, rest mass arises from the integration of trembling-induced curvature over proper-time intervals. That is, mass is not fundamental—it emerges from the cumulative effect of geometric fluctuations experienced along a particle’s worldline. These fluctuations require a finite, nonzero proper-time separation between events to accumulate any physical effect. However, for a null worldline with $d\tau = 0$, there exists no such interval—no physical evolution occurs along the proper-time axis. Trembling modes cannot oscillate in the absence of causal duration, and thus no curvature accumulation or mass generation is possible. The particle is geometrically constrained to propagate along a lightlike filament, without transverse trembling or internal clock cycles. Therefore, any particle whose worldline satisfies $d\tau^2 = 0$ must have zero rest mass. This is not an assumption, but a rigorous geometric consequence of the causal structure of spacetime and the definition of mass in TSRT. Such particles—later identified empirically as photons or other massless excitations—move along null geodesics and exhibit only longitudinal trembling, if any. Their lack of inertial mass is not postulated but emerges from the requirement that proper time remain null. This completes the classification of particle types in TSRT: only timelike worldlines admit trembling structures that generate mass; null worldlines are necessarily massless by construction.

S23.3 Longitudinal-Only Trembling for Massless Particles

For particles traveling along null trajectories, the proper time τ identically vanishes: $d\tau^2 = 0$ at every point along the path. In the TSRT framework, this imposes strong geometric constraints on allowable trembling deviations. In particular, any fluctuation $\Xi^\mu(\tau)$ added to the base trajectory must preserve the null condition:

$$g_{\mu\nu}(x) \left(\frac{dx^\mu}{d\lambda} + \frac{d\Xi^\mu}{d\lambda} \right) \left(\frac{dx^\nu}{d\lambda} + \frac{d\Xi^\nu}{d\lambda} \right) = 0. \quad (\text{S651})$$

This condition can only be satisfied if $\Xi^\mu(\tau)$ is aligned with the direction of the original null tangent vector $dx^\mu/d\lambda$. In other words, the fluctuation must be proportional to the null direction

itself:

$$\Xi^\mu(\tau) = \alpha(\tau) \frac{dx^\mu}{d\lambda}, \quad (\text{S652})$$

where $\alpha(\tau)$ is an arbitrary scalar function. Such fluctuations deform the affine parametrization along the lightlike worldline without introducing any transverse displacement. These are known as longitudinal deviations, and they shift the phase of the trajectory (i.e., the location along the null generator) without pushing the worldline off the light cone.

This has profound consequences. In TSRT, the trembling modes that give rise to inertial mass originate from transverse oscillations in spacetime. These fluctuations accumulate curvature and generate proper-time cycles. Through geometric action over finite τ , they encode the internal clock of the particle—responsible for building rest mass from local curvature. However, in the case of lightlike motion, the absence of a proper-time interval $d\tau = 0$ forbids such transverse oscillations. Any attempt to perturb a null trajectory in a transverse direction would violate the causal condition $d\tau^2 \geq 0$, producing either a spacelike (acausal) displacement or a timelike one (which would imply rest mass and contradict the null constraint).

Hence, massless particles are restricted to a degenerate class of trembling behavior: their motion is constrained to longitudinal fluctuations along a null filament, with no transverse structure and no internal phase accumulation. Trembling occurs, but only in the direction of propagation, and its influence is limited to local phase-like shifts rather than curvature cycles. This is not merely a heuristic argument. From the variational perspective, consider an infinitesimal fluctuation of the worldline that perturbs the line element $d\tau^2$. If $\delta d\tau^2 > 0$, the perturbed trajectory becomes timelike rather than null. In TSRT, timelike trajectories allow for finite proper-time intervals, enabling the trembling oscillations to cycle transversely through spacetime. These cycles locally curve the geometry along the worldline and accumulate geometric action—effectively storing curvature over time. This accumulated curvature is what gives rise to rest mass in the TSRT framework. Therefore, any such deviation from the null condition would imply the spontaneous emergence of inertial mass through proper-time-based curvature accumulation, which contradicts the defining structure of massless particles. If, on the other hand, $\delta d\tau^2 < 0$, the path becomes spacelike, violating causality. Thus, the only physically allowable trembling in the null case satisfies:

$$\delta d\tau^2 = 0, \quad (\text{S653})$$

which can only occur for longitudinal deviations along the null direction itself. This preserves the causal boundary of the light cone and ensures that trembling remains confined to null geometries.

Therefore, massless particles cannot develop rest mass via trembling: their geometric structure is too rigid. The absence of a proper-time interval prevents the phase-based accumulation of curvature, and the restriction to longitudinal trembling forbids internal cyclic behavior. The result is a particle that moves strictly at the maximal causal speed c , along a lightlike geodesic, with no internal clock and no inertial mass. This conclusion is not postulated—it emerges directly from the causal constraints imposed by TSRT. In other words, in TSRT, longitudinal-only trembling is not merely a feature—it is a geometric necessity for massless particles. It enforces

causal propagation exactly along the light cone and explains the absence of inertial structure in null trajectories. The trembling field of a massless particle becomes a degenerate longitudinal field that transports energy and momentum along a direction fixed by the spacetime causal structure but cannot deviate from it. This directly ties the absence of mass to the vanishing of proper-time curvature cycles in TSRT.

S23.4 Photons as the Extremal Limit of Trembling Dynamics

In the TSRT framework, as developed in Section S9.2 and formalized through the action integral in Equation (S665), mass arises from the accumulation of geometric action along timelike worldlines. These worldlines support trembling deviations transverse to the direction of motion, which generate local perturbations in the metric and induce spacetime curvature. When integrated over finite proper-time intervals, these fluctuations yield a net geometric contribution to the action, interpreted as inertial rest energy. This principle is encapsulated in the quantization relation derived in Equation (S517):

$$\mathcal{S} = E \cdot T = \hbar, \quad (\text{S654})$$

where $T = 2\pi/\omega$ denotes the proper-time duration of one complete trembling cycle, and $E \sim \epsilon^2\omega^2$ reflects the effective energy stored in the curvature induced by the trembling amplitude ϵ and frequency ω . As derived more fully in Equation (S517), this action quantization arises from the high-frequency, low-amplitude limit where $E \sim \epsilon^2\omega^2$ and $T \sim 1/\omega$, so that $\mathcal{S} \sim \epsilon^2\omega \rightarrow \hbar$. This energy encapsulates the localized geometric deviation from flatness caused by the particle's internal motion and determines its inertial mass via curvature accumulation.

However, in the special case of massless particles—whose worldlines satisfy $d\tau^2 = 0$, as shown in Equation (S645), there exists no proper-time interval over which trembling can cyclically evolve. That is, the condition $d\tau = 0$ forbids any finite causal separation along the trajectory, eliminating the possibility of accumulating geometric phase or curvature through internal oscillations. Despite this, TSRT requires that such particles still carry energy and momentum, and thus must admit a limiting form of trembling dynamics compatible with null motion.

To resolve this, we consider the extremal limit in which the trembling amplitude $\epsilon \rightarrow 0$ and frequency $\omega \rightarrow \infty$, while ensuring that the action per mode remains finite. In TSRT, the effective energy scale associated with trembling is $E \sim \epsilon^2\omega^2$, and the proper-time period of one oscillation is $T = 2\pi/\omega$. Therefore, the action accumulated over one trembling cycle is:

$$\mathcal{S} = E \cdot T \sim \epsilon^2\omega^2 \cdot \frac{1}{\omega} = \epsilon^2\omega. \quad (\text{S655})$$

To maintain a finite action in the limit $d\tau \rightarrow 0$, we require:

$$\epsilon^2\omega \rightarrow \hbar. \quad (\text{S656})$$

This condition ensures that even in the null limit, where proper-time vanishes and oscillations collapse into a geometrically degenerate path, the residual trembling retains an effective

tive action of $\hbar$. This quantized unit links null dynamics to Planck-scale processes without invoking external quantum postulates; it defines a geometric limit in which the trembling structure remains energetically nontrivial but is compressed into a vanishingly small proper-time domain. Oscillatory content persists in a limiting sense—sufficient to transmit energy and momentum—but without the temporal resolution needed to sustain internal trembling cycles or curvature accumulation. Consequently, mass does not arise in this limit. The particle follows a null worldline and remains massless by construction. This interpretation aligns with the formal proof presented in Subsection S23.2, where it was shown that the rest mass of a particle is directly tied to the nonvanishing of the proper-time interval $d\tau^2$. There, we demonstrated that the four-momentum norm $p^\mu p_\mu = m^2$ becomes ill-defined as $d\tau \rightarrow 0$ unless $m \rightarrow 0$ simultaneously. In other words, sustaining a finite rest mass requires a nonzero proper-time evolution over which the trembling can accumulate geometric curvature. In the limit $d\tau^2 = 0$, the absence of this causal interval precludes the physical mechanisms by which mass is generated in TSRT. Thus, the extremal trembling behavior associated with photons is not only compatible with masslessness but demands it.

In TSRT, photons and other massless excitations are thus not devoid of structure; rather, they occupy the geometric frontier of trembling dynamics, where action is preserved even as proper-time separation vanishes. This limit captures the essential characteristics of null propagation: energy transfer without inertial mass and geometric coherence without internal evolution along τ . This construction applies directly to the photon. In TSRT, the photon is interpreted not as a wavelike excitation of a preexisting field but as a limit case of geometric trembling in which the energy is carried along a lightlike trajectory without internal phase evolution. The photon’s momentum arises from the high-frequency longitudinal trembling, but the absence of a proper-time period prevents the generation of mass. Moreover, this formulation aligns with the Planck–Einstein relation $E = \hbar\omega$, reinterpreted in TSRT as the limit in which the oscillation energy becomes unbounded, while the corresponding period shrinks to zero, keeping the geometric action finite. Unlike conventional field-theoretic models that rely on postulated gauge symmetry and field quantization, the TSRT approach derives the photon’s properties directly from spacetime geometry and causal consistency.

In summary, the photon emerges in TSRT as the boundary configuration of trembling spacetime dynamics: a massless, causally coherent excitation confined to the light cone, carrying quantized action and energy, but lacking proper-time-dependent internal structure. This derivation not only confirms the masslessness of the photon but offers a new geometric interpretation of its propagation, coherence, and quantization within a purely spacetime-based framework.

S23.5 Conclusion

Particles following worldlines with vanishing proper time $d\tau = 0$ arise naturally in TSRT as the geometric boundary between timelike and spacelike propagation. Rather than assuming these represent photons or any specific physical entity, TSRT treats them as a logically distinct class of causal trajectories, characterized by null intervals and subject to strict trembling constraints. Through first-principles analysis, we have shown that trembling along such null worldlines must remain longitudinal, preserving alignment with the light cone and forbidding any transverse

deviation. Most importantly, we have demonstrated that any attempt to associate a nonzero rest mass with a particle moving along a $d\tau^2 = 0$ trajectory leads to inconsistency: proper time would become imaginary, and the mass-energy relation would break down. Thus, the absence of rest mass is a geometric necessity, not a postulate. Only after establishing these results does TSRT identify such null-propagating, massless excitations with physical particles like photons. Their constant speed, causal coherence, and directionally constrained trembling behavior emerge directly from the mathematical structure. These particles define the extremal limit of trembling dynamics and serve as the geometric carriers of maximal causal influence.

In the next section, we show how this limiting behavior leads to a rigorous, derivation of the invariance of the speed of light c in TSRT, without relying on assumptions from special relativity.

S24 Supplementary Material: Derivation of the Invariance of c in TSRT

S24.1 Introduction and Contextual Overview

In Trembling Spacetime Relativity Theory (TSRT), the maximal signal speed—identified with the speed of light in conventional physics—is not postulated, but emerges as a geometric and causal bound from first principles. The purpose of this section is to rigorously derive the invariance of this maximal speed c , showing that all massless excitations must propagate at this speed in every inertial frame.

Unlike in Special Relativity, where the constancy of light speed is postulated as a symmetry principle, TSRT derives this result from the geometric structure of proper-time-parametrized worldlines, the preservation of causal cones under trembling-induced deviations, and the variational coherence of null trajectories. This derivation is deeply interwoven with results established in earlier sections. In Section S3, it was shown that the Lorentzian structure of spacetime arises directly from the requirement that proper time τ be a real-valued, monotonic, and invariant quantity along all physical worldlines. This leads naturally to the identification of null intervals $d\tau^2 = 0$ as causal boundaries. In Section S23, we demonstrated that particles propagating along null trajectories cannot possess rest mass, and that trembling deviations for such particles are restricted to longitudinal components, preventing curvature accumulation. This means null propagation implies and enforces masslessness.

Building on these insights, the present section completes the logical arc: it shows that massless particles must propagate at a fixed maximal speed c , and that this speed is invariant under all frame transformations and trembling-preserving perturbations. The speed of light thus becomes an emergent consequence of trembling spacetime geometry—not an imposed constant.

S24.2 Emergent Causal Cones and the Maximal Speed

The trembling deviation field $\Xi^\mu(\tau)$ perturbs particle trajectories within the allowed causal structure. To maintain causal order, such deviations must preserve the non-negativity of the proper-time interval:

$$d\tau^2 = g_{\mu\nu} dx^\mu dx^\nu \geq 0. \quad (\text{S657})$$

This condition defines the future light cone at each point in spacetime and prohibits trembling-induced excursions into spacelike regions. For massless particles, whose trajectories lie exactly on the light cone, $d\tau^2 = 0$ must hold identically along the entire worldline. TSRT formulates all spacetime evolution relative to proper time τ , as described in Section S3. Accordingly, the spacetime coordinates and the metric are both functions of τ :

$$x^\mu = x^\mu(\tau), \quad g_{\mu\nu} = g_{\mu\nu}(x^\alpha(\tau), \tau). \quad (\text{S658})$$

This dependency encodes the notion that spacetime itself is a dynamical, trembling entity—its geometry evolves alongside the physical processes it governs. Proper time τ serves as the clock not only of particles but of geometry itself.

As shown in Section S23, the condition $d\tau^2 = 0$ uniquely defines the maximal causal propagation limit. Any trajectory exceeding this limit would yield $d\tau^2 < 0$, violating causality and rendering the worldline unphysical. Conversely, any attempt to deviate from a null trajectory into $d\tau^2 > 0$ would generate rest mass, as proven in Subsection S23.2. Thus, null propagation is both a boundary condition and a dynamical attractor for massless particles. From this foundation, we now demonstrate that such null trajectories correspond to propagation at a fixed maximal speed c , and that this speed is invariant under all physical frame changes that preserve the causal structure.

To determine the maximal causal propagation speed in TSRT, we focus specifically on null trajectories, which lie on the boundary of the causality cone and define the limit of causal influence. For these trajectories, the proper-time interval satisfies:

$$g_{\mu\nu} dx^\mu dx^\nu = 0. \quad (\text{S659})$$

Since all four coordinates x^μ are functions of the experienced proper time τ , we can write $dx^\mu = \frac{dx^\mu}{d\tau} d\tau$, and thus the null condition becomes:

$$g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0. \quad (\text{S660})$$

This is the key condition characterizing the limiting propagation speed of causal signals in TSRT. However, it is not automatically preserved in a dynamically trembling geometry. To ensure that a null trajectory remains null over time, we impose the requirement that this condition holds for all τ , and investigate the conditions under which this requirement is satisfied. Differentiating Equation (S660) with respect to τ , we obtain:

$$\frac{d}{d\tau} \left(g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} \right) = 0, \quad (\text{S661})$$

which expands as:

$$\frac{dg_{\mu\nu}}{d\tau} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} + 2g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{d^2 x^\nu}{d\tau^2} = 0. \quad (\text{S662})$$

This relation defines a constraint between the local metric fluctuations (first term) and the acceleration of the massless particle (second term). It tells us that for the null condition to remain preserved along the trajectory, the trembling-induced variation in the metric must be exactly compensated by the acceleration of the trajectory. That is,

$$\frac{dg_{\mu\nu}}{d\tau} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = -2g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{d^2 x^\nu}{d\tau^2}. \quad (\text{S663})$$

This is not an identity but a consistency condition. In the following subsection, we derive the generalized geodesic equation from the variational principle applied to a proper-time-dependent metric. We then show that this geodesic equation yields precisely the compensating term on the right-hand side of Equation (S663). Therefore, the preservation of the null condition is not assumed, but proven as a dynamical consequence of TSRT.

S24.3 Frame Independence and Causal Structure Preservation

Trembling paths are constructed to be Lorentz-covariant. The deviation field $\Xi^\mu(\tau)$ transforms as a four-vector under Lorentz boosts, preserving the causal relation $d\tau^2 \geq 0$ in all frames. If a massless excitation propagated at a speed greater than c in one inertial frame, it would appear to propagate backward in time in another. This contradicts the foundational constraint of proper-time monotonicity:

$$\frac{d\tau}{d\lambda} > 0. \quad (\text{S664})$$

To ensure forward time evolution across all frames, the maximal signal speed must be invariant under Lorentz transformations. This enforces the universality of c without postulating it. Subsection S24.2, Equation (S662) in particular, demonstrated that the preservation of the null condition in trembling spacetime requires a specific dynamical compensation between metric fluctuations and the particle's proper-time acceleration. This was formulated as a necessary condition for causal consistency but left open whether such a compensation mechanism actually arises from the dynamics of TSRT. In this subsection, we prove that this condition is indeed satisfied by deriving the full geodesic equation from first principles, using a variational principle applied to the trembling spacetime metric. The result confirms that the required acceleration term naturally emerges from the structure of the equation of motion, thus validating the dynamical preservation of null trajectories.

In Trembling Spacetime Relativity Theory (TSRT), the motion of a particle through spacetime is governed by the geometry of a dynamically fluctuating metric $g_{\mu\nu}(x, \tau)$, which depends explicitly on both spacetime position and the particle's experienced proper time τ . Unlike conventional general relativity, which assumes a fixed causal structure and postulates the invariance of the speed of light c , TSRT allows the maximum propagation speed to emerge naturally from the underlying trembling geometry. To determine the dynamical trajectory of a particle in this setting, we adopt a variational approach rooted in the principle of extremal proper time. The particle's worldline is parametrized by proper time τ , and the action functional governing its motion is given by:

$$S = \int d\tau \mathcal{L}(x^\mu, \dot{x}^\mu, \tau), \quad \text{with} \quad \mathcal{L} := \sqrt{g_{\mu\nu}(x(\tau), \tau) \dot{x}^\mu(\tau) \dot{x}^\nu(\tau)}, \quad (\text{S665})$$

the Lagrangian, where $\dot{x}^\mu := \frac{dx^\mu}{d\tau}$, and $g_{\mu\nu}(x, \tau)$ is the proper-time-dependent spacetime metric. This action measures the accumulated proper time along the trajectory and generalizes the relativistic point-particle action to accommodate trembling-induced metric fluctuations. This form of the action functional is consistent with the variational setup introduced in Section S3 (Equation (S665)), where the same proper-time parametrization and trembling-dependent metric structure were used. This ensures that the variational derivation here remains aligned with the geometric foundations of TSRT developed earlier.

To derive the equations of motion, we vary the path $x^\mu(\tau) \rightarrow x^\mu(\tau) + \delta x^\mu(\tau)$, while holding the endpoints fixed: $\delta x^\mu(\tau_1) = \delta x^\mu(\tau_2) = 0$. The variation of the action is:

$$\delta S = \int d\tau \left(\frac{\partial \mathcal{L}}{\partial x^\mu} \delta x^\mu + \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \delta \dot{x}^\mu \right), \quad (\text{S666})$$

where $\delta \dot{x}^\mu = \frac{d}{d\tau}(\delta x^\mu)$. Integrating the second term by parts and discarding the boundary terms (since δx^μ vanishes at the endpoints), we obtain:

$$\delta S = \int d\tau \left[\frac{\partial \mathcal{L}}{\partial x^\mu} - \frac{d}{d\tau} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \right) \right] \delta x^\mu. \quad (\text{S667})$$

Requiring stationarity of the action ($\delta S = 0$) for arbitrary δx^μ yields the Euler–Lagrange equations:

$$\frac{d}{d\tau} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \right) - \frac{\partial \mathcal{L}}{\partial x^\mu} = 0. \quad (\text{S668})$$

To evaluate these terms explicitly, we compute:

$$\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} = \frac{1}{2\mathcal{L}} \cdot 2g_{\mu\nu}(x, \tau) \dot{x}^\nu = \frac{g_{\mu\nu}(x, \tau) \dot{x}^\nu}{\mathcal{L}}. \quad (\text{S669})$$

Similarly, using the chain rule:

$$\frac{\partial \mathcal{L}}{\partial x^\mu} = \frac{1}{2\mathcal{L}} \cdot \frac{\partial g_{\alpha\beta}}{\partial x^\mu} \dot{x}^\alpha \dot{x}^\beta. \quad (\text{S670})$$

Substituting into Equation (S668), we find:

$$\frac{d}{d\tau} \left(\frac{g_{\mu\nu}(x, \tau) \dot{x}^\nu}{\mathcal{L}} \right) - \frac{1}{2\mathcal{L}} \cdot \frac{\partial g_{\alpha\beta}}{\partial x^\mu} \dot{x}^\alpha \dot{x}^\beta = 0. \quad (\text{S671})$$

Now, observe that:

$$\frac{d}{d\tau} \left(\frac{g_{\mu\nu} \dot{x}^\nu}{\mathcal{L}} \right) = \frac{1}{\mathcal{L}} \frac{d}{d\tau} (g_{\mu\nu} \dot{x}^\nu) - \frac{1}{\mathcal{L}^2} \frac{d\mathcal{L}}{d\tau} g_{\mu\nu} \dot{x}^\nu. \quad (\text{S672})$$

To proceed with a variational principle in the TSRT framework, it is crucial to understand the role of the proper-time parametrization in determining the form of the Lagrangian $\mathcal{L}$. Specifically, we emphasize that along any worldline parametrized by proper time τ , the Lagrangian is identically equal to unity. This is not an arbitrary normalization but follows directly from the geometric definition of proper time in a Lorentzian manifold.

Let us begin with the action functional introduced in TSRT:

$$S = \int d\tau \mathcal{L}, \quad (\text{S673})$$

where τ is proper time, and the Lagrangian $\mathcal{L}$ is defined as the line element per unit proper time:

$$\mathcal{L} := \sqrt{g_{\mu\nu}(x(\tau), \tau) \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}}. \quad (\text{S674})$$

This definition corresponds to the standard arc-length Lagrangian in general relativity and follows classical treatments of proper-time actions (see, e.g., [31, 57]). For a curve parametrized by an arbitrary monotonic parameter λ , the Lagrangian takes the form:

$$\mathcal{L}_\lambda = \sqrt{g_{\mu\nu}(x(\lambda)) \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}}. \quad (\text{S675})$$

In this general case, $\mathcal{L}_\lambda$ is not necessarily unity—it represents the rate at which proper time

accumulates relative to λ , i.e., $d\tau/d\lambda$.

However, when the parameter λ is chosen to be the proper time τ itself, this expression simplifies dramatically. Since the line element is defined by:

$$d\tau^2 = g_{\mu\nu}(x(\tau))dx^\mu dx^\nu, \quad (\text{S676})$$

and since $dx^\mu = (dx^\mu/d\tau)d\tau$, we substitute into the above to obtain:

$$d\tau^2 = g_{\mu\nu}(x(\tau)) \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} d\tau^2. \quad (\text{S677})$$

Cancelling $d\tau^2$ on both sides, we find:

$$g_{\mu\nu}(x(\tau)) \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 1. \quad (\text{S678})$$

This condition is known as the unit-speed constraint or normalization condition for proper-time parametrization. It expresses that the four-velocity vector $u^\mu := dx^\mu/d\tau$ has unit norm:

$$u^\mu u_\mu = g_{\mu\nu} u^\mu u^\nu = 1, \quad (\text{S679})$$

which is standard in the literature on relativistic particle motion (see [34], Sec. 2.2; [42], Chap. 1). Thus, the square root defining the Lagrangian evaluates to:

$$\mathcal{L} = \sqrt{g_{\mu\nu}(x(\tau)) \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}} = \sqrt{1} = 1. \quad (\text{S680})$$

This identity implies that the action in Equation (S665) becomes, as in Equation (S673):

$$S = \int d\tau, \quad (\text{S681})$$

which is simply the total proper time between two events—a geometric invariant under reparametrizations, and the natural quantity to extremize when deriving geodesic equations via variational principles.

The condition $\mathcal{L} = 1$ encapsulates the fact that along a proper-time-parametrized curve, the rate of accumulation of spacetime interval per unit of proper time is exactly one. In physical terms, this corresponds to the normalization of a particle's worldline such that its own clock measures unit elapsed time per unit evolution—a natural and frame-invariant condition that ensures consistent physical interpretation across observers. This simplification has practical implications. Because $\mathcal{L} = 1$, the variation of the action no longer needs to account for a non-trivial Lagrangian function—it becomes a variation of the path geometry alone. This simplifies the derivation of the geodesic equation and allows the trembling corrections in TSRT to be clearly distinguished from the classical variational structure. It is important to note that this unity of the Lagrangian is not an imposed constraint, but an automatic consequence of choosing the proper-time parametrization. Had we retained an arbitrary λ , then $\mathcal{L}_\lambda \neq 1$, and the corresponding Euler–Lagrange equations would contain additional structure (such as Lagrange multipliers) to account for the non-unit speed. In proper time, this ambiguity vanishes, and the

dynamics take a manifestly covariant and geometrically natural form.

Therefore, the identity $\mathcal{L} = 1$ along proper-time-parametrized worldlines is not a triviality but a geometric axiom of TSRT. It follows directly from the definition of proper time as the invariant interval in a Lorentzian manifold and underpins the variational consistency of the theory. As such, it plays a critical role in ensuring the causal and geometric integrity of TSRT's trembling geodesic dynamics.

As a consequence, in proper-time parametrization, the Lagrangian simplifies and introduces no additional dynamical weight, allowing us to compute variations without rescaling. Thus, since $\mathcal{L} = 1$ along any proper-time parametrized worldline, all derivatives of $\mathcal{L}$ vanish, and Equation (S672) simplifies to:

$$\frac{d}{d\tau} (g_{\mu\nu} \dot{x}^\nu) - \frac{1}{2} \frac{\partial g_{\alpha\beta}}{\partial x^\mu} \dot{x}^\alpha \dot{x}^\beta = 0. \quad (\text{S682})$$

This is the Euler–Lagrange equation in a trembling geometry, balancing kinetic evolution with spacetime gradients. It explicitly incorporates the τ -dependence of the metric through both direct coordinate derivatives and the implicit evolution of $g_{\mu\nu}$ along the path $x^\mu(\tau)$. Equation (S682) confirms that the proper-time variational principle yields equations of motion compatible with the TSRT framework and captures the influence of both curvature and trembling in a unified form. This Euler–Lagrange equation also appears in General Relativity when the proper time integral is varied to obtain the geodesic equation for a free particle [34]. However, in GR, the metric $g_{\mu\nu}(x)$ is a static field defined over spacetime and does not evolve along the particle's worldline. In contrast, TSRT generalizes this structure by allowing the metric to depend explicitly on the experienced proper time τ , encoding dynamic geometric fluctuations. This trembling dependence introduces new dynamical behavior not present in GR, while preserving the formal structure of the variational principle.

For interpretive clarity, the Euler–Lagrange Equation (S682),

$$\frac{d}{d\tau} \left(g_{\mu\nu}(x, \tau) \frac{dx^\nu}{d\tau} \right) - \frac{1}{2} \frac{\partial g_{\alpha\beta}(x, \tau)}{\partial x^\mu} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0, \quad (\text{S683})$$

can be rewritten in a more familiar geometric form—the generalized geodesic equation. This reformulation separates the inertial (acceleration) part of the motion from the curvature-induced effects of the background geometry, making the spacetime influence on the path more transparent.

We begin by applying the chain rule to the first term in Equation (S683):

$$\frac{d}{d\tau} \left(g_{\mu\nu} \frac{dx^\nu}{d\tau} \right) = \frac{\partial g_{\mu\nu}}{\partial x^\lambda} \frac{dx^\lambda}{d\tau} \frac{dx^\nu}{d\tau} + g_{\mu\nu} \frac{d^2 x^\nu}{d\tau^2}. \quad (\text{S684})$$

Substituting this into Equation (S683), we get:

$$\left(\frac{\partial g_{\mu\nu}}{\partial x^\lambda} \frac{dx^\lambda}{d\tau} \frac{dx^\nu}{d\tau} \right) + g_{\mu\nu} \frac{d^2 x^\nu}{d\tau^2} - \frac{1}{2} \frac{\partial g_{\alpha\beta}}{\partial x^\mu} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0. \quad (\text{S685})$$

Rearranging terms:

$$g_{\mu\nu} \frac{d^2 x^\nu}{d\tau^2} = \frac{1}{2} \left[\frac{\partial g_{\alpha\beta}}{\partial x^\mu} - 2 \frac{\partial g_{\mu\beta}}{\partial x^\alpha} \right] \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau}. \quad (\text{S686})$$

Now, we multiply both sides by the inverse metric $g^{\mu\rho}$ and relabel indices to isolate the acceleration term $\frac{d^2 x^\nu}{d\tau^2}$. The resulting equation takes the form:

$$\frac{d^2 x^\nu}{d\tau^2} + \tilde{\Gamma}_{\alpha\beta}^\nu(x, \tau) \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} = 0, \quad (\text{S687})$$

where $\tilde{\Gamma}_{\alpha\beta}^\nu(x, \tau)$ is the generalized connection coefficient, defined as:

$$\tilde{\Gamma}_{\alpha\beta}^\nu := \frac{1}{2} g^{\nu\sigma} \left(\frac{\partial g_{\sigma\alpha}}{\partial x^\beta} + \frac{\partial g_{\sigma\beta}}{\partial x^\alpha} - \frac{\partial g_{\alpha\beta}}{\partial x^\sigma} \right). \quad (\text{S688})$$

These coefficients encode how the coordinate grid "bends" due to the curvature and trembling of spacetime. In general relativity, the $\Gamma_{\alpha\beta}^\nu$ are the Christoffel symbols, derived from a spacetime metric that depends only on coordinates x^μ . In TSRT, however, the metric $g_{\mu\nu}(x, \tau)$ also depends explicitly on the particle's proper time τ . As a result, $\tilde{\Gamma}_{\alpha\beta}^\nu$ inherits this time dependence, reflecting how the underlying geometry trembles as the particle moves along its worldline. This time dependence is crucial: it reflects that the geometry is not a fixed background but a dynamically evolving structure tied to the proper-time progression of the particle. The geodesic equation (S687) thus describes the natural path of a particle through trembling spacetime, where both curvature and internal fluctuations are unified into a single geometric evolution.

Importantly, this formulation makes no assumption about the existence or value of a maximum speed such as the speed of light. In conventional relativity, the constant c is built into the structure of spacetime through Lorentz invariance. By contrast, TSRT derives the geodesic equation directly from a variational principle applied to a proper-time-dependent geometry. The resulting dynamics are therefore more general and allow for causal structure—including speed limits—to emerge from the geometry itself rather than being externally imposed. If a universal limiting speed does exist—i.e., a maximal velocity for causal propagation—its emergence must follow from the structure of Equation (S687) together with the requirement that proper time remain real and positive for physically admissible worldlines. TSRT thereby offers a conceptual shift: the constancy and invariance of c are not embedded axioms but derived consequences of the trembling structure of spacetime. This result confirms the internal consistency of the causal structure in TSRT. Specifically, we now revisit the constraint derived in Section S24.2, which required that for a null trajectory $d\tau^2 = 0$ to remain preserved throughout proper time evolution, the trembling-induced variation of the metric must be dynamically compensated by the acceleration of the particle. The condition was expressed as:

$$\frac{dg_{\mu\nu}}{d\tau} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = -2g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{d^2 x^\nu}{d\tau^2}. \quad (\text{S689})$$

To prove that this compensation condition indeed arises from TSRT dynamics, we now

contract the geodesic equation (S687) with the four-velocity $u^\mu := \frac{dx^\mu}{d\tau}$, yielding:

$$u_\nu \frac{d^2 x^\nu}{d\tau^2} = -u_\nu \tilde{\Gamma}_{\alpha\beta}^\nu u^\alpha u^\beta, \quad (\text{S690})$$

where indices are lowered using the metric: $u_\nu = g_{\nu\lambda} u^\lambda$. Next, recall the definition of the generalized connection coefficient from Equation (S688):

$$\tilde{\Gamma}_{\alpha\beta}^\nu = \frac{1}{2} g^{\nu\sigma} \left(\frac{\partial g_{\sigma\alpha}}{\partial x^\beta} + \frac{\partial g_{\sigma\beta}}{\partial x^\alpha} - \frac{\partial g_{\alpha\beta}}{\partial x^\sigma} \right). \quad (\text{S691})$$

Substituting this into the right-hand side of Equation (S690) and simplifying, we find:

$$u_\nu \tilde{\Gamma}_{\alpha\beta}^\nu u^\alpha u^\beta = \frac{1}{2} \left(\frac{\partial g_{\mu\nu}}{\partial x^\lambda} u^\mu u^\nu u^\lambda - \frac{1}{2} \frac{\partial g_{\mu\nu}}{\partial x^\sigma} u^\mu u^\nu u^\sigma \right) = \frac{1}{2} \frac{dg_{\mu\nu}}{d\tau} u^\mu u^\nu, \quad (\text{S692})$$

We now make the connection with the consistency condition previously derived in Section S24.2. From the geodesic equation (S687), we contracted with $u^\mu := \frac{dx^\mu}{d\tau}$ to obtain:

$$g_{\mu\nu} u^\mu \frac{d^2 x^\nu}{d\tau^2} = -\frac{1}{2} \frac{dg_{\mu\nu}}{d\tau} u^\mu u^\nu. \quad (\text{S693})$$

This equation expresses the fact that the particle's proper-time acceleration $\frac{d^2 x^\nu}{d\tau^2}$ must balance the metric's proper-time variation $\frac{dg_{\mu\nu}}{d\tau}$, projected along the particle's four-velocity u^μ , in order to maintain consistency. To rewrite Equation (S693) in the form of the null preservation condition, we proceed step by step.

First, recall that $u^\mu := \frac{dx^\mu}{d\tau}$, so we can write Equation (S693) as:

$$g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{d^2 x^\nu}{d\tau^2} = -\frac{1}{2} \frac{dg_{\mu\nu}}{d\tau} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}. \quad (\text{S694})$$

Multiplying both sides of the equation by 2 gives:

$$2g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{d^2 x^\nu}{d\tau^2} = -\frac{dg_{\mu\nu}}{d\tau} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}. \quad (\text{S695})$$

Finally, rearranging terms yields:

$$\frac{dg_{\mu\nu}}{d\tau} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} + 2g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{d^2 x^\nu}{d\tau^2} = 0, \quad (\text{S696})$$

which is exactly the condition required to preserve the null condition dynamically under trembling evolution, originally stated in Equation (S689).

This derivation complements the causal constraints outlined in Section S23, where it was shown that the proper-time interval $d\tau^2 \geq 0$ defines the causal boundary separating timelike and null trajectories. The current result demonstrates that this boundary is not only kinematically defined but also dynamically preserved by the equations of motion derived from the TSRT variational principle. Thus, we have shown explicitly that the geodesic equation derived from the TSRT variational principle satisfies the consistency relation needed to maintain the causal

null structure:

$$g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0 \quad \text{remains preserved for all } \tau. \quad (\text{S697})$$

This completes the proof that TSRT dynamics preserve lightlike trajectories through internal geometric consistency, rather than external imposition. The geodesic equation derived from the variational principle enforces the precise compensation required to maintain the null condition dynamically: trembling-induced variations in the metric are inherently counterbalanced by the acceleration term in the equation of motion. As a result, null trajectories remain null throughout evolution, even in a fluctuating spacetime background. Thus, the consistency condition derived in Section S24.2 is not a conjecture but a dynamical consequence of the TSRT framework. The light cone structure is preserved under proper-time evolution, and the existence and invariance of a universal limiting speed—later shown to be the speed of light—emerges not from imposed symmetry principles, but from the internal causal geometry of trembling spacetime itself.

S24.4 Derivation of the Invariance of c

From the structure of TSRT, proper-time monotonicity requires $d\tau^2 \geq 0$, null worldlines $d\tau^2 = 0$ define the extremal case, and Lorentz symmetry guarantees frame independence of this constraint. These principles necessitate the existence of a maximal, invariant speed, denoted c . It is not imposed—it follows from the demand that no trembling-induced deviation violates causality in any frame. All massless excitations propagate at this maximal speed. This emergence of c requires no external assumptions about electromagnetic fields, photons, or quantum mechanics—it is a purely geometric and causal result of TSRT.

We now extend this framework to demonstrate how classical field equations, such as Maxwell's equations, emerge from abelian trembling structures. To determine the universal causal speed, we analyze the implications of the self-consistent causal structure of TSRT. The fundamental postulate is that causal influences cannot propagate outside the causality cone, defined dynamically by the metric structure. We introduce the general relation between spatial and temporal displacements:

$$\frac{dx^i}{d\tau} = v(\tau) \frac{dx^0}{d\tau}, \quad (\text{S698})$$

where $v(\tau)$ represents the maximum causal speed. This relation between spatial and temporal derivatives arises naturally from the general form of the line element

$$d\tau^2 = g_{\mu\nu} dx^\mu dx^\nu, \quad (\text{S699})$$

and serves as a coordinate-level description of the local causal structure. The function $v(\tau)$ represents the maximal permissible speed that keeps the worldline on or within the light cone, and is therefore directly constrained by the metric geometry. Unlike traditional approaches, we do not assume this is the speed of light a priori. We define $v(\tau) := \frac{dx^i}{dx^0}$ along the trajectory. Differentiating both sides of Equation (S698) with respect to proper time τ , we obtain:

$$\frac{dv(\tau)}{d\tau} = \frac{d}{d\tau} \left(\frac{dx^i}{dx^0} \right) = \frac{d}{d\tau} \left(\frac{\dot{x}^i}{\dot{x}^0} \right), \quad (\text{S700})$$

where $\dot{x}^\mu = \frac{dx^\mu}{d\tau}$. This expression confirms that any variation in the causal velocity arises from the structure of the proper-time derivatives, and can be constrained directly through the variational principle.

For the causality cone to be well-defined and remain stable over all frames of reference, it must remain dynamically consistent under trembling fluctuations. This imposes the necessary constraint:

$$\frac{dv(\tau)}{d\tau} = 0, \quad (\text{S701})$$

which ensures that $v(\tau)$ is a fundamental constant. The equation will be re-derived further as in Equation (S708) for direct comparison. This conclusion follows from the requirement that the causal structure, as defined by the metric, does not fluctuate in a way that alters the maximum propagation speed of causal influences. We note, however, that this argument is based purely on maintaining the stability of the causality cone and does not incorporate direct interactions with gauge fields.

S24.4.1 Derivation of the Null Condition from Electromagnetic Forces in TSRT

In TSRT, electromagnetic interactions emerge from fluctuations in the metric tensor, as described in Section S10. The vector potential A_μ arises naturally from these fluctuations:

$$\xi_{\mu\nu} = A_\mu A_\nu. \quad (\text{S702})$$

Since charged particles move under the influence of this potential, the equation of motion for a test particle with four-velocity u^μ is given by:

$$\frac{Du^\mu}{d\tau} = \frac{du^\mu}{d\tau} + \Gamma_{\alpha\beta}^\mu u^\alpha u^\beta = F_\nu^\mu u^\nu, \quad (\text{S703})$$

where F_ν^μ is the electromagnetic field tensor, defined by:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (\text{S704})$$

This force equation applies specifically to charged particles interacting with the emergent abelian gauge field. However, for massless neutral particles such as photons, which do not couple directly to the potential A_μ , the dynamics are governed not by this force law but by null geodesics determined by the variational principle. Their propagation follows from the causal structure of the trembling spacetime geometry rather than from electromagnetic coupling.

For a massless particle such as a photon, the motion equation must remain consistent with the underlying gauge structure of electromagnetism. Since light propagation is governed by the same geodesic equation, setting the mass to zero enforces:

$$g_{\mu\nu} u^\mu u^\nu = 0. \quad (\text{S705})$$

This confirms that light must always follow null geodesics in TSRT.

To fully determine the causal structure of TSRT, we return to the fundamental variation of the metric-dependent action. The geodesic equation, derived from extremizing the action,

provides the dynamical evolution of a test particle. By substituting the null condition for light, the equation simplifies to:

$$\frac{d}{d\tau} \left(g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} \right) = 0. \quad (\text{S706})$$

Expanding this derivative explicitly, we obtain:

$$\frac{dg_{\mu\nu}}{d\tau} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} + 2g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{d^2x^\nu}{d\tau^2} = 0. \quad (\text{S707})$$

This is precisely Equation (S663), Equation (S689) or Equation (S696). The first term accounts for metric fluctuations, while the second term corresponds to the acceleration of the photon. Since TSRT dictates that metric fluctuations encode gauge fields, the equation directly relates the variations in spacetime to the propagation of light. Imposing self-consistency of the trembling spacetime structure, we require that the evolution of metric fluctuations must preserve the causal architecture of spacetime at all scales. In particular, fluctuations induced by trembling must not deform the underlying light-cone structure in such a way that the causal limit for signal propagation would vary across spacetime. If such variations were allowed, trembling would introduce preferred directions or speeds, violating both Lorentz invariance and the fundamental requirement that causality be globally preserved.

In TSRT, electromagnetic forces arise from a specific class of metric perturbations corresponding to abelian gauge interactions, as detailed in Section S10. These gauge interactions are mediated by massless excitations—analogue to photons—which propagate along the causal structure defined by the trembling metric. The propagation speed of these gauge fields is therefore not arbitrary: it must be tied to the causal limit enforced by the trembling spacetime itself. For dynamic consistency, the trembling-induced electromagnetic fluctuations must propagate without inducing drift or deformation of the causal cones defined by the underlying metric. Mathematically, this condition requires that the effective local propagation speed $v(\tau)$ of these massless gauge fields remains invariant along the evolution of proper time τ .

If $v(\tau)$ were allowed to vary, the structure of electromagnetic interactions would fluctuate in an uncontrolled manner, leading to causal inconsistency and violation of gauge coherence. Gauge coherence here refers to the requirement that abelian field strengths—derived from longitudinal trembling—must preserve their divergence-free structure across all frames and over proper time. A varying $v(\tau)$ would violate this coherence by altering the fundamental link between field geometry and signal propagation. To prevent such violations, we impose the differential constraint:

$$\frac{dv(\tau)}{d\tau} = 0, \quad (\text{S708})$$

ensuring that the effective maximum speed remains constant along trembling paths.

Unlike the earlier derivation of this result in Equation (S701), where $\frac{dv(\tau)}{d\tau} = 0$ was derived from the requirement of stability of the light cone structure under trembling fluctuations, the present derivation emphasizes its necessity for the consistent transmission of forces governed by abelian gauge interactions.

The trembling perturbations are constrained to lie within the local causal cone defined by proper-time monotonicity. Fluctuations that would project outside this cone would correspond

to retrocausal or acausal propagation, violating the causal structure of spacetime. As a result, the maximum attainable speed for any perturbation or excitation is the limiting speed c defined by the local causal light cone. Trembling excitations approach this limit asymptotically in the absence of rest mass, strictly preserving relativistic causality. Therefore, since these massless excitations define the fastest permissible causal signals in trembling spacetime, the constant propagation speed $v(\tau)$ in Equation (S708) must necessarily be identified with the invariant causal speed c :

$$v(\tau) = c. \quad (\text{S709})$$

Thus, TSRT confirms that the speed of light c emerges dynamically as a universal causal limit, enforced simultaneously by the preservation of the metric light cone and by the self-consistency of gauge interactions derived from trembling spacetime geometry. No external postulate of constant c is required: it is a necessary consequence of causal coherence at both the geometric and interaction levels.

S24.4.2 Independence from Relativity and Metric Assumptions

Unlike special relativity, TSRT does not begin with the Minkowski metric. Instead, the structure of spacetime—and the emergence of light cones—is determined dynamically from the properties of trembling and the requirement that all deviations respect proper-time causality. Thus, TSRT derives the invariance of c without invoking light propagation, Maxwell’s equations, or any prior assumptions of relativistic kinematics. The speed limit emerges purely from internal geometric consistency.

A possible concern is whether the use of a four-dimensional metric tensor $g_{\mu\nu}$ implicitly assumes a predefined causal structure that already contains the speed of light as an upper bound. However, in TSRT, the metric itself is not static but dynamically fluctuating, and its structure is not assumed to be Minkowskian or dictated by Lorentz invariance. Unlike in Special and General Relativity, where the causal limit follows from a postulated spacetime symmetry, in TSRT, the causal structure emerges from the self-consistency of metric fluctuations. The light cone structure, including the existence of null geodesics, is derived from the interaction of massless gauge fields with the evolving metric rather than imposed through an external assumption. Thus, TSRT establishes the speed limit purely from first principles related to the structure of trembling spacetime. Specifically, the invariance of $v(\tau)$ follows from the stability of the causality cone and the self-consistency of metric fluctuations. Furthermore, the requirement that light follows null geodesics is not imposed but emerges naturally from the interaction of massless gauge fields with the dynamic metric. Finally, the causal structure does not require a predefined Minkowski spacetime but is instead determined dynamically through the evolution of metric fluctuations. This ensures that the invariance of the speed of light is a fundamental result of the theory rather than an implicit assumption embedded within the metric formalism itself. Thus, the derivation of $v(\tau) = c$ remains independent of any prior relativity-based assumptions and follows purely from the internal constraints imposed by the trembling spacetime framework.

We now compare the structure and predictions of TSRT with earlier geometric theories of unification, highlighting its key distinctions from Weyl’s scale-invariant theory.

S24.5 Conclusion

The invariance of the speed of light in TSRT is not a foundational postulate but a consequence of causal geometry. The null boundary $d\tau = 0$ defines the fastest permissible propagation of information, and Lorentz symmetry enforces its universality. In this framework, lightlike particles—those with vanishing proper time—trace the extremal trajectories allowed by trembling spacetime, anchoring the causal structure of the universe in a purely geometric foundation. Next, we examine how abelian trembling structures reproduce the equations of electromagnetism, unifying the geometry of trembling with field-theoretic dynamics.

S25 Supplementary Material: Comparison with Weyl's Theory

The idea that geometry could give rise to physical forces is not new. One of the earliest attempts in this direction was Hermann Weyl's theory of scale invariance [100–102]. Weyl sought to unify gravity and electromagnetism by extending Riemannian geometry to include a local scale symmetry—allowing lengths to vary independently at each point in spacetime. In Weyl's theory, the metric tensor $g_{\mu\nu}(x)$ transforms under a local conformal rescaling:

$$g_{\mu\nu}(x) \rightarrow \Omega^2(x)g_{\mu\nu}(x), \quad (\text{S710})$$

where $\Omega(x)$ is a smooth, positive scalar field. To maintain the form of the connection under this rescaling, Weyl introduced a compensating gauge field $A_\mu(x)$, interpreted as the electromagnetic potential. This approach geometrized electromagnetism alongside gravitation but faced severe physical problems: the non-integrability of clock rates implied by Weyl's geometry contradicted experimental evidence (e.g., fixed atomic spectral lines). In contrast, TSRT does not postulate a local scale symmetry or a compensating field. Instead, it derives fundamental forces and quantization from causal constraints imposed on proper-time-parametrized deviations of worldlines. Trembling spacetime in TSRT encodes fine-scale geometric perturbations that obey strict causal monotonicity ($d\tau/d\lambda > 0$) and Lorentz covariance, without requiring arbitrary gauge freedoms.

Where Weyl attempted to unify by symmetry enlargement, TSRT achieves unification by imposing geometric restrictions: trembling structures are not gauge freedoms, but physical degrees of freedom constrained by causal propagation. Moreover, TSRT does not interpret electromagnetism as an imposed symmetry, but as an emergent abelian structure arising from phase-preserving trembling dynamics. Similarly, the quantization of action, energy, and charge arises from the discreteness forced by proper-time coherence, not from the imposition of gauge invariance.

Finally, unlike Weyl's theory, which was incompatible with observed atomic behavior, TSRT naturally explains the emergence of Planck's constant, the quantization of energy, and the upper speed limit c as consequences of its internal structure. No compensating fields, no arbitrary symmetry principles, and no external assumptions are required.

TSRT may thus be viewed as a geometric theory that succeeds where Weyl's theory failed: it unifies interactions through spacetime geometry, not by enlarging symmetry, but by constraining it through causality.

S25.1 Historical Context: Weyl's 1918 Theory of Gravitation and Electromagnetism

In 1918, Hermann Weyl proposed one of the first major attempts to unify gravitation and electromagnetism within a common geometric framework [41]. Motivated by the formal beauty and structural success of general relativity, Weyl introduced the idea that not only directions (as encoded by the affine connection) but also magnitudes (length scales) could be path-dependent. Specifically, he suggested that the metric structure of spacetime should admit an additional

degree of freedom: a non-integrable scaling of length during parallel transport. In Weyl's formulation, the gravitational field was associated with the symmetric part of the affine connection, while the electromagnetic field arose from the antisymmetric scaling field—essentially a one-form field dictating how unit lengths changed along different paths. This led to a theory in which the electromagnetic potential naturally emerged from geometric considerations, without requiring separate postulation. Mathematically, Weyl's theory possessed considerable elegance. It preserved general covariance, extended the idea of local gauge invariance, and suggested a deep unity between gravitational and electromagnetic forces based solely on the properties of spacetime itself. Furthermore, Weyl's notion of local rescaling invariance—later reinterpreted in the context of modern gauge theories—foreshadowed profound developments in field theory and particle physics [103,104].

However, from the outset, Weyl's theory faced severe physical difficulties. The primary objection, raised by Einstein [105], was that Weyl's non-integrable length scaling would imply observable changes in atomic spectra: the size of rods and the rate of clocks would vary depending on their histories through spacetime. This prediction, which should have produced measurable frequency shifts depending on the path taken by matter through gravitational and electromagnetic fields, was not supported by experimental evidence. Precision measurements of spectral lines showed no such dependence. The non-integrability of clock rates and rulers in Weyl's framework implied a breakdown of absolute measurements, contradicting the empirical constancy of atomic processes. Although Weyl's theory was groundbreaking in introducing the idea of local gauge invariance—a concept that became central to later developments—its specific physical implementation was incompatible with observation. Thus, Weyl's attempt at unification, though mathematically visionary, failed at the fundamental level of empirical validity. The elegance of his geometric extension could not compensate for its contradiction with the causal and stable structure required by physical spacetime. Nonetheless, Weyl's ideas left an enduring legacy, inspiring the conceptual development of modern gauge theories and the deeper geometric formulations that followed.

S25.2 Fundamental Flaws in Weyl's Proposal

Despite its mathematical elegance, Weyl's original unification scheme suffers from deep conceptual flaws that render it physically untenable. The most critical issue lies in Weyl's introduction of a non-integrable length scale. In general relativity, the proper time between two events is a well-defined, path-independent quantity, determined solely by the metric. Weyl's extension breaks this fundamental property: under his theory, the length of a vector after parallel transport depends not only on its endpoints but also on the path taken. Consequently, the ticking rate of clocks and the spatial dimensions of measuring rods would vary depending on their histories through electromagnetic and gravitational fields [105]. This prediction conflicts sharply with empirical reality. Spectroscopic measurements show that atomic transitions have fixed frequencies, independent of the path a clock has traveled through gravitational or electromagnetic fields. No experiment has ever observed the path-dependent time dilation or length contraction that Weyl's theory implies. As Einstein pointed out, if Weyl's geometry were correct, no stable atoms, no consistent measurements, and no reproducible standards of length and time could

exist [105]. The absence of such effects decisively falsified Weyl’s proposal.

Another fundamental problem concerns causality and stability. In Weyl’s framework, the scaling degree of freedom would behave like a massless scalar field coupled to matter, potentially inducing instabilities in the propagation of light and particles. No such instabilities are observed in nature. Electromagnetic waves propagate in vacuum with a sharply defined speed c , and matter evolves along stable, causal trajectories. Allowing the scale of lengths to fluctuate freely would disrupt this coherent structure, leading to inconsistencies with the observed causal evolution of physical systems.

Finally, while Weyl’s notion of local scale invariance presaged important developments in gauge theory, his specific physical identification of the electromagnetic potential with a scaling connection failed to produce the correct field equations for electromagnetism. Maxwell’s equations require a specific structure of antisymmetric field strengths and dynamical propagation properties that Weyl’s geometric construction did not naturally recover [103].

In summary, Weyl’s 1918 theory, although mathematically profound and conceptually inspiring, fails at three decisive levels: It violates empirical observations regarding atomic stability and reproducible standards of measurement, it disrupts the causal and stable evolution of physical processes, and it misrepresents the dynamical structure of electromagnetism. These flaws are not technicalities—they strike at the core requirements for any viable physical theory. A successful unification of gravity and electromagnetism must preserve empirical stability, causal coherence, and the correct dynamical equations. Weyl’s theory, despite its geometric sophistication, does not meet these essential standards.

S25.3 Reappearance of Weyl-Like Ideas: The Case of Lindgren et al.

Draft versions of this work and its follow-up developments were registered for intellectual property purposes in Belgium and France by the author, via the Benelux Office for Intellectual Property (i-Depot 150665 on February 24, 2025; i-Depot 150833 on March 7, 2025; i-Depot 151280 on April 1, 2025) and via France’s Institut National de la Propriété Industrielle (INPI Enveloppe E-Soleau DSO2025003797 on February 25, 2025; DSO2025004471 on March 6, 2025; DSO2025005720 on March 23, 2025; DSO2025008562 on April 25, 2025). Upon completion of the current work, the author became aware, on April 22, 2025, of a paper by Lindgren et al. [106]. Its title and conceptual premise bear superficial resemblance to aspects of Section S10, where electromagnetic gauge fields are shown to arise deterministically from metric fluctuations in trembling spacetime. However, beyond this shared geometric ambition, the two frameworks differ fundamentally in construction, motivation, and internal consistency.

In 2025, Lindgren et al. [106] published a work titled *“Electromagnetism as a purely geometric theory”*, attempting to revive aspects of Weyl’s original program. Their proposal seeks to derive electromagnetic phenomena from purely geometric modifications of spacetime structure, claiming that no additional postulates beyond differential geometry are required.

Lindgren et al.’s approach reintroduces the notion of a path-dependent length connection, similar in spirit to Weyl’s original idea. Their construction modifies the affine connection of spacetime to include additional geometric degrees of freedom interpreted as electromagnetic fields. However, much like Weyl’s attempt, the theory faces critical challenges:

Path-Dependence and Physical Measurements: By modifying the connection in a way that affects the notion of length, the theory risks introducing path-dependent observables, contrary to experimental evidence that the ticking of clocks and the structure of atoms are path-independent.

Violation of Causal Stability: Altering the connection structure without a dynamically constrained trembling framework can lead to ambiguities in the causal structure of spacetime, potentially destabilizing the null cone and allowing non-causal propagation effects.

Absence of Causal Cutoff: In contrast to TSRT, which enforces strict causal constraints on permissible metric fluctuations, the Lindgren et al. model does not inherently regulate the magnitude of geometric modifications, leaving open the possibility of uncontrolled deviations from causal evolution.

Moreover, Lindgren et al. assert that no external postulates—such as the existence of a fundamental length or action scale—are needed in their framework. Yet this absence removes any intrinsic mechanism to impose causality-preserving limits on fluctuations. In TSRT, by contrast, the trembling cutoff directly enforces a finite phase-space resolution and preserves causal order at all scales.

In short, although Lindgren et al.’s work exhibits a commendable effort to explore geometric unification, it ultimately inherits the core weaknesses of Weyl’s proposal: Empirically forbidden path-dependence of fundamental observables, lack of dynamically enforced causal stability, and absence of an intrinsic causal cutoff preventing acausal configurations. Where Weyl’s theory struggled against experimental evidence and internal consistency, Lindgren et al.’s revival faces the same fundamental obstacles, albeit in a modernized mathematical guise. Trembling Spacetime Relativity Theory avoids these pitfalls by deriving gauge structures from causal, curvature-regulated trembling—not by arbitrary connection deformations—and by enforcing strict causal constraints at the deepest geometric level.

S25.4 How TSRT Resolves the Fundamental Problems

Trembling Spacetime Relativity Theory (TSRT) fundamentally differs from both Weyl’s early geometric unification attempts and the modern adaptations by Lindgren et al. [106]. Rather than imposing new geometric structures on spacetime *ad hoc*, TSRT derives all observed physical effects—including electromagnetic gauge fields and uncertainty bounds—directly from the intrinsic causal structure of a deterministically trembling spacetime.

The fundamental improvements introduced by TSRT can be summarized as follows:

1. **Path-Independent Physical Observables.** In TSRT, trembling affects the microscopic structure of spacetime but preserves the path-independence of integrated observables. Proper time, spatial distances, and phase accumulation along worldlines remain consistent and independent of the particular path taken between events. This is guaranteed by the strict enforcement of causal coherence in the trembling fluctuations themselves. No modifications of the length standard occur analogous to Weyl’s path-dependent rescalings, thus preserving agreement with the observed stability of atomic and clock-based phenomena.
2. **Causal Stability and Null Cone Preservation.** The trembling fluctuations in TSRT are con-

strained by the requirement that proper time must advance monotonically everywhere along physical paths:

$$\frac{d\tau}{d\lambda} > 0, \quad (\text{S711})$$

as rigorously established in Section S7. This strict causality constraint dynamically preserves the null cone structure of spacetime and ensures that no superluminal or acausal propagation arises. Fluctuations are geometrically permitted only insofar as they maintain global causal ordering.

3. **Emergence of Gauge Fields from Metric Trembling.** Unlike Weyl and Lindgren et al., who modify connections manually to introduce gauge phenomena, TSRT derives gauge structures naturally from abelian and non-abelian patterns of trembling in the metric tensor. As shown in Section S10, electromagnetic fields emerge as coherent phase patterns of local metric deviations, constrained by causality and curvature—not by arbitrary geometric deformations. This approach ties gauge symmetries directly to spacetime geometry without artificial external postulates.

4. **Existence of an Intrinsic Causal Cutoff.** TSRT imposes an intrinsic cutoff in proper time modulations, ensuring that trembling amplitudes cannot grow beyond limits compatible with causality. This leads to a double-bounded uncertainty relation (Equation (S600)) (fully derived in Section S20) and naturally recovers Planck’s constant $\hbar$ as the minimal action scale arising from the causal geometry itself. Thus, trembling spacetime is self-regulated, preventing pathological behaviors at both the microscopic and macroscopic scales.

5. **Deterministic and Geometric Foundation.** While Weyl’s and Lindgren’s proposals require introducing additional hypotheses (e.g., scale invariance, specific connection modifications, or external dualities), TSRT relies solely on two foundational principles: Spacetime trembles deterministically at all scales, and proper time must remain real and forward-directed under all physical circumstances.

From these, the full structure of gauge interactions, uncertainty relations, black hole thermodynamics, and information bounds emerges as inevitable geometric consequences.

In conclusion, TSRT succeeds where Weyl and Lindgren et al. do not because it derives all physical structure from an internally consistent, causality-preserving trembling of the metric, rather than imposing additional structures by hand. It preserves empirical consistency, maintains the causal stability of spacetime, recovers standard quantum behavior in appropriate limits, and extends naturally into trans-quantum and gravitational regimes without loss of coherence or predictive power.

S25.5 Summary: Lessons from the Weyl–Lindgren–TSRT Comparison

The historical and conceptual analysis presented above leads to several important conclusions:

First, Weyl’s early attempt to unify gravity and electromagnetism through geometric rescaling, though visionary, failed because it introduced path-dependence into fundamental physical observables. This error violated empirical stability (e.g., fixed atomic spectra) and underscored the critical importance of maintaining invariant proper time measures under geometric deformations.

Second, modern works such as that of Lindgren et al. [106] inadvertently revive elements of Weyl’s flawed approach by proposing modifications of spacetime connections to geometrically encode gauge interactions. While their motivations echo the geometric ambitions of TSRT, their framework repeats the historical pitfalls of Weyl: lack of causal constraints, vulnerability to observational inconsistencies, and absence of a natural mechanism to limit or regulate geometric fluctuations.

Third, TSRT emerges as a superior alternative precisely because it avoids these errors at the foundational level. It derives the emergence of forces, uncertainty bounds, and information constraints not by modifying external connections, but by internal geometric trembling subject to strict causal coherence. Proper time remains invariant, causal structure is preserved, and gauge fields emerge deterministically as a natural aspect of trembling spacetime, without any artificial adjustments to connection rules.

Moreover, TSRT reveals that causal stability is not optional: it is the core constraint that governs the admissibility of all geometric structures. Trembling spacetime ensures causal consistency, stabilizes phase-space resolution at the Planck scale, and naturally generates the phenomenology of quantum fields and gravitation, fully integrating them within a single geometric framework.

In short, while Weyl’s and Lindgren’s approaches highlight important ambitions—the geometric unification of forces—TSRT shows how these ambitions can be realized correctly: by grounding geometry itself in deterministic trembling constrained by causality. Rather than modifying how spacetime connects points, TSRT modifies the very way spacetime fluctuates at the most fundamental level, preserving both empirical fidelity and theoretical consistency. Thus, the lessons from Weyl and Lindgren et al. validate the approach of TSRT. Geometry and gauge structure are not to be imposed; they are to be derived from the deeper causal architecture of spacetime itself. Finally, we place the development of TSRT in historical context, tracing how the core concepts evolved from earlier attempts to reconcile gravity, causality, and quantum phenomena.

S26 Supplementary Material: Historical Background

The development of 20th-century physics was dominated by two major frameworks: general relativity and quantum theory. General relativity grounded gravity in the geometry of spacetime [2], while quantum mechanics introduced probabilistic descriptions and operator-based dynamics, culminating in the Standard Model of particle physics. Throughout the century, numerous efforts were made to reconcile these two pillars. Attempts at unification—ranging from Kaluza–Klein theories to string theory and loop quantum gravity—focused either on extending the geometric fabric of spacetime or quantizing the gravitational field itself. Yet none succeeded in producing a fully self-consistent, deterministic, and causally complete framework encompassing both quantum phenomena and general relativistic dynamics. Trembling Space-time Relativity Theory (TSRT) arises from a different trajectory. Instead of quantizing fields or introducing new fundamental particles, TSRT begins from causal geometry itself: it posits that spacetime is not perfectly smooth but exhibits fine-scale deterministic fluctuations. These geometric “tremblings” are constrained by proper-time monotonicity and local Lorentz covariance, and give rise—through coarse-graining—to all familiar physical laws, including quantum dynamics and gauge interactions. Where quantum theory introduces Planck’s constant as a postulate, TSRT derives it as a minimal action per causal oscillation. Where conventional theories impose wave–particle duality and nonlocality as fundamental features, TSRT eliminates these conceptual tensions by showing that all observed quantum behavior arises from local, deterministic metric perturbations. And where symmetry-based approaches assume gauge freedom as primary, TSRT derives gauge interactions from trembling modes classified by their geometric and causal properties.

Historically, this shift marks a return to the geometric realism of Einstein, but with an essential modification: instead of assuming a smooth manifold governed by Einstein’s equations, TSRT treats spacetime as a dynamically trembling medium, where all observed quantization, uncertainty, and interaction emerge from geometry itself. TSRT thus offers a new point of departure—one that neither discards the successes of quantum theory nor inherits its foundational paradoxes. It aims to unify the physical world not by merging quantum and relativistic frameworks, but by subsuming them into a deeper geometric foundation grounded in causal structure and proper time.

S26.1 The Rise of Field Theories and the Separation of Forces

The scientific revolution of the 17th century introduced the concept of force as a fundamental mediator of interactions between objects. Newton’s law of universal gravitation and his laws of motion provided a highly successful, deterministic description of macroscopic phenomena, grounding physics in simple mathematical principles. By the 19th century, the success of Newtonian mechanics was joined by the triumph of Maxwell’s electrodynamics [26], which unified electricity and magnetism into a single set of field equations. The idea that forces could be described not by action-at-a-distance but by local fields became the new standard of physical explanation. Despite these successes, a crucial conceptual separation remained: gravity and electromagnetism were described by entirely different mathematical structures and ontologies.

Gravity acted through instantaneous (in Newton’s view) or long-range forces depending on the gravitational mass, while electromagnetism operated through dynamic vector fields dependent on electric charge and currents.

The 20th century brought a profound deepening of these ideas. Einstein’s general theory of relativity [2] revolutionized the understanding of gravity by replacing Newton’s force model with a geometric picture: mass-energy curves spacetime, and free-falling bodies follow geodesics in this curved geometry. Nevertheless, even after general relativity, the division persisted: gravity was encoded in spacetime curvature, while electromagnetism remained a gauge field over spacetime, treated separately from the geometry itself. No unified theory successfully combined these forces at a fundamental level.

Thus, by the early 20th century, the core framework of physics consisted of separate field theories, each remarkably successful in its domain but fundamentally divided in conceptual structure. The dream of unification—of explaining all forces and interactions as aspects of a single geometric or physical principle—remained tantalizingly out of reach. It was against this backdrop that early geometric unification attempts, such as Weyl’s 1918 theory [41], were born, driven by the hope that spacetime geometry itself might hold the key to explaining not just gravity, but electromagnetism and eventually all forces.

S26.2 Weyl’s Ambition for Geometric Unification

In 1918, Hermann Weyl proposed one of the earliest and most influential attempts to unify gravity and electromagnetism within a single geometric framework [41]. Inspired by the success of Einstein’s general relativity, which had demonstrated that gravitational effects could be attributed to the curvature of spacetime, Weyl sought to extend the geometric interpretation to include electromagnetic interactions. Weyl’s idea was elegant in form: he introduced a new type of geometric transformation, now called a “gauge transformation,” under which not only the coordinates but also the lengths of vectors could vary from point to point in spacetime. In Weyl’s geometry, the connection coefficients—the quantities that determine parallel transport and curvature—were modified by a new vector field. He identified this additional field with the electromagnetic four-potential A_μ . This bold move generalized Riemannian geometry to what is now termed Weyl geometry. In Weyl’s theory, the non-integrability of vector lengths during parallel transport was tied directly to the presence of the electromagnetic field. Gravitation and electromagnetism thus appeared as two different manifestations of spacetime geometry: one related to curvature, the other to a new geometric quantity associated with changes in scale.

However, Weyl’s theory quickly encountered fatal problems. Most notably, it predicted that the rate of clocks and the lengths of rods would vary depending on their histories through spacetime—a prediction grossly inconsistent with empirical observations. Einstein himself pointed out that if Weyl’s theory were correct, the spectral lines of atoms would depend on their prior motion through gravitational and electromagnetic fields, an effect that was not observed [105]. Moreover, Weyl’s identification of the electromagnetic potential with a geometric scaling field lacked a dynamical mechanism capable of reproducing known electromagnetic phenomena with the required accuracy. No explanation was provided for the quantization of electric charge, nor

for the empirical success of Maxwell’s equations in flat spacetime. Despite these shortcomings, Weyl’s work had a profound legacy. His introduction of gauge symmetry foreshadowed the later development of gauge theories in quantum field theory, where local phase invariance would become a central organizing principle. Weyl’s ambition to geometrize all fundamental interactions was visionary, but his specific 1918 proposal failed because it violated key empirical facts about matter and light.

The fundamental lesson of Weyl’s attempt is thus twofold. Geometry may indeed underlie the forces of nature, but not through arbitrary modifications of spacetime structure without regard for physical constraints like causal stability and local proper-time consistency. Successful unification requires a deeper understanding of how fields and spacetime interact—not merely a mathematical analogy but a physically consistent, empirically anchored theory.

Trembling Spacetime Relativity Theory (TSRT) revisits this aspiration, but crucially differs by ensuring that causal integrity, local proper-time monotonicity, and empirical coherence are never violated. TSRT does not alter the basic metric structure arbitrarily; instead, it introduces fine-scale, causal geometric trembling that underlies all forces in a deterministic and testable manner, without the inconsistencies that plagued Weyl’s formulation.

S26.3 The Wave–Particle Dilemma and the Birth of Quantum Mechanics

Following the pioneering work of Einstein on the photoelectric effect [12] and Planck’s quantization of energy [84], Louis de Broglie introduced a bold new idea in 1924: that all particles possess an intrinsic wave character [107]. In his doctoral thesis, de Broglie proposed that a particle of momentum p should be associated with a wave of wavelength λ according to the relation

$$\lambda = \frac{h}{p}, \quad (\text{S712})$$

where h is Planck’s constant.

This revolutionary hypothesis inaugurated the concept of wave–particle duality, suggesting that matter, like light, exhibits both particle-like and wave-like properties depending on the experimental setup. Shortly thereafter, Schrödinger developed a differential equation—the now-famous Schrödinger equation [4]—to describe the evolution of these matter waves:

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi, \quad (\text{S713})$$

where ψ is the wavefunction and $\hat{H}$ is the Hamiltonian operator.

Initially, Schrödinger believed that the wavefunction represented a real, physical wave in space, much like a classical electromagnetic field. However, the emergence of quantum measurement theory, spearheaded by Born [108], reinterpreted ψ as a probability amplitude: $|\psi|^2$ was to be understood as the probability density of finding a particle at a given location. This shift from a physical to a statistical interpretation deepened the conceptual tension between the wave and particle descriptions of reality. The adoption of wave–particle duality created profound philosophical and theoretical ambiguities. Was the wave merely a mathematical tool for predicting measurements, or a real field permeating space? Did particles have definite trajec-

tories, or were they fundamentally indeterminate? These questions led to the birth of quantum mechanics as a probabilistic theory, codified through operator algebras, uncertainty principles, and the abandonment of classical determinism. At the same time, alternative approaches, such as de Broglie’s pilot-wave theory and later Bohmian mechanics [109], attempted to restore determinism by postulating that particles are guided by underlying waves. Yet these models remained marginalized for much of the twentieth century due to the dominant Copenhagen interpretation, which embraced intrinsic randomness and epistemic indeterminacy as fundamental features of nature.

The initial confusion between real waves, probability waves, and particles set the stage for increasingly abstract and complex theoretical developments—including quantum field theory, quantum electrodynamics, and eventually string theory. Each of these frameworks layered additional mathematical structure atop the unresolved paradoxes of wave–particle duality, rather than resolving them at the geometric level. In contrast, Trembling Spacetime Relativity Theory (TSRT) revisits the origin of these paradoxes and offers a unifying resolution: what appears as wave–particle duality is the coarse-grained manifestation of deterministic, fine-scale trembling of spacetime itself. No mystical duality or collapse is required—only the statistical reflection of deeper geometric dynamics. In TSRT, particles remain particles at all times, and their observed statistical properties emerge naturally from the underlying trembling structure, as discussed extensively in Section S20 and Section S19.

S26.4 Heisenberg’s Uncertainty Principle: Revolutionary but Incomplete

In 1927, Werner Heisenberg formulated the uncertainty principle [3], one of the cornerstones of modern quantum mechanics. He demonstrated that there is a fundamental limit to how precisely the position x and momentum p of a particle can be simultaneously known:

$$\Delta x \Delta p \geq \frac{\hbar}{2}. \quad (\text{S714})$$

Here, Δx and Δp denote the standard deviations of position and momentum measurements, and $\hbar$ is Planck’s constant divided by 2π .

Heisenberg’s insight was revolutionary: it implied that uncertainty is not merely a limitation of measurement technology, but an intrinsic property of nature itself. Unlike classical mechanics, where exact knowledge of initial conditions determines future evolution uniquely, quantum mechanics replaced determinism with an irreducible probabilistic framework. However, this triumph came at a conceptual cost. By attributing uncertainty to an intrinsic feature of physical reality, rather than to hidden variables or incomplete knowledge, the theory abandoned classical notions of causality and continuity. Microscopic phenomena were deemed fundamentally random, and the smooth trajectories of Newtonian physics were replaced by probabilistic wavefunctions evolving under operator dynamics. Moreover, Heisenberg’s uncertainty principle, though powerful, provided no explanation for why such a limit should exist. It was treated as an axiom, deeply tied to the non-commutative structure of quantum observables, but without a deeper physical cause grounded in spacetime geometry or dynamics.

This surrender of determinism became codified in the Copenhagen interpretation, champi-

oned by Bohr and Heisenberg, where measurement itself was elevated to a special, fundamentally non-unitary process, and the observer played an essential role in the definition of physical reality. Later interpretations and reformulations—such as the many-worlds interpretation [110], objective collapse models [111], and relational quantum mechanics [112]—attempted to address the discomfort with this loss of classical causality, but none fully restored a realist, deterministic description without introducing new paradoxes or complexities.

In contrast, Trembling Spacetime Relativity Theory (TSRT) provides a geometric explanation for the uncertainty principle without relinquishing determinism or causality. As shown in Section S20, the uncertainty bounds arise naturally from the causal and curvature constraints imposed by the trembling of spacetime itself. In TSRT, uncertainty is no longer an arbitrary limitation—it is the inevitable result of the requirement that proper time remains real, forward-directed, and causally connected, even under fine-scale geometric fluctuations. Thus, while Heisenberg’s principle was a major advance, it represented only a partial understanding of a deeper, geometric truth: uncertainty is not a property of measurement, but a necessary consequence of the causal structure of spacetime.

S26.5 Quantum Field Theory and the Inflation of Complexity

Following the initial success of quantum mechanics, physicists sought to reconcile the theory with the demands of special relativity. This effort culminated in the development of quantum field theory (QFT), beginning with quantum electrodynamics (QED) [113–115], which accurately described electromagnetic interactions through the exchange of virtual photons. QFT introduced the revolutionary idea that particles are excitations of underlying fields permeating spacetime. While extraordinarily successful in matching experimental results, QFT came at the price of vastly increased mathematical complexity. Concepts such as vacuum fluctuations, infinite zero-point energies, and renormalization procedures—where infinities are formally subtracted to yield finite results—became necessary to make the theory predictive.

The Standard Model of particle physics, building on these ideas, achieved an unprecedented unification of electromagnetic, weak, and strong forces through intricate gauge symmetry structures. However, gravity stubbornly resisted quantization within this framework, and numerous unresolved issues—such as the hierarchy problem, the cosmological constant problem, and the measurement problem—persisted.

To address these and to achieve a full unification, even more elaborate theories emerged. String theory [7,8], for instance, replaced point particles with vibrating one-dimensional strings, requiring ten or more dimensions of spacetime, supersymmetry, and a landscape of possible vacua. While mathematically rich and inspiring, these frameworks introduced vast new layers of assumptions and abstractions, moving further away from empirical verifiability. Each step in this progression—QFT, the Standard Model, supersymmetry, string theory—added layers of mathematical sophistication but also obscured physical intuition. Rather than simplifying our understanding of the universe, theories became increasingly baroque, patching over foundational gaps rather than resolving them. In contrast, TSRT offers a profound simplification. It shows that the phenomena traditionally associated with quantum mechanics—uncertainty, superposition-like behavior, energy quantization—and those associated with general relativ-

ity—spacetime curvature, causal horizons, gravitational redshift—can emerge from a single geometric principle: the deterministic trembling of spacetime constrained by causal structure. No renormalization, no extra dimensions, and no ad hoc postulates are required. Thus, where conventional physics expanded complexity to fit observation, TSRT contracts explanation to its geometric core, achieving unity with elegance and conceptual economy.

S26.6 The Persistent Gaps: Unification, Gravity, and Causality

Despite extraordinary achievements over the past century, mainstream physics has not succeeded in closing several of its deepest conceptual gaps. Most notably, the unification of the four fundamental forces—gravitational, electromagnetic, weak, and strong—remains incomplete.

General relativity describes gravity as the curvature of spacetime due to mass-energy, while quantum field theory portrays the other forces as mediated by particle exchanges in flat or perturbatively curved backgrounds. Attempts to quantize gravity—whether through canonical quantization, path integral methods, loop quantum gravity [116], or string theory—have faced profound difficulties. None has produced an experimentally verified theory of quantum gravity. Worse, many approaches abandon or compromise on essential features like local causality, background independence, or predictive uniqueness. Even apart from gravity, quantum mechanics itself leaves open the problem of measurement and the emergence of classicality. The standard Copenhagen interpretation, many-worlds theory, and objective collapse models each introduce additional postulates that are external to the unitary evolution of the Schrödinger equation. Uncertainty is treated as an axiom rather than derived from deeper physical structure.

Moreover, the origin of Planck’s constant $\hbar$, the meaning of probabilistic outcomes, and the reason for a universal maximum speed c remain unexplained in standard frameworks. These constants are simply inserted into the theories, rather than being derived as necessary consequences of more fundamental principles. All existing mainstream approaches, in one form or another, leave causality as an assumption rather than a theorem. While general relativity guarantees causal structure for classical trajectories, quantum theory generally allows nonlocal correlations and superpositions that obscure causal order at the fundamental level. Thus, despite extraordinary progress, modern physics has largely accepted a fragmented view of nature, with gravity and quantum phenomena treated as separate domains, linked only by incomplete, heuristic bridges.

In contrast, TSRT directly addresses these persistent gaps. As demonstrated throughout this work:

- Uncertainty is not postulated but emerges from causal constraints on trembling spacetime (see Section S20).
- The speed of light c is not assumed but derived from the preservation of causal cones under trembling metric fluctuations (see Section S24).
- Planck’s constant $\hbar$ emerges naturally from the action threshold imposed by proper time stability (see Section S19).

- The necessity of spacetime trembling itself is proven as a consequence of preserving global causal order (see Section S21).

TSRT thus offers not merely another alternative—it provides a conceptual closure long sought but never attained within previous frameworks. It restores geometric unity, causal determinism, and foundational simplicity to the heart of physical law.

S26.7 The Paradigm Shift Introduced by TSRT

Trembling Spacetime Relativity Theory (TSRT) marks a decisive break from the traditional trajectory of modern physics. Instead of multiplying fields, dimensions, or probabilistic axioms to explain different phenomena, TSRT reveals that the fundamental structure of spacetime itself—when allowed to tremble around the Minkowski background—is sufficient to account for the essential features of both gravitation and quantum phenomena.

This paradigm shift is characterized by several fundamental achievements:

- **Geometric Simplicity:** TSRT retains a purely geometric foundation, avoiding the introduction of external fields or arbitrary degrees of freedom. Metric fluctuations themselves, governed by causality and proper-time stability, produce all emergent phenomena traditionally associated with quantum uncertainty, field forces, and relativistic limits.
- **Causal Determinism:** Unlike quantum theories that surrender determinism at the fundamental level, TSRT preserves strict causal evolution at every scale. Proper time τ advances monotonically along trembling paths, and all uncertainty or statistical behavior arises from unresolved geometric oscillations, not intrinsic randomness.
- **Unified Force Emergence:** Trembling spacetime naturally gives rise to all known interactions without the need to postulate distinct fundamental forces. Abelian fluctuations reproduce electromagnetism (see Section S10), while non-Abelian structures generate the weak and strong interactions (see Section S9.4). Gravitation arises as the coherent limit of metric fluctuations, Section S13.
- **Origin of Fundamental Constants:** Planck's constant $\hbar$ and the speed of light c are no longer arbitrary inputs but necessary consequences of the geometric and causal structure (see Sections S19 and S24).
- **Resolution of Conceptual Paradoxes:** TSRT eliminates foundational paradoxes such as wavefunction collapse, action-at-a-distance, and black hole information loss, replacing them with local geometric evolution that preserves causal structure (see Sections S22 and S21).

Unlike prior approaches that attempt to stitch quantum theory and general relativity together, TSRT shows that both phenomena are manifestations of a single deeper principle: the causal trembling of spacetime itself. Thus, TSRT is not just a new model. It is a profound reorganization of physical thought: a return to geometric unity, causal determinism, and conceptual

economy, fulfilling Einstein’s dream of a coherent and causally complete understanding of nature—but realized not through additional assumptions, but through the necessary trembling of spacetime.

S26.8 Summary: Historical Lessons and the New Path Forward

The historical trajectory of theoretical physics—from Newton’s fields to Maxwell’s unification, from Einstein’s curved spacetime to the probabilistic framework of quantum mechanics—has been one of extraordinary creativity but also increasing fragmentation. Each new discovery introduced new mathematical layers, new entities, and new postulates, often at the cost of deep conceptual unity.

Weyl’s 1918 attempt at geometric unification was a pivotal moment, demonstrating the enduring appeal of grounding all forces in spacetime structure. Yet its internal contradictions—chiefly, the inability to preserve causality and stable clocks—forewarned the dangers of extending geometry without causal safeguards. Quantum mechanics, sparked by De Broglie’s insight and formalized through Heisenberg and Schrödinger’s work, brilliantly captured experimental data but did so at the price of surrendering determinism, causality, and geometric intuition.

Quantum field theory and its later extensions, from QED to string theory, further multiplied complexity. While these frameworks achieved remarkable calculational success, they often relied on an ever-growing edifice of assumptions: fields atop fields, dimensions atop dimensions, symmetries atop symmetries, with no clear geometric or causal origin.

Throughout this evolution, three fundamental challenges remained unresolved:

1. The geometric unification of all interactions.
2. The causal and deterministic foundation of uncertainty.
3. The integration of gravitation with quantum phenomena.

Trembling Spacetime Relativity Theory (TSRT) offers a radical resolution. It returns to the oldest dream of physics: that nature’s laws are simple, geometric, causal, and universal. TSRT shows that if spacetime is allowed to tremble—subject to strict causal constraints—then all forces, uncertainties, and constants of nature emerge naturally from the local structure of the metric itself. Planck’s constant, the speed of light, the uncertainty principle, gauge interactions, mass, spin, gravitation, black hole entropy, and even the limit of information resolution—all of these are not postulates to be imposed, but consequences to be derived from the trembling of spacetime geometry. This geometric return does not reject the achievements of twentieth-century physics; it subsumes and completes them. It explains why quantum mechanics worked where it did, why general relativity succeeded where it did, and why apparent paradoxes arise when each is applied outside its natural regime.

In summary, TSRT restores unity to the fractured edifice of modern physics. It shows that behind the complexity of fields and particles lies a simple, deterministic reality: the causal trembling of spacetime itself. It invites a new scientific era—not one built on ever more elaborate constructs, but one grounded in the fundamental, causal, and geometric nature of reality. Thus,

the future of physics does not lie in escaping spacetime, but in understanding the fine structure of its trembling fabric.

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